



NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

FINAL

# FLIGHT PLAN APOLLO 8

AS-503/CSM-103

NOV 22, 1968

PREPARED BY  
FLIGHT PLANNING BRANCH  
FLIGHT CREW SUPPORT DIVISION

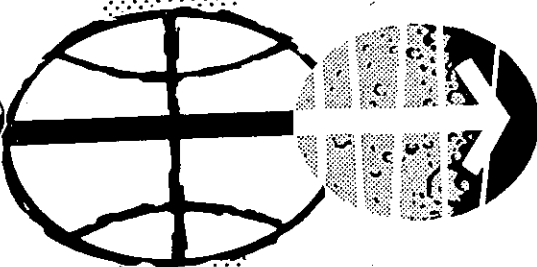
SECTION I

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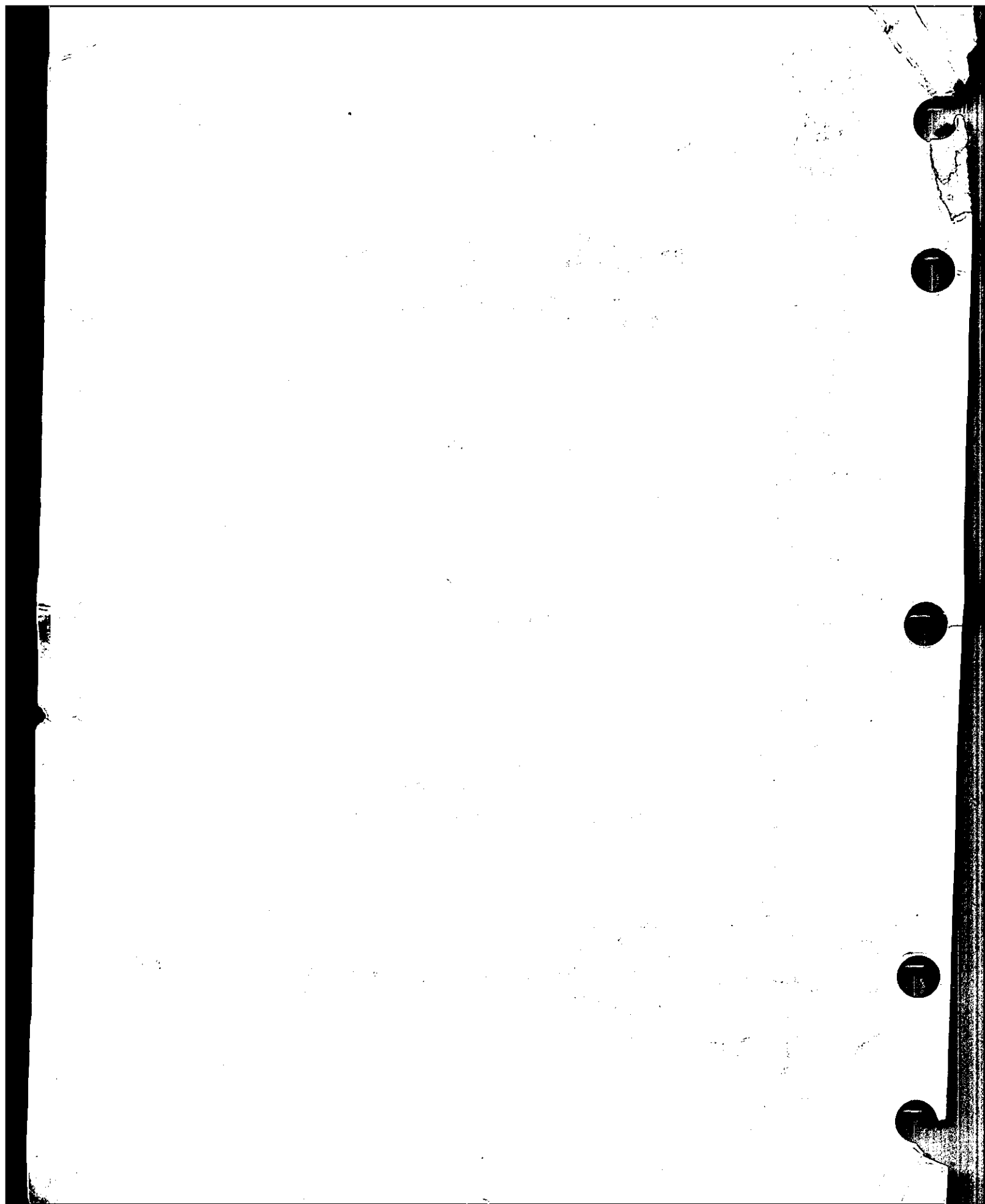
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MANNED SPACECRAFT CENTER  
HOUSTON, TEXAS



APOLLO AS503/CSM 103

FINAL FLIGHT PLAN

November 22, 1968

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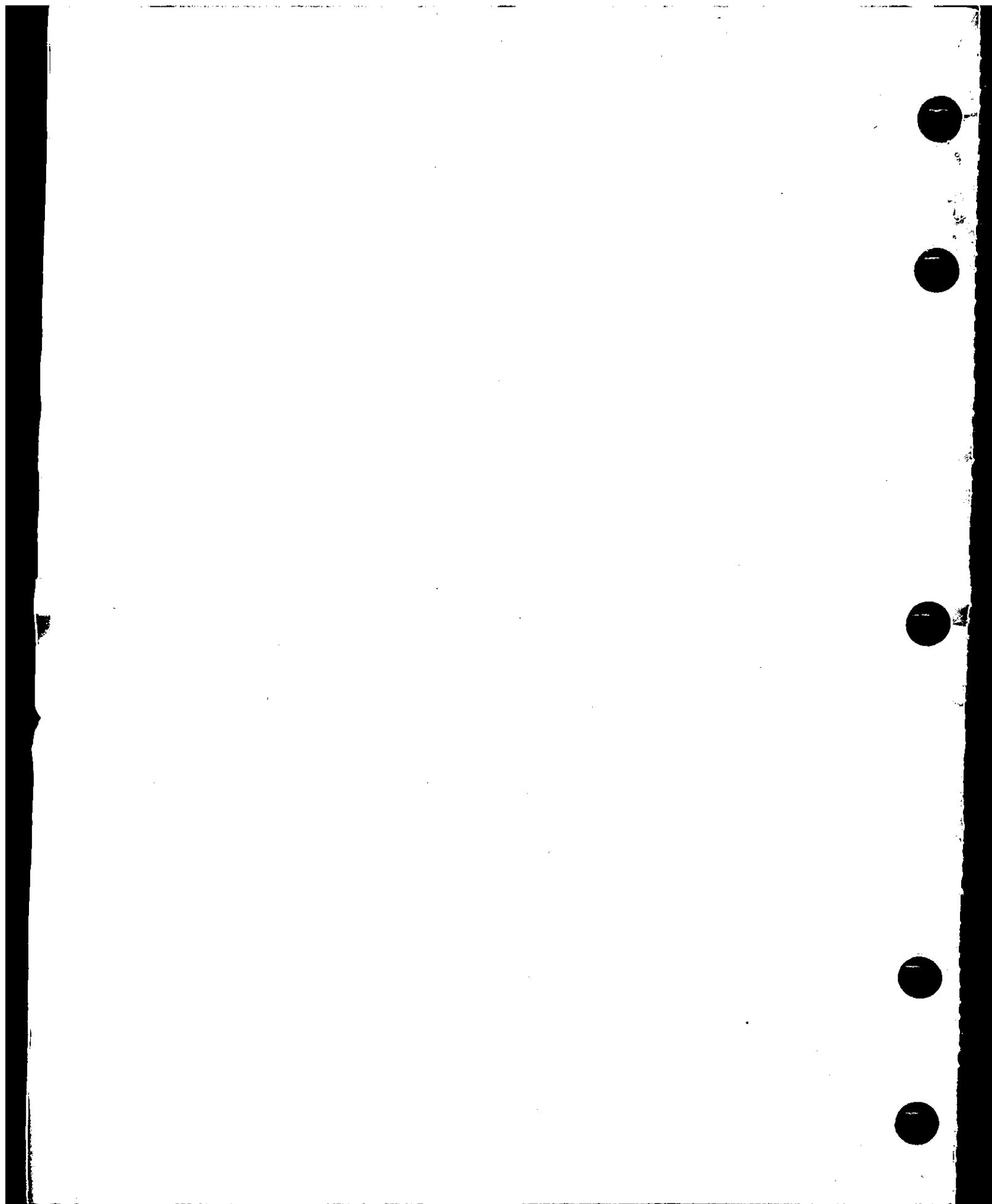
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## INTRODUCTION

This Flight Plan has been prepared by the Flight Planning Branch, Flight Crew Support Division, with technical support by TRW Systems.

This document schedules the AS503/CSM103 operations and crew activities to fulfill, when possible, the test objectives defined in the Mission Requirements, SA503/CSM103, C' Type Mission, (Lunar Orbit).

The trajectory parameters used in this Flight Plan are for a December 21, 1968 launch, with a 72° launch azimuth and were supplied by Mission Planning and Analysis Division as defined by the Apollo Mission C' Spacecraft Operational Trajectory.

The Apollo 8 Flight Plan is under the configuration control of the Crew Procedures Control Board (CPCB). All proposed changes to this document that fall in the following categories should be submitted to the CPCB via a Crew Procedures Change Request:

1. Items that impose additional crew training or impact crew procedures.
2. Items that impact the accomplishment of detailed test objectives.
3. Items that result in a significant RCS or EPS budget change.
4. Items that result in moving major activities to a different activity day in the Flight Plan.
5. Items that require a change to the flight data file.

The Chief, Flight Planning Branch (FCSD) will determine what proposed changes fall in the above categories.

Mr. T. A. Guillory will act as co-ordinator for all proposed changes to the Apollo 8 Flight Plan.

This Flight Plan is not to be reproduced without the written approval of the Chief, Flight Crew Support Division.

# ABBREVIATIONS

|      |                                    |                             |        |                                         |
|------|------------------------------------|-----------------------------|--------|-----------------------------------------|
| F.F. | ACCEL                              | Accelerometer               | CDH    | Constant Delta Altitude                 |
|      | ACT                                | Activation                  | CDR    | Commander                               |
|      | ACQ                                | Acquisition                 | CDU    | Control Data Unit                       |
|      | AEA                                | Abort Electronics Assembly  | CIRC   | Circularization                         |
|      | AGS                                | Abort Guidance Sybsystem    | CK     | Check                                   |
|      | AH                                 | Ampere Hours                | CM     | Command Module                          |
|      | ALIGN                              | Alignment                   | CMC    | Command Module Computer                 |
|      | ALT                                | Altitude                    | CONT   | Continue                                |
|      | AMP or amp                         | Ampere                      | CMD    | Command                                 |
|      | AMPL                               | Amplifier                   | CMP    | Command Module Pilot                    |
|      | ANG                                | Antigua                     | CNTL   | Control                                 |
|      | Ant                                | Antenna                     | CNTLS  | Controls                                |
|      | AOS                                | Acquisition of Signal       | C.O.   | Cut off                                 |
|      | AOT                                | Alignment Optical Telescope | C/O    | Check out                               |
|      | APS                                | Ascent Propulsion Subsystem | COAS   | Crew Optical Alignment Sight            |
|      | ARS                                | Atmosphere Revitalization   | COMM   | Communications                          |
|      | ASC                                | Ascension                   | COMP   | Computational                           |
|      | ASCT                               | Ascent                      | CONFIG | Configuration                           |
|      | ATT                                | Attitude                    | CP     | Control Point                           |
|      | AUX                                | Auxiliary                   | CRO    | Carnarvon, Australia                    |
|      | AZ                                 | Azimuth                     | CRYO   | Cryogenic                               |
| BAT  | Battery                            |                             | CSI    | Coelliptic Sequence Initiation Maneuver |
| BDA  | Bermuda                            |                             | CSM    | Command Service Module                  |
| BP   | Barber Pole                        |                             | C&WS   | Caution and Warning System              |
| BT   | Burn time                          |                             | CYI    | Grand Canary Island                     |
| Bio  | Bio-Medical Data on Voice Downlink |                             | DAP    | Digital Auto Pilot                      |
| BiW  | Black & white                      |                             | D/B    | Deadband                                |
| BU   | Backup                             |                             | DEDA   | Data Entry and Display Assembly         |
| BRKT | Bracket                            |                             | DEGS   | Degrees                                 |
|      |                                    |                             | DEPL   | Depletion                               |
| CAL  | Pt. Arquillo, California           |                             | DET    | Determination or Digital Event Timer    |
| CAL  | Calibration Angle                  |                             | DIFF   | Difference                              |
| CAM  | Camera                             |                             | DK     | Docked                                  |
| CB   | Circuit Breaker                    |                             | DOI    | Descent Orbit Insertion                 |

# ABBREVIATIONS (Cont'd)

|          |                                    |            |                                    |
|----------|------------------------------------|------------|------------------------------------|
| DPS      | Descent Propulsion Subsystem       | GET        | Ground Elapsed Time                |
| DSE      | Data Storage Equipment             | GETI       | Ground Elapsed Time of Ignition    |
| DSKY     | Display and Key Board              | GLY        | Glycol                             |
| DTO      | Detailed Test Objective            | GMT        | Greenwich Mean Time                |
| DUA      | Digital Uplink Assembly            | G&N        | Guidance and Navigation            |
| DWN      | Down                               | GNCS       | Guidance Navigation Control System |
|          |                                    | GWM        | Guam                               |
| E        | Erasable                           | GYM        | Guaymas, Mexico                    |
| ECS      | Environmental Control Subsystem    |            |                                    |
| EPH      | Earth Far Horizon                  | HA         | Apogee Altitude                    |
| EI       | Earth (atmosphere) Interface       | HAW        | Hawaii                             |
| ELDMK    | Earth Landmark                     | HBR        | High Bit Rate (TLM)                |
| EMS      | Entry Monitor System               | HD         | Highly Desirable                   |
| ENH      | Earth Near Horizon                 | HGA        | High Gain Antenna                  |
| EPO      | Earth Parking Orbit                | HI         | High                               |
| EPS      | Electrical Power Subsystem         | Hp         | Perigee Altitude                   |
| EQUIP    | Equipment                          | HSK        | Honeysuckle (Canberra, Australia)  |
| EST      | Eastern Standard Time              | HTR        | Heater                             |
| EVAP     | Evaporator                         | HTV        | USNS Huntsville                    |
| EVT      | Extravehicular Transfer            |            |                                    |
| EXT      | External                           | ID         | Identification                     |
|          |                                    | IGN        | Ignition                           |
| f        | F Stop                             | IMU        | Inertial Measurement Unit          |
| FC       | Fuel Cell                          | INIT       | Initialization                     |
| FDAI     | Flight Director Attitude Indicator | INT        | Intervalometer                     |
| FLT      | Flight                             | IP         | Initial Point                      |
| FM       | Frequency Modulated                | IU         | Instrumentation Unit               |
| FOV      | Field of View                      | IVT        | Intravehicular Transfer            |
| fps      | Feet per second                    |            |                                    |
| FQ       | Flight Qualification               | JETT       | Jettison                           |
| FT or ft | Feet                               |            |                                    |
| FTP      | Full Throttle Position             | kwh        | Kilowatt Hour                      |
|          |                                    |            |                                    |
| GBI      | Grand Bahama                       | LAT        | Latitude                           |
| GDC      | Gyro Display Coupler               | LBR        | Low Bit Rate (TLM)                 |
| GDS      | Goldstone, California              | LBS or lbs | Pounds                             |



# ABBREVIATIONS (Cont'd)

|       |                                  |        |                                             |
|-------|----------------------------------|--------|---------------------------------------------|
| LCG   | Liquid Cooled Garment            | MEAS   | Measurement                                 |
| LDG   | Landing                          | MER    | USNS Mercury                                |
| LDMK  | Landmark                         | MET    | Mission Event Timer                         |
| LEB   | Lower Equipment Bay              | M/I    | Minimum Impulse                             |
| LFH   | Lunar Far Horizon                | MIN    | Minimum                                     |
| LGC   | LM Guidance Computer             | MLA    | Merritt Island                              |
| LH    | Left-hand                        | MNVR   | Maneuver                                    |
| L/H   | Local Horizontal                 | MON    | Monitor                                     |
| LHEB  | Left-hand Equipment Bay          | MSFN   | Manned Space Flight Network                 |
| LHFEB | Left-hand Forward Equipment Bay  | MTVC   | Manual Thrust Vector Control                |
| LIOH  | Lithium Hydroxide                |        |                                             |
| LLM   | Lunar Landing Mission            | NAV    | Navigation                                  |
| LLOS  | Landmark Line of Sight           | NCC    | Corrective Combination Maneuver             |
| LM    | Lunar Module                     | nm     | Nautical Miles                              |
| LMP   | Lunar Module Pilot               | NOM    | Nominal                                     |
| LNH   | Lunar Near Horizon               | NSR    | Nominal Slow Rate                           |
| LOI   | Lunar Orbit Insertion            | NXX    | Noun XX                                     |
| LONG  | Longitude                        |        |                                             |
| LOS   | Loss of Signal                   | OBS    | Observation                                 |
| LPO   | Lunar Parking Orbit              | O/F    | Oxidizer to Fuel                            |
| LR    | Landing Radar                    | OPER   | Operate                                     |
| LT    | Light                            | ORB    | Orbital                                     |
| LTC   | Lighting                         | ORDEAL | Orbit Rate Display Earth and Lunar          |
| LV    | Launch Vehicle                   | ORIENT | Orientation                                 |
| L/V   | Local Vertical                   | OVHD   | Overhead                                    |
| LVPD  | Launch Vehicle Pressure Display  |        |                                             |
|       |                                  | P      | Pitch                                       |
| M     | Mandatory                        | PAD    | Voice Update                                |
| MAD   | Madrid, Spain                    | PCM    | Pulse Code Modulation                       |
| MAN   | Manual                           | PC     | Pericyntian                                 |
| MAX   | Maximum                          | PGA    | Pressure Garment Assembly                   |
| MAX Q | Maximum Dynamic Pressure         | PGNCS  | Primary Guidance Navigation Control Section |
| MCC   | Midcourse Correction             | PIPA   | Pulse Integrating Pendulous Accelerometer   |
| MCC-H | Mission Control Center - Houston | PM     | Phase Modulated                             |
| MDC   | Main Display Console             | POL    | Polarity or Polarizing                      |

# ABBREVIATIONS (Cont'd)

|          |                                          |       |                               |
|----------|------------------------------------------|-------|-------------------------------|
| PREF     | Preferred                                | S     | Shaft                         |
| PREP     | Preparation                              | SA    | Shaft Angle                   |
| PRESS    | Pressure                                 | S/C   | Spacecraft                    |
| PRIM     | Primary                                  | SCE   | Signal Conditioning Equipment |
| PT       | Point                                    | SCS   | Stabilization Control System  |
| PRN      | Pseudo-Random Noise                      | SCT   | Scanning Telescope            |
| PROP     | Proportional                             | SEC   | Secondary                     |
| PU       | Propellant Utilization                   | SECO  | S-IVB Engine Cut-off          |
| PUGS     | Propellant Utilization and Gaging System | SEP   | Separate                      |
| PTC      | Passive Thermal Control                  | SEQ   | Sequence                      |
| PWR      | Power                                    | SLA   | Service Module LM Adapter     |
| Pxx      | Program XX                               | SLOS  | Star Line-of-Sight            |
|          |                                          | SM    | Service Module                |
| Qty      | Quantity                                 | SPOT  | Spot Meter                    |
|          |                                          | SPS   | Service Propulsion System     |
| TA       |                                          | SR    | Sunrise                       |
| R        | Roll                                     | SRX   | S-Band Receiver Mode No. X    |
| R&B      | Red & Blue                               | SS    | Sunset                        |
| RAD      | Radiator                                 | STX   | S-Band Transmit Mode No. X    |
| RCDR     | Recorder                                 | STBY  | Standby                       |
| RCS      | Reaction Control Subsystem               | Sw    | Switch                        |
| RCU      | Remote Control Unit                      | SXT   | Sextant                       |
| RCV      | Receiver                                 | SYNC  | Synchronization               |
| RED      | USNS Redstone                            |       |                               |
| REFSMMAT | Reference Stable Member Matrix           | T     | Trunnion                      |
| REG      | Regulator                                | T     | Time of Ephemeris Update      |
| REQD     | Required                                 | EPHEM |                               |
| RH       | Right-hand                               | TA    | Trunnion Angle                |
| RNDZ     | Rendezvous                               | TAN   | Tananarive                    |
| RR       | Rendezvous Radar                         | TCA   | Time of Closest Approach      |
| RSI      | Roll Stability Indicator                 | TBD   | To Be Determined              |
| RT       | Real Time                                | TEC   | Trans Earth Coast             |
| RTC      | Real-Time Command                        | TEI   | Transearch Insertion          |
| Rxx      | Routine XX                               |       |                               |

ABBREVIATIONS (Cont'd)

|       |                           |
|-------|---------------------------|
| TEMP  | Temperature               |
| TERM  | Terminate                 |
| TEX   | Corpus Christi, Texas     |
| T&D   | Transposition and Docking |
| TGT   | Target                    |
| TIG   | Time of Ignition          |
| TLC   | Trans Lunar Coast         |
| TLI   | Translunar Insertion      |
| TLM   | Telemetry                 |
| TPF   | Terminal Phase Final      |
| TPI   | Terminal Phase Initiation |
| TPM   | Terminal Phase Midcourse  |
| T/R   | Transmitter/Receiver      |
| TRANS | Translation               |
| TV    | Television                |
| TVC   | Thrust Vector Control     |
| TWR   | Tower                     |

|      |                    |
|------|--------------------|
| UMB  | Umbilical          |
| UNDK | Undock             |
| US   | United States Pass |

|     |                     |
|-----|---------------------|
| VHF | Very High Frequency |
| VLV | Valve               |
| Vxx | Verb xx             |

|     |                 |
|-----|-----------------|
| W/O | Without         |
| WRT | With Respect to |
| WTN | USNS Watertown  |

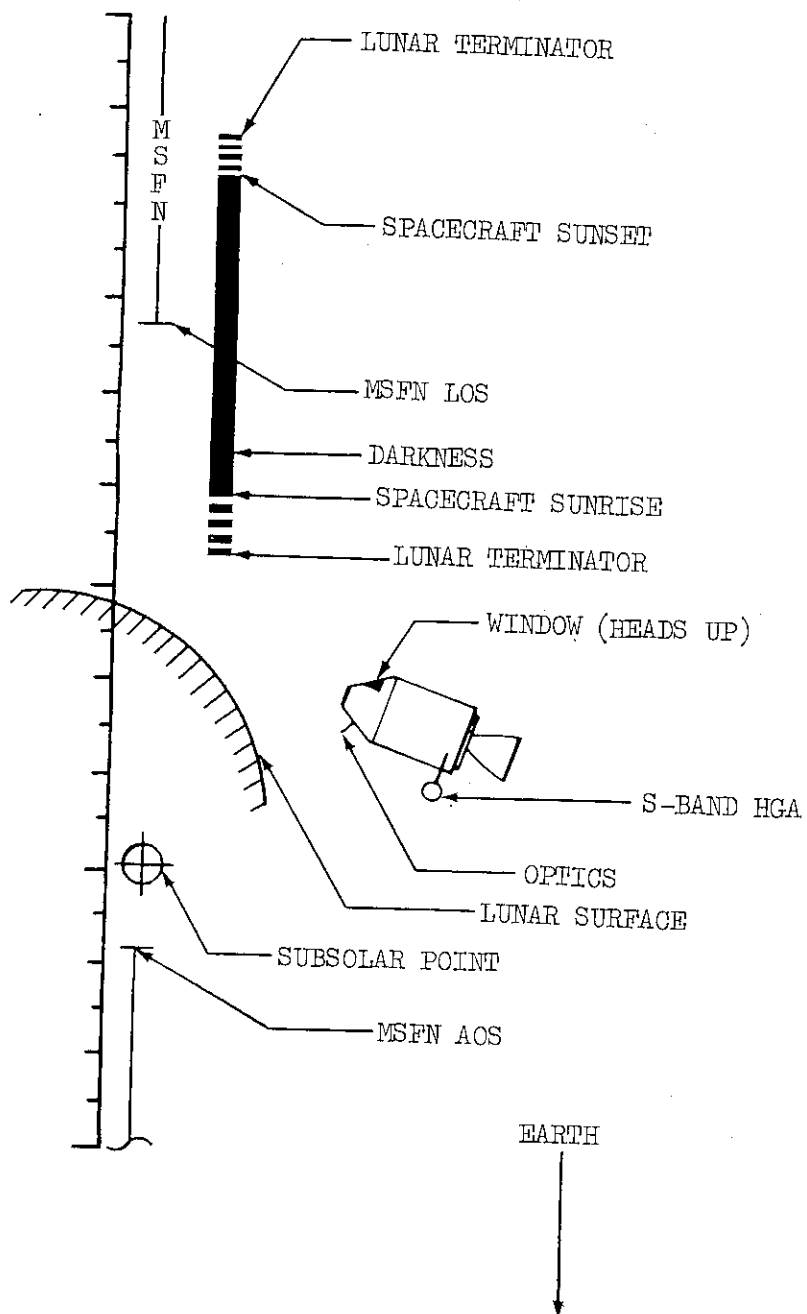
|         |                         |
|---------|-------------------------|
| XFER    | Transfer                |
| XMIT    | Transmit or Transmitter |
| XPONDER | Transponder             |

Y                      Yaw

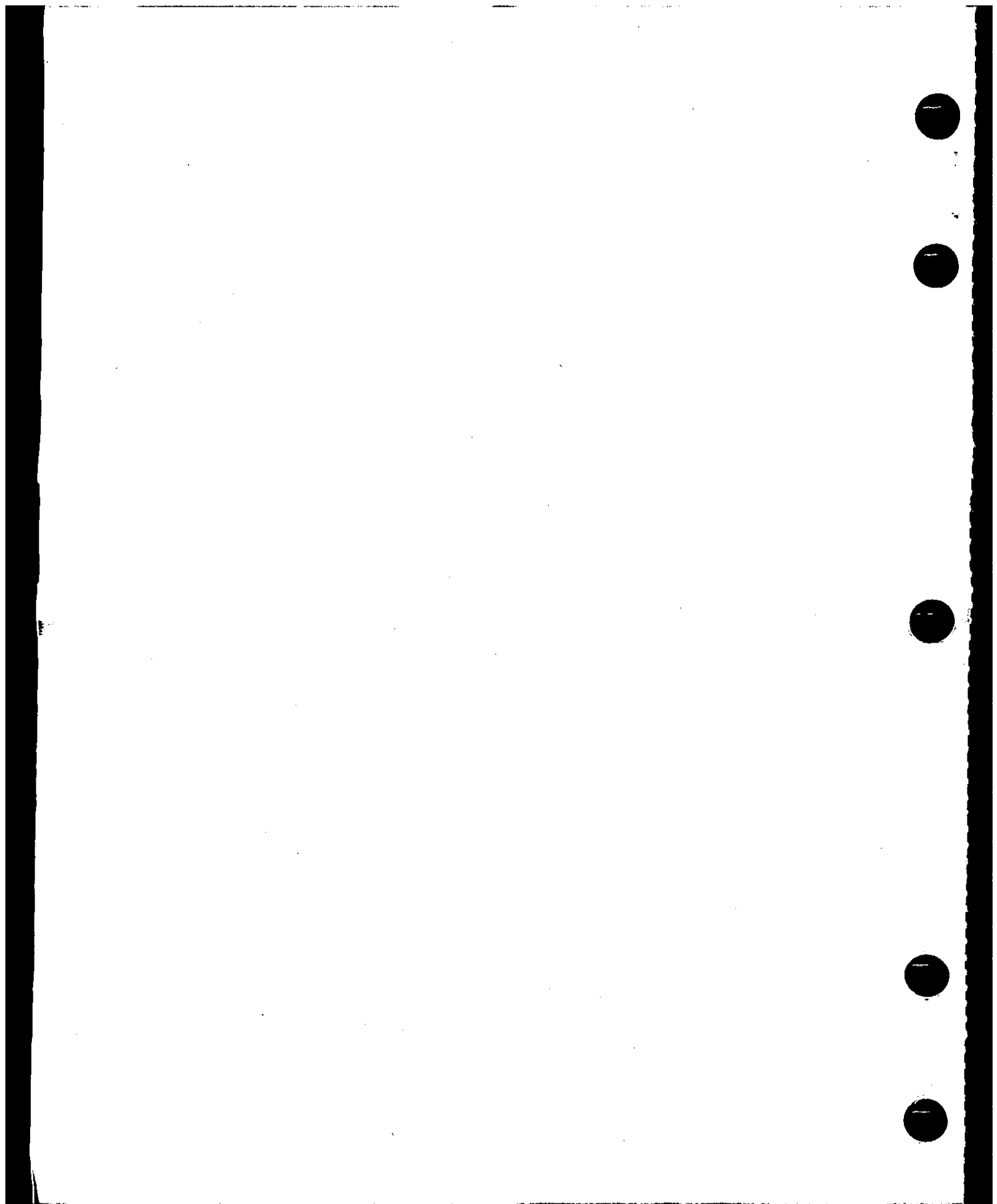
|             |                                           |
|-------------|-------------------------------------------|
| $\Delta V$  | Velocity Change (Differential)            |
| $\Delta VC$ | Velocity Change at Engine Cutoff          |
| $\Delta R$  | Position Change (Differential)            |
| 8-balls     | Flight Director Attitude Indicator (FDAI) |
| LBS or lbs  | Pounds                                    |

## CAMERA NOMENCLATURE

2/80/B3 - Hasselblad No. 2/80mm Lens/Stowage Location  
16/18/20 - 16mm Camera/18mm Lens/20 Frames Per Second  
1/150/New B&W/INT - Hasselblad No. 1/150mm Lens/VSE New  
B&W Magazine/with interferometer  
C-EX - color film - external  
POL FILTER 1/4 - Polarization filter, Shutter Speed 1/4  
SO368 - Type of External Film  
2/80/H.S. - Hasselblad No. 2/80mm Lens/High Speed Film  
Cable/BRRT/R&B - Use Power Cable, Camera Mounted on Bracket  
with Red & Blue Filter



SECTION I - GENERAL



## FLIGHT PLAN DESCRIPTION

### DAY 1

The spacecraft is launched into a 100-nm circular orbit with a launch azimuth that varies between 72° and 108° during the daily launch window. The lift-off time varies during the monthly launch window, but is constrained to daylight launches. The trajectory for this flight plan is a 72° launch azimuth December 21, 1968 with a 12:51:00 GMT (7:51:00 AM, EST) lift-off. Both the variable lift-off time and launch azimuth effect the earth-orbit station coverage, IMU alignment time, translunar and trans-earth coast time, and lunar surface lighting conditions.

The spacecraft remains attached to the S-IVB for approximately 4 hours. During this time, the crew checks out the CSM systems, performs an optics check, IMU realignment to the pad REFSMAT, monitors the TLI maneuver, and performs transposition and separation from the S-IVB. The crewmen doff and stow their helmets and gloves prior to TLI and they are not donned again. The TLI maneuver places the S/C in a free return lunar trajectory.

Following TLI, the CMP doffs his PGA, performs an IMU realign, and a series of star/earth horizon sightings. All translunar and transearth sightings contain three marks per set. After these sightings, the CDR and LMP doff and stow their PGA's. At this point, the first eat period is scheduled.

At TLI + 6 hours (09:00 GET), the first MCC decision point is scheduled. Before this point and prior to each midcourse, there is a MCC-H state vector update, target load, maneuver pad update, an IMU alignment, and systems checks. For the nominal trajectory, the midcourses are nominally zero.

After MCC<sub>1</sub>, a series of star/earth landmark sightings is scheduled at which time the CDR sleep period begins (11:00 GET) followed by the CMP and LMP sleep period (18:00 GET). During the CDR sleep period, the CMP performs a series of star/earth horizon sightings.

### DAY 2

Crew activity on day 2 begins at approximately 24:00 GET. At 26:30 GET, the CMP performs a series of star/earth horizon sightings, then MCC<sub>2</sub> at TLI + 25 hours (28:00 GET), followed by another series of star/earth horizon sightings. Then the CDR sleep period begins (29:00 GET) followed by the CMP and LMP sleep period (36:00 GET). During the CDR sleep period, TV is scheduled for 15 minutes (31:15 GET) and a series of star/earth horizon sightings is performed at (34:15 GET). During day 2, MCC<sub>3</sub> occurs at LOI -22 hours (47:00 GET). The maneuver is preceded by a series of star/earth horizon sightings.

### DAY 3

Crew activity on day 3 begins at approximately 48:00 GET. During the CDR sleep period (52:00 GET), the CMP performs a series of star/lunar horizon sightings (52:15 GET) and TV is scheduled for 15 minutes (55:15 GET). At LOI - 8 hours (61:00 GET), MCC<sub>4</sub> is scheduled. The CMP and LMP sleep period (62:00 GET) is scheduled between MCC<sub>4</sub> and LOI<sub>1</sub>.

At 69:07 GET, the LOI<sub>1</sub> maneuver places the S/C into a 60 x 170-nm lunar orbit. Scheduled during the first two lunar revolutions are camera preparation, eat period, COAS ground track determination, control point and pseudo landing site, observations, photographs of targets of opportunity, TV transmission, and LOI<sub>2</sub> preparation. The IMU is realigned once during each dark period in lunar orbit.

### DAY 4

Crew activity on day 4 begins at approximately 72:00 GET. At 73:31 GET, the LOI<sub>2</sub> circularization maneuver places the S/C into a 60-nm circular orbit. Scheduled during revolutions 3 and 4 are a 2-hour CMP rest period, landmark training photography, vertical stereo photography, and landmark lighting evaluation.

Scheduled during revolutions 5 and 6 are a 3-hour CDR rest period, one control point landmark tracking, and a pseudo landing site tracking during each daylight period. Each tracking consists of 4 marks.

Scheduled during revolutions 7 and 8 are a 2-hour LMP rest period, three control point landmark trackings, and a pseudo landing site tracking during each daylight period. Each tracking consists of 4 marks.

Scheduled during revolutions 9 and 10 are a 2-hour CMP rest period, convergent stereo photography, an eat period, TEI preparation, and the TEI maneuver at 89:15 GET.

After TEI, the CDR sleep period begins (89:20 GET) followed by the CMP and the LMP sleep period (94:00 GET). During the CDR sleep period, the CMP performs a series of star/lunar horizon sightings.

### DAY 5

Crew activity on day 5 begins at approximately 96:00 GET. After the sleep period, the CMP performs a series of star/lunar horizon and star/earth horizon sightings; then MCC<sub>5</sub> at TEI + 15 hours (104:00 GET), followed by TV transmission at 104:15 GET and a series of star/earth horizon sightings (105:15 GET).

During the CDR sleep period (105:00 GET), the CMP performs a series of star/lunar horizon and star/earth horizon sightings. The CMP and LMP sleep periods begin at 112:00 GET.



#### DAY 6

Crew activity on day 6 begins at approximately 120:00 GET. After a series of star/earth horizon sightings, MCC<sub>6</sub> is scheduled at TEI + 33 hours

(122:00 GET) followed by a series of star/lunar horizon and star/earth horizon sightings. Also scheduled is the CDR sleep period (127:00) and both the CMP and LMP sleep periods (134:00). During the CDR sleep period, TV transmission is scheduled (128:00 GET), and the CMP performs a series of star/earth horizon and star/lunar horizon sightings. After the sleep period, the CMP does a series of star/earth horizon sightings.

#### DAY 7

Crew activity on day 7 begins at approximately 144:00 GET. At EI minus 2 hours (144:50 GET), MCC<sub>7</sub> is scheduled. CM/SM separation nominally occurs at 146:35 GET with splashdown in the Pacific Ocean approximately 20 minutes later.

## FLIGHT PLAN NOTES

### A. CREW

1. Crew designation is as follows:

| <u>Designation</u>         | <u>Couch Postion</u> | <u>Prime</u> | <u>Backup</u> |
|----------------------------|----------------------|--------------|---------------|
| Commander (CDR)            | Left                 | Borman       | Armstrong     |
| Command Module Pilot (CMP) | Center               | Lovell       | Aldrin        |
| Lunar Module Pilot (LMP)   | Right                | Anders       | Haise         |

2. The crew will nominally follow a 17-hour work/7-hour rest cycle where possible. One crewman will be awake at all times with all crewmen awake during major burns. The crew will eat together when possible with additional activities held to a minimum during eat periods. The eat period is normally one-hour duration.
3. The pressure suits will be doffed after TLI and donned prior to entry. Helmets and gloves will be doffed prior to TLI and will remain off until prior to entry.
4. A crew status report will be made twice a day on each crewman.
5. General flight plan updates containing changes to the following day's activity schedule, will be voiced up once per day.
6. Negative reporting will be used in reporting completion of each checklist.

### B. COMMUNICATIONS AND INSTRUMENTATION

1. S-band will be prime for voice, ranging and PCM with the VHF used as backup for voice during near earth phases ( $< 4000$  nm). The S-band backup voice system will be checked during the first orbit and the high gain antenna checked after S-IVB/CSM separation.
2. The Flight Qual Recorder will be used as follows:
  - a. Lift-off -45 seconds through 0:12:00 GET
  - b. TLI -2 minutes through TLI cutoff +1 minute
  - c. CSM/S-IVB Sep -2 minutes through Sep +1 minute
  - d. LOI<sub>1</sub> -2 minutes to end of tape

3. The DSE will normally be dumped by real time command (RTC).
4. During communications, the spacecraft will be referred to as "Apollo 8" and the ground as "Houston". The crew member call sign will be their assigned crew position.

#### C. CSM SYSTEMS

1. The spacecraft (S/C) lift-off switch positions are listed in Apollo (CSM 103) Operations Handbook (AOH), Volume 2, August 1, 1968.
2. Fuel cell purging for  $H_2$  and  $O_2$  will be scheduled R/T.
3. The S/C will remain fully poweredup throughout the mission (IMU, CMC and SCS always in operate) with the optics on as required.
4. IMU drift checks are scheduled after TLI, before  $LOI_1$ , on revolutions 2 and 4 through 10 in lunar orbit, and prior to entry.
5. The potable water is chlorinated at 24-hour intervals.
6. All onboard gage readings will be read directly from the spacecraft gages and not corrected by the appropriate factors.

# SCHEDULED COMMUNICATION TESTS

| GET                           | ALTITUDE<br>nm | CSM<br>ANTENNA | SIGNAL<br>COMBINATION | GROUND<br>ANT | COMMUNICATIONS MODE                   |
|-------------------------------|----------------|----------------|-----------------------|---------------|---------------------------------------|
| 03:20                         |                | OMNI           | 4.2                   | 85'           | GDS Carrier, PRN, voice, 51.2 KBPS TM |
| 05:45                         | 24,000         | HI GAIN        | 4.2                   | 85'           | GDS Carrier, PRN, voice, HBR TM       |
|                               | 24,000         | HI GAIN        | 6.2                   | 85'           | GDS Carrier, PRN, Voice, HBR TM       |
|                               | 24,000         | HI GAIN        | .2                    | 85'           | GDS Plybk of Recorded Voice & LBR TM  |
| 07:00                         | 35,000         | HI GAIN        | 4.3                   | 85'           | GDS Carrier, PRN, voice, LBR TM       |
| to                            | 35,000         | HI GAIN        | 6.3                   | 85'           | GDS Carrier, PRN, voice, LBR TM       |
| 08:00                         | 35,000         | HI GAIN        | 5.2                   | 85'           | GDS Carrier, PRN, voice HBR TM        |
|                               | 35,000         | HI GAIN        | 8.1                   | 85'           | GDS Carrier, voice and LBR TM         |
| 29:00                         | 110,000        | HI GAIN        | 4                     | 30'           | ASC Carrier, PRN, Voice               |
| to                            | 110,000        | HI GAIN        | 5                     | 30'           | ASC Carrier, PRN, Updata              |
| 30:00                         | 110,000        | HI GAIN        | 6                     | 30'           | ASC Carrier, PRN, Voice, Updata       |
| or                            | 110,000        | HI GAIN        | 8                     | 30'           | ASC Carrier, BU Voice                 |
| 125,000                       | 110,000        | HI GAIN        | 4.2                   | 30'           | ASC Carrier, PRN, Voice, HBR TM       |
| to                            | 110,000        | HI GAIN        | 5.2                   | 30'           | ASC Carrier, PRN, Voice, HBR TM       |
| 126,000                       | 110,000        | HI GAIN        | 4.3                   | 30'           | ASC Carrier, PRN, Voice, LBR TM       |
|                               | 110,000        | HI GAIN        | 8.1                   | 30'           | ASC Carrier, Voice, HBR TM            |
|                               | 110,000        | OMNI           | 1                     | 30'           | ASC Carrier, PRN                      |
|                               | 110,000        | OMNI           | .5                    | 30'           | ASC Carrier, LBR TM                   |
|                               | 110,000        | OMNI           | .6                    | 30'           | ASC Carrier, Key Subcarrier           |
|                               | 110,000        | OMNI           | 1.7                   | 30'           | ASC Carrier, PRN Ranging              |
|                               | 110,000        | OMNI           | .8                    | 30'           | ASC Carrier, BU Voice, LBR TM         |
|                               | 110,000        | OMNI           | .10                   | 30'           | ASC Carrier, BU Voice                 |
| 31:15                         | 120,000        | HI GAIN        | .4                    | 85'           | GDS CSM Television                    |
| <u>NORMAL OPERATING MODES</u> |                |                |                       |               |                                       |
| 69:00                         | 205,000        | HI GAIN        | 6.2                   | 85'           | Carrier, PRN, Voice, HBR TM           |
|                               | 205,000        | HI GAIN        | 6.3                   | 85'           | Carrier, PRN, Voice, LBR TM           |
|                               | 205,000        | HI GAIN        | .2                    | 85'           | Plybk of Recorded voice & LBR TM      |
| 89:00                         | 205,000        | HI GAIN        | .4                    | 85'           | CSM television                        |

Pages 1-8 thru 1-21 contain the following voice update forms and the explanation for the abbreviations and symbols used.

|                   |                            |
|-------------------|----------------------------|
| Pg 1-8 and 1-9    | TLI PAD/EXPLANATION        |
| Pg 1-10 and 1-11  | P27 UPDATE PAD/EXPLANATION |
| Pg 1-12 thru 1-14 | MANEUVER PAD/EXPLANATION   |
| Pg 1-15 thru 1-17 | ENTRY PAD/EXPLANATION      |
| Pg 1-18 thru 1-21 | MAP UPDATE PAD/EXPLANATION |

| TLI     |         |                    |
|---------|---------|--------------------|
| X : :   | X : :   | TB6p               |
| X X X   | X X X   | R                  |
| X X X   | X X X   | P                  |
| X X X   | X X X   | Y                  |
| X X X : | X X X : | BT                 |
| +       | +       | $\Delta VC'$<br>VI |
| X X X   | X X X   | R SEP              |
| X X X   | X X X   | P SEP              |
| X X X   | X X X   | Y SEP              |

TLI PAD

|       |                      |                                                                                               |
|-------|----------------------|-----------------------------------------------------------------------------------------------|
| TB6p  | X:XX:XX(HRS:MIN:SEC) | PREDICTED TIME OF BEGINNING OF<br>S-IVB RESTART PREPARATION FOR<br>TLI (TB6 = TLI IGN -9 MIN) |
| R     | XXX (DEG)            | PREDICTED SPACECRAFT IMU                                                                      |
| P     | XXX (DEG)            | GIMBAL ANGLES AT TLI                                                                          |
| Y     | XXX (DEG)            | IGNITION                                                                                      |
| BT    | XX:XX (MIN:SEC)      | DURATION OF TLI BURN                                                                          |
| ΔVC'  | XXXXX.X (fps)        | NOMINAL TLI ΔV SET INTO<br>EMS ΔV CONTROL                                                     |
| VI    | +XXXXXX (fps)        | NOMINAL INERTIAL VELOCITY<br>DISPLAYED ON DSKY AT TLI<br>CUTOFF                               |
| R SEP | XXX (DEG)            | PREDICTED SPACECRAFT IMU                                                                      |
| P SEP | XXX (DEG)            | GIMBAL ANGLES AT COMPLETION                                                                   |
| Y SEP | XXX (DEG)            | OF S-IVB MNVR TO CSM/S-IVB<br>SEP ATTITUDE                                                    |

| P27 UPDATE |      |       |   |   |       |   |  |       |   |   |   |
|------------|------|-------|---|---|-------|---|--|-------|---|---|---|
| PURP       |      | V     |   |   | V     |   |  | V     |   |   |   |
| GET        |      | :     | : |   | :     | : |  | :     | : |   |   |
| 304        | 01   | INDEX |   |   | INDEX |   |  | INDEX |   |   |   |
|            | 02   |       |   |   |       |   |  |       |   |   |   |
|            | 03   |       |   |   |       |   |  |       |   |   |   |
|            | 04   |       |   |   |       |   |  |       |   |   |   |
|            | 05   |       |   |   |       |   |  |       |   |   |   |
|            | 06   |       |   |   |       |   |  |       |   |   |   |
|            | 07   |       |   |   |       |   |  |       |   |   |   |
|            | 10   |       |   |   |       |   |  |       |   |   |   |
|            | 11   |       |   |   |       |   |  |       |   |   |   |
|            | 12   |       |   |   |       |   |  |       |   |   |   |
|            | 13   |       |   |   |       |   |  |       |   |   |   |
|            | 14   |       |   |   |       |   |  |       |   |   |   |
|            | 15   |       |   |   |       |   |  |       |   |   |   |
|            | 16   |       |   |   |       |   |  |       |   |   |   |
|            | 17   |       |   |   |       |   |  |       |   |   |   |
|            | 20   |       |   |   |       |   |  |       |   |   |   |
|            | 21   |       |   |   |       |   |  |       |   |   |   |
|            | 22   |       |   |   |       |   |  |       |   |   |   |
|            | 23   |       |   |   |       |   |  |       |   |   |   |
|            | 24   |       |   |   |       |   |  |       |   |   |   |
| N34        | HRS  | X     | X | X |       |   |  | X     | X | X |   |
|            | MIN  | X     | X | X | X     |   |  | X     | X | X | X |
| NAV CHECK  | SEC  | X     | X |   |       |   |  | X     | X |   |   |
| N43        | LAT  |       | 0 |   |       |   |  |       | 0 |   |   |
|            | LONG |       |   |   |       |   |  |       |   |   |   |
|            | ALT  | +     | 0 |   |       |   |  | +     | 0 |   |   |



P27 UPDATE

|              |                       |                                                               |
|--------------|-----------------------|---------------------------------------------------------------|
| PURP         | XXX                   | TYPE OF DATA TO BE RECEIVED<br>(SUCH AS: NAV - LIFT-OFF TIME) |
| V            | XX                    | TYPE OF COMMAND LOAD<br>(70 - 71 - 72 - 73)                   |
| GET          | XXX:XX:XX(HR:MIN:SEC) | TIME DATA RECORDED                                            |
| 01           | XX (OCTAL)            | INDEX NO. OF COMMAND WORDS<br>IN LOAD                         |
| 02-24        | XXXXX                 | NO. OF CORRECTION COMMAND<br>WORDS                            |
| NAV<br>CHECK |                       | TO CONFIRM POINT ABOVE GROUND<br>TRACK FOR A GIVEN TIME       |
| T            | XX:XX:XX(HRS:MIN:SEC) | TIME                                                          |
| LAT          | XX:XX (DEG)           | LATITUDE                                                      |
| LONG         | XXX:XX (DEG)          | LONGITUDE                                                     |
| ALT          | XXX.X (nm)            | ALTITUDE                                                      |

| MANEUVER                   |       |   |   |   |   |   |                      |                     |           |
|----------------------------|-------|---|---|---|---|---|----------------------|---------------------|-----------|
| NORTH & SOUTH<br>SET STARS |       |   |   |   |   |   | PURPOSE<br>PROP/GUID |                     |           |
| R<br>P<br>Y                | ALIGN | — | — | — | + |   |                      |                     | WT N47    |
|                            | ALIGN | — | — | — |   | O | O                    | .                   | PTRIM N48 |
|                            | ALIGN | — | — | — |   | O | O                    | .                   | YTRIM     |
|                            | ALIGN | — | — | — | + | O | O                    |                     | HRS GETI  |
|                            |       |   |   |   | + | O | O                    | O                   | MIN N33   |
|                            |       |   |   |   | + | O |                      |                     | SEC       |
| ULLAGE                     |       |   |   |   |   |   | .                    | ΔV <sub>X</sub> N61 |           |
|                            |       |   |   |   |   |   | .                    | ΔV <sub>Y</sub>     |           |
|                            |       |   |   |   |   |   | .                    | ΔV <sub>Z</sub>     |           |
|                            |       |   |   | X | X | X |                      | R                   |           |
|                            |       |   |   | X | X | X |                      | P                   |           |
|                            |       |   |   | X | X | X |                      | Y                   |           |
|                            |       |   |   | + |   |   | .                    | ΔA N44              |           |
|                            |       |   |   |   |   |   | .                    | ΔP                  |           |
|                            |       |   |   | + |   |   | .                    | ΔVT                 |           |
| HORIZON/WINDOW             |       |   |   | X | X | X | .                    | BT                  |           |
|                            |       |   |   | X |   |   | .                    | ΔVC                 |           |
|                            |       |   |   | X | X | X | X                    | SXTS                |           |
|                            |       |   |   | + |   |   | .                    | SFT                 |           |
|                            |       |   |   | + |   |   | .                    | TRN                 |           |
|                            |       |   |   | X | X | X |                      | BSS                 |           |
|                            |       |   |   | X | X |   | .                    | SPA                 |           |
|                            |       |   |   | X | X | X | .                    | SXP                 |           |
| OTHER                      |       |   |   |   | O |   | .                    | LAT N61             |           |
|                            |       |   |   |   |   |   | .                    | LONG                |           |
|                            |       |   |   | + |   |   | .                    | RTGO EMS            |           |
|                            |       |   |   | + |   |   | .                    | VIO                 |           |
|                            |       |   |   |   | . | . | .                    | GET .05G            |           |

MANEUVER PAD

|                |                           |                                                                          |
|----------------|---------------------------|--------------------------------------------------------------------------|
| PURPOSE        | XXXXXX                    | TYPE OF MNVR TO BE PERFORMED                                             |
| PROP/GUID      |                           | PROPULSION SYSTEM (SPS/RCS)/<br>GUIDANCE (SCS/G&N)                       |
| WT             | XXXXX (lbs)               | PREMANEUVER VEHICLE WEIGHT                                               |
| P TRIM         | X.XX (DEG)                | SPS PITCH GIMBAL OFFSET TO<br>PLACE THRUST                               |
| Y TRIM         | X.XX (DEG)                | SPS YAW GIMBAL OFFSET TO<br>PLACE THRUST                                 |
| GETI           | XX:XX:XX<br>(HRS:MIN:SEC) | TIME OF MNVR IGNITION                                                    |
| $\Delta V_X$   | XXXX.X (fps)              | P30 VELOCITY TO BE GAINED<br>COMPONENTS IN LOCAL VERTICAL<br>COORDINATES |
| $\Delta V_Y$   | XXXX.X (fps)              |                                                                          |
| $\Delta V_Z$   | XXXX.X (fps)              |                                                                          |
| R              | XXX (DEG)                 | IMU GIMBAL ANGLES OF<br>MANEUVER ATTITUDE                                |
| P              | XXX (DEG)                 |                                                                          |
| Y              | XXX (DEG)                 |                                                                          |
| H <sub>A</sub> | XXXX.X (nm)               | PREDICTED APOGEE ALTITUDE<br>AFTER MANEUVER                              |
| H <sub>P</sub> | XXXX.X (nm)               | PREDICTED PERIGEE ALTITUDE<br>AFTER MANEUVER                             |
| $\Delta V_T$   | XXXX.X                    | TOTAL VELOCITY OF MANEUVER                                               |
| BT             | X:XX (MIN:SEC)            | MANEUVER DURATION                                                        |
| $\Delta V_C$   | XXXX.X (fps)              | PREMANEUVER $\Delta V$ SETTING IN<br>EMS $\Delta V$ COUNTER              |
| SXTS           | XX (OCTAL)                | SEXTANT STAR FOR MANEUVER<br>ATTITUDE CK                                 |
| SFT            | XXX.X (DEG)               | SEXTANT SHAFT SETTING FOR<br>MANEUVER ATTITUDE CK                        |
| TRN            | XX.X (DEG)                | SEXTANT TRUNNION SETTING FOR<br>MANEUVER ATTITUDE CK                     |
| BSS            | XXX (OCTAL)               | BORESIGHT STAR FOR MANEUVER<br>ATTITUDE CK USING THE COAS                |
| SPA            | XX.X (DEG)                | BSS PITCH ANGLE ON COAS                                                  |

MANEUVER PAD (cont'd)

|                               |              |                                                                             |
|-------------------------------|--------------|-----------------------------------------------------------------------------|
| SXP                           | X.X (DEG)    | BSS X POSITION ON COAS                                                      |
| LAT                           | XX.XX        | LATITUDE AND LONGITUDE OF THE                                               |
| LONG                          | XXX.XX       | LANDING POINT FOR ENTRY                                                     |
|                               |              | GUIDANCE                                                                    |
| RTGO                          | XXXX.X       | RANGE TO GO FOR EMS                                                         |
|                               |              | INITIALIZATION                                                              |
| VIO                           | XXXXXX (fps) | INERTIAL VELOCITY AT .05G FOR                                               |
|                               |              | EMS INITIALIZATION                                                          |
| GET(.05G)                     | XX:XX:XX     | TIME OF .05G                                                                |
| NORTH &<br>SOUTH<br>SET STARS |              | STARS FOR TELESCOPE FOR<br>BACKUP GDC ALIGN                                 |
| R, P, Y<br>(ALIGN)            |              | ATTITUDE TO BE SET IN<br>ATTITUDE SET TW FOR BACKUP<br>GDC ALIGN            |
| ULLAGE                        |              | NO. OF SM RCS JETS USED AND<br>LENGTH OF TIME OF ULLAGE                     |
| HORIZON<br>WINDOW             |              | WINDOW MARKING AT WHICH<br>HORIZON IS PLACED AT A<br>SPECIFIED TIG (ATT CK) |

| ENTRY |   |   |   |  |   |   |   |   |   |   |   | AREA                |                   |
|-------|---|---|---|--|---|---|---|---|---|---|---|---------------------|-------------------|
| X     | X | X |   |  |   |   |   | X | X | X |   | R                   | .05G              |
| X     | X | X |   |  |   |   |   | X | X | X |   | P                   | .05G              |
| X     | X | X |   |  |   |   |   | X | X | X |   | Y                   | .05G              |
|       |   |   | . |  | . |   |   |   | . |   | . | GET                 | HOR               |
| X     | X | X |   |  |   |   |   | X | X | X |   | P                   | CK                |
|       | O |   |   |  | . |   |   |   | O |   | . | LAT                 | N61               |
|       |   |   |   |  | . |   |   |   |   |   | . | LONG                |                   |
| X     | X | X |   |  | . |   |   | X | X | X |   | MAX                 | G                 |
| +     |   |   |   |  | . |   |   | + |   |   |   | V400K               | N60               |
| -     | O | O |   |  | . |   |   | - | O | O |   | T400K               |                   |
| +     |   |   |   |  | . |   |   | + |   |   |   | RTGO                | EMS               |
| +     |   |   |   |  | . |   |   | + |   |   |   | VIO                 |                   |
|       |   |   | . |  | . |   |   |   | . |   | . | RRT                 |                   |
| X     | X |   |   |  | . |   |   | X | X |   |   | RET                 | .05G*             |
| +     | O | O |   |  | . |   |   | + | O | O |   | D <sub>L</sub> MAX* |                   |
| +     | O | O |   |  | . |   |   | + | O | O |   | D <sub>L</sub> MIN* | N69               |
| +     |   |   |   |  | . |   |   | + |   |   |   | V <sub>L</sub> MAX* |                   |
| +     |   |   |   |  | . |   |   | + |   |   |   | V <sub>L</sub> MIN* |                   |
| X     | X | X |   |  | . |   |   | X | X | X |   | D <sub>O</sub>      |                   |
| X     | X |   |   |  | . |   |   | X | X |   |   | RET                 | V <sub>CIRC</sub> |
| X     | X |   |   |  | . |   |   | X | X |   |   | RETBBO              |                   |
| X     | X |   |   |  | . |   |   | X | X |   |   | RETEBO              |                   |
| X     | X |   |   |  | . |   |   | X | X |   |   | RETDRO              |                   |
| X     | X | X | X |  |   |   |   | X | X | X | X | SXTS                |                   |
| +     |   |   |   |  | . |   | O | + |   |   |   | SFT                 |                   |
| +     |   |   |   |  | . | O | O | + |   |   | O | TRN                 |                   |
| X     | X | X |   |  | . |   |   | X | X | X |   | BSS                 |                   |
| X     | X |   |   |  | . |   |   | X | X |   |   | SPA                 |                   |
| X     | X | X |   |  | . |   |   | X | X | X |   | SXP                 |                   |
| X     | X | X | X |  | . |   |   | X | X | X | X | LIFT VECTOR         |                   |

ENTRY PAD

|                    |                           |                                                                         |
|--------------------|---------------------------|-------------------------------------------------------------------------|
| AREA               | XXX                       | SPLASHDOWN AREA DEFINED BY<br>TARGET LINE                               |
| R .05G             | XXX(DEG)                  | SPACECRAFT IMU GIMBAL ANGLES                                            |
| P .05G             | XXX(DEG)                  | REQUIRED FOR AERODYNAMIC                                                |
| Y .05G             | XXX(DEG)                  | TRIM AT .05G                                                            |
| GET<br>(HOR CK)    | XX:XX:XX<br>(HRS:MIN:SEC) | TIME OF ENTRY ATTITUDE<br>HORIZ CHECK AT EI -17 MIN.                    |
| P<br>(HOR CK)      | XXX(DEG)                  | PITCH ATTITUDE FOR HORIZON<br>CHECK AT EI -17 MIN                       |
| LAT                | ±XX.XX(DEG)               | LATITUDE OF TARGET POINT                                                |
| LONG               | ±XXX.XX(DEG)              | LONGITUDE OF TARGET POINT                                               |
| MAX G              | XX.X (G's)                | PREDICTED MAXIMUM REENTRY<br>ACCELERATION                               |
| V <sub>400K</sub>  | XXXXX (fps)               | INERTIAL VELOCITY AT ENTRY<br>INTERFACE                                 |
| γ <sub>400K</sub>  | X.XX(DEG)                 | INERTIAL FLIGHT PATH ANGLE AT<br>ENTRY INTERFACE                        |
| RTGO               | +XXXX.X(nm)               | RANGE TO GO FROM .05G TO TARGET<br>FOR EMS INITIALIZATION               |
| VIO                | +XXXXX.(fps)              | INERTIAL VELOCITY AT .05G<br>FOR EMS INITIALIZATION                     |
| RRT                | XX:XX:XX<br>(HRS:MIN:SEC) | REENTRY REFERENCE TIME BASED<br>ON GET OF PREDICTED 400K<br>(DET START) |
| RET .05G           | XX:XX<br>(MIN:SEC)        | TIME OF .05G FROM 400K (RRT)                                            |
| D <sub>L</sub> MAX | X.XX (G's)                | MAXIMUM ACCEPTABLE VALUE OF<br>PREDICTED DRAG LEVEL (FROM CMC)          |
| D <sub>L</sub> MIN | X.XX (G's)                | MINIMUM ACCEPTABLE VALUE OF<br>PREDICTED DRAG LEVEL (FROM CMC)          |
| V <sub>L</sub> MAX | XXXXX (fps)               | MAXIMUM ACCEPTABLE VALUE OF<br>EXIT VELOCITY (FROM CMC)                 |
| V <sub>L</sub> MIN | XXXXX (fps)               | MINIMUM ACCEPTABLE VALUE OF<br>EXIT VELOCITY (FROM CMC)                 |

ENTRY PAD (cont'd)

|                       |                    |                                                           |
|-----------------------|--------------------|-----------------------------------------------------------|
| DO                    | X.XX (G's)         | PLANNED DRAG LEVEL DURING<br>CONSTANT G                   |
| RET V <sub>CIRC</sub> | XX:XX<br>(MIN:SEC) | TIME FROM EI THAT S/C VELOCITY<br>BECOMES CIRCULAR        |
| RETBBO                | XX:XX<br>(MIN:SEC) | TIME FROM EI TO THE BEGINNING<br>OF BLACKOUT              |
| RETEBO                | XX:XX<br>(MIN:SEC) | TIME FROM EI TO THE END OF<br>BLACKOUT                    |
| RETDRO                | XX:XX<br>(MIN:SEC) | TIME FROM EI TO DROGUE DEPLOY                             |
| SXTS                  | XX(OCTAL)          | SEXTANT STAR FOR ENTRY ATTITUDE<br>CHECK                  |
| SFT                   | XXX.X(DEG)         | SEXTANT SHAFT SETTING FOR<br>ENTRY ATTITUDE CHECK         |
| TRN                   | XX.X(DEG)          | SEXTANT TRUNNION SETTING FOR<br>ENTRY ATTITUDE CHECK      |
| BSS                   | XXX(OCTAL)         | BORESIGHT STAR FOR ENTRY<br>ATTITUDE CHECK USING THE COAS |
| SPA                   | XX.X(DEG)          | BSS PITCH ANGLE ON COAS                                   |
| SXP                   | X.X(DEG)           | BSS X POSITION ON COAS                                    |
| LIFT VECTOR           | XX                 | LIFT VECTOR DESIRED AT .05G's<br>BASED ON ENTRY CORRIDOR  |

# MAP UPDATE

REV 1/2 REMARKS

GET (hrs:min:sec)

LOS : :

PM : :

AOS : :

SS : :

LOS : :

SR : :

PM : :

AOS : :

SS : :

REV 2/3 REMARKS

GET (hrs:min:sec)

LOS : :

SR : :

PM : :

AOS : :

SS : :

REV 3/4 . REMARKS

GET (hrs:min:sec)

LOS : :

SR : :

PM : :

AOS : :

SS : :



| MAP UPDATE        |         |  |
|-------------------|---------|--|
| REV 7/8           | REMARKS |  |
| GET (hrs:min:sec) |         |  |
| LOS    :    :     |         |  |
| SR       :    :   |         |  |
| PM       :    :   |         |  |
| AOS       :    :  |         |  |
| SS        :    :  |         |  |
|                   |         |  |
| REV 8/9           | REMARKS |  |
| GET (hrs:min:sec) |         |  |
| LOS    :    :     |         |  |
| SR       :    :   |         |  |
| PM       :    :   |         |  |
| AOS       :    :  |         |  |
| SS        :    :  |         |  |
|                   |         |  |
| REV 9/10          | REMARKS |  |
| GET (hrs:min:sec) |         |  |
| LOS    :    :     |         |  |
| SR       :    :   |         |  |
| PM       :    :   |         |  |
| AOS       :    :  |         |  |
| SS        :    :  |         |  |
|                   |         |  |
| REV 10            | REMARKS |  |
| GET (hrs:min:sec) |         |  |
| LOS    :    :     |         |  |
| SR       :    :   |         |  |
| AOS       :    :  |         |  |
| SS        :    :  |         |  |

| MAP UPDATE        |         |
|-------------------|---------|
| REV 4/5           | REMARKS |
| GET (hrs:min:sec) |         |
| LOS   :   :       |         |
| SR     :   :      |         |
| PM     :   :      |         |
| AOS   :   :       |         |
| SS     :   :      |         |
|                   |         |
| REV 5/6           | REMARKS |
| GET (hrs:min:sec) |         |
| LOS   :   :       |         |
| SR     :   :      |         |
| PM     :   :      |         |
| AOS   :   :       |         |
| SS     :   :      |         |
|                   |         |
| REV 6/7           | REMARKS |
| GET (hrs:min:sec) |         |
| LOS   :   :       |         |
| SR     :   :      |         |
| PM     :   :      |         |
| AOS   :   :       |         |
| SS     :   :      |         |
|                   |         |
|                   |         |

MAP UPDATE

|     |                       |                                                   |
|-----|-----------------------|---------------------------------------------------|
| LOS | XX:XX:XX(HRS:MIN:SEC) | TIME OF MSFN LOSS OF SIGNAL IN LUNAR ORBIT        |
| AOS | XX:XX:XX(HRS:MIN:SEC) | TIME OF MSFN ACQUISITION OF SIGNAL                |
| SS  | XX:XX:XX(HRS:MIN:SEC) | TIME OF SPACECRAFT SUNSET IN LUNAR ORBIT          |
| SR  | XX:XX:XX(HRS:MIN:SEC) | TIME OF SPACECRAFT SUNRISE IN LUNAR ORBIT         |
| PM  | XX:XX:XX(HRS:MIN:SEC) | TIME OF CROSSING OF MOON'S PRIME MERIDIAN (150°W) |



SECTION II - DETAILED TIMELINE



# FLIGHT PLAN

| TIME                   | EVENT                                                 | REMARKS                            |
|------------------------|-------------------------------------------------------|------------------------------------|
| -00:45                 | LMP: FLT RCDR - RECORD                                |                                    |
| -00:09                 | LCC: IGNITION COMMAND                                 |                                    |
| -00:01                 | L/V ENGINE LTS (5) - OUT                              |                                    |
| 00:00                  | LCC:CDR: <u>REPORT</u> LIFT-OFF<br>P11 AUTO           | LIFT-OFF LT - ON, MET STARTS COUNT |
| 00:02                  | CDR: <u>REPORT</u> YAW MNVR                           |                                    |
| 00:11                  | CDR: <u>REPORT</u> ROLL AND PITCH PROGRAM<br>INITIATE |                                    |
| 00:28                  | CDR <u>REPORT</u> ROLL COMPLETE                       |                                    |
| 00:42                  | MCC-H:CDR: <u>REPORT</u> MARK MODE 1B                 |                                    |
| 00:50                  | LMP: <u>REPORT</u> CABIN PRESS DECREASE               |                                    |
| 01:17                  | CDR: <u>REPORT</u> MAX Q                              |                                    |
| 01:50                  | MCC-H:CDR: <u>REPORT</u> MARK MODE 1C                 |                                    |
| 02:00                  | MCC-H:CDR: <u>REPORT</u> GO/NO GO FOR STAGING         |                                    |
| 02:05                  | CDR: <u>REPORT</u> INBOARD ENGINE CUTOFF              |                                    |
| 02:31                  | CDR: <u>REPORT</u> OUTBOARD ENGINE CUTOFF             | LTS 1, 2, 3, & 4 - ON              |
| 02:32                  | CDR: <u>REPORT</u> S-IC/S-II STAGING                  | LTS OFF                            |
| MISSION AS503/103      |                                                       | EDITION FINAL                      |
| DATE November 22, 1968 |                                                       | PAGE 2-i                           |

# FLIGHT PLAN

| TIME              | EVENT                                                 | REMARKS                        |
|-------------------|-------------------------------------------------------|--------------------------------|
| 03:00             | CDR: <u>REPORT</u> 2ND PLANE SEP                      | >65% THRUST-S-II SEP LIGHT OUT |
| 03:07             | CDR: <u>REPORT</u> TWR JETT & MODE II                 |                                |
| 03:25             | CDR: <u>REPORT</u> GUIDANCE INITIATE                  |                                |
| 03:53             | MCC-H: <u>REPORT</u> TRAJECTORY AND GUID.<br>GO/NO GO |                                |
| 04:00             | CMP: <u>REPORT</u> S/C GO/NO GO                       | IF LAUNCH AZIMUTH <90°         |
| 05:00             | LMP: <u>REPORT</u> S/C GO/NO GO                       |                                |
| 05:53             | MCC-H:CDR: <u>REPORT</u> S-IVB TO ORBIT<br>CAPABILITY |                                |
| 06:00             | CDR: <u>REPORT</u> S/C GO/NO GO                       |                                |
| 06:15             | LMP: OMNI ANT-D                                       |                                |
| 07:00             | CDR: <u>REPORT</u> S/C GO/NO GO                       |                                |
| 08:00             |                                                       |                                |
| 08:20             | MCC-H:CDR: <u>REPORT</u> GO/NO GO FOR STAGING         |                                |
| 08:40             | CDR: <u>REPORT</u> S-II CUTOFF,<br>S-II STAGING       |                                |
| MISSION AS503/103 |                                                       | EDITION FINAL                  |
|                   |                                                       | DATE November 22, 1968         |
|                   |                                                       | PAGE 2-ii                      |

MSC FORM 2114C (JUL 67)



# FLIGHT PLAN

| TIME              | EVENT                                                 | REMARKS                |
|-------------------|-------------------------------------------------------|------------------------|
| 08:45             | CDR: <u>REPORT</u> S-IVB IGNITION                     |                        |
| 09:00             | CDR: <u>REPORT</u> S/C GO/NO GO                       |                        |
|                   | MCC-H: <u>REPORT</u> TRAJECTORY AND GUID.<br>GO/NO GO |                        |
| 09:50             | MCC-H:CDR: <u>REPORT</u> MARK MODE IV                 |                        |
| 10:00             | MCC-H:CDR: <u>REPORT</u> GO/NO GO FOR ORBIT           |                        |
|                   | MCC-H: <u>REPORT</u> PREDICTED TIME OF SECO           |                        |
| 11:21             | CDR: <u>REPORT</u> SECO AND HP                        |                        |
| 11:31             | MCC-H:CDR: <u>REPORT</u> ORBITAL GO/NO GO             |                        |
| 12:00             | LMP: FLT RCDR - OFF                                   |                        |
| MISSION AS503/103 |                                                       | EDITION FINAL          |
|                   |                                                       | DATE November 22, 1968 |
|                   |                                                       | PAGE 2-iii             |

# FLIGHT PLAN

|       | CDR                       | CMP                        | LMP                       | MCC-H                        |
|-------|---------------------------|----------------------------|---------------------------|------------------------------|
| 00:00 | POST INSERTION CONFIG     | POST INSERTION CONFIG      | POST INSERTION CONFIG     |                              |
|       |                           | SM RCS CK                  |                           |                              |
|       |                           | CM RCS CK                  |                           |                              |
| 00:15 | REMOVE HELMET & GLOVES    | C&W CK                     | REMOVE HELMET & GLOVES    |                              |
|       |                           | REMOVE HELMET & GLOVES     |                           |                              |
|       | ECS POST INSERTION CONFIG |                            | ECS POST INSERTION CONFIG |                              |
|       | GDC ALIGN TO IMU          | INGRESS LEB                | EPS PERIODIC MONITOR      |                              |
|       |                           | O <sub>2</sub> MAIN REG CK | ECS MONITOR CK            |                              |
| 00:30 | MOUNT & INITIALIZE ORDEAL |                            | SPS PERIODIC MONITOR      |                              |
|       | INSTALL COAS              | JETTISON OPTICS COVER      | PUGS TEST                 |                              |
|       | COAS HORIZON CK           | RECORD ΔAZ CORRECTION      | ECS REDUNDANT COMP CK     |                              |
|       |                           |                            | FC PURGE CK               | VOICE UPDATE: ΔAZ CORRECTION |
| 00:45 |                           |                            |                           |                              |
|       |                           | OPTICS CK                  |                           |                              |
|       |                           | IMU REALIGN P52            |                           |                              |
|       |                           | OPTION 3 - REFSMMAT        |                           |                              |
|       |                           | STAR ID _____              | BIOMED Sw - CENTER        |                              |
|       |                           | (cont'd)                   |                           |                              |
| 01:00 |                           |                            |                           |                              |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 00:00 - 01:00 | 1/LPO   | 2-1  |

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FLIGHT PLANNING BRANCH

# FLIGHT PLAN

CDR

CMP

LMP

MCC-H

01:00

H  
S  
K

STAR ANGLE DIFF

TORQUE ANGLES:

X ———  
Y ———  
Z ———

01:15

01:30

SCS ATT REFERENCE  
COMPARISON CK

REPORT: GYRO TORQUE  
ANGLES

BACKUP COMM CK

GIVE GO FOR  
COMM CK

01:45

U  
S

RECORD ABORT BLOCK PAD  
(TLI +90 MIN AND TLI  
+4 HOUR)

VOICE  
UPDATE:  
BLOCK  
DATA

C  
Y  
I

RECORD TLI PAD

BIOMED Sw - RIGHT

VOICE  
UPDATE:  
TLI PAD

P27 UPDATE:  
STATE  
VECTORS

02:00

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 01:00 - 02:00 | 1/LPO   | 2-2  |

# BURN STATUS REPORT

X X ☐ :  $\Delta$ TIG  
 X X : BT  
☐  $V_{gx}$

TRIM

X X X R  
 X X X P  
 X X X Y

☐  $V_i$   
☐ h  
☐ h  
☐  $\Delta V_c$

X X X FUEL

X X X OX

X X X UNBALANCE

REMARKS:

## TLI BURN CHART

|     | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME                | RESIDUALS |
|-----|---------------------|------------------|---------------------------------|-----------|
| TLI | 10°/SEC<br>SHUTDOWN | +45°<br>SHUTDOWN | B/T+6 SEC &<br>$V_i$ =PAD VALVE | NO TRIM   |

| TLI PREMATURE SHUTDOWN |                                                                   |
|------------------------|-------------------------------------------------------------------|
| ha >60,000 nm          | LUNAR ORBIT OR FLYBY (DEPENDING ON $\Delta V$ REQD)               |
| ha >22,000 nm          | TWO PHASING MANEUVERS TO SEMI-SYNCHRONOUS ORBIT. DIRECT ENTRY     |
| ha >41,000 nm          | HIGH ALTITUDE ORBITS FOLLOWED BY DEBOOST TO 400 nm APOGEE         |
| ha 100 - 41,000 nm     | EITHER HI ALTITUDE (41,000 ha) OR LOW ALT, DEPENDING ON LANDMARKS |

2-2a

# FLIGHT PLAN

|       | CDR                                                                      | CMP                        | LMP               | MCC-H    |
|-------|--------------------------------------------------------------------------|----------------------------|-------------------|----------|
| 02:00 |                                                                          |                            |                   |          |
| 02:15 | TLI PREP<br>EMS ΔV TEST                                                  | TLI PREP<br>TRANS TO COUCH | TLI PREP          |          |
| 02:30 | GO/NO-GO FOR PYRO ARM<br>GO/NO-GO FOR TLI<br>GDC ALIGN AND DRIFT CK      |                            | BIOMED Sw - LEFT  | GO/NO-GO |
| 02:45 | TB-6<br>P47 BURN ATT CK<br>TLI                                           | GETI = 2:50:37             | FLT RCDR - RECORD |          |
| 03:00 | SECO S-IVB INERTIAL<br>SECO +20 SEC S-IVB TO<br>LH, ORB RATE, HEADS DOWN |                            | FLT RCDR - OFF    |          |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 02:00 - 03:00 | 1/LPO   | 2-3  |

# FLIGHT PLAN

|       | CDR                                                                                                 | CMP                                                                                                                                  | LMP                                                                                                                                   | MCC-H                                       |
|-------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 03:00 | TLI BURN STATUS REPORT                                                                              | RECORD GET SEP MNVR<br>INIT                                                                                                          |                                                                                                                                       | VOICE<br>UPDATE:<br>GET OF SEP<br>MNVR INIT |
|       | S-IVB MNVR TO SEP ATT                                                                               | TRANS CSM STATE VECTOR<br>TO LM SLOT                                                                                                 |                                                                                                                                       |                                             |
| 03:15 | GO/NO-GO FOR 90-MIN<br>ABORT                                                                        | UNSTOW PHOTO EQUIP<br>B3 16mm DAC<br>18mm LENS<br>RT ANG MIRROR<br>16mm C-EX MAG<br>PWR CABLE<br>70mm CAM<br>80mm LENS<br>70mm C-MAG | S-BAND XPONDER - SEC<br>FLT RCDR - RECORD                                                                                             | GO/NO-GO<br>COMM TEST<br>MODE: 4.2          |
|       | TRANSPOSITION FROM<br>S-IVB +X FOR 1 fps,<br>COAST FOR 1 MIN, -X<br>FOR 0.5 fps, PITCH UP<br>4°/SEC | R13 SPOTMETER                                                                                                                        | FLT RCDR - OFF<br>NONESS BUS - OFF<br>PHOTOGRAPH S-IVB<br>16/18/C-EX, 1/250, f11,<br>6 fps (1 MAG)<br>2/80/C, 1/250, SPOT<br>(10 EXP) |                                             |
| 03:30 | FLY FORMATION                                                                                       | DOFF & STOW PGA                                                                                                                      |                                                                                                                                       |                                             |
| 03:45 | MNVR TO LOCAL VERTICAL<br>-X RADIALLY UPWARD<br>1.5 fps                                             | TRANS CSM STATE VECTOR<br>TO LM SLOT                                                                                                 |                                                                                                                                       |                                             |
| 04:00 |                                                                                                     |                                                                                                                                      |                                                                                                                                       |                                             |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 03:00 - 04:00 | 1/TLC   | 2-4  |

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FLIGHT PLANNING BRANCH

# FLIGHT PLAN

CDR

CMP

LMP

MCC-H

04:00

04:15

04:30

04:45

05:00

M  
S  
F  
N

GDC ALIGN TO IMU

IMU REALIGN P52  
OPTION 3 - REFSMMAT  
STAR ID \_\_\_\_\_  
STAR ANGLE DIFF \_\_\_\_\_

TORQUE ANGLES: \_\_\_\_\_  
X \_\_\_\_\_  
Y \_\_\_\_\_  
Z \_\_\_\_\_

CISLUNAR NAVIGATION P23

TRN BIAS  
1. STAR 14 ENH  
STAR \_\_\_\_\_ E \_\_\_\_\_ H  
1 SET - 3 MARKS  
EACH

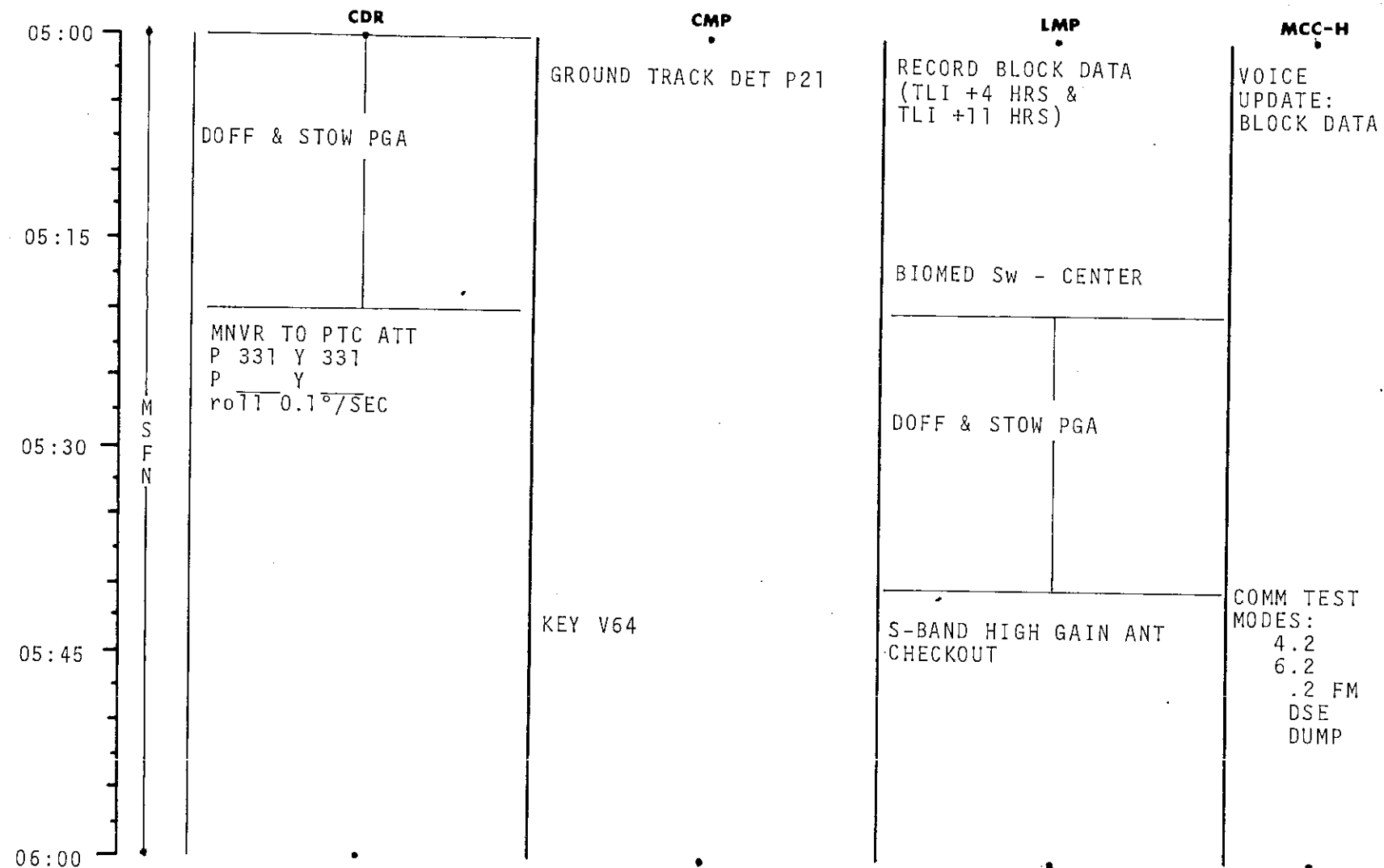
2. STAR 15 EFH  
STAR \_\_\_\_\_ E \_\_\_\_\_ H  
2 SETS - 3 MARKS  
EACH

3. STAR 16 EFH  
STAR \_\_\_\_\_ E \_\_\_\_\_ H  
2 SETS - 3 MARKS  
EACH

REPORT PERSONAL  
RADIATION DOSIMETER  
READINGS

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 04:00 - 05:00 | 1/TLC   | 2-5  |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 05:00 - 06:00 | 1/TLC   | 2-6  |

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FLIGHT PLANNING BRANCH



# FLIGHT PLAN

|       | CDR        | CMP        | LMP        | MCC-H |
|-------|------------|------------|------------|-------|
| 06:00 | EAT PERIOD | EAT PERIOD | EAT PERIOD |       |
| 06:15 |            |            |            |       |
| 06:30 |            |            |            |       |
| 06:45 |            |            |            |       |
| 07:00 |            |            |            |       |

MSFN

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 06:00 - 07:00 | 1/TLC   | 2-7  |

# FLIGHT PLAN

|       | CDR              | CMP                                                                                                                             | LMP | MCC-H                                                    |
|-------|------------------|---------------------------------------------------------------------------------------------------------------------------------|-----|----------------------------------------------------------|
| 07:00 |                  |                                                                                                                                 |     | PERFORM<br>COMM TESTS<br>MODES: 4.3<br>6.3<br>5.2<br>8.1 |
| 07:15 |                  |                                                                                                                                 |     |                                                          |
| 07:30 | M<br>S<br>F<br>N | L10H CANISTER CHANGE<br>(CARTRIDGE NO. 3 FROM<br>A3 INTO CANISTER A)                                                            |     |                                                          |
| 07:45 |                  | OPEN COOLANT CONTROL<br>ATTENUATION PANEL<br>EVAP WATER CONT SEC<br>VLV - OFF<br><br>CLOSE COOLANT CONTROL<br>ATTENUATION PANEL |     |                                                          |
| 08:00 |                  | WASTE STOWAGE VENT -<br>CLOSED<br>BAT VENT VLV - OPEN<br>(UNTIL SERVICE<br>METER = 0)<br>BAT VENT VLV - CLOSED                  |     | P27 UPDATE:<br>STATE<br>VECTOR<br>TGT LOAD               |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
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FLIGHT PLANNING BRANCH

# BURN STATUS REPORT

|                          |   |                          |   |                 |
|--------------------------|---|--------------------------|---|-----------------|
| X                        | X | <input type="checkbox"/> | : | ΔTIG            |
| X                        | X | <input type="checkbox"/> | : | BT              |
| <input type="checkbox"/> |   |                          | . | V <sub>gx</sub> |

|      |   |   |   |
|------|---|---|---|
| TRIM |   |   |   |
| X    | X | X | R |
| X    | X | X | P |
| X    | X | X | Y |

|                          |  |  |                 |
|--------------------------|--|--|-----------------|
| <input type="checkbox"/> |  |  | V <sub>gx</sub> |
| <input type="checkbox"/> |  |  | V <sub>gy</sub> |
| <input type="checkbox"/> |  |  | V <sub>gz</sub> |
| <input type="checkbox"/> |  |  | ΔV <sub>c</sub> |

|   |   |   |      |
|---|---|---|------|
| X | X | X | FUEL |
|---|---|---|------|

|   |   |   |    |
|---|---|---|----|
| X | X | X | OX |
|---|---|---|----|

|   |   |   |           |
|---|---|---|-----------|
| X | X | X | UNBALANCE |
|---|---|---|-----------|

REMARKS:

MCC'S

## BURN CHART

|          | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME | RESIDUALS       |
|----------|---------------------|------------------|------------------|-----------------|
| MCC(ALL) | 10°/SEC<br>TAKEOVER | 10°<br>TAKEOVER  | B/T +1 SEC       | TRIM TO 0.2 fps |

2-8a

# FLIGHT PLAN

|            | CDR                                                                                                  | CMP                                                                                              | LMP               | MCC-H                        |
|------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------|------------------------------|
| 08:00      | MNVR TO P52 ATT                                                                                      | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>AND DRIFT CK<br>STAR ID _____<br>STAR ANGLE DIFF _____ | RECORD MNVR PAD   | VOICE<br>UPDATE:<br>MNVR PAD |
| 08:15      |                                                                                                      | TORQUE ANGLES _____<br>X _____<br>Y _____<br>Z _____                                             |                   |                              |
| 08:30      | V47 TRANS LM STATE<br>VECTOR TO CSM SLOT<br>EXT ΔV P30<br>SPS/RCS THRUST P40/P41<br>MNVR TO BURN ATT | SXT STAR CK<br>TRANS TO COUCH                                                                    | BIOMED Sw - RIGHT | PIPA BIAS<br>CK              |
| 08:45      | EMS ΔV TEST                                                                                          |                                                                                                  |                   |                              |
| TLI +6 HRS | GDC ALIGN                                                                                            |                                                                                                  |                   |                              |
| 09:00      | MCC <sub>1</sub> ΔV=NOMINALLY ZERO                                                                   |                                                                                                  |                   |                              |

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|-----------|---------|-------------------|---------------|---------|------|
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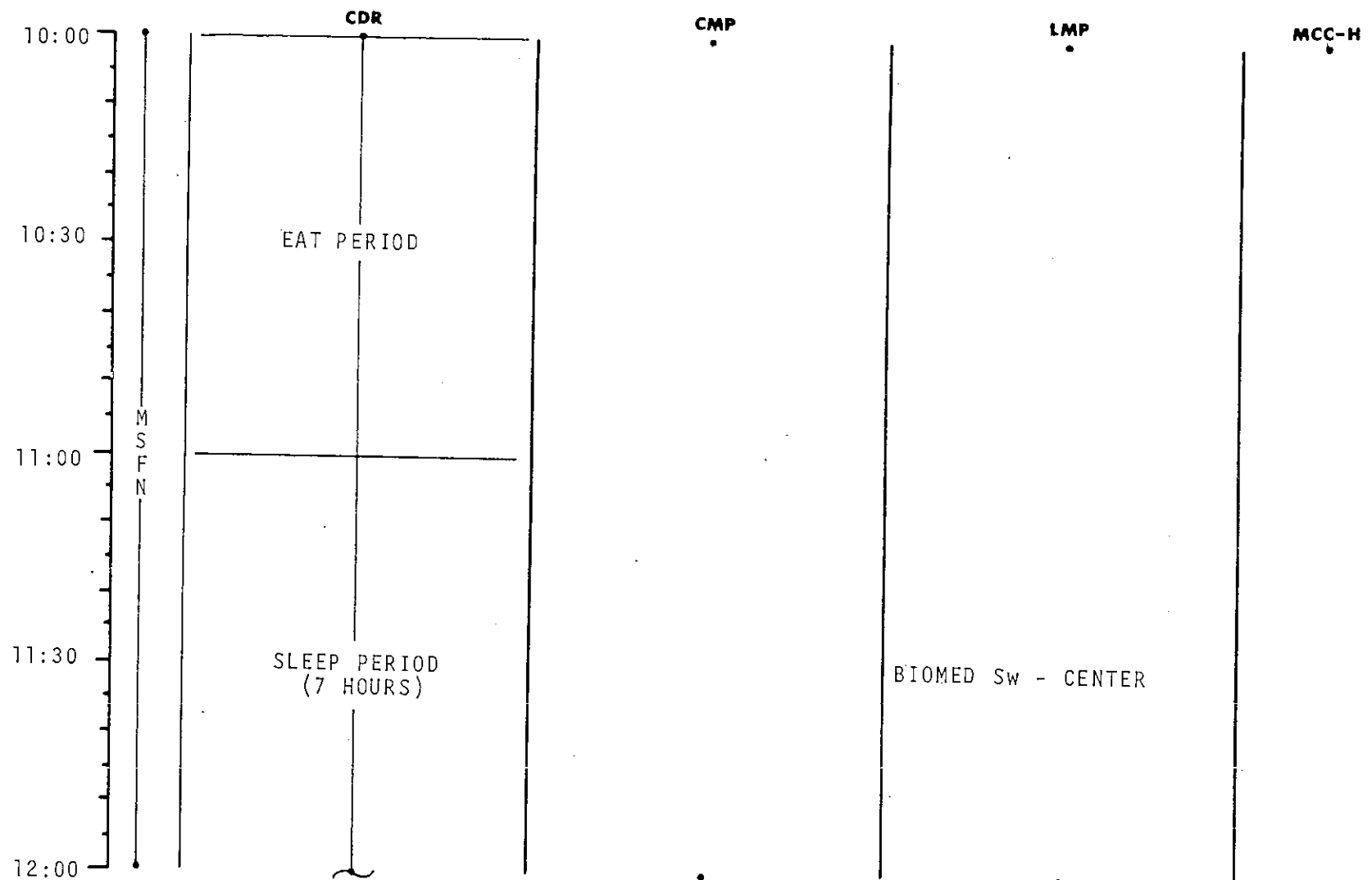
FLIGHT PLANNING BRANCH

# FLIGHT PLAN

|       | CDR                                                                                                     | CMP                                                                                                                                                                                                                                                                                                                               | LMP                                   | MCC-H |
|-------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------|
| 09:00 | V66 TRANS CSM STATE<br>VECTOR TO LM SLOT<br>MCC <sub>1</sub> BURN STATUS REPORT<br>MNVR TO SIGHTING ATT | SM RCS CK<br>TRN BIAS<br><div>CISLUNAR NAVIGATION P23</div><br>1. STAR 15 ELD MK 10<br>LAT 28.876°N<br>LONG/2 56.292°W<br>ALT 000.01<br>STAR ELD MK<br>LAT<br>LONG/2<br>ALT<br>2 SETS<br><br>2. STAR 16 ELD MK 10<br>LAT 28.876°N<br>LONG/2 56.292°W<br>ALT 000.01<br>STAR ELD MK<br>LAT<br>LONG/2<br>ALT<br>GROUND TRACK DET P21 | SPS MONITOR CK<br>INITIATE BAT CHARGE |       |
| 09:30 | M<br>S<br>F<br>N                                                                                        |                                                                                                                                                                                                                                                                                                                                   |                                       |       |
| 10:00 | MNVR TO PTC ATT<br>P 331 Y 331<br>P Y<br>ROLL 0.1°/SEC                                                  |                                                                                                                                                                                                                                                                                                                                   |                                       |       |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 09:00 - 10:00 | 1/TLC   | 2-10 |

# FLIGHT PLAN

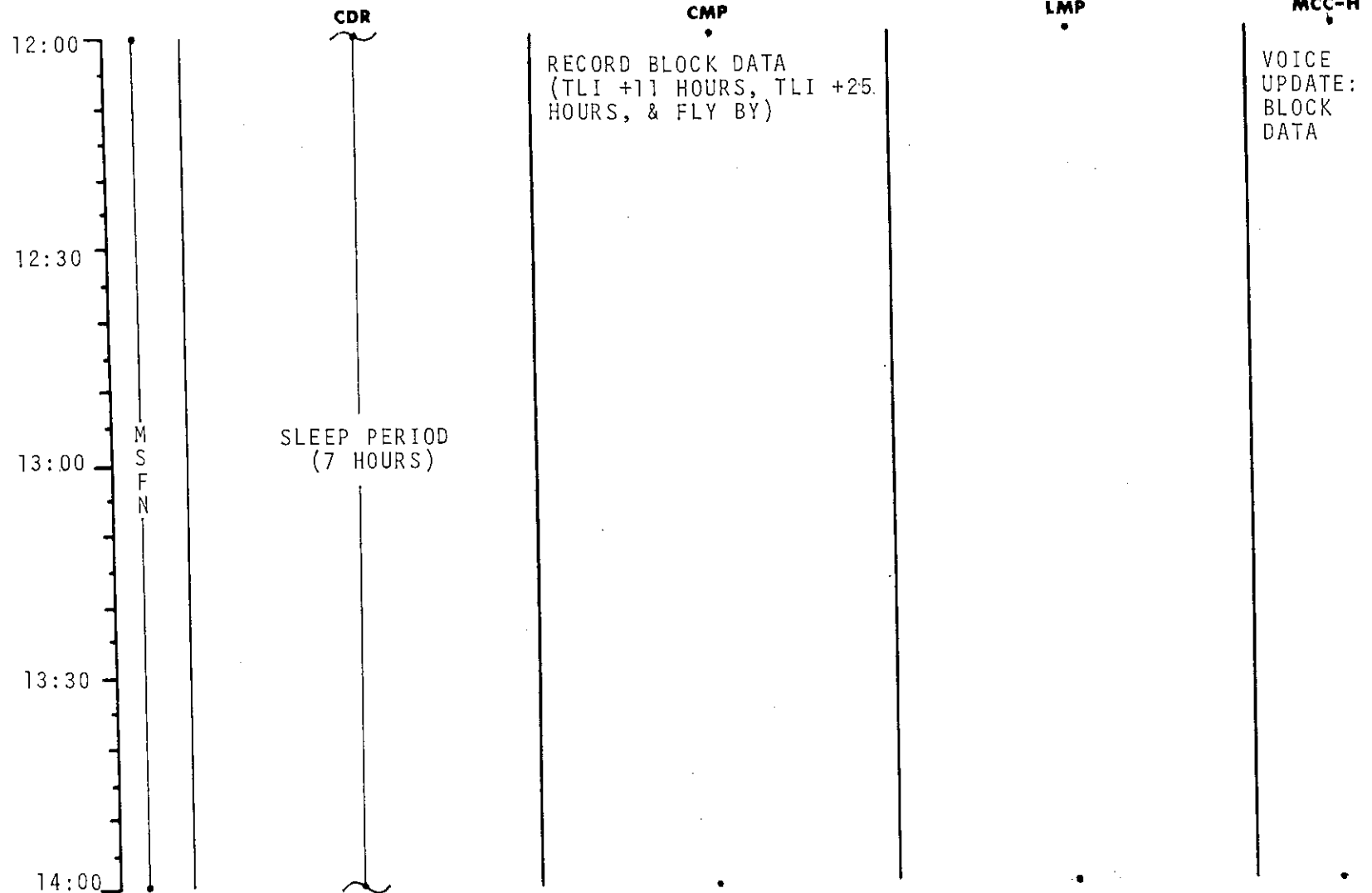


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE  |
|-----------|---------|-------------------|---------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 10:00 - 12:00 | 1/TLC   | 2-11. |

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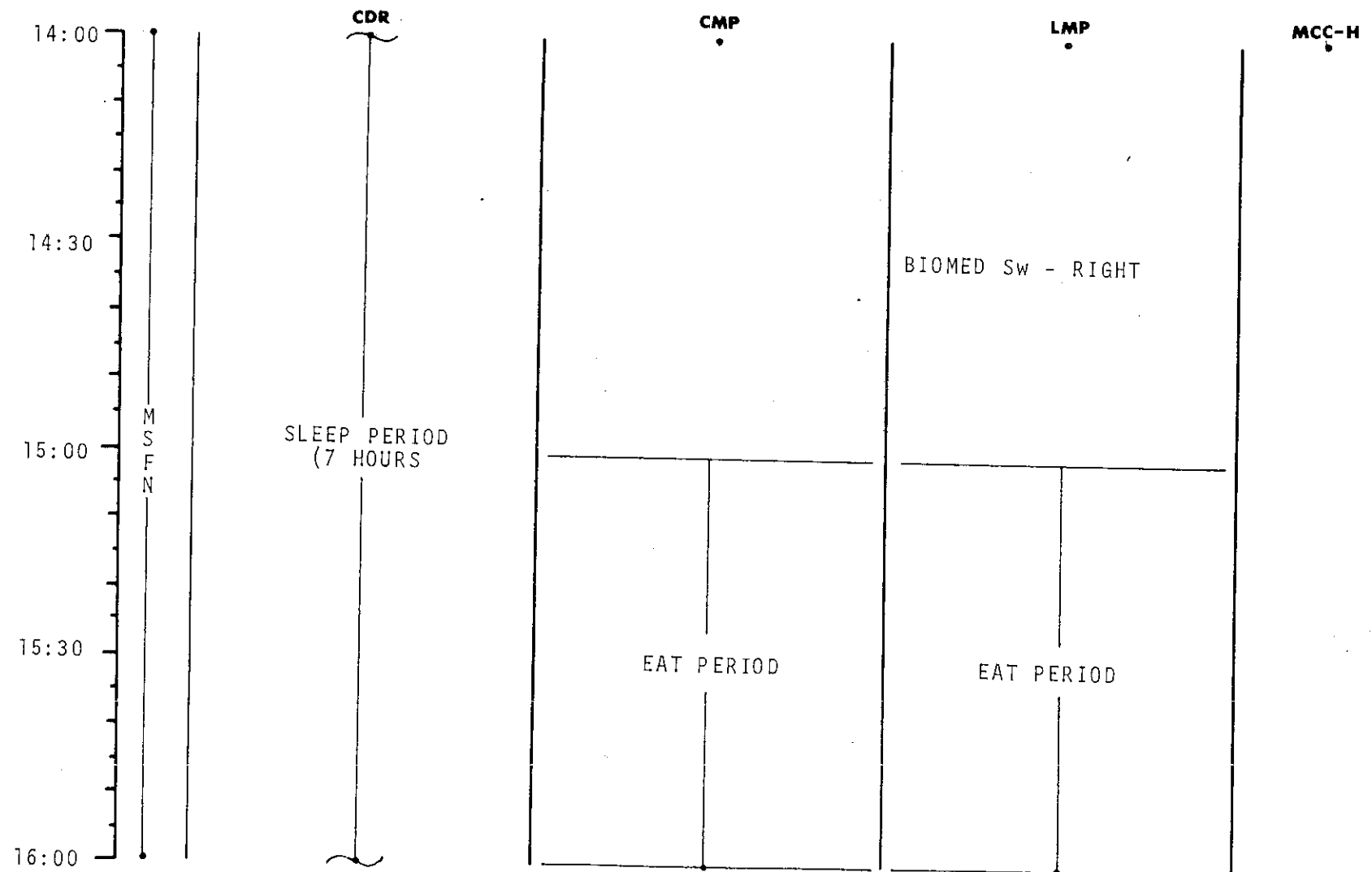
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 12:00 - 14:00 | 1/TLC   | 2-12 |

# FLIGHT PLAN



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|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 14:00 - 16:00 | 1/TLC   | 2-13 |

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FLIGHT PLANNING BRANCH



# FLIGHT PLAN

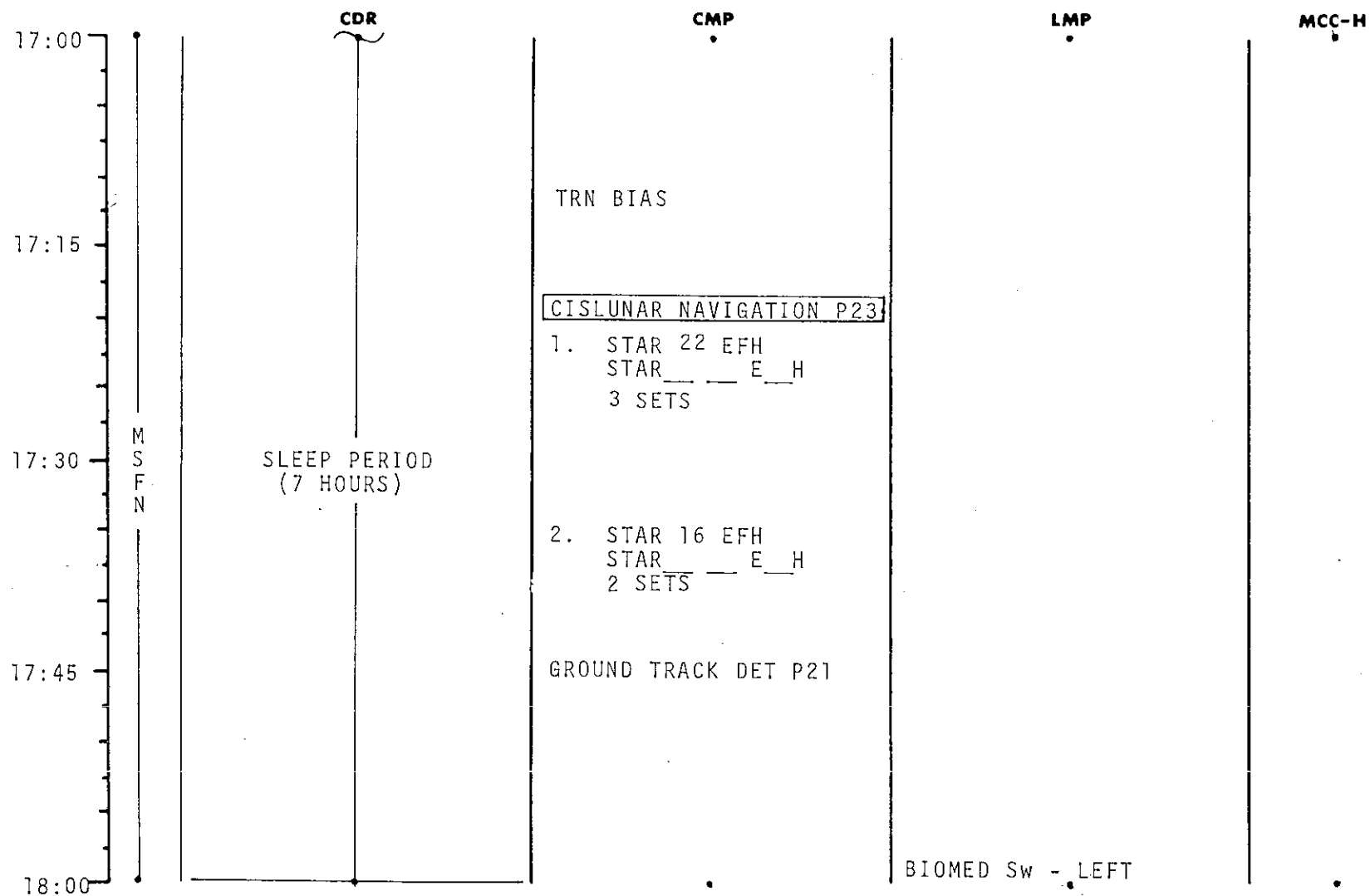
|       | CDR                       | CMP                                                                                                                                                                      | LMP             | MCC-H |
|-------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------|
| 16:00 | SLEEP PERIOD<br>(7 HOURS) | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID _____<br>STAR ANGLE DIFF _____<br><br>TORQUE ANGLES:<br>X _____<br>Y _____<br>Z _____<br><br>SCT STAR VISIBILITY _____ | MNVR TO P52 ATT |       |
| 16:15 |                           |                                                                                                                                                                          |                 |       |
| 16:30 |                           |                                                                                                                                                                          |                 |       |
| 16:45 |                           |                                                                                                                                                                          |                 |       |
| 17:00 |                           |                                                                                                                                                                          |                 |       |

MSFN

GDC ALIGN TO IMU  
MNVR TO SIGHTING ATT

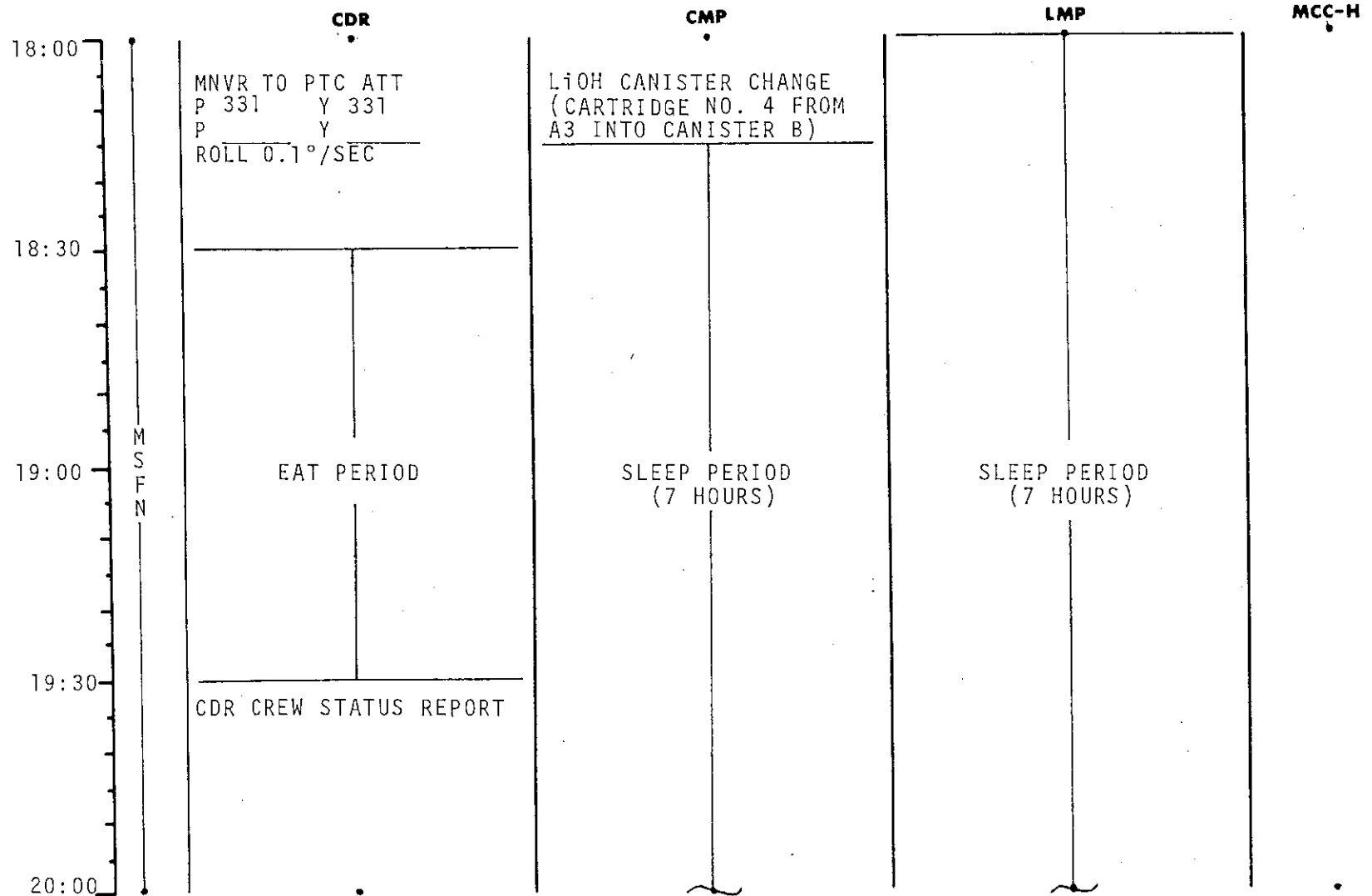
| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 16:00 - 17:00 | 1/TLC   | 2-14 |

# FLIGHT PLAN



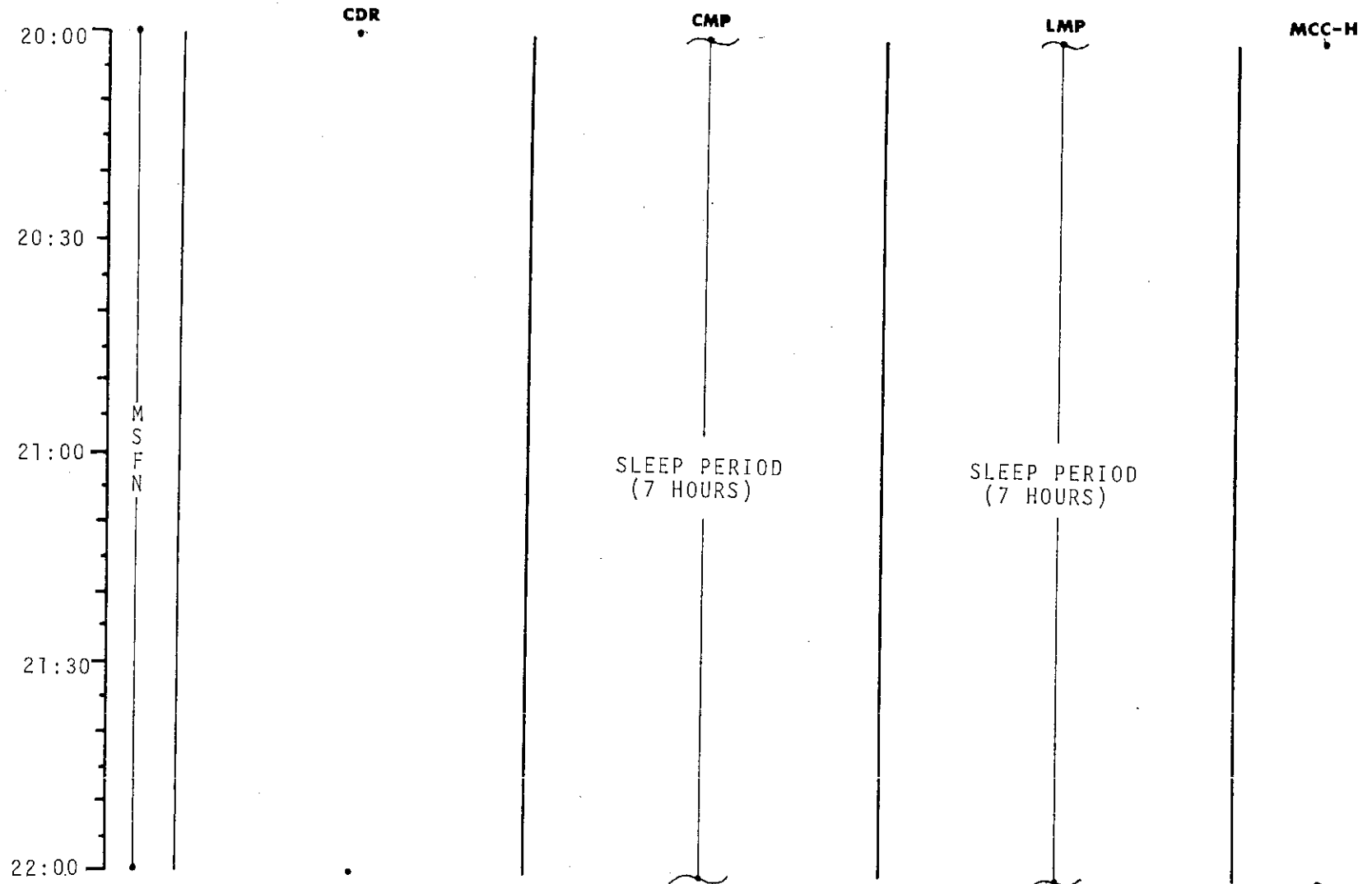
| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 17:00 - 18:00 | 1/TLC   | 2-15 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 18:00 - 20:00 | 1/TLC   | 2-16 |

# FLIGHT PLAN

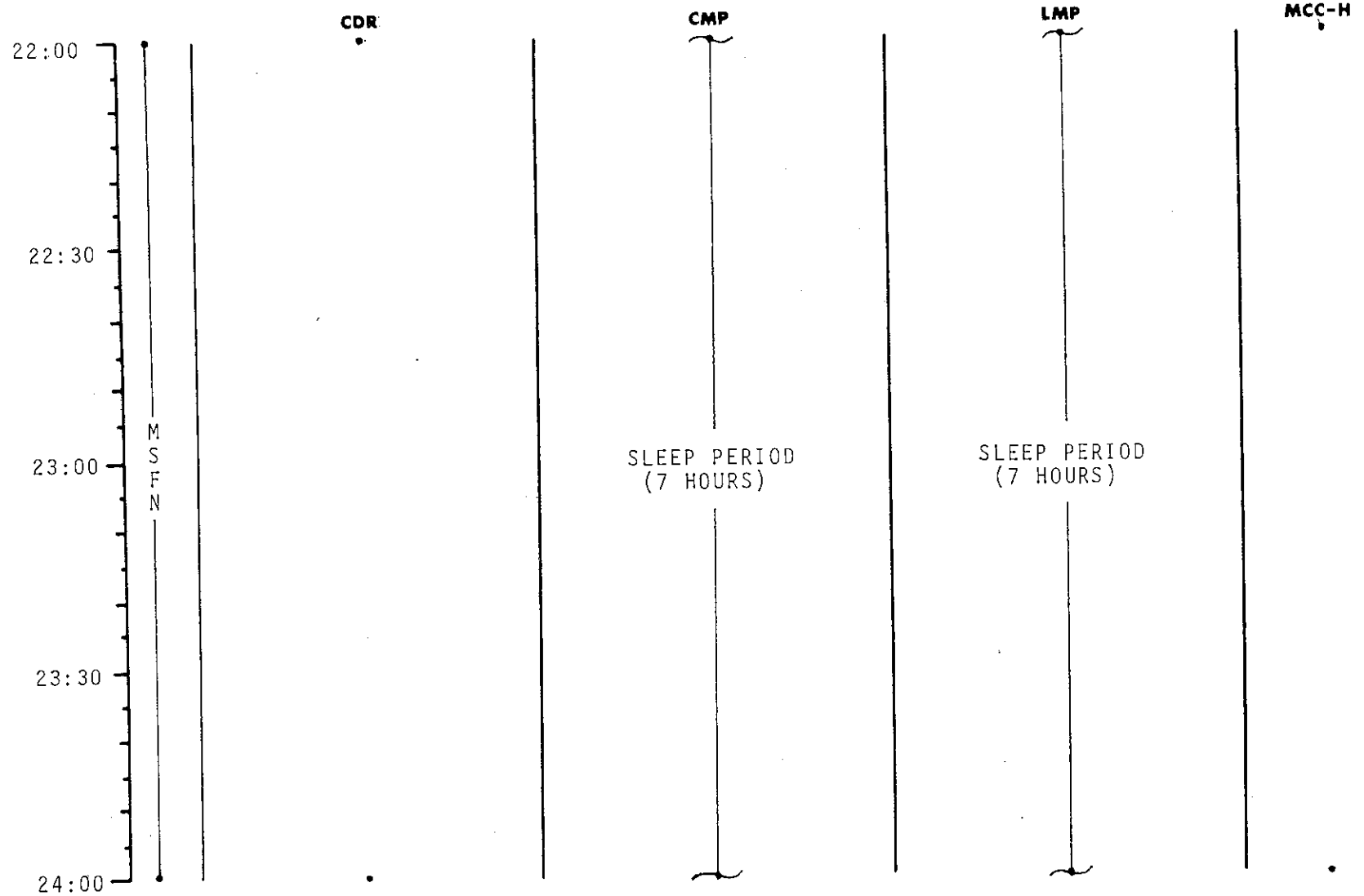


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 20:00 - 22:00 | 1/TLC   | 2-17 |

MSC Form 1910 (Nov 68)

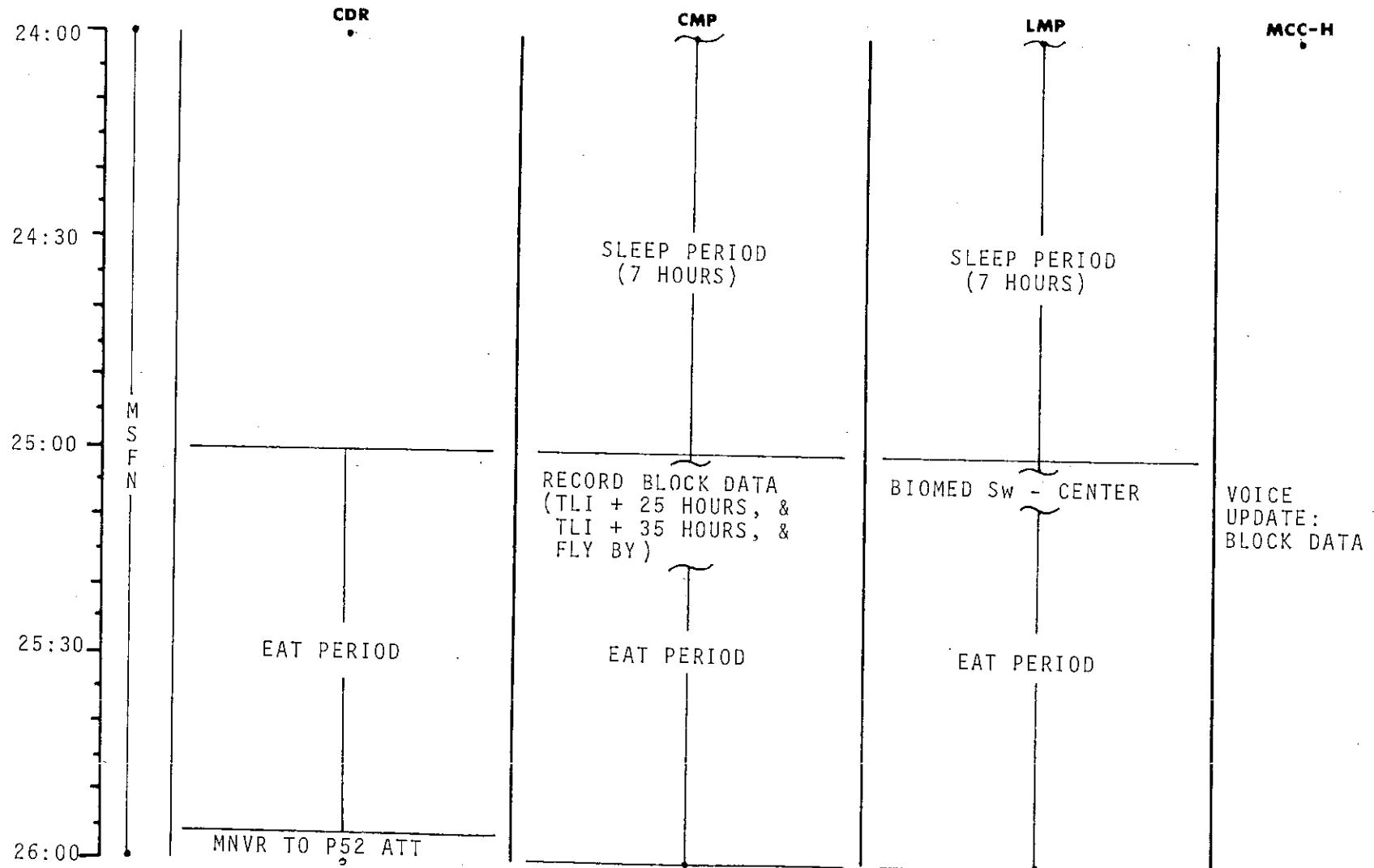
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 22:00 - 24:00 | 1/TLC   | 2-18 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 24:00 - 26:00 | 2/TLC   | 2-19 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

# FLIGHT PLAN

CDR

CMP

LMP

MCC-H

26:00  
26:15  
26:30  
26:45  
27:00

M  
S  
F  
N

GDC ALIGN TO IMU  
MNVR TO SIGHTING  
ATTITUDE

IMU REALIGN P52  
OPTION 3 - REFSMMAT  
STAR ID  
STAR ANGLE DIFF

TORQUE ANGLES:  
X  
Y  
Z

TRN BIAS

CISLUNAR NAVIGATION P23

1. STAR 16 EFH  
STAR E H  
1 SET
2. STAR 22 EFH  
STAR E H  
1 SET
3. STAR 26 ENH  
STAR E H  
1 SET

CMP/LMP CREW STATUS  
REPORT

RECORD MNVR PAD

VOICE  
UPDATE:  
MNVR PAD

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 26:00 - 27:00 | 2/TLC   | 2-20 |

|   |   |  |   |   |                 |
|---|---|--|---|---|-----------------|
| X | X |  | • |   | ΔTIG            |
| X | X |  | • |   | BT              |
|   |   |  |   | • | V <sub>gx</sub> |

| TRIM |   |   |                 |
|------|---|---|-----------------|
| X    | X | X | R               |
| X    | X | X | P               |
| X    | X | X | Y               |
|      |   |   | V <sub>gx</sub> |
|      |   |   | V <sub>gy</sub> |
|      |   |   | V <sub>gz</sub> |
|      |   |   | $\Delta V_c$    |
| X    | X | X | FUEL            |
| X    | X | X | OX              |
| X    | X | X | UNBALANCE       |

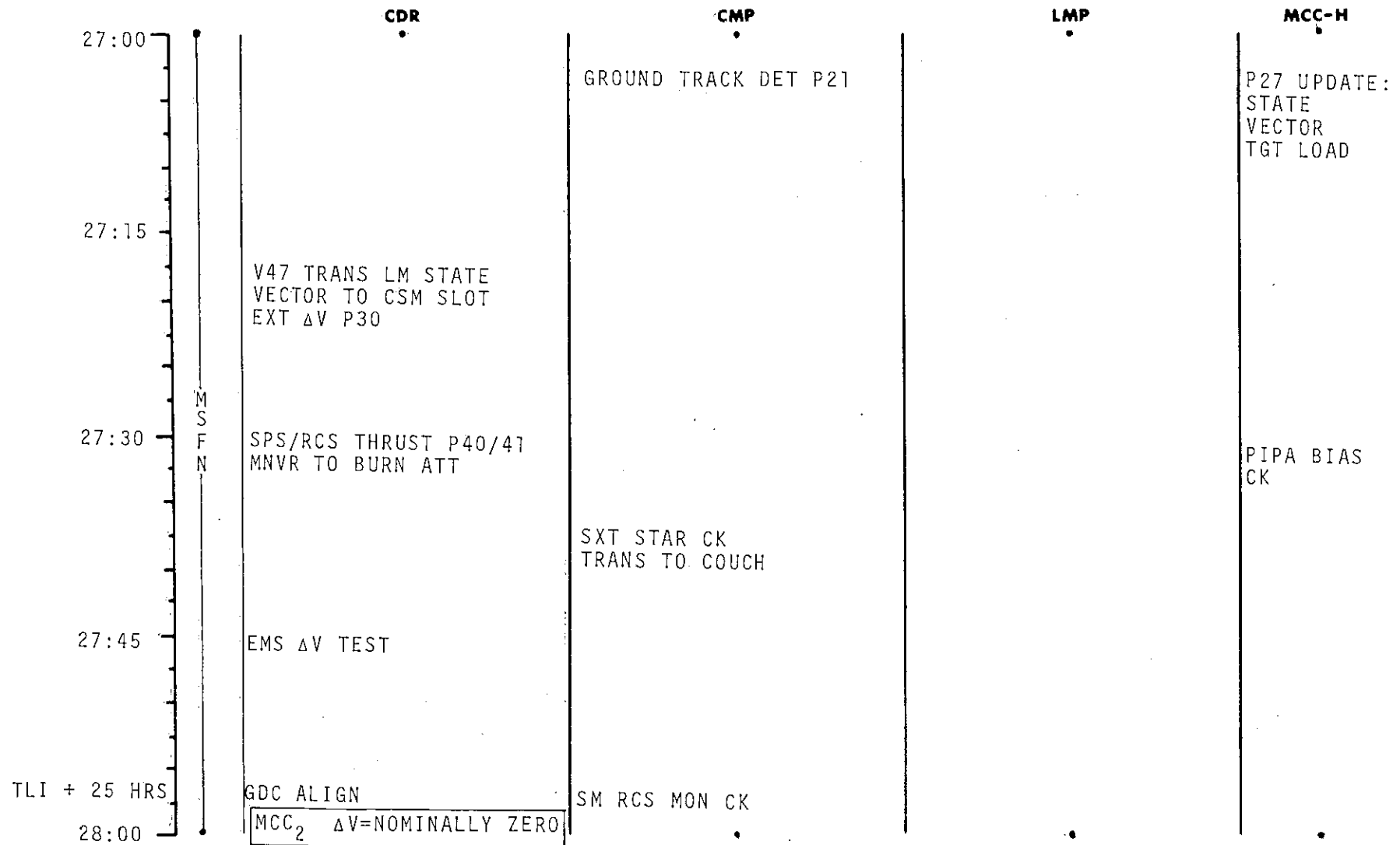
REMARKS:

BURN CHART

|           |                     |                  |                  |                 |
|-----------|---------------------|------------------|------------------|-----------------|
|           | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME | RESIDUALS       |
| MCC (ALL) | 10°/SEC<br>TAKEOVER | 10°<br>TAKEOVER  | B/T +1 SEC       | TRIM TO 0.2 fps |



# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 27:00 - 28:00 | 2/TLC   | 2-21 |

# FLIGHT PLAN

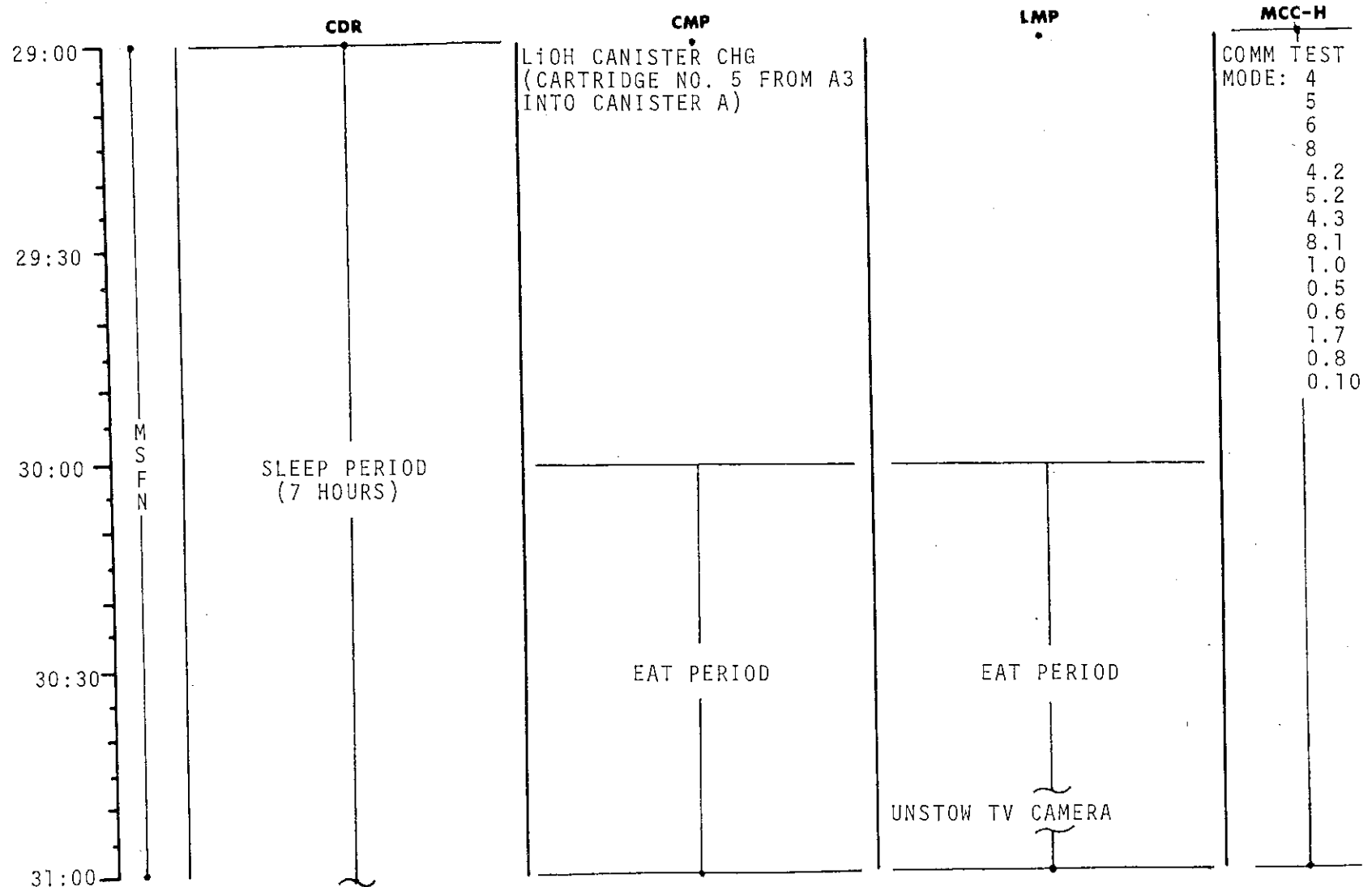
|       | CDR                                                                                                         | CMP                                                                          | LMP                                                            | MCC-H |
|-------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------|-------|
| 28:00 | V66 TRANS CSM STATE<br>VECTOR TO LM SLOT<br><br>MNVN TO SIGHTING ATT<br>MCC <sub>2</sub> BURN STATUS REPORT | SM RCS MONITOR CK<br>TRN BIAS                                                | SPS MONITOR CK<br>INITIATE BAT CHARGE<br>BIOMED Sw - RIGHT     |       |
| 28:15 |                                                                                                             | <div>CISLUNAR NAVIGATION P23</div> 1. STAR 16 EFH<br>STAR__ _ E__ H<br>1 SET |                                                                |       |
| 28:30 |                                                                                                             | 2. STAR 22 EFH<br>STAR__ _ E__ H<br>1 SET                                    |                                                                |       |
| 28:45 |                                                                                                             | 3. STAR 21 EFH<br>STAR__ _ E__ H<br>1 SET                                    |                                                                |       |
| 29:00 | CDR CREW STATUS REPORT                                                                                      | 4. STAR 26 ENH<br>STAR__ _ E__ H<br>1 SET<br><br>GROUND TRACK DET P21        | MNVR TO PTC ATT<br>P 331 Y 331<br>P__ _ Y__ _<br>ROLL 0.1°/SEC |       |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 28:00 - 29:00 | 2/TLC   | 2-22 |

MSC Form 1910 (Nov 68)

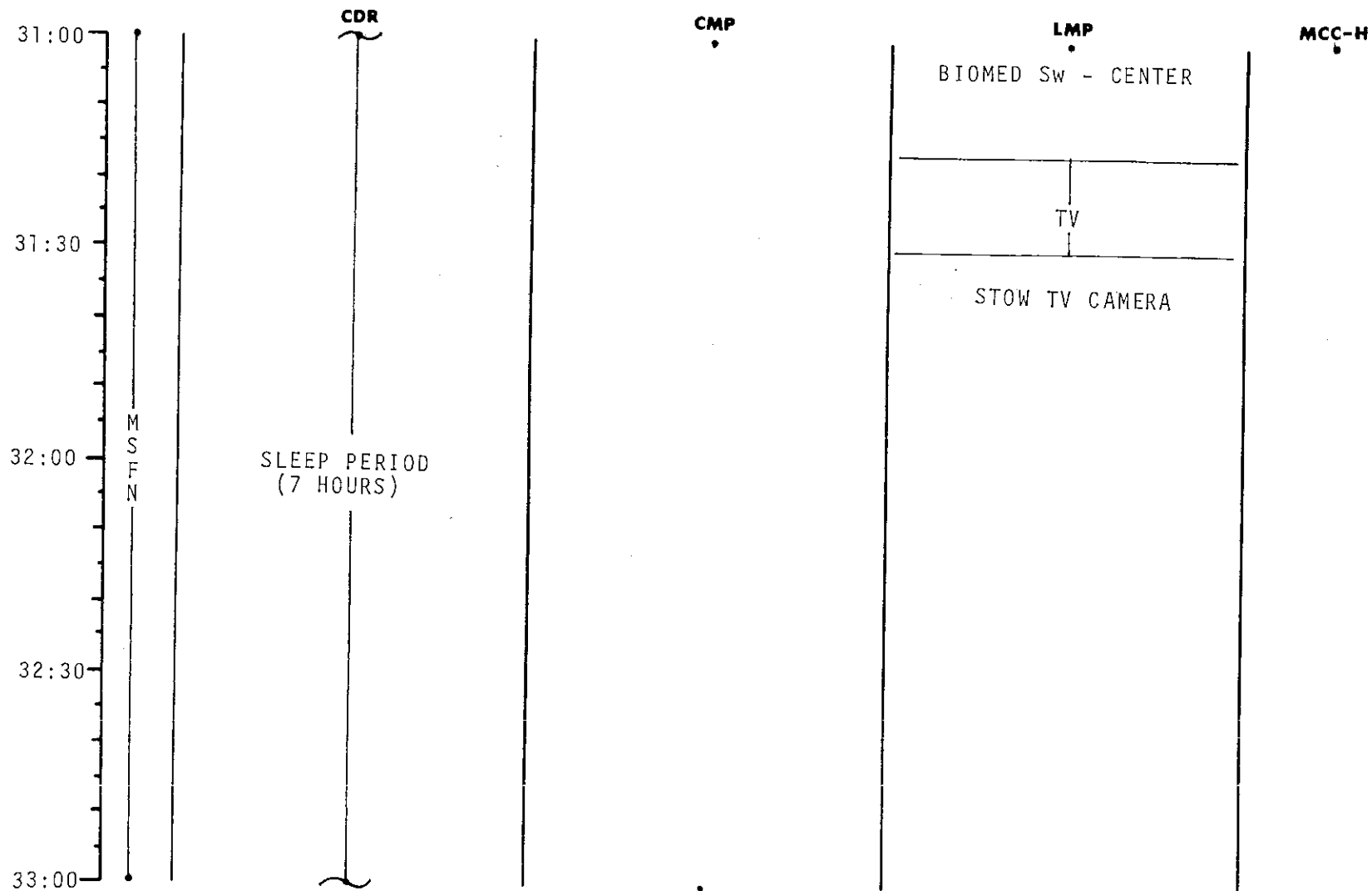
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 29:00 - 31:00 | 2/TLC   | 2-23 |

# FLIGHT PLAN

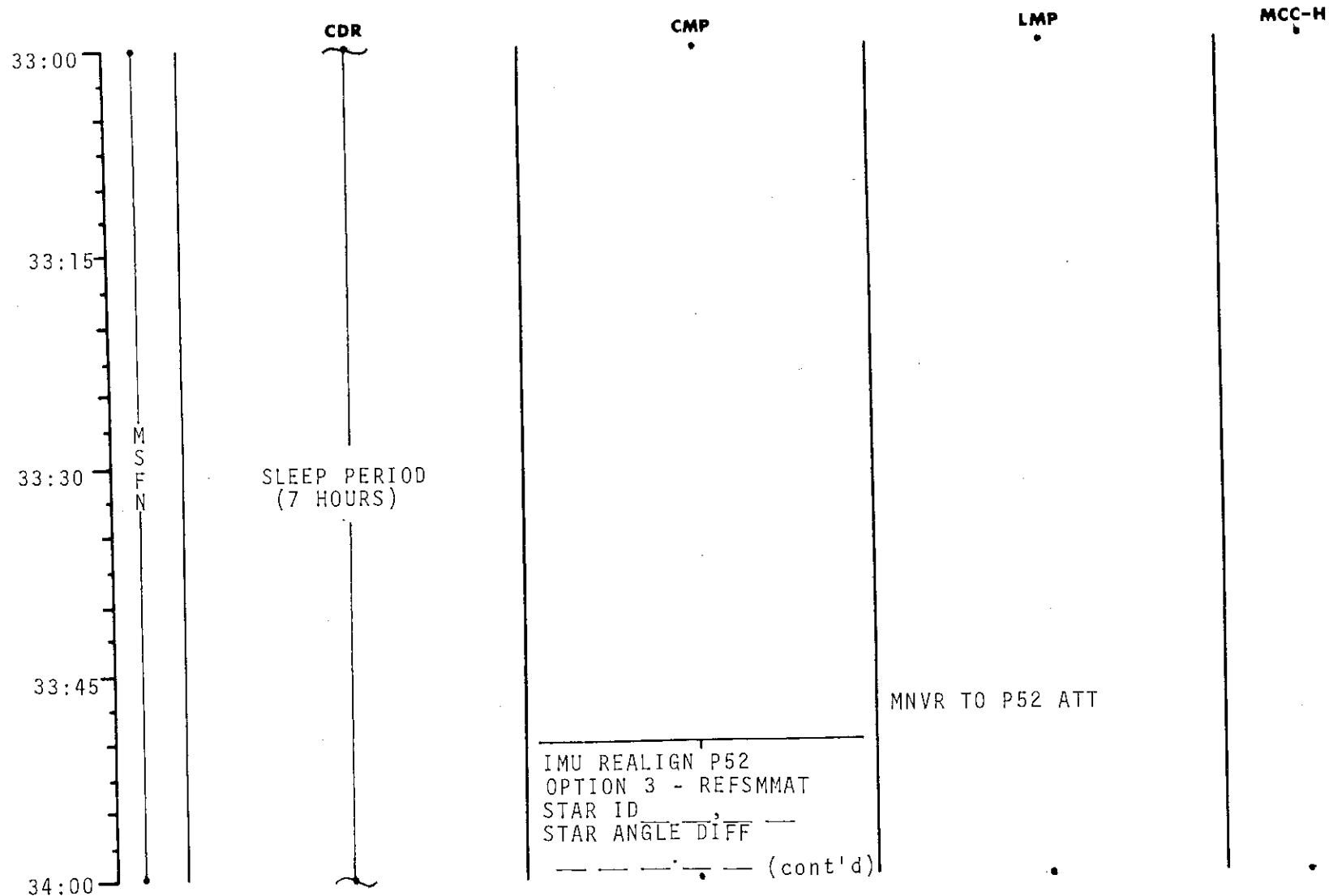


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 31:00 - 33:00 | 2/TLC   | 2-24 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 33:00 - 34:00 | 2/TLC   | 2-25 |

# FLIGHT PLAN

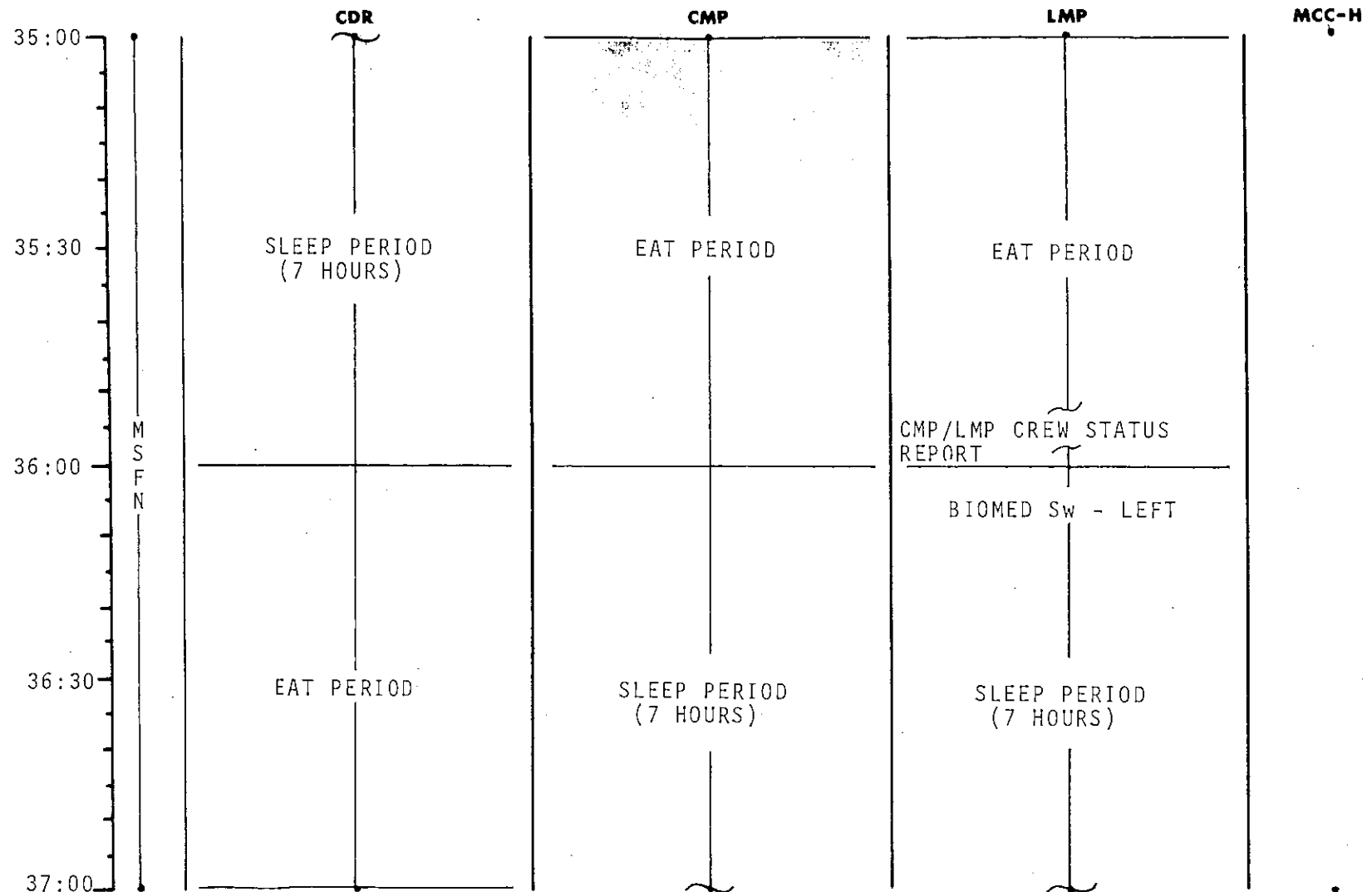
|       |                           |                                                                      |                                                        |                                |
|-------|---------------------------|----------------------------------------------------------------------|--------------------------------------------------------|--------------------------------|
| 34:00 | CDR                       | CMP                                                                  | LMP                                                    | MCC-H                          |
|       |                           | TORQUE ANGLES:<br>X _____<br>Y _____<br>Z _____                      | BIOMED SW - RIGHT                                      |                                |
| 34:15 |                           | SCT STAR VISIBILITY<br>TRN BIAS                                      | MNVR TO SIGHTING ATT                                   |                                |
|       |                           | CISLUNAR NAVIGATION P23                                              |                                                        |                                |
| 34:30 | SLEEP PERIOD<br>(7 HOURS) | 1. STAR 16 EFH<br>STAR _____ E _____ H<br>1 SET                      |                                                        |                                |
|       |                           | 2. STAR 22 EFH<br>STAR _____ E _____ H<br>1 SET                      |                                                        |                                |
| 34:45 |                           | 3. STAR 26 ENH<br>STAR _____ E _____ H<br>1 SET                      |                                                        |                                |
|       |                           | GROUND TRACK DET P21                                                 | MNVR TO PTC ATT<br>P 331 Y 331<br>P Y<br>ROLL 0.1°/SEC | VOICE<br>UPDATE:<br>BLOCK DATA |
| 35:00 |                           | RECORD BLOCK DATA<br>(TLI + 35 HOURS, TLI +<br>44 HOURS, AND FLY BY) |                                                        |                                |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 34:00 - 35:00 | 2/TLC   | 2-26 |

MSC Form 1910 (Nov 68)

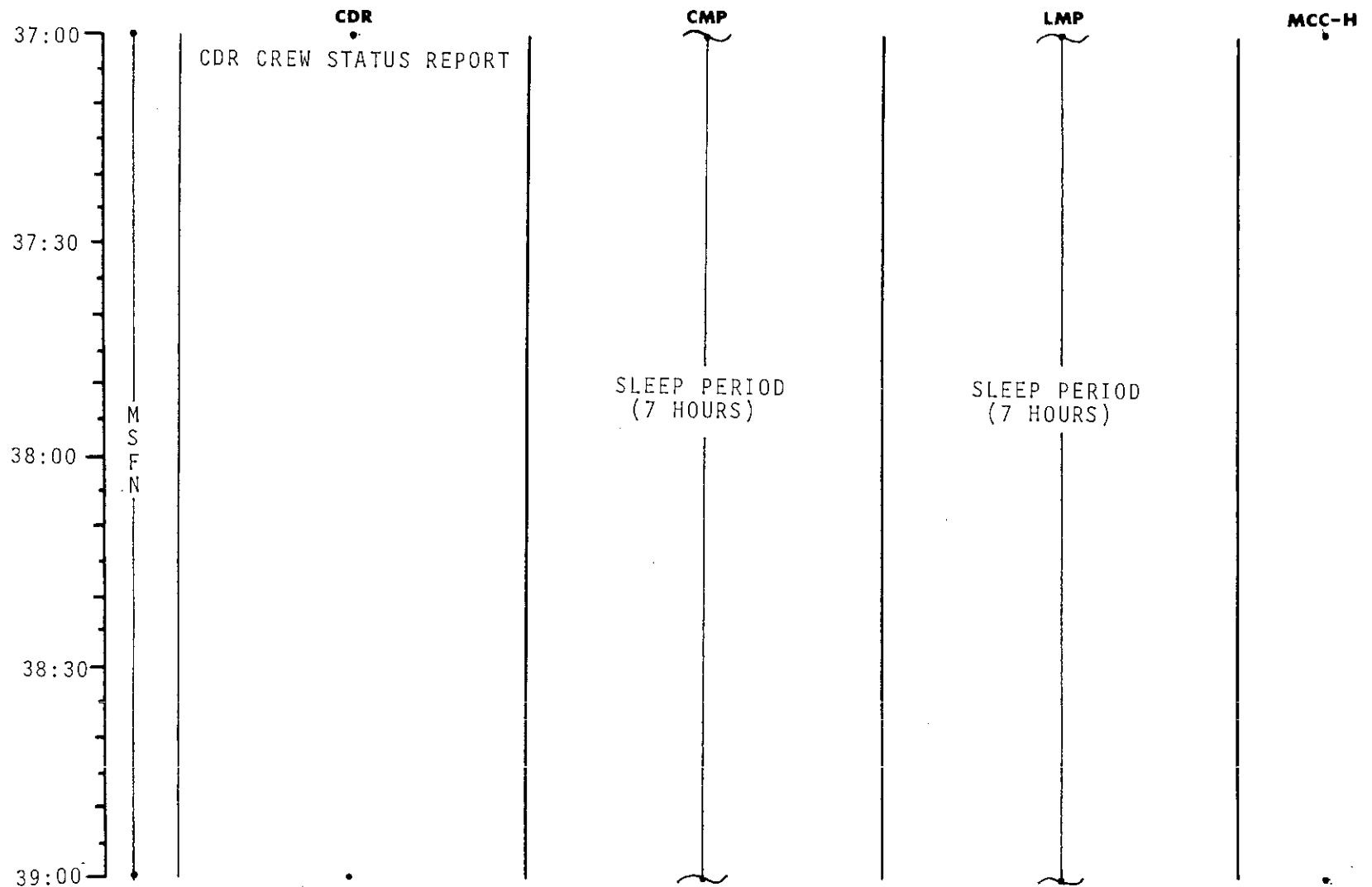
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 35:00 - 37:00 | 2/TLC   | 2-27 |

# FLIGHT PLAN



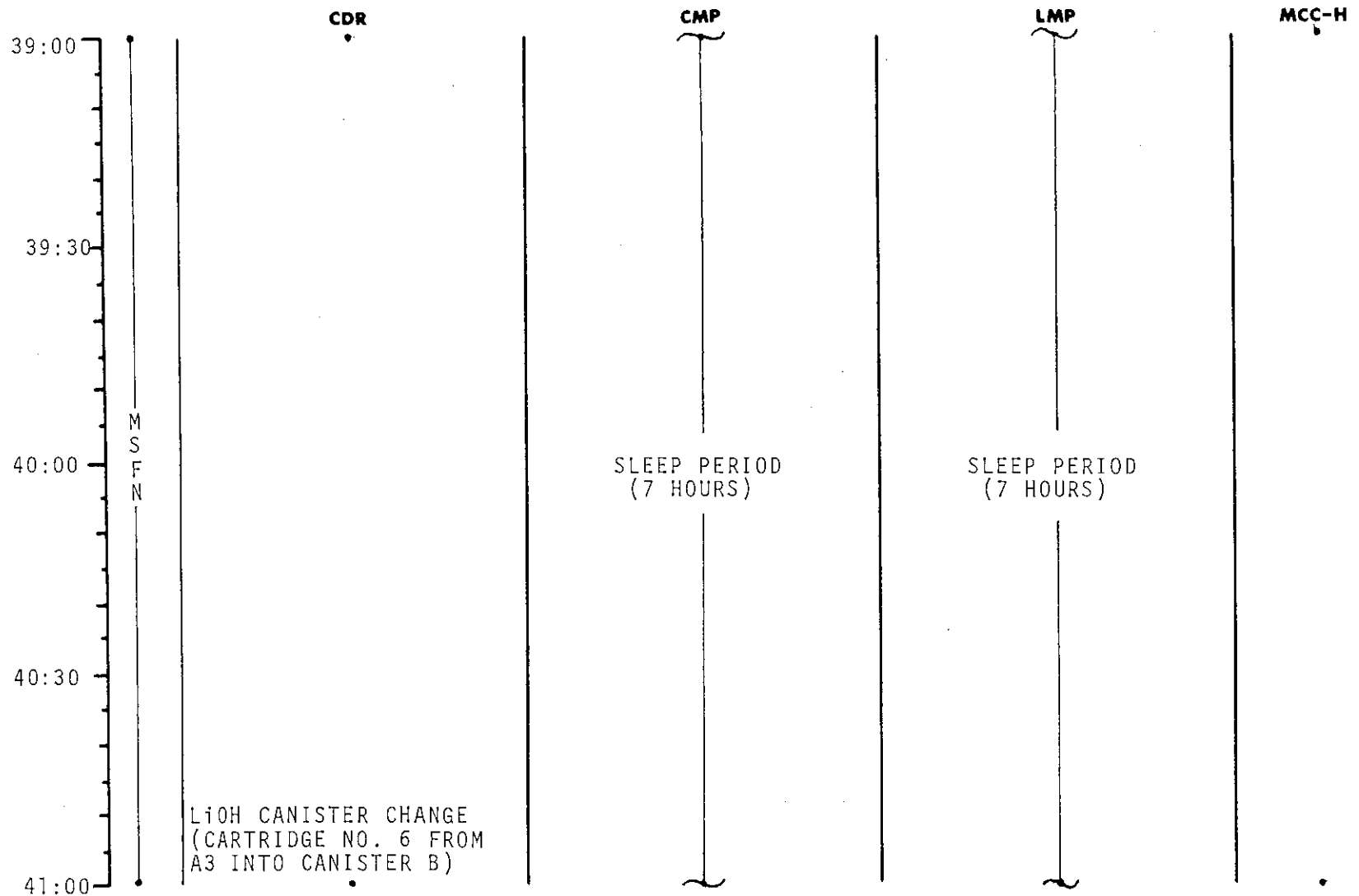
| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 37:00 - 39:00 | 2/TLC   | 2-28 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

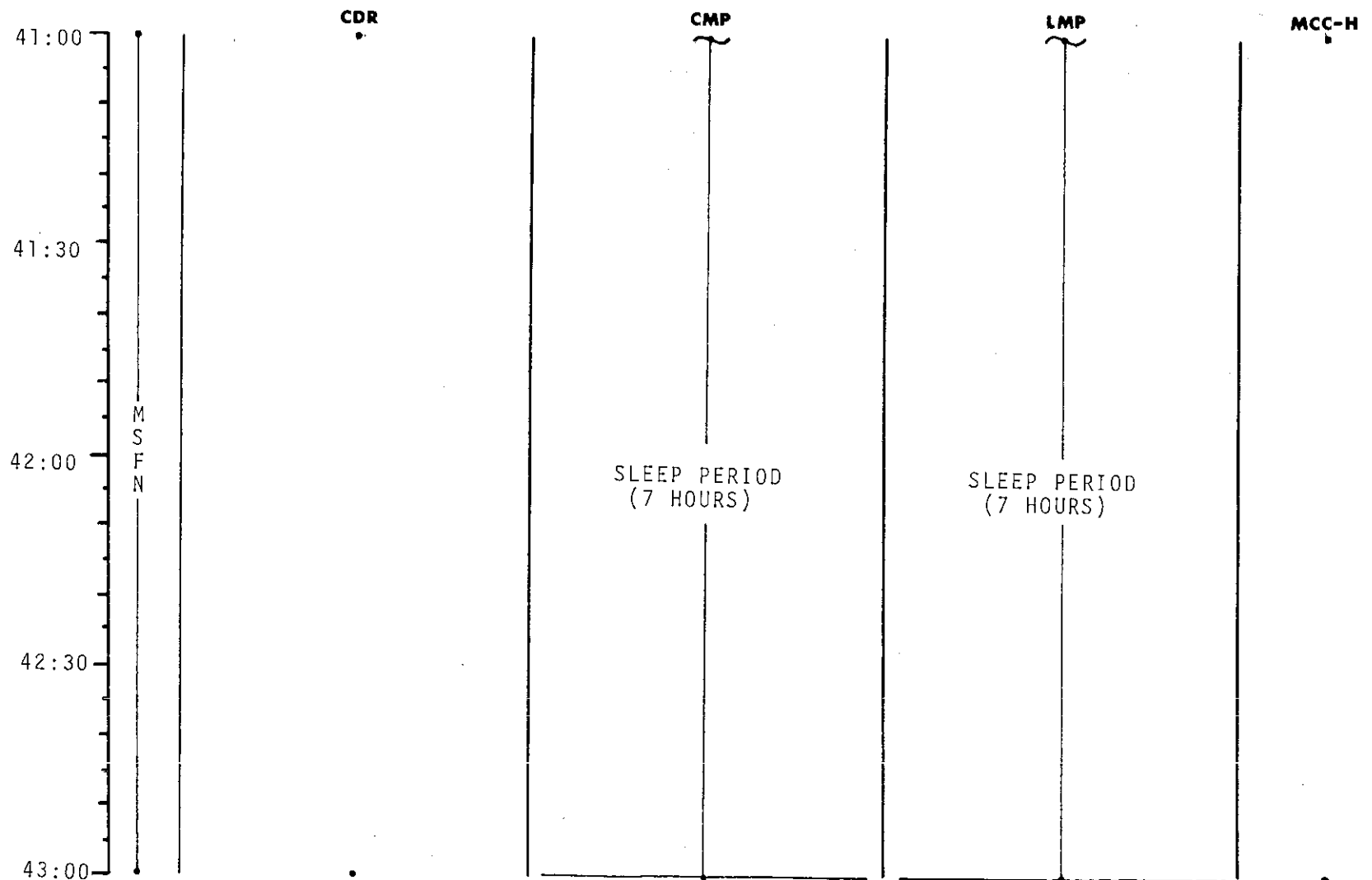


# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 39:00 - 41:00 | 2/TLC   | 2-29 |

# FLIGHT PLAN

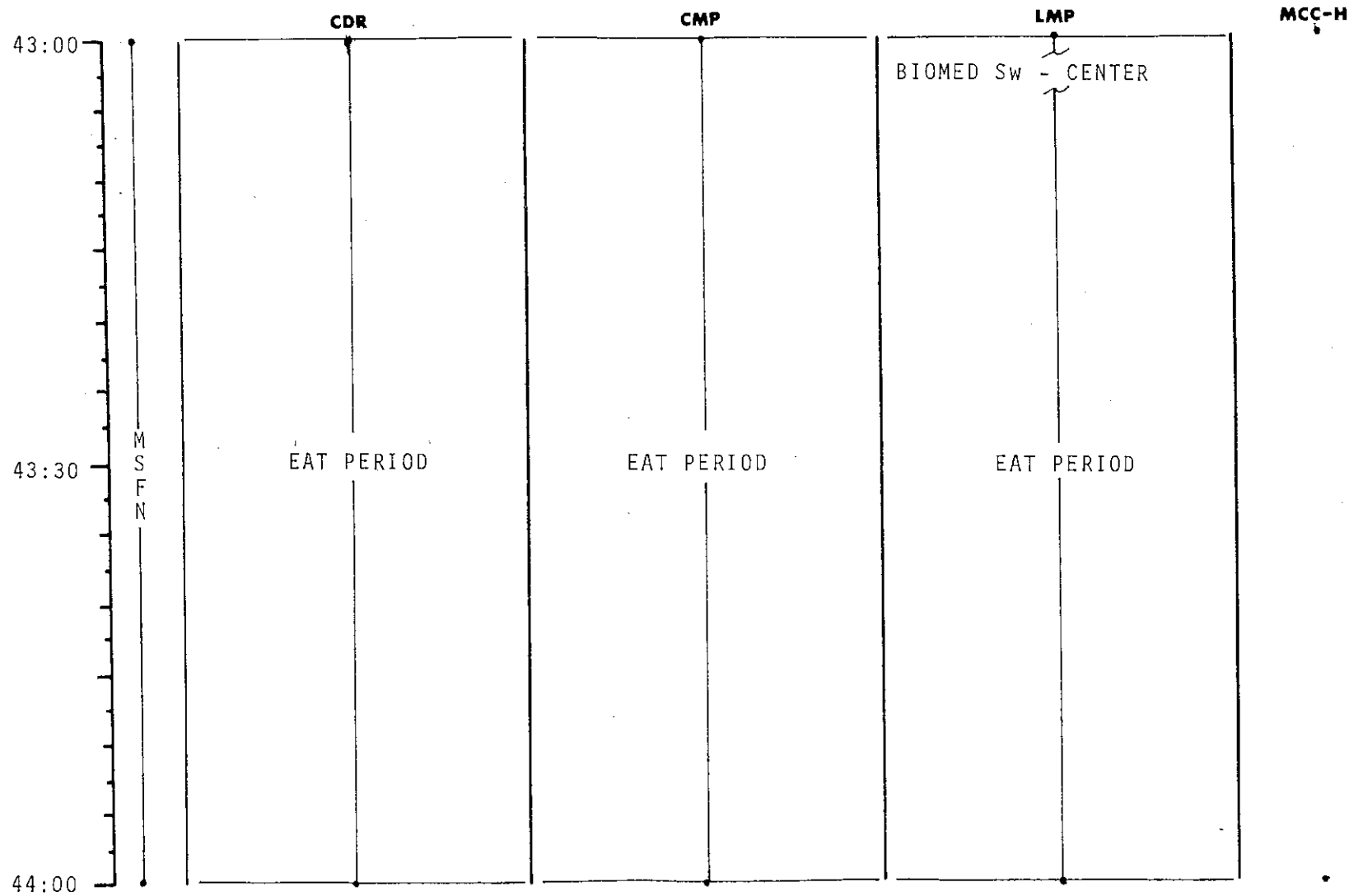


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 41:00 - 43:00 | 2/TLC   | 2-30 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 43:00 - 44:00 | 2/TLC   | 2-31 |

# FLIGHT PLAN

|       | CDR                                     | CMP                                                                                                                                           | LMP                           | MCC-H                          |
|-------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------|
| 44:00 |                                         | RECORD BLOCK DATA<br>(TLI + 44 HOURS, FLY BY,<br>PC + 2 (QUICK RETURN),<br>& PC + 2)                                                          | CMP/LMP CREW STATUS<br>REPORT | VOICE<br>UPDATE:<br>BLOCK DATA |
| 44:30 | M<br>S<br>F<br>N<br><br>MNVR TO P52 ATT | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID _____<br>STAR ANGLE DIFF _____<br><br>TORQUE ANGLES: _____<br>X _____<br>Y _____<br>Z _____ |                               |                                |
| 45:00 | GDC ALIGN TO IMU                        |                                                                                                                                               |                               |                                |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 44:00 - 45:00 | 2/TLC   | 2-32 |

# FLIGHT PLAN

|       | CDR                  | CMP                                        | LMP | MCC-H |
|-------|----------------------|--------------------------------------------|-----|-------|
| 45:00 | MNVR TO SIGHTING ATT | TRN BIAS                                   |     |       |
|       |                      | CISLUNAR NAVIGATION P23                    |     |       |
| 45:15 |                      | 1. STAR 33 LNH<br>STAR__ __ L__H<br>2 SETS |     |       |
|       |                      |                                            |     |       |
| 45:30 | M<br>S<br>F<br>N     | 2. STAR 37 LNH<br>STAR__ __ L__H<br>1 SET  |     |       |
|       |                      |                                            |     |       |
| 45:45 |                      | 3. STAR 45 LFH<br>STAR__ __ L__H<br>1 SET  |     |       |
|       |                      |                                            |     |       |
| 46:00 |                      | 4. STAR 42 LFH<br>STAR__ __ L__H<br>1 SET  |     |       |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 45:00 - 46:00 | 2/TLC   | 2-33 |

# BURN STATUS REPORT

|      |   |   |   |  |                 |
|------|---|---|---|--|-----------------|
| X    | X |   | • |  | ΔTIG            |
| X    | X |   | • |  | BT              |
|      |   |   | • |  | V <sub>gx</sub> |
| TRIM |   |   |   |  |                 |
| X    | X | X |   |  | R               |
| X    | X | X |   |  | P               |
| X    | X | X |   |  | Y               |
|      |   |   | • |  | V <sub>gx</sub> |
|      |   |   | • |  | V <sub>gy</sub> |
|      |   |   | • |  | V <sub>gz</sub> |
|      |   |   | • |  | ΔV <sub>c</sub> |
| X    | X | X |   |  | FUEL            |
| X    | X | X |   |  | OX              |
| X    | X | X |   |  | UNBALANCE       |

REMARKS:

MCC'S

## BURN CHART

|          | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME | RESIDUALS       |
|----------|---------------------|------------------|------------------|-----------------|
| MCC(ALL) | 10°/SEC<br>TAKEOVER | 10°<br>TAKEOVER  | B/T +1 SEC       | TRIM TO 0.2 fps |

2-33a

# FLIGHT PLAN

|                      | CDR                                                    | CMP                               | LMP               | MCC-H                                                                      |
|----------------------|--------------------------------------------------------|-----------------------------------|-------------------|----------------------------------------------------------------------------|
| 46:00                |                                                        |                                   | BIOMED Sw - RIGHT |                                                                            |
|                      |                                                        | GROUND TRACK DET P21              | RECORD MNVR PAD   | P27 UPDATE:<br>STATE<br>VECTOR<br>TGT LOAD<br>VOICE<br>UPDATE:<br>MNVR PAD |
| 46:15                | V47 TRANS LM STATE<br>VECTOR TO CSM SLOT<br>EXT ΔV P30 |                                   |                   |                                                                            |
| 46:30                | SPS/RCS THRUST P40/41<br><br>MNVR TO BURN ATT          | SXT STAR CK<br><br>TRANS TO COUCH |                   | PIPA BIAS<br>CK                                                            |
| 46:45                | EMS ΔV TEST                                            |                                   |                   |                                                                            |
| LOI -22 HRS<br>47:00 | GDC ALIGN<br><br>MCC <sub>3</sub> ΔV=NOMINALLY ZERO    | SM RCS MON CK                     |                   |                                                                            |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 46:00 - 47:00 | 2/TLC   | 2-34 |

# FLIGHT PLAN

|       | CDR                                                    | CMP                                    | LMP                               | MCC-H |
|-------|--------------------------------------------------------|----------------------------------------|-----------------------------------|-------|
| 47:00 | MNVR TO SIGHTING ATT                                   | SM RCS MON CK                          | SPS MON CK<br>INITIATE BAT CHARGE |       |
|       | V66 TRANS CSM STATE<br>VECTOR TO LM SLOT               |                                        |                                   |       |
| 47:15 | MCC <sub>3</sub> BURN STATUS REPORT                    | TRN BIAS                               |                                   |       |
|       |                                                        | <u>CISLUNAR NAVIGATION P23</u>         |                                   |       |
|       |                                                        | 1. STAR 16 EFH<br>STAR__ E__H<br>1 SET |                                   |       |
| 47:30 |                                                        | 2. STAR 22 EFH<br>STAR__ E__H<br>1 SET |                                   |       |
|       |                                                        | 3. STAR 26 ENH<br>STAR__ E__H<br>1 SET |                                   |       |
| 47:45 | MNVR TO PTC ATT<br>P 331 Y 331<br>P Y<br>ROLL 0.1°/SEC | GROUND TRACK DET P27                   |                                   |       |
| 48:00 |                                                        |                                        |                                   |       |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 47:00 - 48:00 | 2/TLC   | 2-35 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH



# FLIGHT PLAN

|       | CDR        | CMP        | LMP                | MCC-H |
|-------|------------|------------|--------------------|-------|
| 48:00 |            |            |                    |       |
| 48:30 |            |            |                    |       |
| 49:00 |            |            | BIOMED Sw - CENTER |       |
| 49:30 | EAT PERIOD | EAT PERIOD | EAT PERIOD         |       |
| 50:00 |            |            |                    |       |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 48:00 - 50:00 | 3/TLC   | 2-36 |

# FLIGHT PLAN

|       | CDR                    | CMP                                                                              | LMP                                          | MCC-H                          |
|-------|------------------------|----------------------------------------------------------------------------------|----------------------------------------------|--------------------------------|
| 50:00 |                        |                                                                                  |                                              |                                |
| 50:30 |                        |                                                                                  |                                              |                                |
| 51:00 |                        |                                                                                  |                                              |                                |
| 51:30 | MNVR TO P52 ATT        |                                                                                  | RECORD BLOCK DATA<br>(FLY BY & PC + 2 HOURS) | VOICE<br>UPDATE:<br>BLOCK DATA |
| 52:00 | CDR CREW STATUS REPORT | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID<br>STAR ANGLE DIFF<br>(cont'd) |                                              |                                |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 50:00 - 52:00 | 3/TLC   | 2-37 |

MSC Form 1910 (Nov 68)

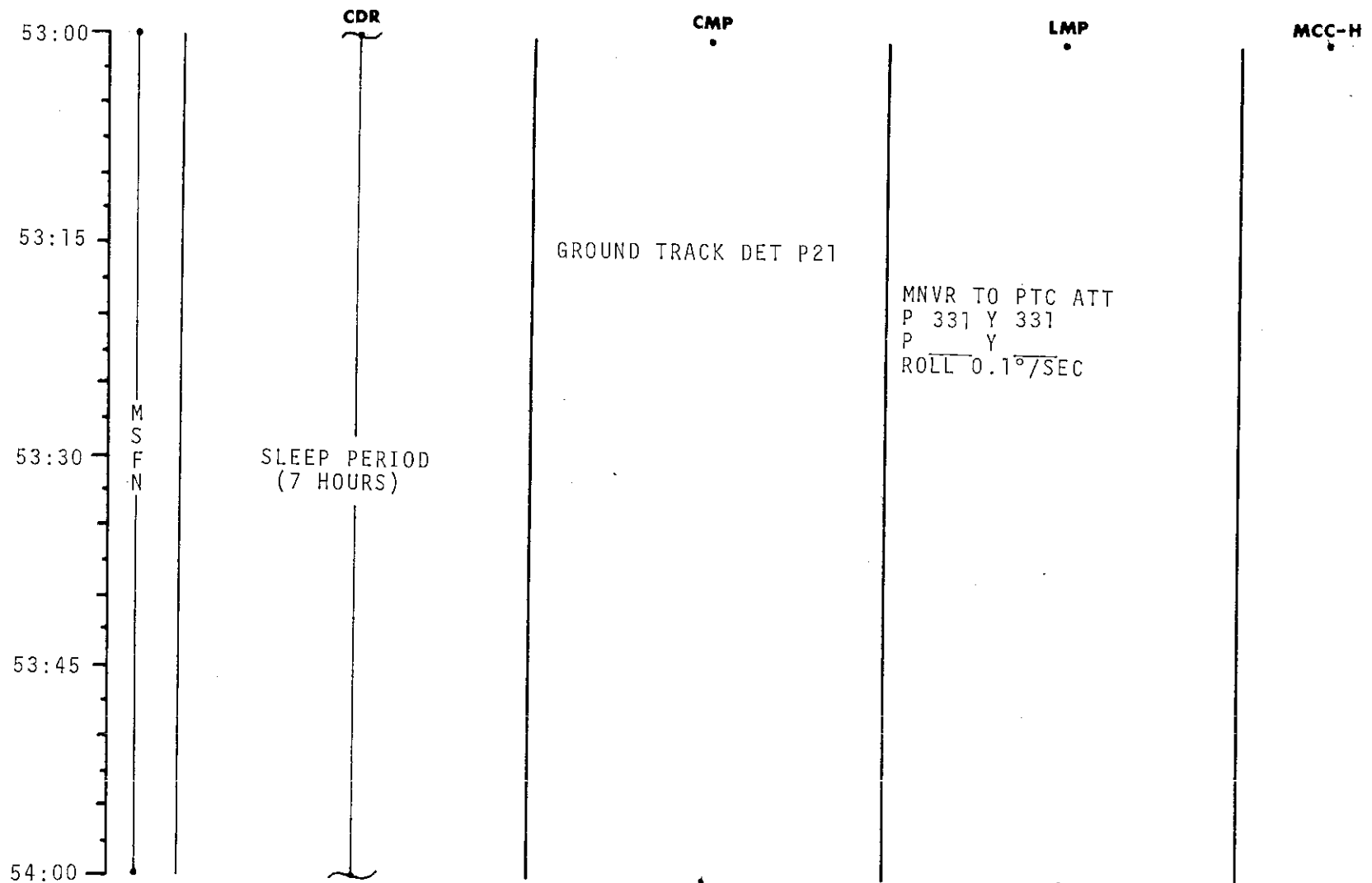
FLIGHT PLANNING BRANCH

# FLIGHT PLAN

|       | CDR                       | CMP                                              | LMP                                                           | MCC-H |
|-------|---------------------------|--------------------------------------------------|---------------------------------------------------------------|-------|
| 52:00 | SLEEP PERIOD<br>(7 HOURS) | TORQUE ANGLES:<br>X _____<br>Y _____<br>Z _____  | BIOMED Sw - RIGHT                                             |       |
| 52:15 |                           | TRN BIAS<br>CISLUNAR NAVIGATION P23              | TRANS TO LH COUCH<br>MNVR TO SIGHTING ATT<br>GDC ALIGN TO IMU |       |
| 52:30 |                           | 1. STAR 37 LNH<br>STAR _____ L _____ H<br>2 SETS |                                                               |       |
| 52:45 |                           | 2. STAR 33 LNH<br>STAR _____ L _____ H<br>3 SETS |                                                               |       |
| 53:00 |                           |                                                  |                                                               |       |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 52:00 - 53:00 | 3/TLC   | 2-38 |

# FLIGHT PLAN

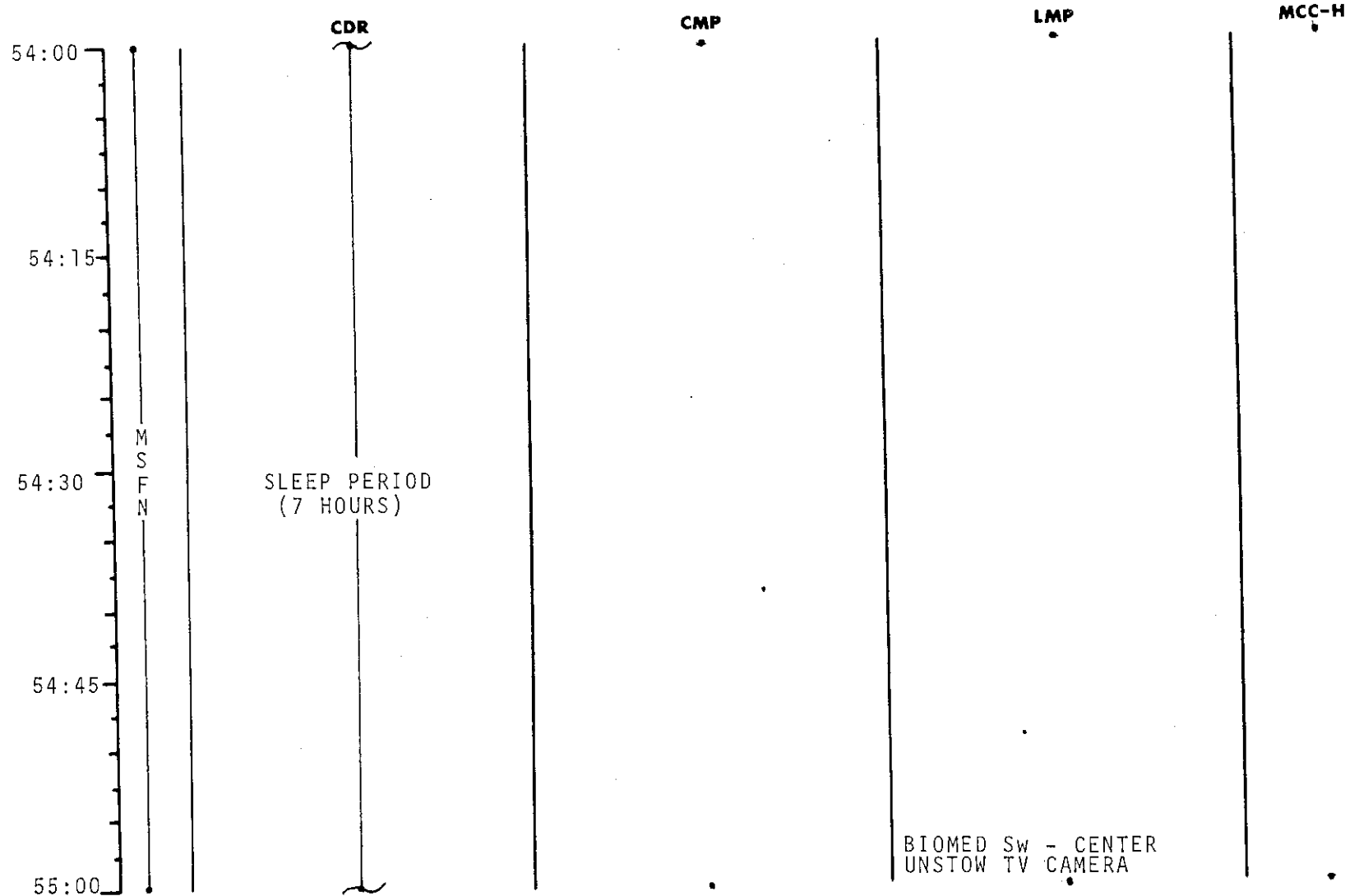


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 53:00 - 54:00 | 3/TLC   | 2-39 |

MSC Form 1910 (Nov 68)

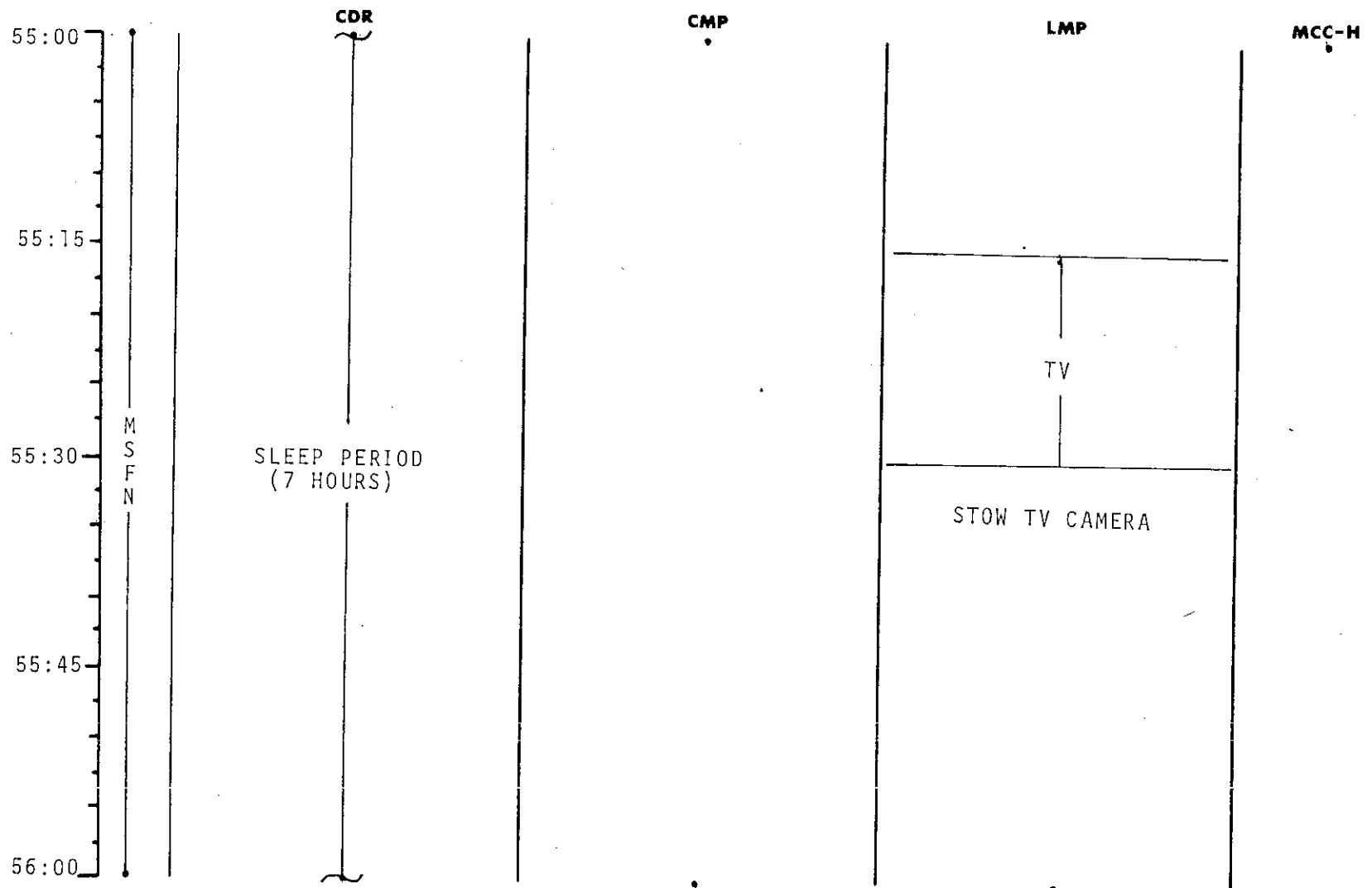
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 54:00 - 55:00 | 3/TLC   | 2-40 |

# FLIGHT PLAN

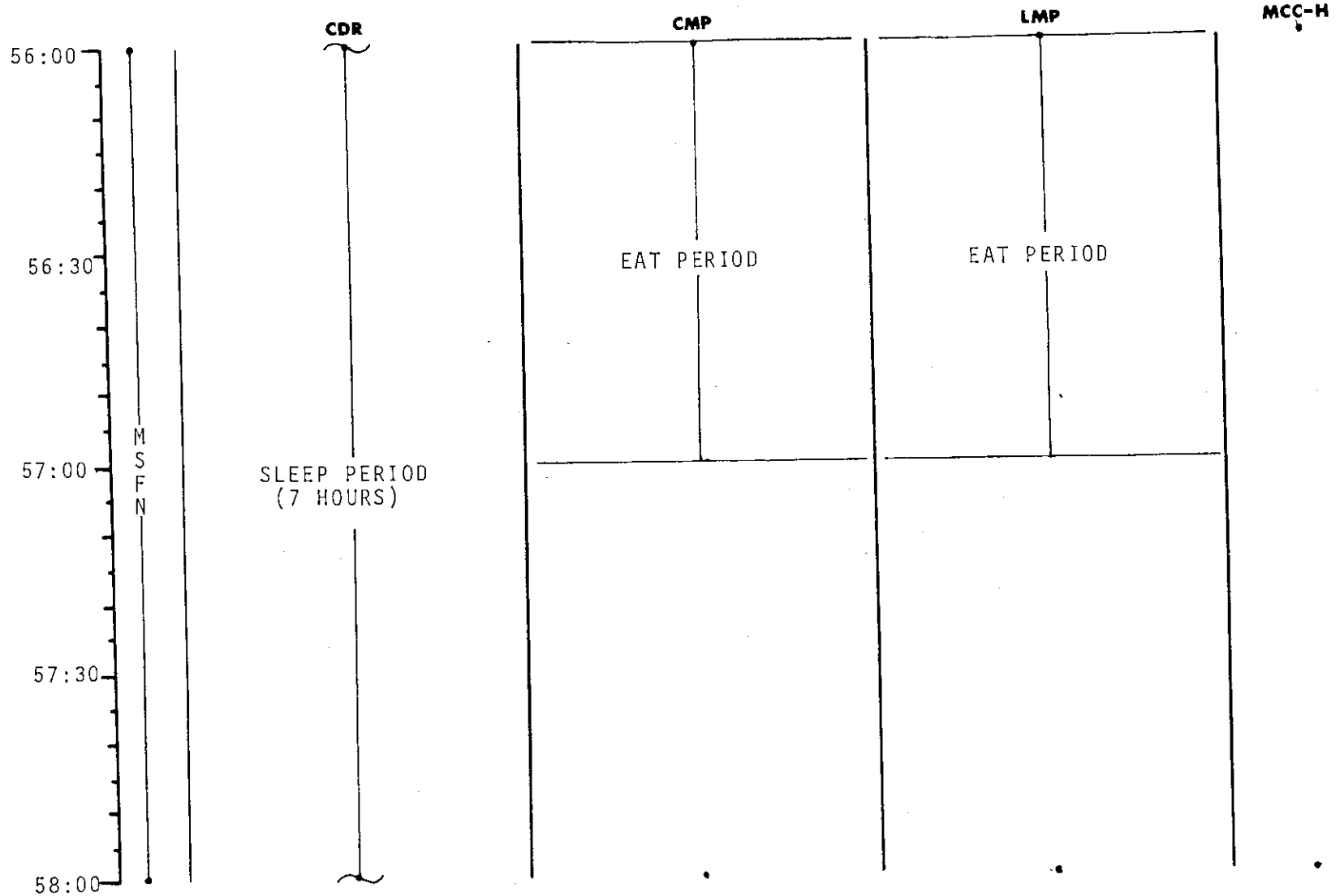


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 55:00 - 56:00 | 3/TLC   | 2-41 |

MSC Form 1910 (Nov 68)

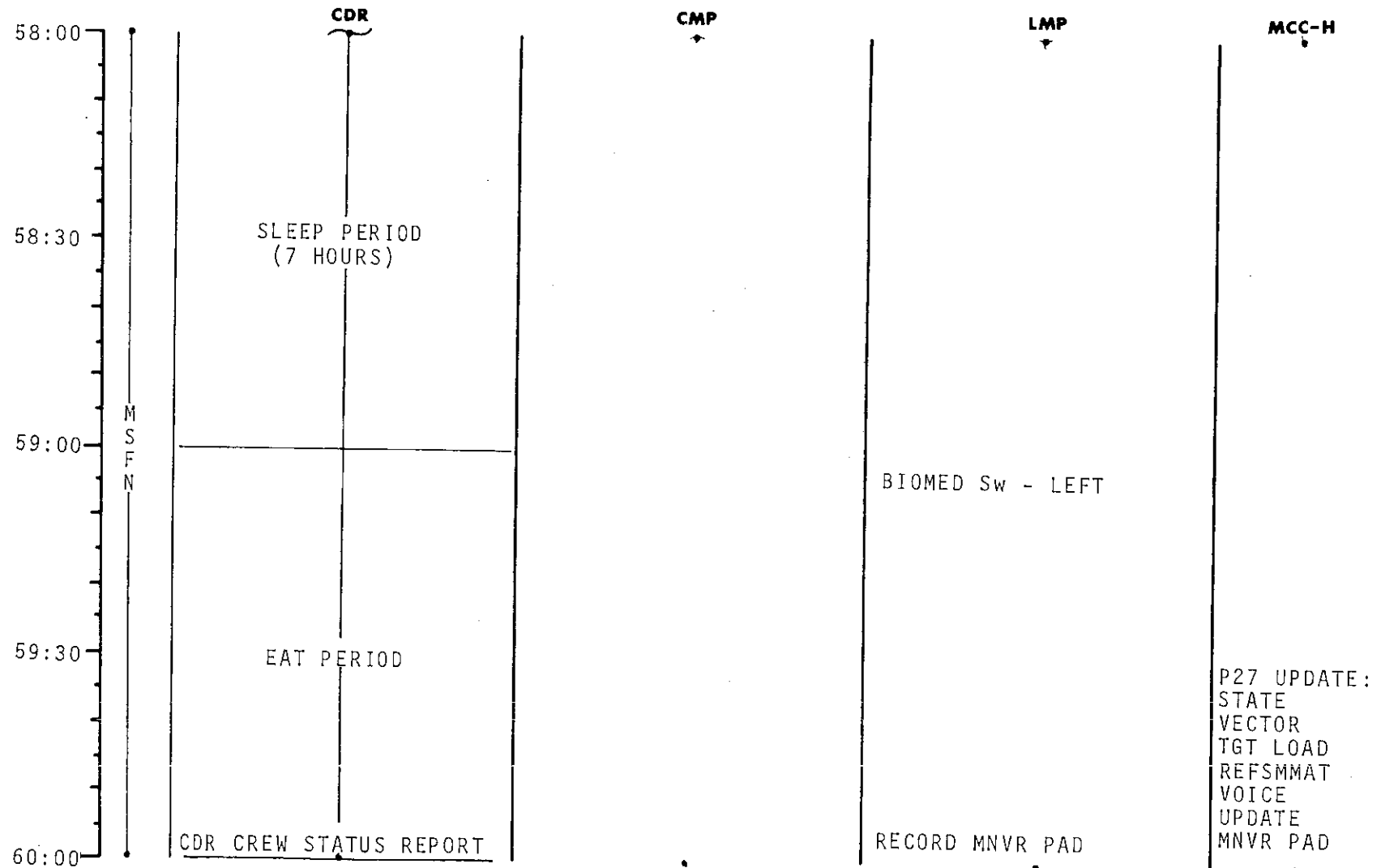
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 56:00 - 58:00 | 3/TLC   | 2-42 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 58:00 - 60:00 | 3/TLC   | 2-43 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH



# BURN STATUS REPORT

X X ☐ : ΔTIG  
 X X : BT  
☐ : V<sub>gx</sub>

TRIM  
 X X X R  
 X X X P  
 X X X Y

☐ V<sub>gx</sub>  
☐ V<sub>gy</sub>  
☐ V<sub>gz</sub>  
☐ ΔV<sub>c</sub>

X X X FUEL

X X X OX

X X X UNBALANCE

REMARKS:

2-43a

## MCC'S

### BURN CHART

|          | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME | RESIDUALS       |
|----------|---------------------|------------------|------------------|-----------------|
| MCC(ALL) | 10°/SEC<br>TAKEOVER | 10°<br>TAKEOVER  | B/T +1 SEC       | TRIM TO 0.2 fns |

# FLIGHT PLAN

|       | CDR                                                      | CMP                                                                               | LMP | MCC-H        |
|-------|----------------------------------------------------------|-----------------------------------------------------------------------------------|-----|--------------|
| 60:00 | MNVR TO P52 ATT                                          | IMU REALIGN P52<br>OPTION 1 - PREFERRED<br>STAR ID _____<br>STAR ANGLE DIFF _____ |     |              |
| 60:15 |                                                          | TORQUE ANGLES:<br>X _____<br>Y _____<br>Z _____                                   |     |              |
| 60:30 | EXT ΔV P30<br>SPS/RCS THRUST P40/41<br>MNVR TO BURN ATT  | SXT STAR CK                                                                       |     | PIPA BIAS CK |
| 60:45 | EMS ΔV TEST                                              | TRANS TO COUCH                                                                    |     |              |
| 61:00 | GDC ALIGN TO IMU<br>MCC <sub>4</sub> ΔV = NOMINALLY ZERO | SM RCS MON CK                                                                     |     |              |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 60:00 - 61:00 | 3/TLC   | 2-44 |

MSC Form 1910 (Nov 68)

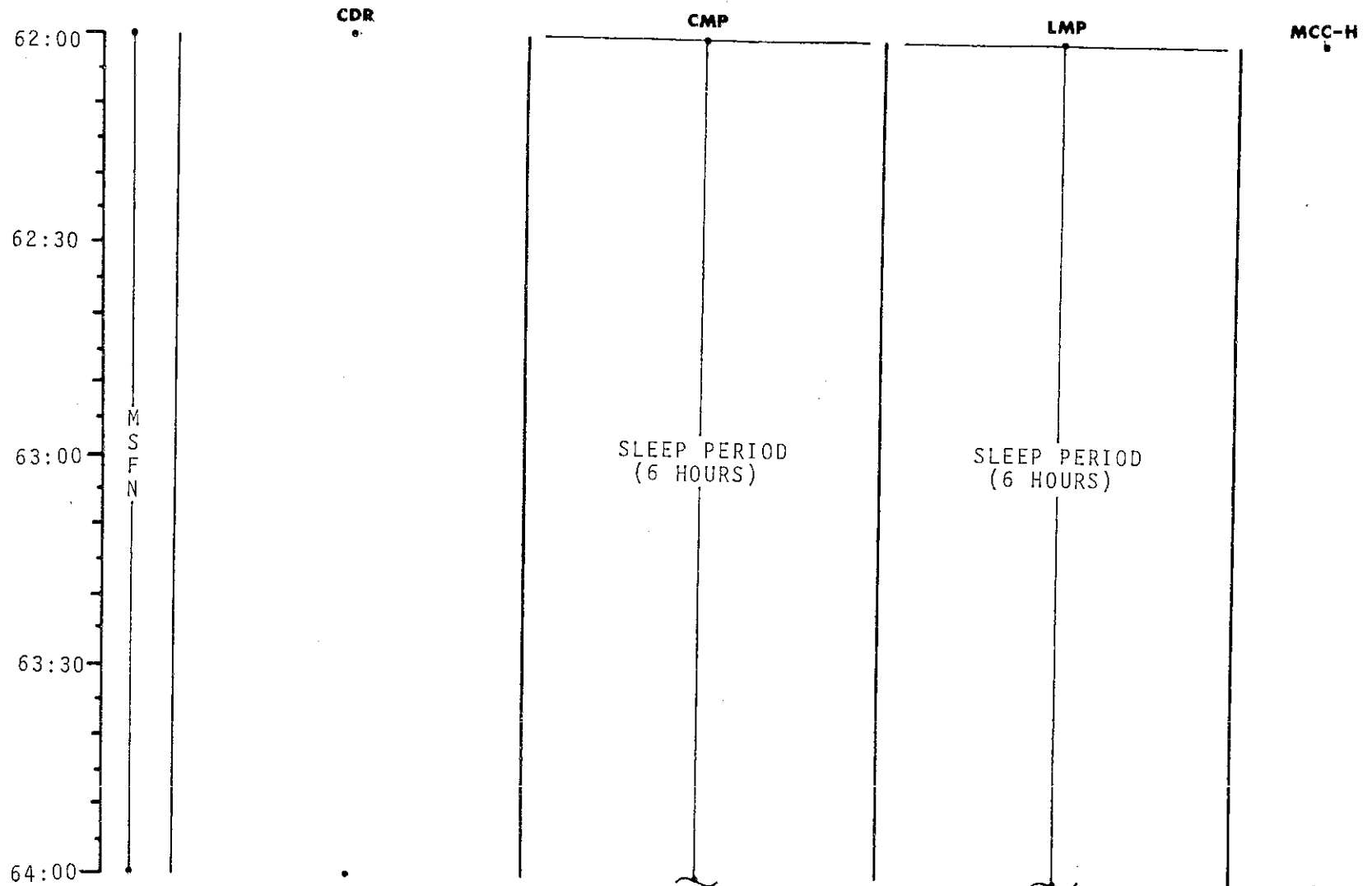
FLIGHT PLANNING BRANCH

# FLIGHT PLAN

|       | CDR                                                     | CMP                                                        | LMP                                                        | MCC-H |
|-------|---------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|-------|
| 61:00 | V66 TRANS CSM STATE<br>VECTOR TO LM SLOT                | SM RCS MON CK<br>GROUND TRACK DET P21<br>(STATE VECTOR CK) | SPS MON CK<br>INITIATE BAT CHARGE<br>ECS REDUNDANT COMP CK |       |
| 61:15 | MNVR TO PTC ATT<br>P 122 Y 315<br>P Y<br>ROLL 0.1 °/SEC |                                                            |                                                            |       |
| 61:30 | MSFN                                                    |                                                            |                                                            |       |
| 61:45 |                                                         |                                                            |                                                            |       |
| 62:00 |                                                         |                                                            | CMP/LMP CREW STATUS<br>REPORT                              |       |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 61:00 - 62:00 | 3/TLC   | 2-45 |

# FLIGHT PLAN

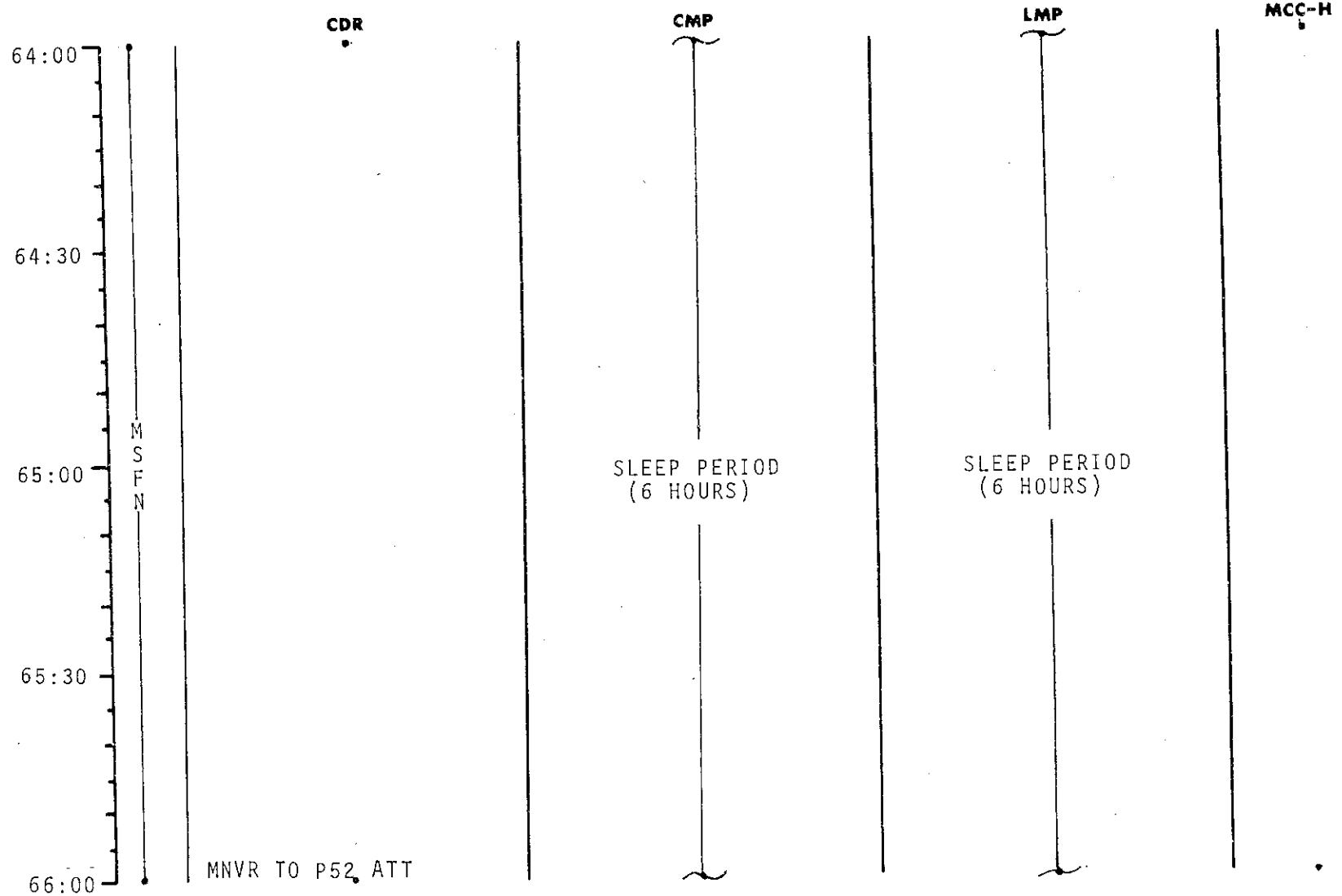


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 62:00 - 64:00 | 3/TLC   | 2-46 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 64:00 - 66:00 | 3/TLC   | 2-47 |

# FLIGHT PLAN

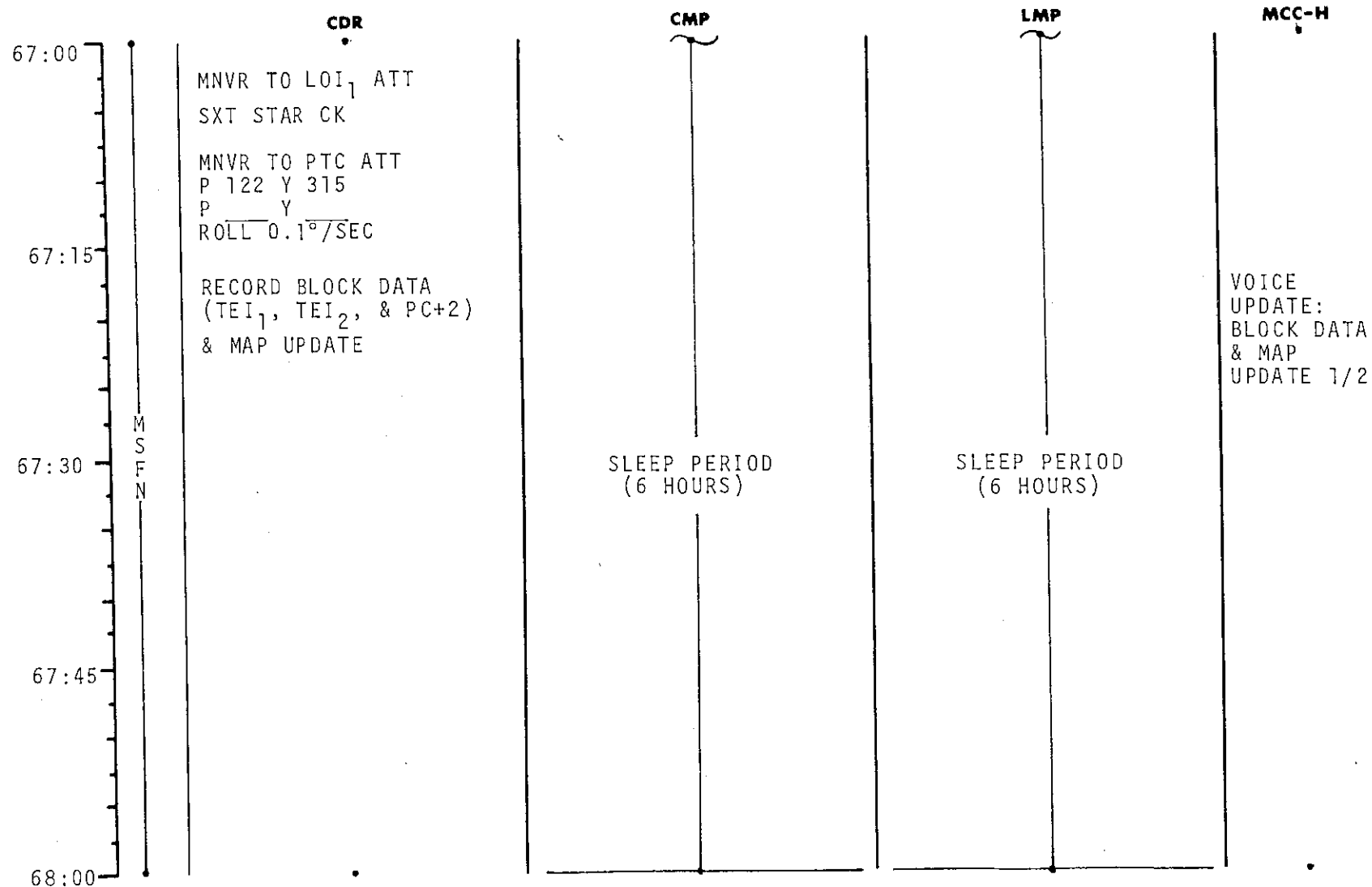
|       | CDR                                                                                                                                                                                                                                                                        | CMP                       | LMP                       | MCC-H                        |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|------------------------------|
| 66:00 | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID _____<br>STAR ANGLE DIFF _____<br><br>TORQUE ANGLES: _____<br>X _____<br>Y _____<br>Z _____<br><br>GDC ALIGN TO IMU<br>MNVR TO PTC ATT<br>P 122 Y 315<br>P _____ Y _____<br>ROLL 0.1°/SEC<br><br>RECORD MNVR PAD FOR LOI | SLEEP PERIOD<br>(6 HOURS) | SLEEP PERIOD<br>(6 HOURS) | VOICE<br>UPDATE:<br>MNVR PAD |
| 66:15 |                                                                                                                                                                                                                                                                            |                           |                           |                              |
| 66:30 |                                                                                                                                                                                                                                                                            |                           |                           |                              |
| 66:45 |                                                                                                                                                                                                                                                                            |                           |                           |                              |
| 67:00 |                                                                                                                                                                                                                                                                            |                           |                           |                              |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 66:00 - 67:00 | 3/TLC   | 2-48 |

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FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 67:00 - 68:00 | 3/TLC   | 2-49 |

# FLIGHT PLAN

|       | CDR                                                             | CMP                                                                                        | LMP                                                                                                                              | MCC-H                                      |
|-------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 68:00 | GO/NO GO FOR LOI <sub>1</sub>                                   |                                                                                            | CMP/LMP CREW STATUS REPORT                                                                                                       | GO/NO GO                                   |
|       |                                                                 | GROUND TRACK DET - P21<br>(LOI ALTITUDE DET)                                               | RECORD MANEUVER PAD                                                                                                              | P27 UPDATE:<br>STATE VECTOR<br>TARGET LOAD |
|       | MNVR TO P52 ATT                                                 | IMU REALIGN P52<br>OPTION 3 - REFMMAT AND<br>GYRO DRIFT TEST<br>STAR ID<br>STAR ANGLE DIFF |                                                                                                                                  | VOICE UPDATE:<br>MNVR PAD                  |
| 68:30 | EXTERNAL ΔV P30<br>EMS CK<br>SPS THRUST P40<br>MNVR TO BURN ATT | TORQUE ANGLES:<br>X — — — —<br>Y — — — —<br>Z — — — —                                      | PRE LOI SYSTEMS CKS:<br>C&W CK<br>CM RCS CK<br>SM RCS CK<br>SPS PERIODIC MONITOR<br>EPS PERIODIC MONITOR<br>ECS PERIODIC MONITOR |                                            |
|       | GDC ALIGN TO IMU                                                | SXT STAR CK<br>TRANSFER TO COUCH                                                           |                                                                                                                                  |                                            |
| 68:57 |                                                                 |                                                                                            |                                                                                                                                  |                                            |
| 69:00 |                                                                 |                                                                                            |                                                                                                                                  |                                            |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 68:00 - 69:00 | 3/TLC   | 2-50 |

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FLIGHT PLANNING BRANCH



# BURN STATUS REPORT

X X ☐ :  $\Delta TIG$   
 X X : BT  
☐ .  $V_{gx}$

## TRIM

X X X R  
 X X X P  
 X X X Y  
☐  $V_{gx}$   
☐  $V_{gy}$   
☐  $V_{gz}$   
☐  $\Delta V_c$   
 X X X FUEL  
 X X X OX  
 X X X UNBALANCE

REMARKS:

# LOI<sub>1</sub> BURN CHART

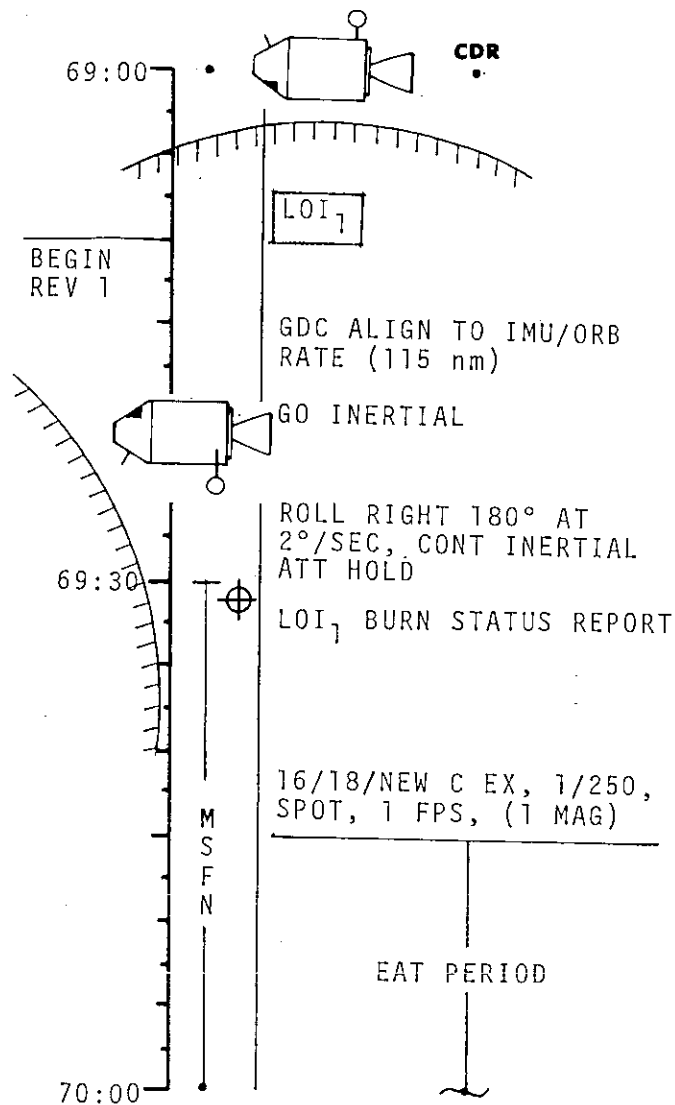
|                  | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME | RESIDUALS |
|------------------|---------------------|------------------|------------------|-----------|
| LOI <sub>1</sub> | 10°/SEC<br>TAKEOVER | 10°<br>TAKEOVER  | B/T+6 SEC        | NO TRIM   |

## LOI<sub>1</sub> ABORT MODES

| LOI $V_{go}$ | B/T       | TRAJECTORY  | ABORT MODE                    |
|--------------|-----------|-------------|-------------------------------|
| 3050-2100    | 0 -1:20   | HYPERBOLIC  | COAST OUT OF SPHERE-P37       |
| 2100-1650    | 1:20-2:00 | UNSTABLE    | 5 HR COAST. MODE I ABORT      |
| 1650-0       | 2:00-4:06 | LUNAR ORBIT | MODE III ABORT AFTER<br>1 REV |

2-50a

# FLIGHT PLAN



SM RCS MONITOR CK

GETI = 69:07:30  
 $\Delta V$  = 2991 fps  
 BT = 4 MIN 5.8 SEC

V66 TRANS CSM STATE VECTOR TO LM SLOT

SM RCS MONITOR CK

2/80/B3 + 1/150/A8, R&B FILTER/U4, SPOT/R13, INT/U4, 2 B&W FILM PACKS 1A8

UNSTOW ORBITAL CHARTS, 16/18/NEW C-EX/CABLE/BRKT/U3

2/80/NEW B&W

EAT PERIOD

EAT PERIOD

LMP  
 BIOMED SW - CENTER

SPS MONITOR CK

INITIATE BAT CHARGE  
 CAMERA PREP

V64 ACQUIRE MSFN ON HGA

2/150/NEW B&W 1/250

1/80/B&W, 1/250

EAT PERIOD

MCC-H

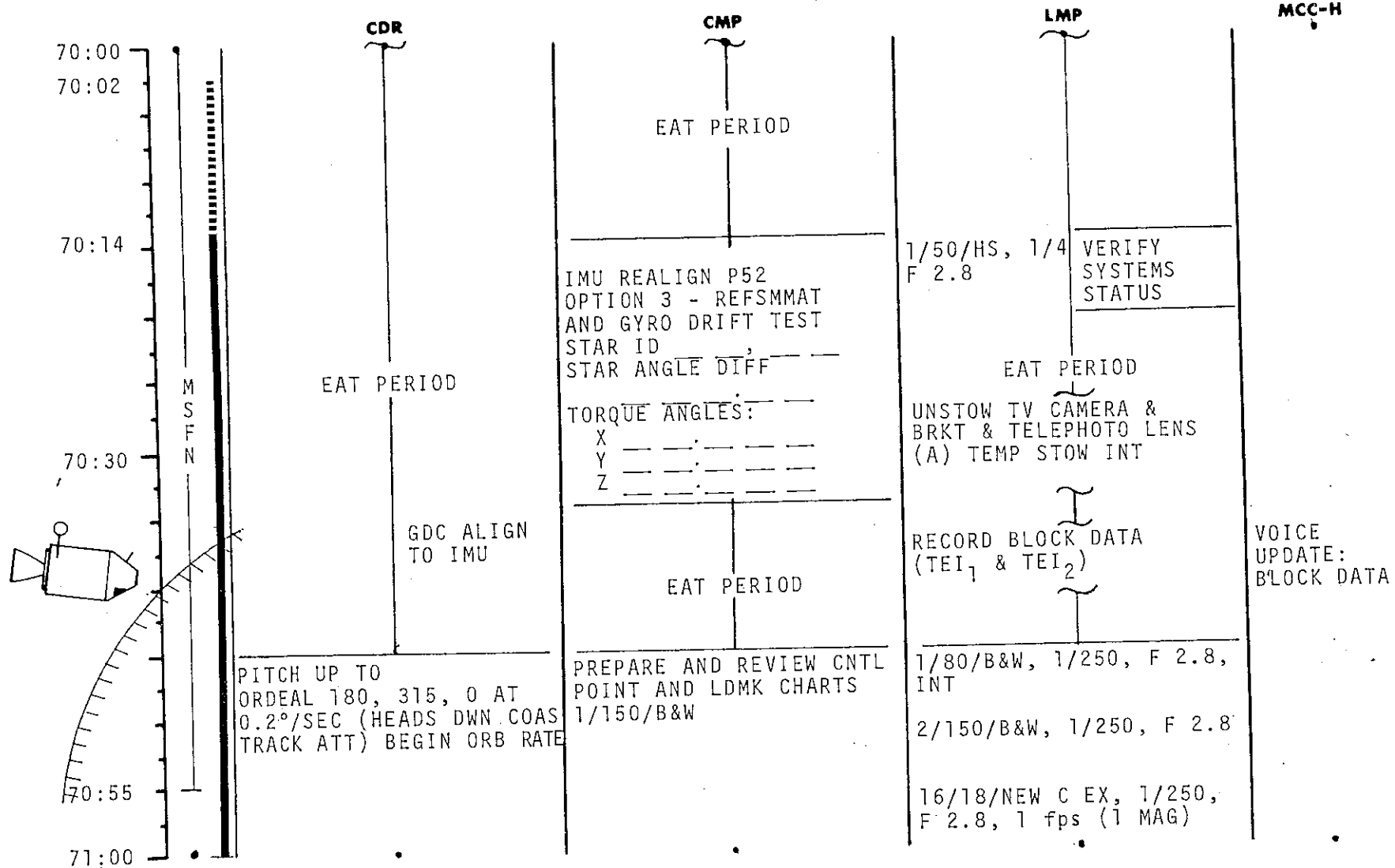
ACQUIRE S/C

DUMP DSE

PIPA BIAS CK

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 69:00 - 70:00 | 3/LPO   | 2-51 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 70:00 - 71:00 | 3/LPO   | 2-52 |

# FLIGHT PLAN

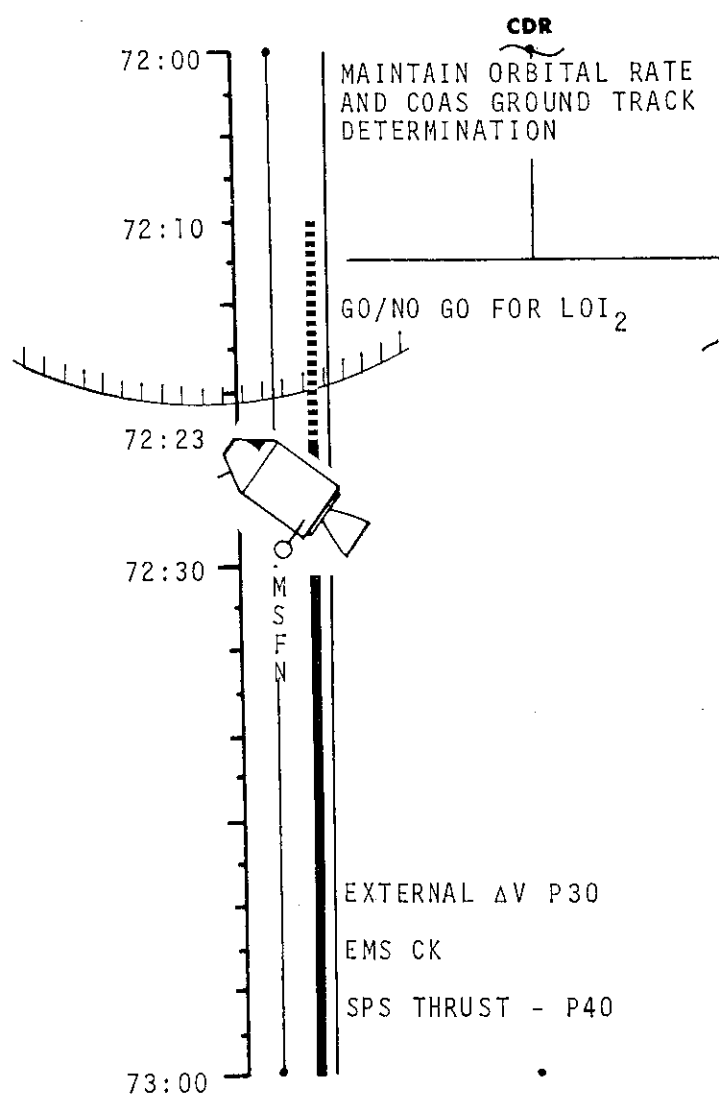
|                | CDR                                                               | CMP                                                                                                  | LMP                                                                                                        | MCC-H                                                        |
|----------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 71:00          | MAINTAIN ORB RATE AND<br>BEGIN COAS GROUND TRACK<br>DETERMINATION | CONTROL PT (CP)<br>SIGHTING (THRU<br>HATCH WINDOW)                                                   | START 16 MM CAMERA<br>AT TERMINATOR<br><br>SPOT METER READINGS<br><br>PHOTOGRAPH TARGETS<br>OF OPPORTUNITY |                                                              |
| 71:09          |                                                                   | CP1 7 1: 1 3: 4 6<br>TCA : : : :<br>CP1 NO. 504<br>LAT -05.250°S<br>LONG -162.700°W<br>ALT 000.00 nm |                                                                                                            |                                                              |
| BEGIN<br>REV 2 |                                                                   | CP2 7 1: 2 7: 4 0<br>TCA : : : :<br>CP2 NO. 526<br>LAT -10.200°S<br>LONG +155.100°E<br>ALT 000.00 nm |                                                                                                            |                                                              |
| 71:30          | YAW RT 45° FOR TV<br>ORDEAL 180, 315, 45                          |                                                                                                      | CONFIGURE FOR TV                                                                                           |                                                              |
| 71:38          | TV<br><br>YAW LEFT 45° TO<br>ORDEAL 180, 315, 0                   | CP3 7 1: 4 6: 3 8<br>TCA : : : :<br>CP3 NO. 334<br>LAT -09.100°S<br>LONG +95.900°E<br>ALT 000.00 nm  | V64 ACQUIRE MSFN ON HGA                                                                                    | ACQUIRE S/C<br>P27 UPDATE:<br>STATE<br>VECTOR<br>TARGET LOAD |
| 72:00          |                                                                   | PSEUDO LDG SITE LDMK<br>SIGHTING (THRU HATCH<br>WINDOW)                                              | RECORD MAP UPDATE                                                                                          | VOICE<br>UPDATE:<br>MAP UPDATE<br>2/3                        |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 71:00 - 72:00 | 3/LPO   | 2-53 |

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FLIGHT PLANNING BRANCH

# FLIGHT PLAN



**CMP**

B-1 | 7 2 : 0 9 : 1 0  
TCA | : : : : : :  
LDMK NO. B1  
LAT +02.675°N  
LONG +35.025°E  
ALT -000.99 nm

LiOH CANISTER CHANGE  
(CARTRIDGE 9 FROM B5  
INTO CANISTER A)

IMU REALIGN P52  
OPTION 3 - REFSMMAT  
AND GYRO DRIFT TEST  
STAR ID : :  
STAR ANGLE DIFF : : : : :

TORQUE ANGLES: : : : :  
X : : : : :  
Y : : : : :  
Z : : : : :

**LMP**

BIOMED Sw - LEFT  
STOP 16 MM CAMERA  
RECORD MNVR PAD, BLOCK  
DATA (TEI<sub>3</sub>, TEI<sub>3</sub> NO  
LOI<sub>2</sub>).

1/80/HS, 1/4, F 2.8

PRE LOI SYSTEMS CKS:

C&W CK  
CM RCS CK  
SPS PERIODIC MONITOR  
EPS PERIODIC MONITOR  
ECS PERIODIC MONITOR

**MCC-H**

GO/NO GO  
VOICE  
UPDATE:  
MNVR PAD  
BLOCK DATA

DUMP DSE

PIPA BIAS  
CK

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 72:00 - 73:00 | 4/LPO   | 2-54 |

# BURN STATUS REPORT

|   |   |   |   |   |   |                 |
|---|---|---|---|---|---|-----------------|
| X | X |   | . | . | . | ΔTIG            |
| X | X |   | . | . | . | BT              |
|   |   |   |   |   |   | V <sub>gx</sub> |
|   |   |   |   |   |   |                 |
|   |   |   |   |   |   | TRIM            |
| X | X | X |   |   |   | R               |
| X | X | X |   |   |   | P               |
| X | X | X |   |   |   | Y               |
|   |   |   |   |   | . | V <sub>gx</sub> |
|   |   |   |   |   | . | V <sub>gy</sub> |
|   |   |   |   |   | . | V <sub>gz</sub> |
|   |   |   |   |   | . | ΔV <sub>c</sub> |
| X | X | X |   |   |   | FUEL            |
| X | X | X |   |   |   | OX              |
| X | X | X |   |   |   | UNBALANCE       |

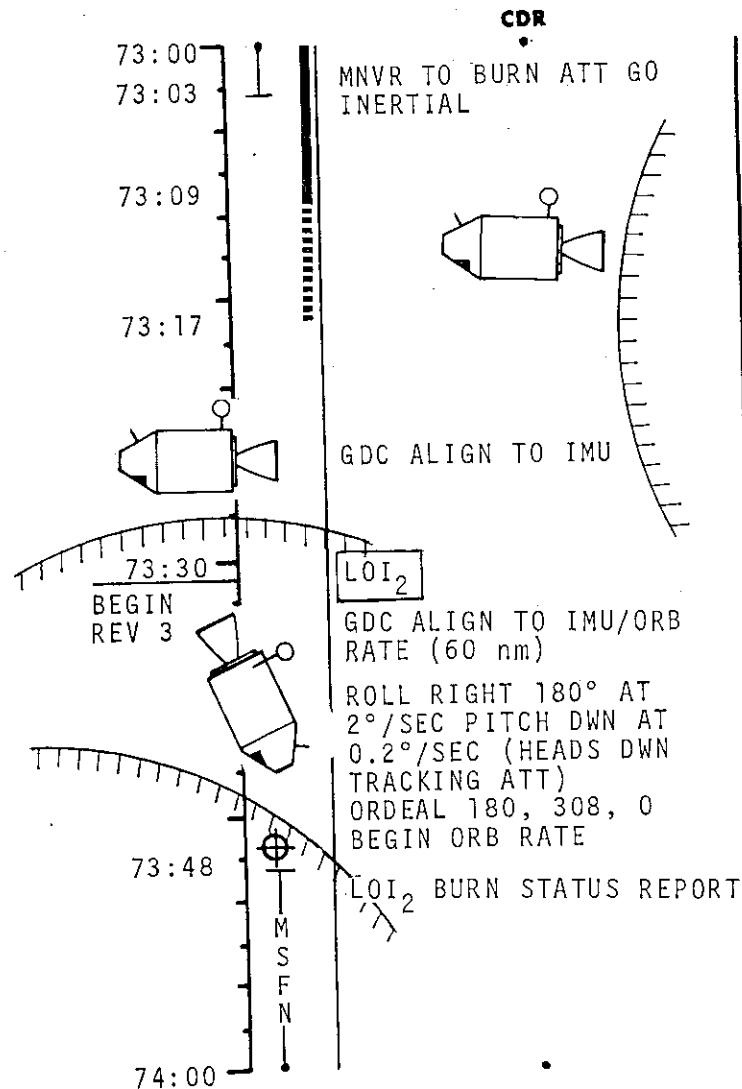
REMARKS:

## LOI<sub>2</sub> BURN CHART

|                  | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME | RESIDUALS |
|------------------|---------------------|------------------|------------------|-----------|
| LOI <sub>2</sub> | 10°/SEC<br>TAKEOVER | 10°<br>TAKEOVER  | B/T +1 SEC       | NO TRIM   |

2-54a

# FLIGHT PLAN



**CMP**

SXT STAR CK

TRANSFER TO COUCH

SM RCS MONITOR CK

GETI = 73:30:54  
ΔV = 138.5 fps  
BT = 9.7 SEC

V66 TRANS CSM STATE VECTOR TO LM SLOT

SM RCS MONITOR CK

REST PERIOD  
(2 HOURS)

**LMP**

2/80/B&W, 1/250  
16/18/C EX, 1/250  
F CHART, 6 fps, BRACK  
1/150/B&W, 1/60,  
CHART, BRKT  
PRE LOI SYSTEMS CKS

SPS MONITOR CK

INITIATE BAT CHARGE

V64 ACQUIRE MSFN ON HGA

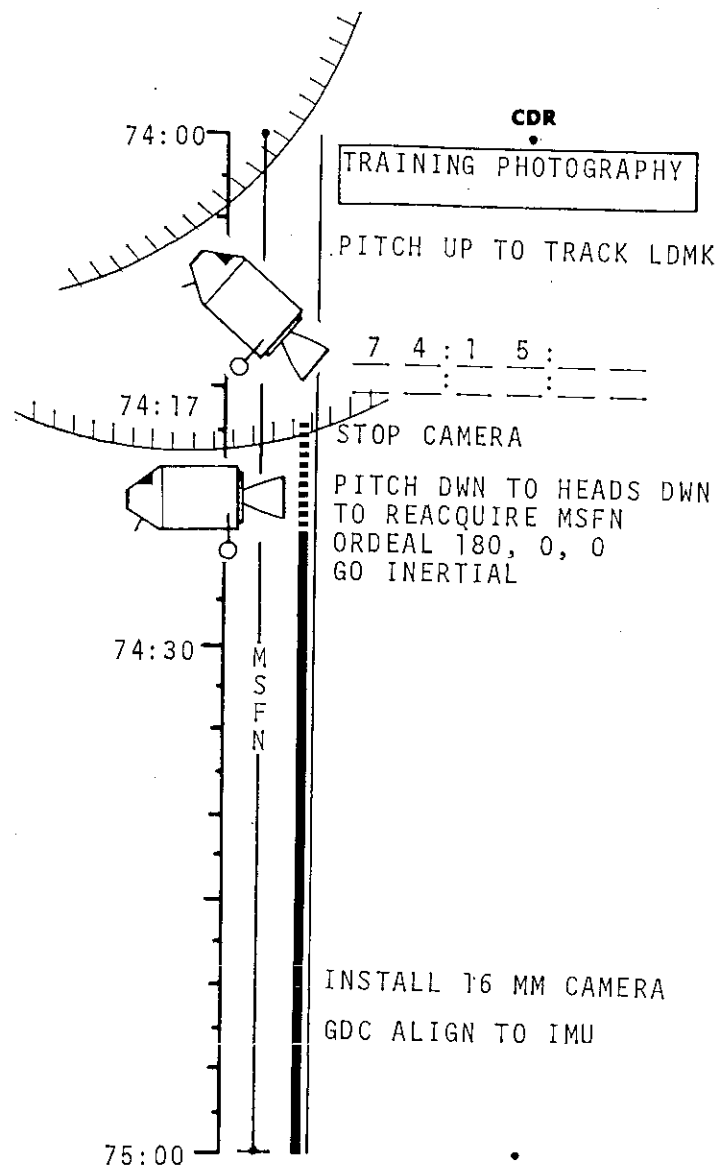
BIOMED SW - RIGHT

**MCC-H**

ACQUIRE S/C  
PIPA BIAS CK

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 73:00 - 74:00 | 4/LPO   | 2-55 |

# FLIGHT PLAN



**CMP**

REST PERIOD  
(2 HOURS)

**LMP**

TRAINING PHOTOGRAPHY  
70mm & 16mm

START CAMERAS 7 MIN PRIOR  
LDMK 70mm - 20 SEC INT  
UNTIL TRACKING - THEN  
5 SEC F STOP FROM CHART

STOP CAMERAS

1/80/HS, 1/4, F 2.8

VERIFY SYSTEMS STATUS

RECORD MAP UPDATE

IMU REALIGN P52  
OPTION 3 - REFSMMAT  
AND GYRO DRIFT TEST  
STAR ID  
STAR ANGLE DIFF

TORQUE ANGLES:

X  
Y  
Z

RECORD BLOCK DATA  
(TEI<sub>3</sub> & TEI<sub>4</sub>)

**MCC-H**

VOICE  
UPDATE:  
MAP UPDATE  
3/4

DUMP DSE

VOICE  
UPDATE:  
BLOCK DATA

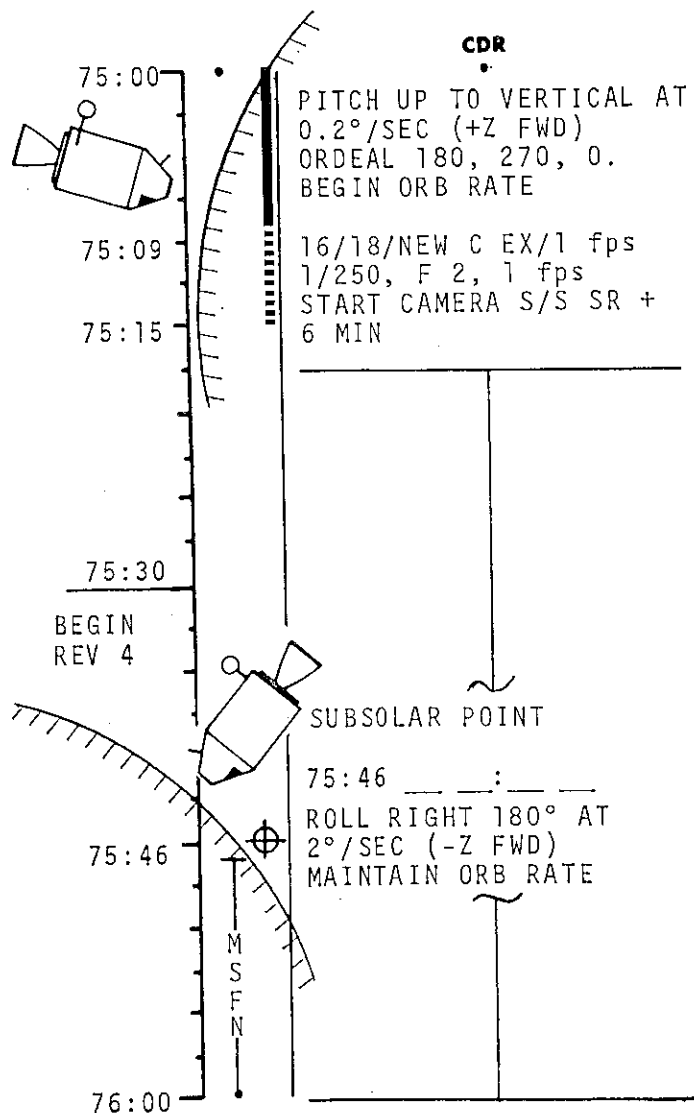
| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 74:00 - 75:00 | 4/LPO   | 2-56 |

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FLIGHT PLANNING BRANCH



# FLIGHT PLAN



**CMP**

REST PERIOD  
(2 HOURS)

INSTALL SXT BRKT & 16mm CAMERA. TAKE PHOTOS AS TIME PERMITS.

SA 0° TA 0°

**LMP**

1/80/NEW B&W, 1/250, F 2.8 BRKT/INT

2/150/B&W

START CAMERA 1 S/C  
SR + 6 MIN

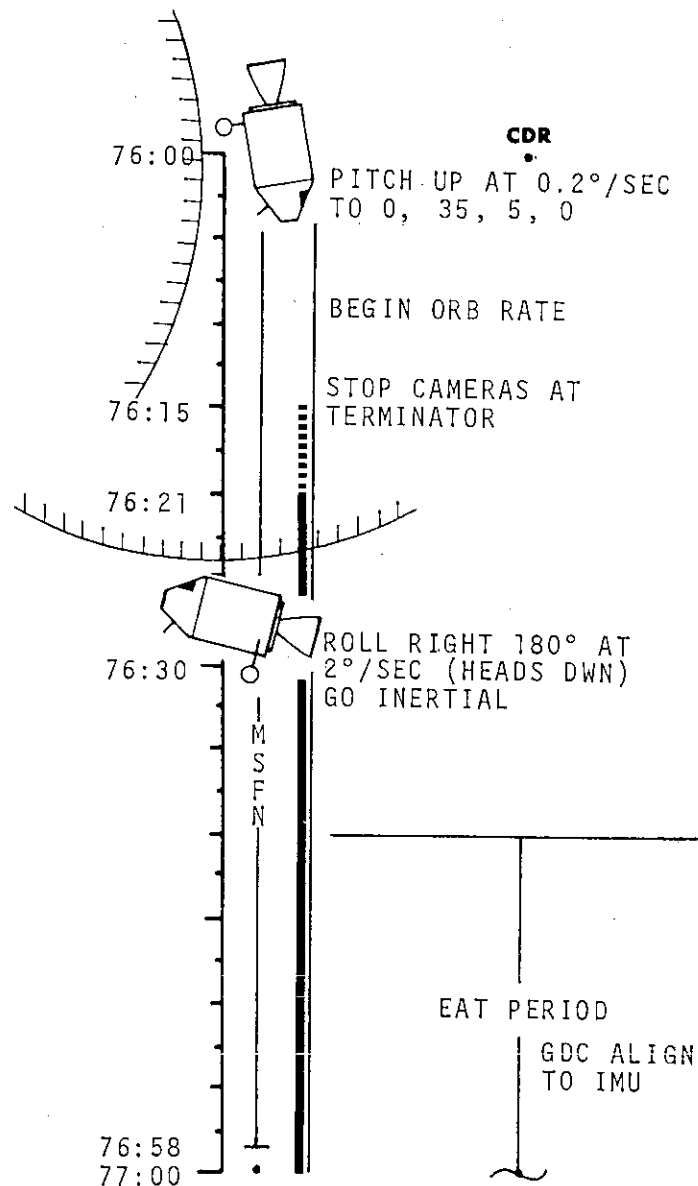
VERTICAL STEREO PHOTOGRAPHY  
EXTRA EXP EACH  
5 MIN - NOTE TIME

ACQUIRE MSFN ON OMNI ANT

**MCC-H**

ACQUIRE S/C

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 75:00 - 76:00 | 4/LPO   | 2-57 |



## FLIGHT PLAN

CMP

LMP

MCC-H

IPI  
ACQ  
IPII  
ACQ

LDMK LIGHTING EVALUATION  
(THRU OPTICS) OF LDMK B1

IMU REALIGN P52  
OPTION 3 - REFSMMAT  
AND GYRO DRIFT TEST  
STAR ID  
STAR ANGLE DIFF

TORQUE ANGLES:  
X  
Y  
Z

AT PITCH UP 1/80/B&W,  
1/250, F 11

STOP CAMERA 1 AT TERM

RECORD MAP UPDATE

V64 ACQUIRE MSFN ON HGA  
1/80/HS, 1/4, F 2.8  
(20 SEC STRIP - 20 EXP)

VERIFY SYSTEMS STATUS

RECORD BLOCK DATA:  
(TEI<sub>4</sub> & TEI<sub>5</sub>)

TRANS TO LEFT COUCH

1/80/B&W, 1/250, F 2.8  
INT (CDR)

BIOMED Sw - CENTER

VOICE  
UPDATE:  
MAP UPDATE  
4/5  
P27 UPDATE:  
STATE  
VECTOR

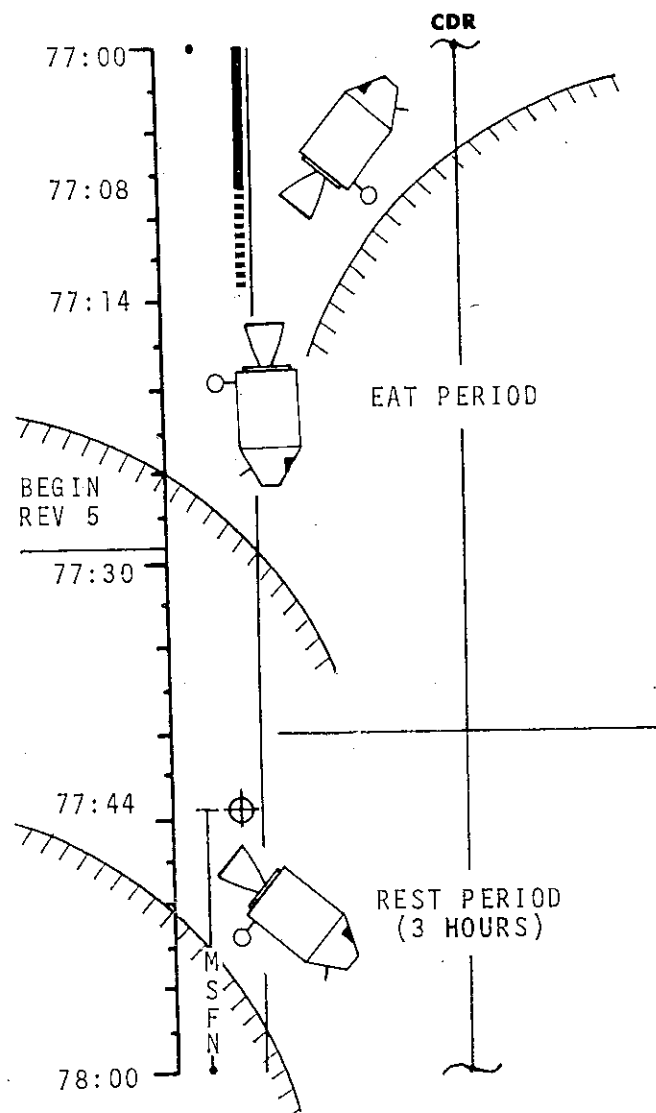
ACQUIRE S/C

DUMP DSE

VOICE  
UPDATE:  
BLOCK DATA

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 76:00 - 77:00 | 4/LPO   | 2-58 |

# FLIGHT PLAN



CMP

MAP REVIEW

P22 MANUAL ACQ  
SA 0°, TA 10°

CNTL POINT SIGHTING

IPI | — — : — — : — —  
ACQ | — — : — — : — —

CP2 NO. 526  
LAT -10.200°S  
LONG/2 +77.550°  
ALT +000.00 nm

NEW COORDINATES  
LAT — — : — — : — —°  
LONG/2 — — : — — : — —°  
ALT — — : — — : — — nm

REVIEW LDMK MAP

LMP

ROLL LEFT 180° AT  
2°/SEC PITCH DWN AT  
0.2°/SEC TO  
ORDEAL 0, 5, 0. BEGIN  
ORB RATE.  
2/150/B&W, 1/250 F 2.8

2/150/B&W, 1/250 (CDR)  
PITCH DWN AT 0.3°/SEC  
AT ACQ

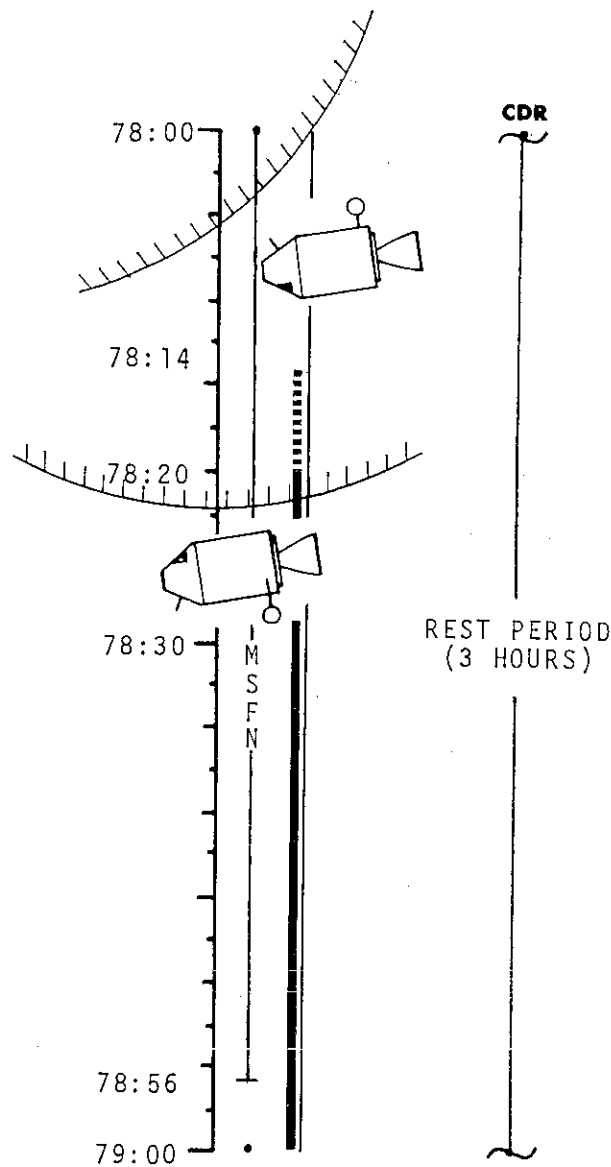
ACQUIRE MSFN ON OMNIS  
AT LOSS PITCH UP AT  
0.2°/SEC TO  
ORDEAL 0, 5, 0. BEGIN  
ORB RATE.  
REPORT NEW CP COORDINATES

MCC-H

ACQUIRE S/C

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 77:00 - 78:00 | 4/LPO   | 2-59 |

# FLIGHT PLAN



## PSEUDO LDG SITE SIGHTINGS

P22 - AUTO OPTICS

IPI | ---:---:---

TCA | ---:---:---

LDMK NO. B1

LAT +02.675°N

LONG/2 +17.512°

ALT -000.99 nm

NEW COORDINATES

LAT +---:---:---°

LONG/2 +---:---:---°

ALT ---:---:--- nm

IMU REALIGN P52

OPTION 3 - REFSMMAT

AND GYRO DRIFT TEST

STAR ID

STAR ANGLE DIFF ---

TORQUE ANGLES: ---

X ---:---:---

Y ---:---:---

Z ---:---:---

PITCH DWN 0.3°/SEC AT ACQ

AT LOSS ROLL RIGHT 180°

AT 2°/SEC TO HEADS DWN

GO INERTIAL

V64 ACQUIRE MSFN ON HGA

1/80/HS, 1/4, F 2.8

REPORT NEW COORDINATES

RECORD BLOCK DATA

(TEI<sub>6</sub>), & MAP UPDATE

VERIFY SYSTEMS STATUS

GDC ALIGN TO IMU

16/18/ C EX, 1/250, 1 fps

(--- MAGS)

2/150/CI21/1/250 SPOT

GENERAL OBSERVATIONS

ACQUIRE S/C

P27 UPDATE:

STATE

VECTOR

VOICE

UPDATE:

BLOCK DATA

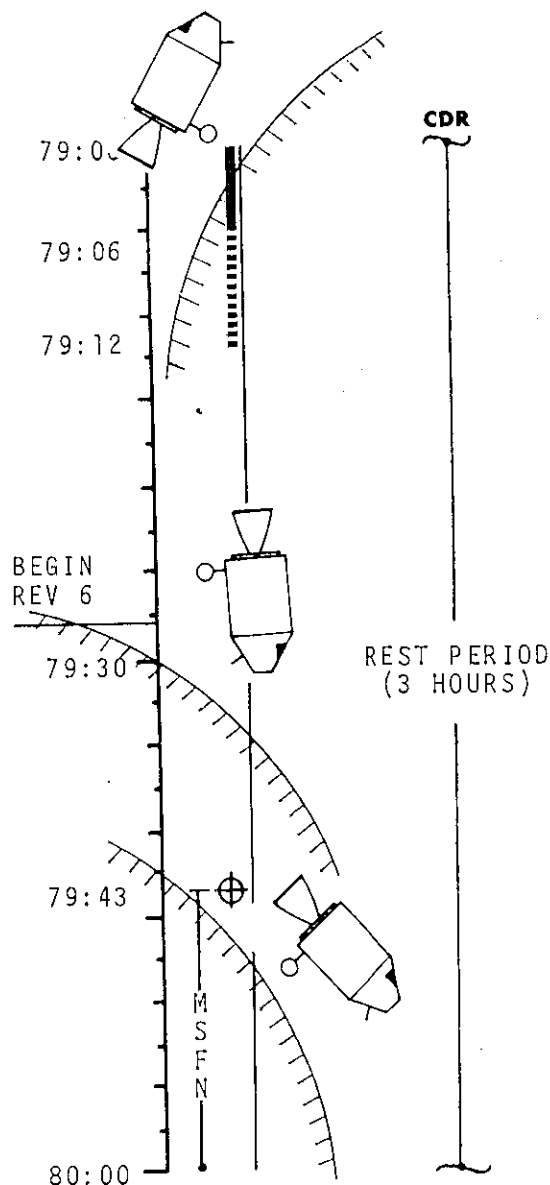
& MAP

UPDATE

5/6

DUMP DSE

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 78:00 - 79:00 | 4/LPO   | 2-60 |



## FLIGHT PLAN

CMP

LMP

MCC-H

REVIEW CONTROL POINT MAP

ROLL 180° AT 2°/SEC  
PITCH DWN AT 0.2°/SEC TO  
ORDEAL 0, 5, 0.  
BEGIN ORB RATE

### CNTL POINT SIGHTING

P22 - AUTO OPTICS  
IPI — — — — —  
TCA — — — — —  
CP2 NO. 526  
LAT -10.200°S  
LONG/2 +77.550°  
ALT +000.00 nm

NEW COORDINATES  
LAT — — — — —°  
LONG/2 — — — — —°  
ALT — — — — —nm

REVIEW LDMK MAP

PITCH DWN 0.3°/SEC AT ACC

AT LOSS PITCH UP AT  
0.2°/SEC TO  
ORDEAL 0, 5, 0. BEGIN  
ORB RATE

ACQUIRE MSFN ON OMNIS

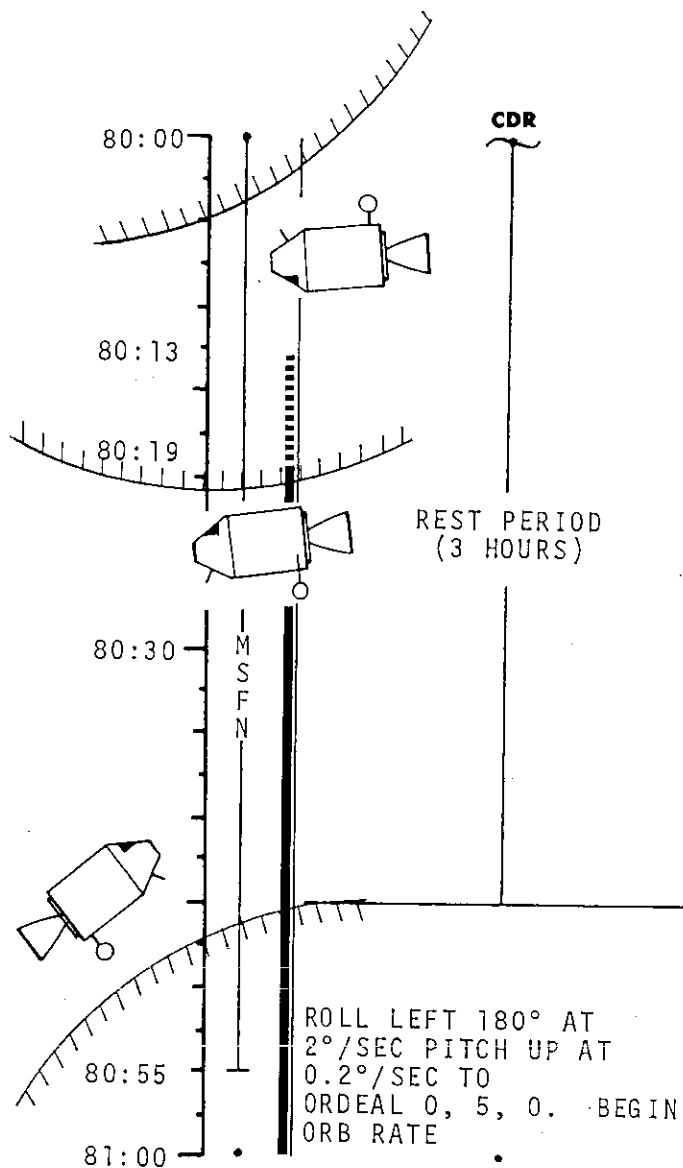
ACQUIRE S/C

REPORT NEW COORDINATES

### PSEUDO LDG SITE SIGHTING

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 79:00 - 80:00 | 4/LPO   | 2-61 |

# FLIGHT PLAN



P22 AUTO OPTICS  
 IPI | ---:---:---  
 TCA | ---:---:---  
 LDMK NO. B1  
 LAT +02.675°N  
 LONG/2 +17.512°  
 ALT -000.99 nm  
 NEW COORDINATES  
 LAT ---:---:---  
 LONG/2 ---:---:---  
 ALT ---:---:--- nm

IMU REALIGN P52  
 OPTION 3 - REFSMMAT  
 AND GYRO DRIFT TEST  
 STAR ID ---  
 STAR ANGLE DIFF ---

TORQUE ANGLES: ---  
 X ---  
 Y ---  
 Z ---

16/SXT/NEW C XT, ---  
 6 fps

PITCH DWN 0.3°/SEC AT ACQ

AT LOSS ROLL RIGHT 180°  
 AT 2°/SEC. GO INERTIAL

V64 ACQUIRE MSFN ON HGA

REPORT NEW COORDINATES  
 RECORD BLOCK DATA  
 (TEI7) & MAP UPDATE

VERIFY SYSTEMS STATUS

1/80/B&W, 1/250 F 2.8 INT

2/150/B&W, 1/250 (R&B)

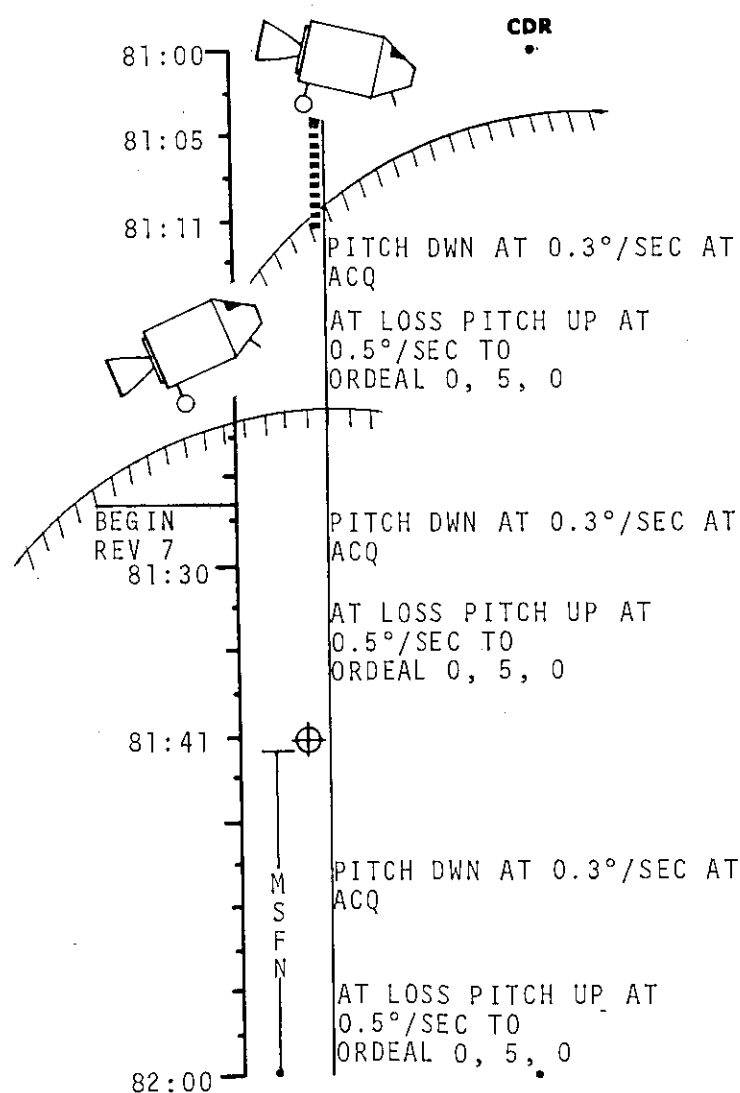
TRANS TO RIGHT COUCH

EAT PERIOD

ACQUIRE S/C  
 P27 UPDATE:  
 STATE  
 VECTOR  
 VOICE  
 UPDATE:  
 BLOCK DATA  
 & MAP  
 UPDATE 6/7  
 DUMP DSE

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 80:00 - 81:00 | 4/LPO   | 2-62 |

# FLIGHT PLAN



| CMP                      |            |
|--------------------------|------------|
| CNTL PT SIGHTINGS (3)    |            |
| P22 MAN ACQ SA 0° TA 10° |            |
| IPI                      | ---        |
| ACQ                      | ---        |
| CP1 NO.                  | 504        |
| LAT                      | -05.250°S  |
| LONG/2                   | -81.350°   |
| ALT                      | +000.00 nm |
| P22 AUTO OPTICS          |            |
| IPI                      | ---        |
| TCA                      | ---        |
| CP2 NO.                  | 526        |
| LAT                      | -10.200°S  |
| LONG/2                   | +77.550°   |
| ALT                      | +000.00 nm |
| P22 MAN ACQ SA 0° TA 10° |            |
| IPI                      | ---        |
| ACQ                      | ---        |
| CP3 NO.                  | 334        |
| LAT                      | -09.100°S  |
| LONG/2                   | +47.950°   |
| ALT                      | +000.00 nm |
| PSEUDO LDG SITE SIGHTING |            |
| P22 AUTO OPTICS          |            |

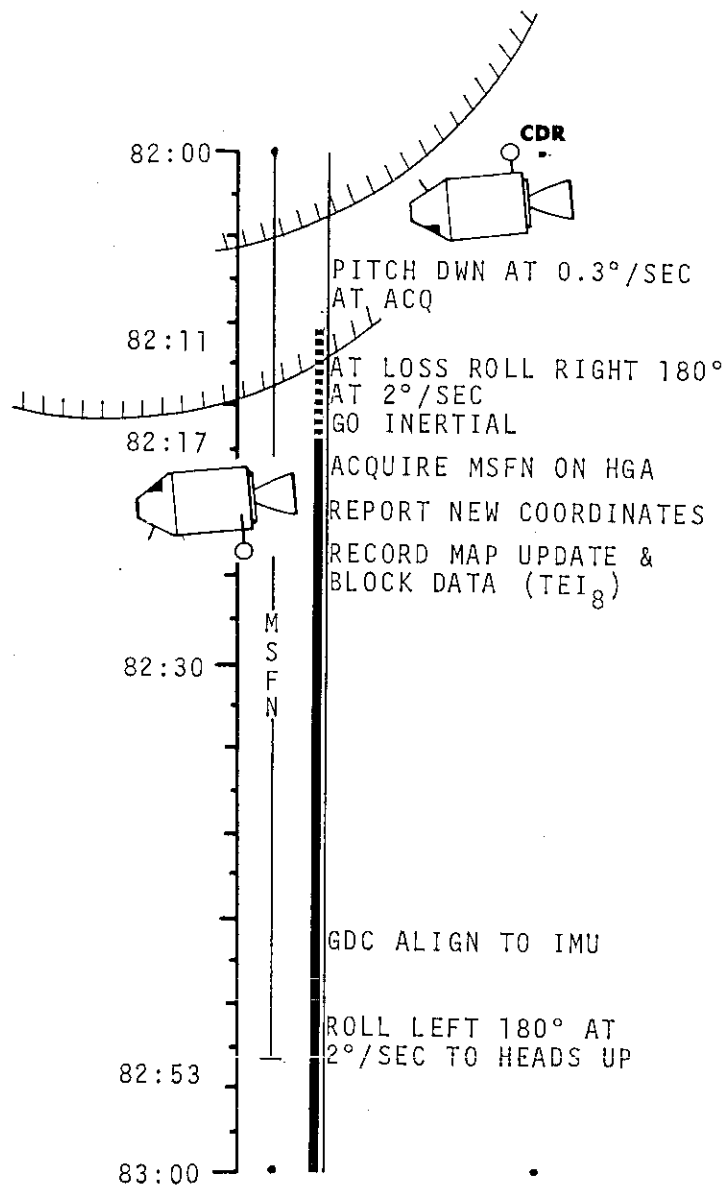
| LMP              |     |
|------------------|-----|
| BIOMED SW - LEFT |     |
| EAT PERIOD       |     |
| NEW COORDINATES  |     |
| LAT              | --- |
| LONG/2           | --- |
| ALT              | --- |
| EAT PERIOD       |     |
| NEW COORDINATES  |     |
| LAT              | --- |
| LONG/2           | --- |
| ALT              | --- |
| EAT PERIOD       |     |
| NEW COORDINATES  |     |
| LAT              | --- |
| LONG/2           | --- |
| ALT              | --- |

MCC-H

ACQUIRE S/C

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 81:00 - 82:00 | 4/LPO   | 2-63 |

# FLIGHT PLAN



**CMP**

IPI \_\_\_\_\_

TCA \_\_\_\_\_

LDG SITE NO. B1

LAT +02.675°N

LONG/2 +17.512°

ALT -000.99 nm

NEW COORDINATES

LAT \_\_\_\_\_ °

LONG/2 \_\_\_\_\_ °

ALT \_\_\_\_\_ nm

IMU REALIGN P52

OPTION 3 - REFSMMAT

AND GYRO DRIFT TEST

STAR ID \_\_\_\_\_

STAR ANGLE DIFF \_\_\_\_\_

TORQUE ANGLES:

X \_\_\_\_\_

Y \_\_\_\_\_

Z \_\_\_\_\_

16/SXT/CXT, \_\_\_\_\_, 6 fps

**LMP**

REST PERIOD  
(2 HOURS)

**MCC-H**

ACQUIRE S/C

P27 UPDATE:

STATE

VECTOR

VOICE

UPDATE:

MAP UPDATE

7/8

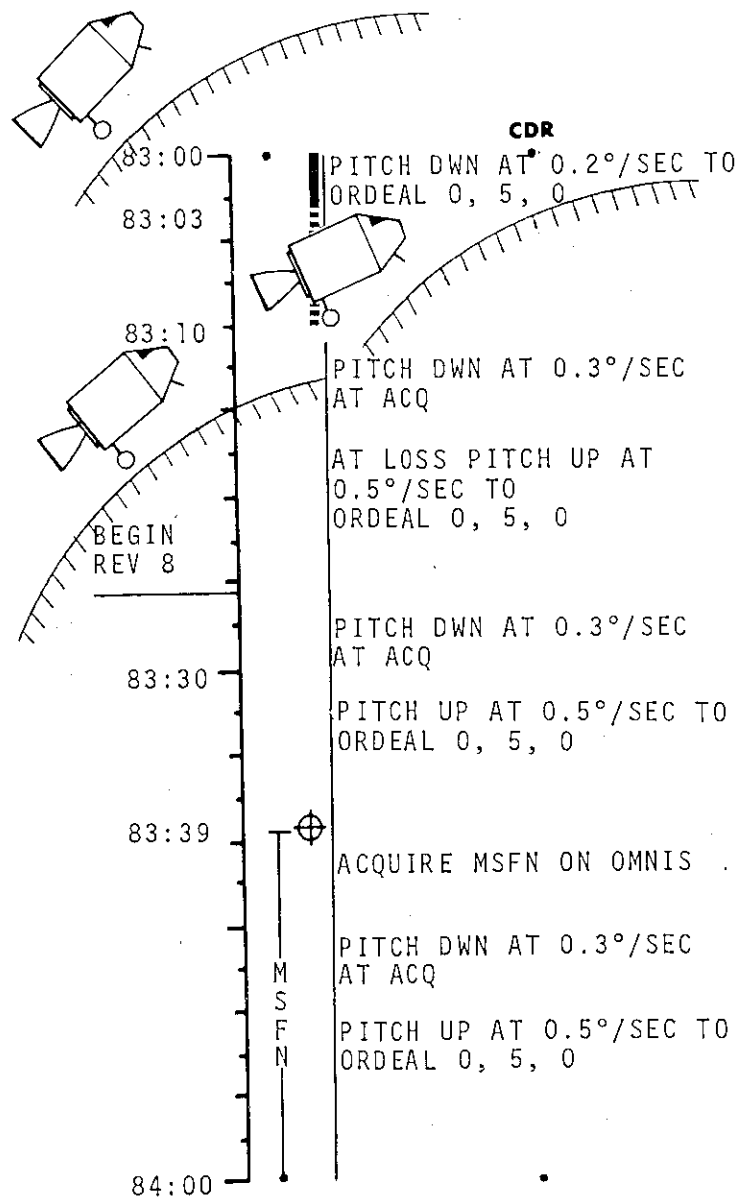
BLOCK DATA

DUMP DSE

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 82:00 - 83:00 | 4/LPO   | 2-64 |



# FLIGHT PLAN



**CMP**

CNTL PT SIGHTINGS (3)

P22 AUTO OPTICS

IPI \_\_\_\_\_  
TCA \_\_\_\_\_  
CP1 NO. 504  
LAT -05.250°S  
LONG/2 -81.350°  
ALT +000.00 nm

P22 AUTO OPTICS

IPI \_\_\_\_\_  
TCA \_\_\_\_\_  
CP2 NO. 526  
LAT -10.200°S  
LONG/2 +77.550°  
ALT +000.00 nm

P22 AUTO OPTICS

IPI \_\_\_\_\_  
TCA \_\_\_\_\_  
CP3 NO. 334  
LAT -09.100°S  
LONG/2 +47.950°  
ALT +000.00 nm

PSEUDO LDG SITE SIGHTING

P22 AUTO OPTICS

**LMP**

REST PERIOD  
(2 HOURS)

NEW COORDINATES

LAT \_\_\_\_\_°  
LONG/2 \_\_\_\_\_°  
ALT \_\_\_\_\_nm

NEW COORDINATES

LAT \_\_\_\_\_°  
LONG/2 \_\_\_\_\_°  
ALT \_\_\_\_\_nm

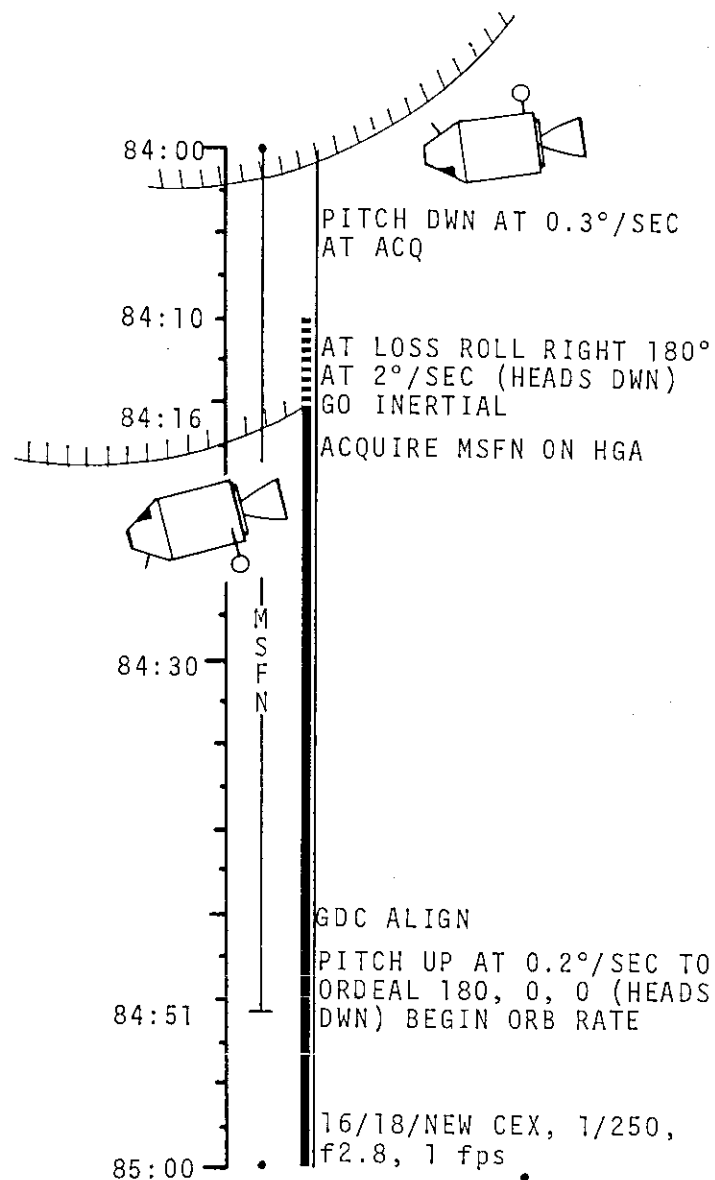
NEW COORDINATES

LAT \_\_\_\_\_°  
LONG/2 \_\_\_\_\_°  
ALT \_\_\_\_\_nm

**MCC-H**

ACQUIRE S/C

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 83:00 - 84:00 | 4/LP0   | 2-65 |



# FLIGHT PLAN

**CMP**

IPI  
TCA  
LDMK NO. B1  
LAT +02.512°N  
LONG/2 +17.512°  
ALT +000.99 nm

L10H CANISTER CHANGE  
(CARTRIDGE 10 FROM B5  
INTO CANISTER B)

IMU REALIGN P52  
OPTION 3 - REFSMMAT  
AND GYRO DRIFT TEST  
STAR ID  
STAR ANGLE DIFF

TORQUE ANGLES:

X  
Y  
Z

REST PERIOD  
(2 HOURS)

**LMP**

NEW COORDINATES  
LAT  
LONG/2  
ALT nm

REST PERIOD  
(2 HOURS)

2/150/NEW B&W, 1/250  
(R&B)

BIOMED Sw - RIGHT

VERIFY SYSTEMS STATUS

RECORD MAP UPDATE &  
BLOCK DATA (TEI<sub>9</sub>)

1/80/H.S. B, f2.8, POL

SR -12 MIN START DARK

1/80 B&W, 1/4, f4, POL

SR -4 MIN START CORONA

**MCC-H**

ACQUIRE S/C

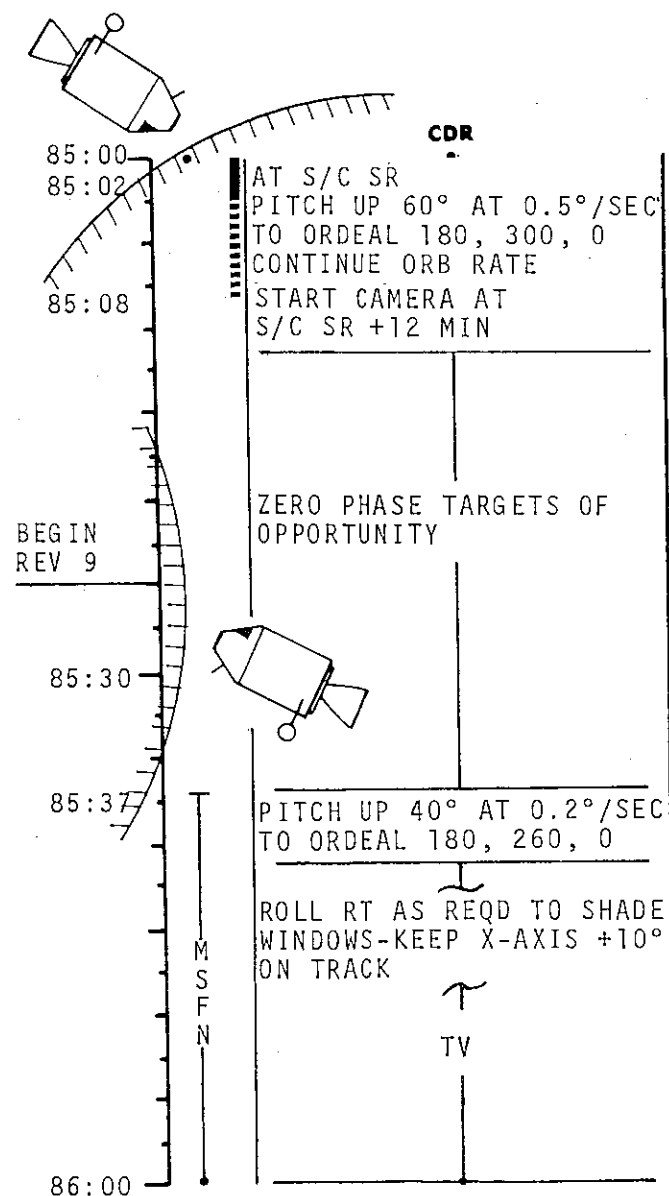
P27 UPDATE:  
STATE  
VECTOR

DUMP DSE

VOICE  
UPDATE:  
MAP UPDATE  
8/9  
BLOCK DATA

| MISSION   | DITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|--------|-------------------|---------------|---------|------|
| AS503/103 | FINAL  | November 22, 1968 | 84:00 - 85:00 | 4/LPO   | 2-66 |

# FLIGHT PLAN



CMP

REST PERIOD  
(2 HOURS)

LMP

1/80/NEW C BRKT INT

START CAMERA 1 AT  
SR +12 MIN

CONVERGENT STEREO  
PHOTOGRAPHY-EXTRA  
EXP EACH 5 MIN -  
NITE TIME

ACQUIRE MSFN ON HGA

RECORD MAP UPDATE

2/80 B&W/R&B  
4 STOPS WIDER

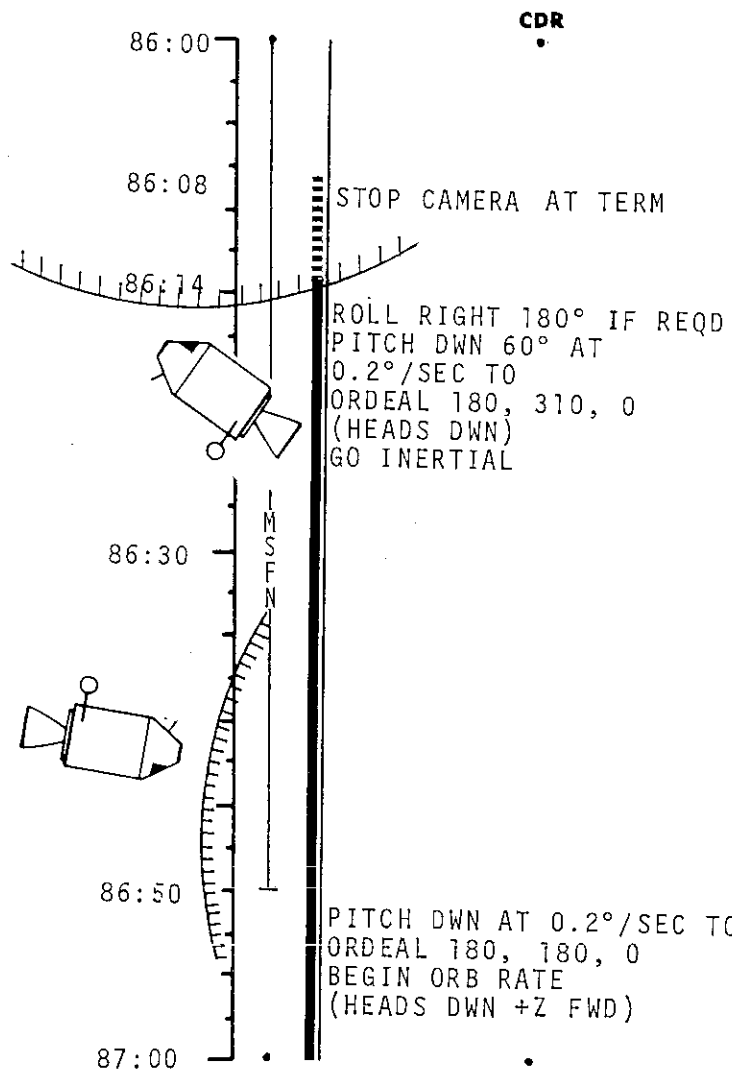
MCC-H

ACQUIRE S/C

VOICE  
UPDATE:  
MAP UPDATE  
9/10

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 85:00 - 86:00 | 4/LP0   | 2-67 |

# FLIGHT PLAN



CDR

CMP

LMP

MCC-H

REST PERIOD  
(2 HOURS)

STOP CAMERA AT TERM  
2/80/HS, 1/4 f2.8

RECORD BLOCK DATA  
(TEI<sub>10</sub>) & MNVR PAD

VERIFY SYSTEMS STATUS

1/80/NEW B&W 1/250,  
f2.8 INT (CDR)  
2/150/C, 1/250

P27 UPDATE:  
STATE  
VECTOR

VOICE  
UPDATE:  
BLOCK DATA  
MNVR PAD

DUMP DSE

IMU REALIGN P52  
OPTION 3 - REFSMMAT  
AND GYRO DRIFT TEST  
STAR ID \_\_\_\_\_  
STAR ANGLE DIFF \_\_\_\_\_

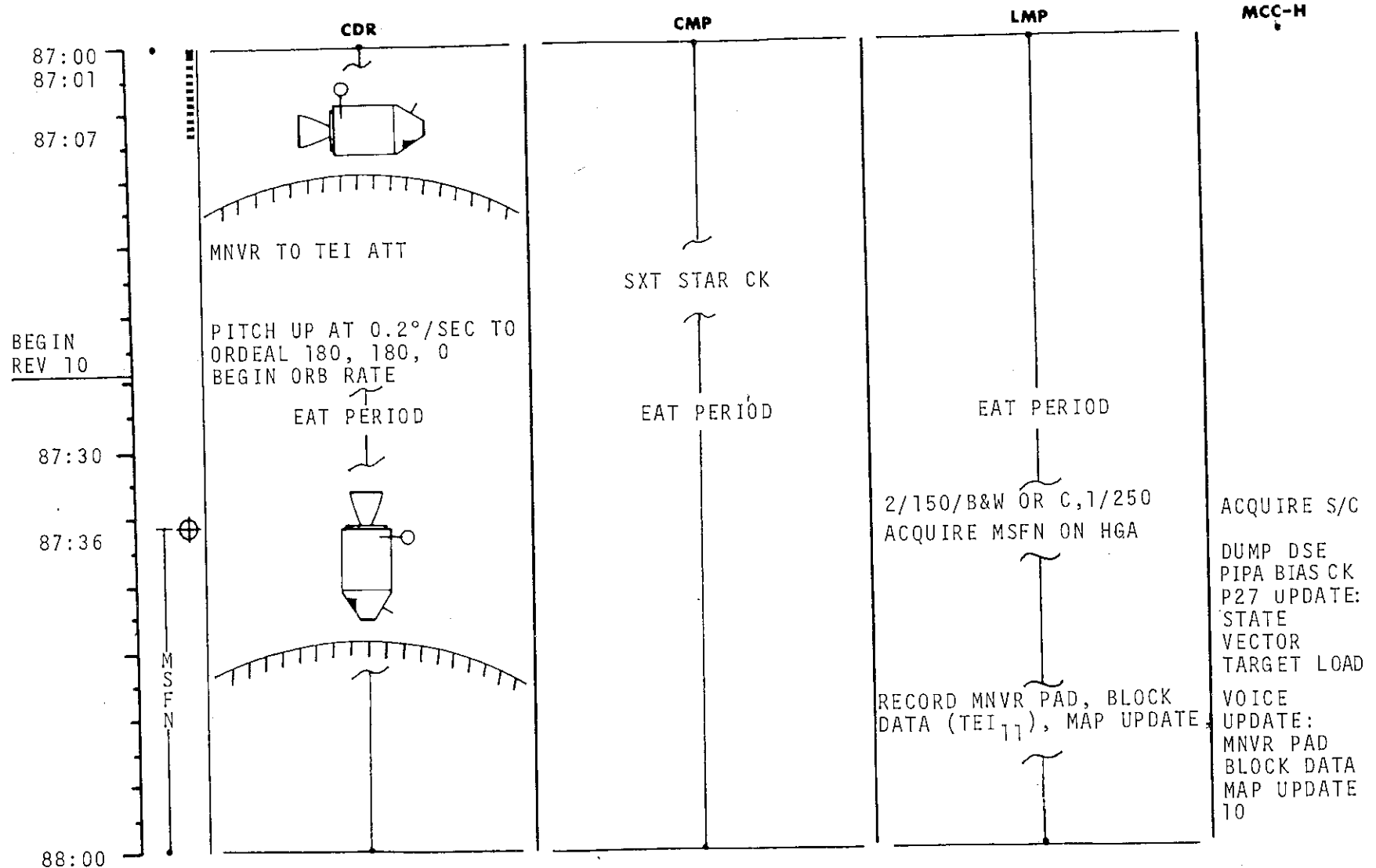
TORQUE ANGLES: \_\_\_\_\_  
X \_\_\_\_\_  
Y \_\_\_\_\_  
Z \_\_\_\_\_

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 86:00 - 87:00 | 4/LPO   | 2-68 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 87:00 - 88:00 | 4/LPO   | 2-69 |

# FLIGHT PLAN

|       | CDR                                                               | CMP                                                                                         | LMP                                                                                                                             | MCC-H    |
|-------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------|
| 88:00 | GO/NO GO FOR TEI                                                  |                                                                                             | BIOMED Sw - LEFT                                                                                                                | GO/NO GO |
| 88:07 |                                                                   |                                                                                             |                                                                                                                                 |          |
| 88:13 |                                                                   |                                                                                             |                                                                                                                                 |          |
|       | MSFN                                                              | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>AND GYRO DRIFT TEST<br>STAR ID<br>STAR ANGLE DIFF | 2/150/C SPOT<br>16/75/CEX/1 fps/SPOT                                                                                            |          |
| 88:30 | EXTERNAL ΔV P30<br>EMS TEST<br>SPS THRUST P40<br>MNVR TO BURN ATT | TORQUE ANGLES:<br>X<br>Y<br>Z                                                               | PRE TEI SYSTEMS CKS<br>C&W CK<br>CM RCS CK<br>SM RCS CK<br>SPS PERIODIC MONITOR<br>EPS PERIODIC MONITOR<br>ECS PERIODIC MONITOR |          |
| 88:48 | PUGS MODE SW - AUX                                                | SXT STAR CK<br>TRANSFER TO COUCH                                                            |                                                                                                                                 |          |
| 89:00 |                                                                   |                                                                                             |                                                                                                                                 |          |

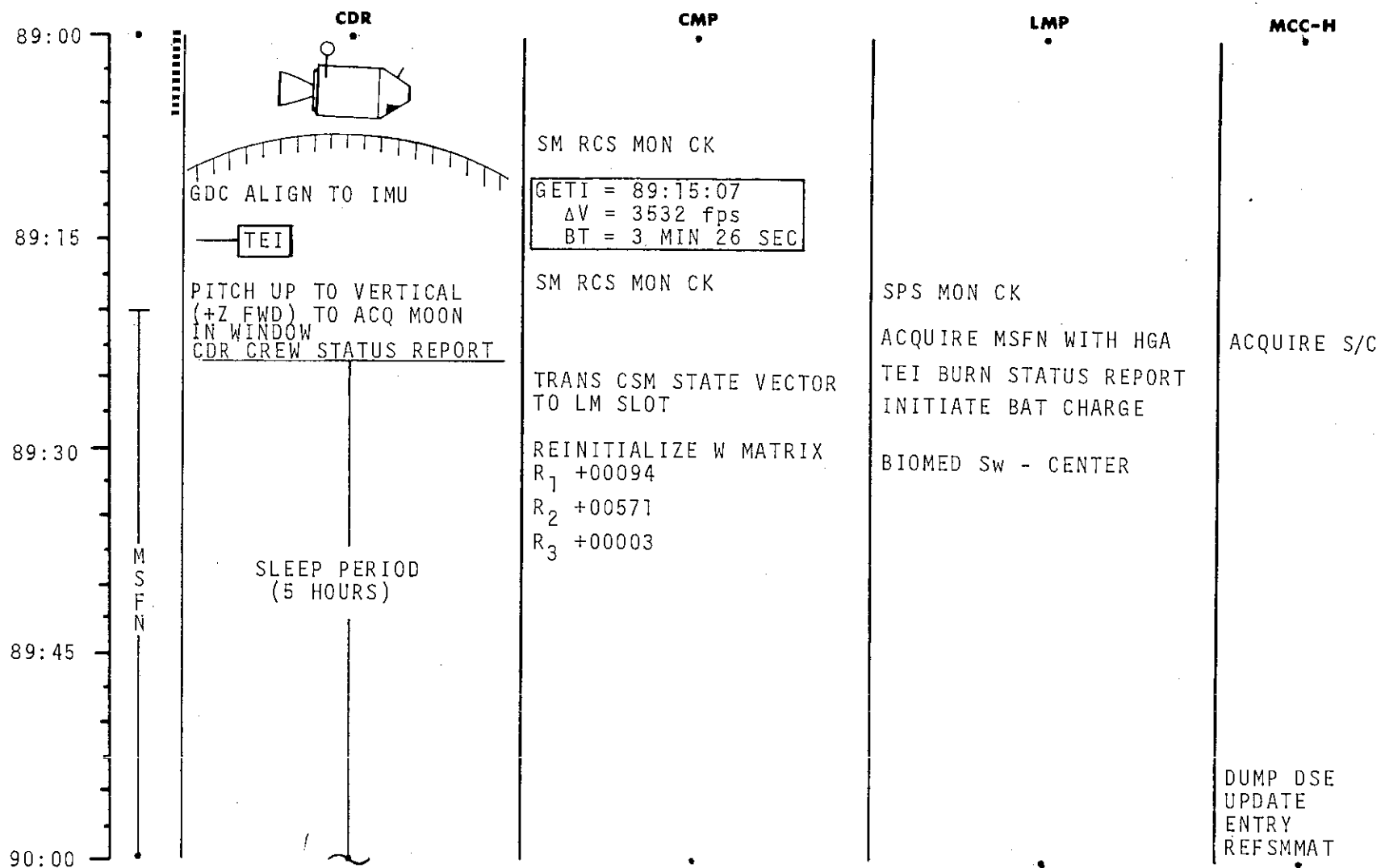
| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 88:00 - 89:00 | 4/LPO   | 2-70 |

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FLIGHT PLANNING BRANCH



# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 89:00 - 90:00 | 4/TEC   | 2-71 |

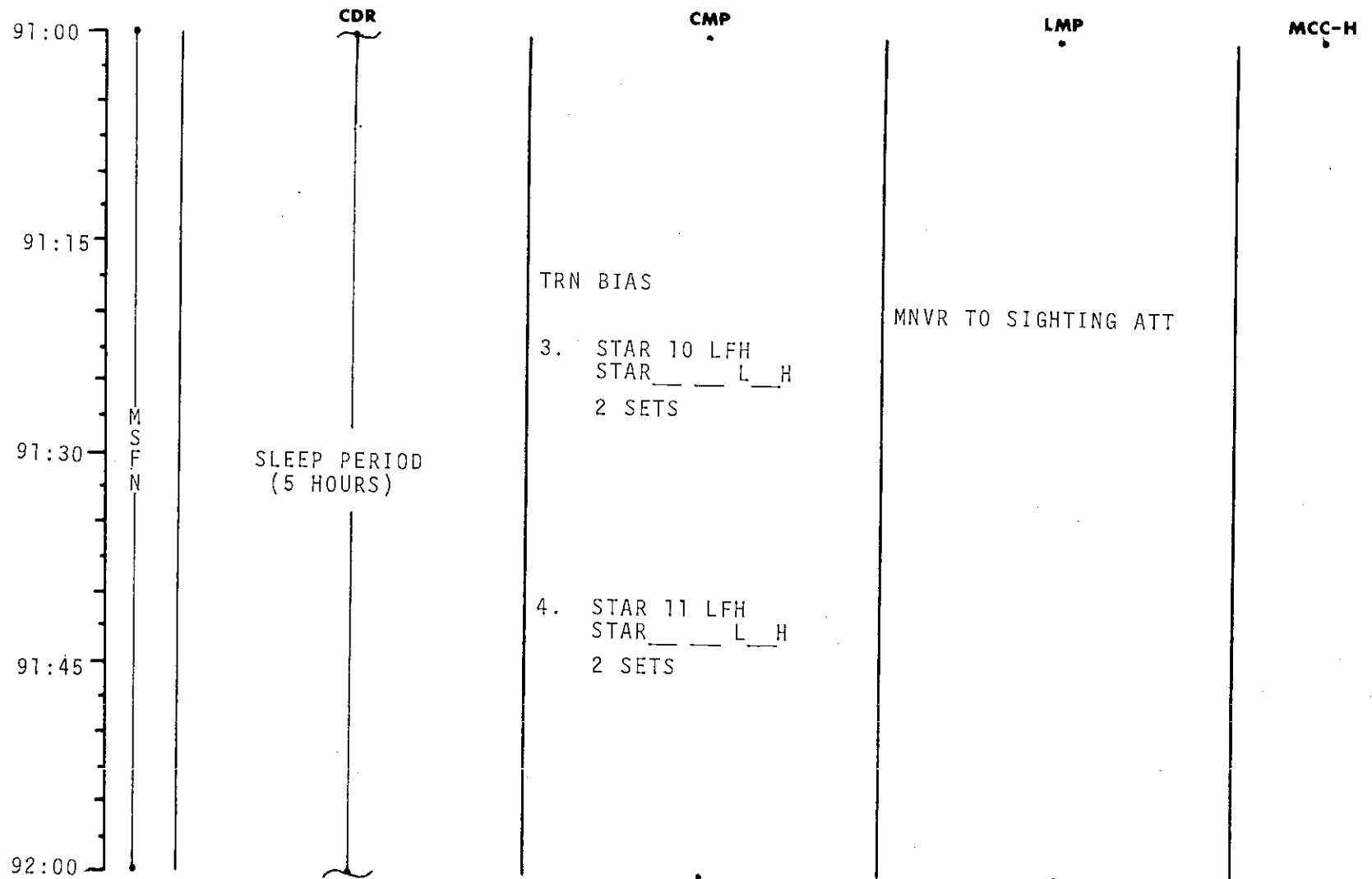


# FLIGHT PLAN

|       | CDR                       | CMP                                                                   | LMP                  | MCC-H |
|-------|---------------------------|-----------------------------------------------------------------------|----------------------|-------|
| 90:00 | SLEEP PERIOD<br>(5 HOURS) | IMU REALIGN P52<br>OPTION 1 - PREFERRED<br>STAR ID<br>STAR ANGLE DIFF | MNVR TO P52 ATT      |       |
| 90:15 |                           | TORQUE ANGLES:<br>X<br>Y<br>Z                                         |                      |       |
| 90:30 |                           | TRN BIAS                                                              | MNVR TO SIGHTING ATT |       |
| 90:45 |                           | CISLUNAR NAVIGATION P23<br>1. STAR 01 LNH<br>STAR L H<br>2 SETS       |                      |       |
| 91:00 |                           | 2. STAR 02 LNH<br>STAR L H                                            |                      |       |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 90:00 - 91:00 | 4/TEC   | 2-72 |

# FLIGHT PLAN

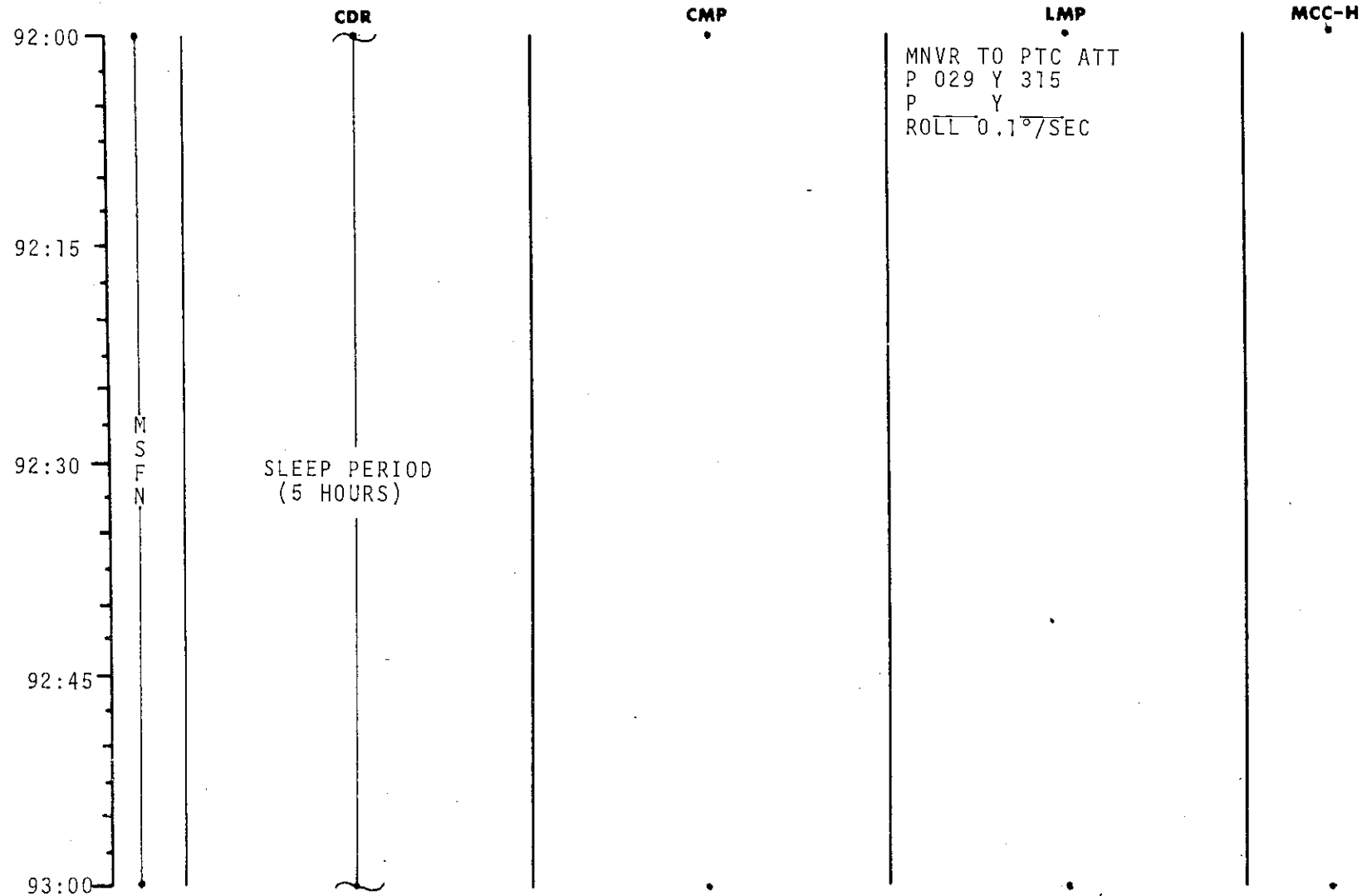


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 91:00 - 92:00 | 4/TEC   | 2-73 |

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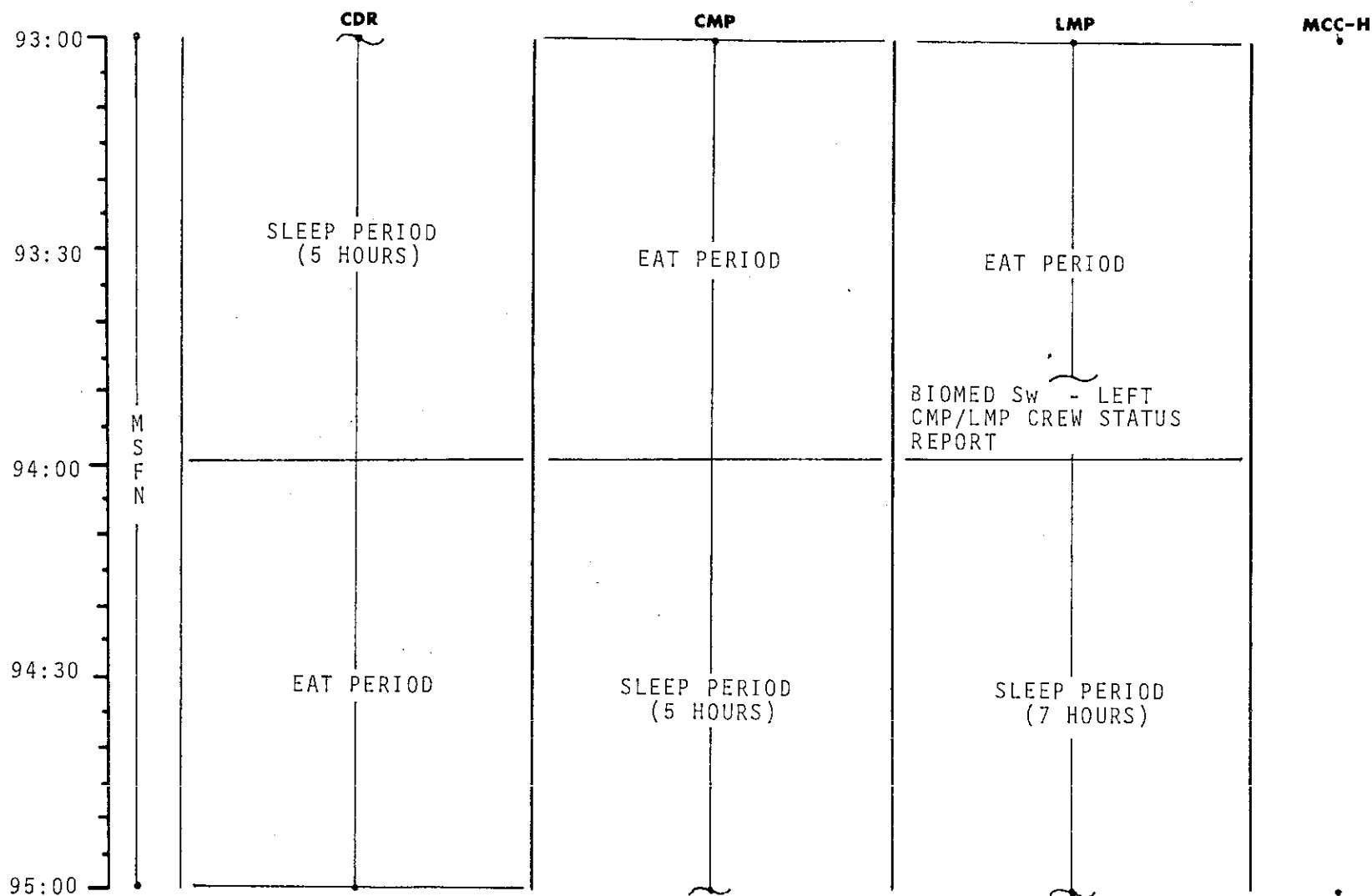
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 92:00 - 93:00 | 4/TEC   | 2-74 |

# FLIGHT PLAN

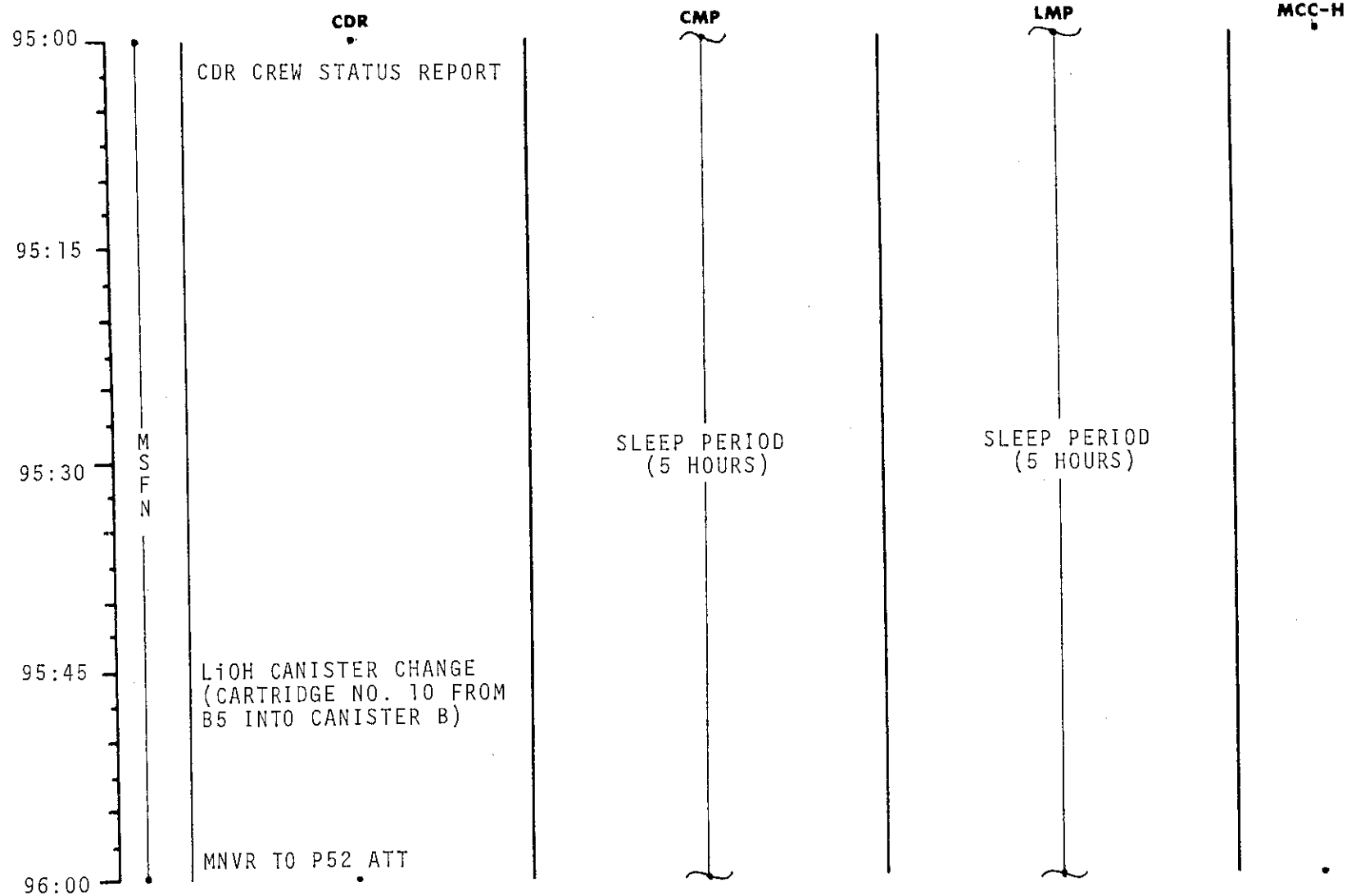


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 93:00 - 95:00 | 4/TEC   | 2-75 |

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FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 95:00 - 96:00 | 4/TEC   | 2-76 |

# FLIGHT PLAN

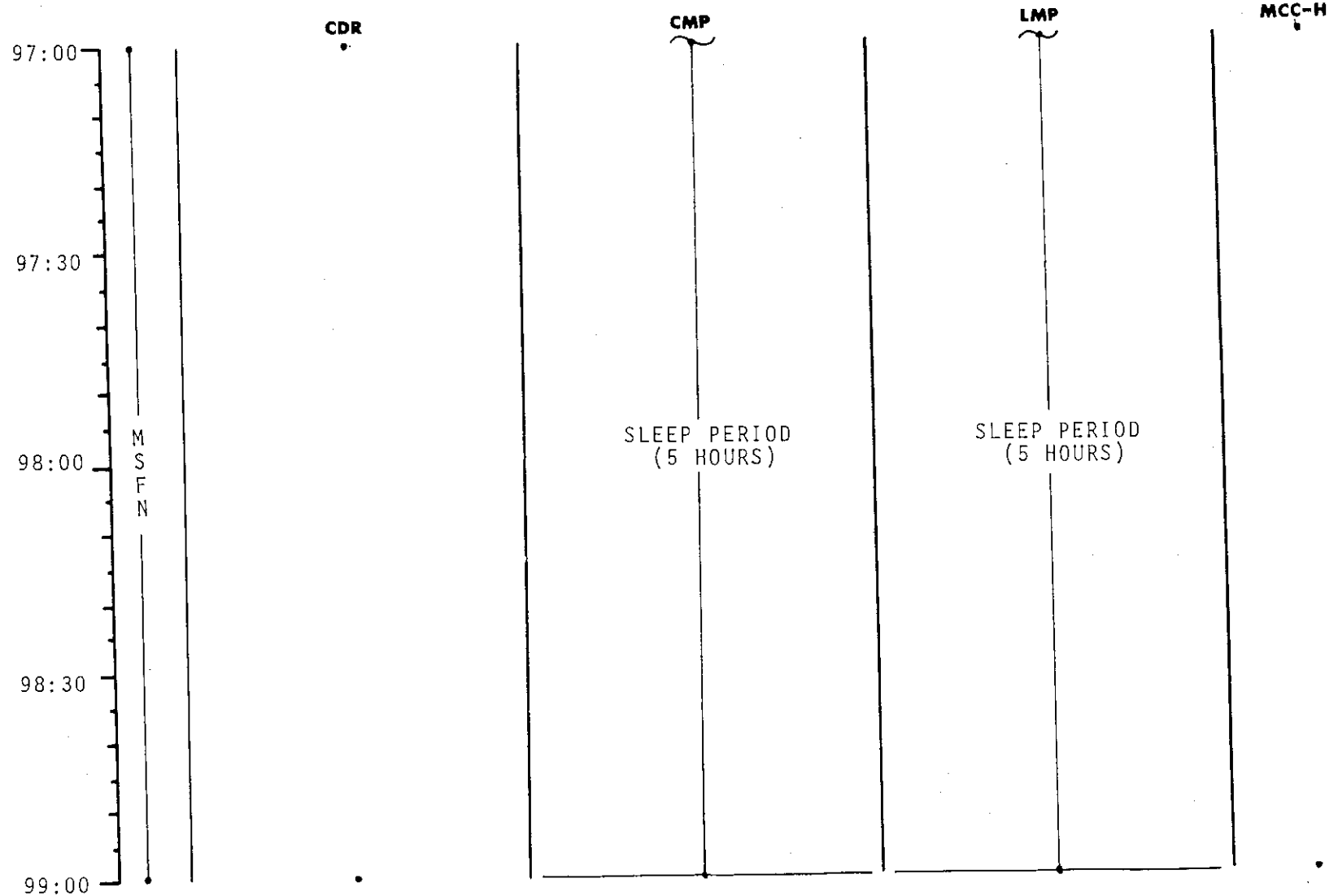
|       | CDR                                                                  | CMP                       | LMP                       | MCC-H |
|-------|----------------------------------------------------------------------|---------------------------|---------------------------|-------|
| 96:00 | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID<br>STAR ANGLE DIFF |                           |                           |       |
| 96:15 | TORQUE ANGLES:<br>X<br>Y<br>Z                                        |                           |                           |       |
| 96:30 | MNVR TO PTC ATT<br>P 029 Y 315<br>P Y<br>ROLL 0.1°/SEC               | SLEEP PERIOD<br>(5 HOURS) | SLEEP PERIOD<br>(5 HOURS) |       |
| 96:45 |                                                                      |                           |                           |       |
| 97:00 |                                                                      |                           |                           |       |

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 96:00 - 97:00 | 5/TEC   | 2-77 |

MSC Form 1910 (Nov 68)

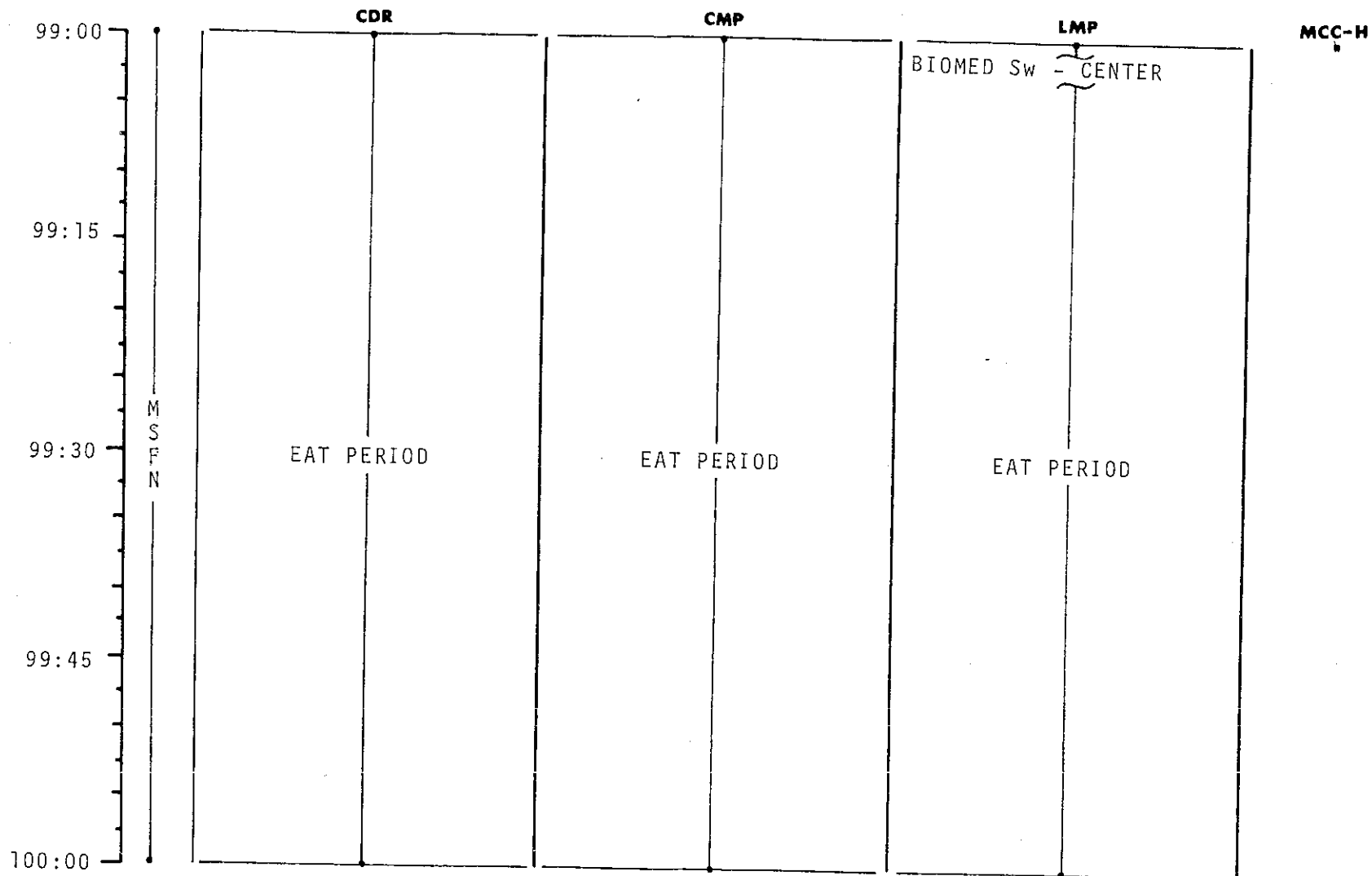
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 97:00 - 99:00 | 5/TEC   | 2-78 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME           | DAY/REV | PAGE |
|-----------|---------|-------------------|----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 99:00 - 100:00 | 5/TEC   | 2-79 |

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FLIGHT PLANNING BRANCH



# FLIGHT PLAN

|        | CDR                  | CMP                                                                                                                                                                                                                                | LMP                        | MCC-H |
|--------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------|
| 100:00 | MNVR TO P52 ATT      | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID _____<br>STAR ANGLE DIFF _____<br><br>TORQUE ANGLES: _____<br>X _____<br>Y _____<br>Z _____                                                                                      | CMP/LMP CREW STATUS REPORT |       |
| 100:15 |                      |                                                                                                                                                                                                                                    |                            |       |
|        | MNVR TO SIGHTING ATT | TRN BIAS<br><br><div>CISLUNAR NAVIGATION P23</div> 1. STAR 02 LNH<br>STAR _____ L _____ H<br>1 SET _____<br><br>2. STAR 17 LFH<br>STAR _____ L _____ H<br>1 SET _____<br><br>3. STAR 01 LNH<br>STAR _____ L _____ H<br>1 SET _____ |                            |       |
| 100:30 | M S F N              |                                                                                                                                                                                                                                    |                            |       |
| 100:45 |                      |                                                                                                                                                                                                                                    |                            |       |
| 101:00 | MNVR TO SIGHTING ATT |                                                                                                                                                                                                                                    |                            |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 100:00 - 101:00 | 5/TEC   | 2-80 |

# FLIGHT PLAN

|        | CDR | CMP                                                                                                                                                                                                                                                                                                      | LMP               | MCC-H |
|--------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------|
| 101:00 |     |                                                                                                                                                                                                                                                                                                          |                   |       |
| 101:15 |     |                                                                                                                                                                                                                                                                                                          |                   |       |
| 101:30 |     | TRN BIAS<br><br><div style="border: 1px solid black; padding: 2px;">             Cislunar Navigation P23<br/>             1. STAR 22 EFH<br/>             STAR__ E__H<br/>             2 SETS<br/><br/>             2. STAR 26 ENH<br/>             STAR__ E__H<br/>             2 SETS           </div> |                   |       |
| 101:45 |     |                                                                                                                                                                                                                                                                                                          |                   |       |
| 102:00 |     |                                                                                                                                                                                                                                                                                                          | BIOMED Sw - RIGHT |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 101:00 - 102:00 | 5/TEC   | 2-81 |

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FLIGHT PLANNING BRANCH

# FLIGHT PLAN

CDR

CMP

LMP

MCC-H

102:00

102:15

102:30

102:45

103:00

M  
S  
F  
N

MNVR TO P52 ATT

3. STAR 31 ENH  
STAR \_\_\_ E \_\_\_ H  
2 SETS

IMU REALIGN P52  
OPTION 3 - RESFMMAT  
STAR ID \_\_\_\_\_  
STAR ANGLE DIFF \_\_\_\_\_

TORQUE ANGLES: \_\_\_\_\_  
X \_\_\_\_\_  
Y \_\_\_\_\_  
Z \_\_\_\_\_

RECORD MNVR PAD

P27 UPDATE  
STATE  
VECTOR  
TGT LOAD  
REFSMMAT

VOICE  
UPDATE:  
MNVR PAD

| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 102:00 103:00 | 5/TEC   | 2-82 |

# BURN STATUS REPORT

X X ☐ :  $\Delta$ TIG  
 X X : BT  
☐ :  $V_{gx}$

## TRIM

X X X R  
 X X X P  
 X X X Y  
☐  $V_{gx}$   
☐  $V_{gy}$   
☐  $V_{gz}$   
☐  $\Delta V_c$

X X X FUEL  
 X X X OX  
 X X X UNBALANCE

REMARKS:

# MCC'S

## BURN CHART

|          | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME | RESIDUALS       |
|----------|---------------------|------------------|------------------|-----------------|
| MCC(ALL) | 10°/SEC<br>TAKEOVER | 10°<br>TAKEOVER  | B/T +1 SEC       | TRIM TO 0.2 fps |

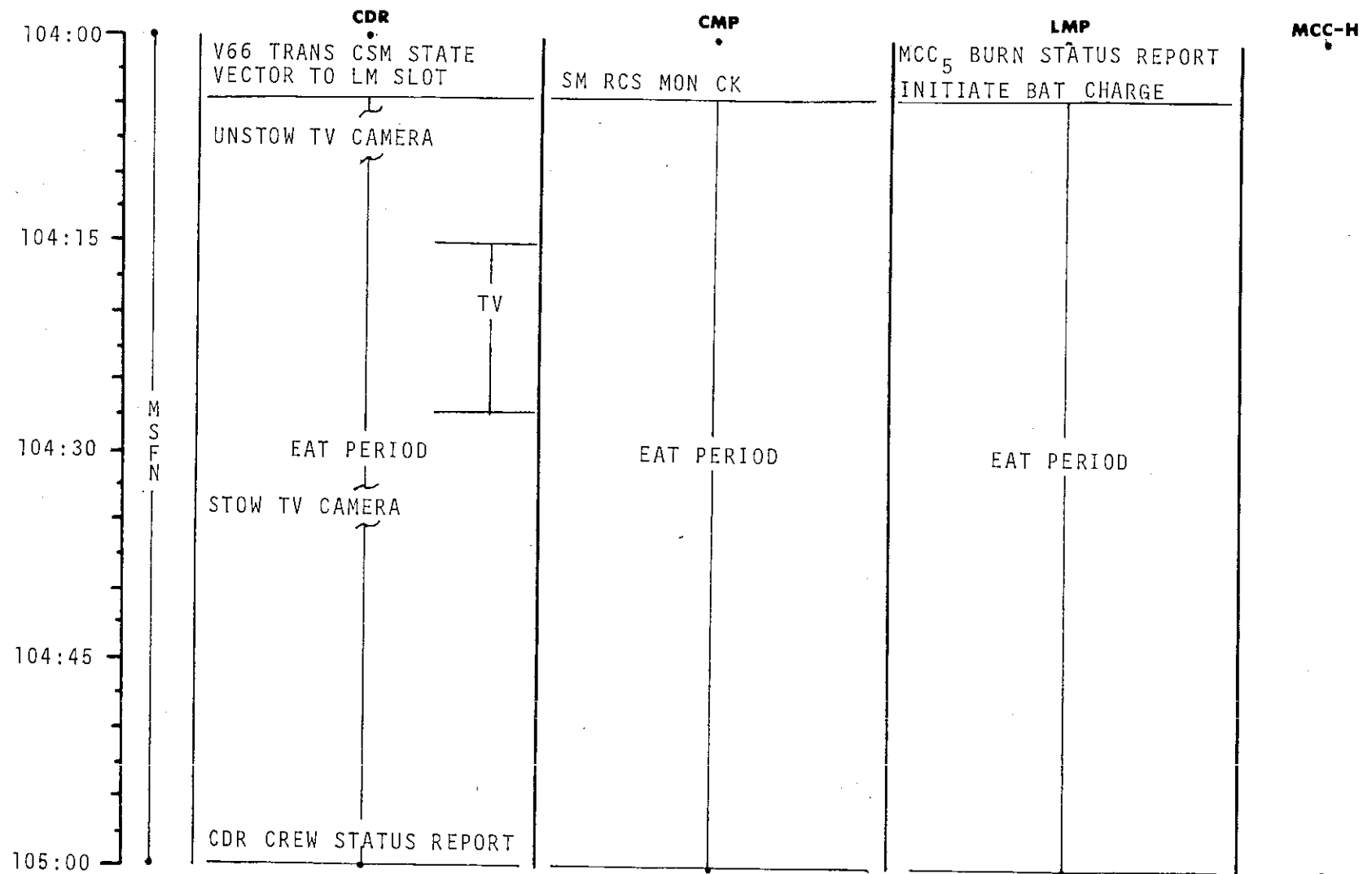
2-82a

# FLIGHT PLAN

|        | CDR                                      | CMP            | LMP | MCC-H           |
|--------|------------------------------------------|----------------|-----|-----------------|
| 103:00 |                                          |                |     |                 |
| 103:15 | V47 TRANS LM STATE<br>VECTOR TO CSM SLOT |                |     |                 |
|        | EXT ΔV P30                               |                |     |                 |
| 103:30 | SPS/RCS THRUST P40/41                    |                |     |                 |
|        | MNVR TO BURN ATT                         | SXT STAR CK    |     |                 |
| 103:45 | EMS TEST                                 | TRANS TO COUCH |     | PIPA BIAS<br>CK |
| TEI+15 | GDC ALIGN TO IMU                         | SM RCS MON CK  |     |                 |
| 104:00 | MCC <sub>5</sub> ΔV=NOMINALLY ZERO       |                |     |                 |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 103:00 - 104:00 | 5/TEC   | 2-83 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 104:00 - 105:00 | 5/TEC   | 2-84 |

MSC Form 1910 (Nov 68)

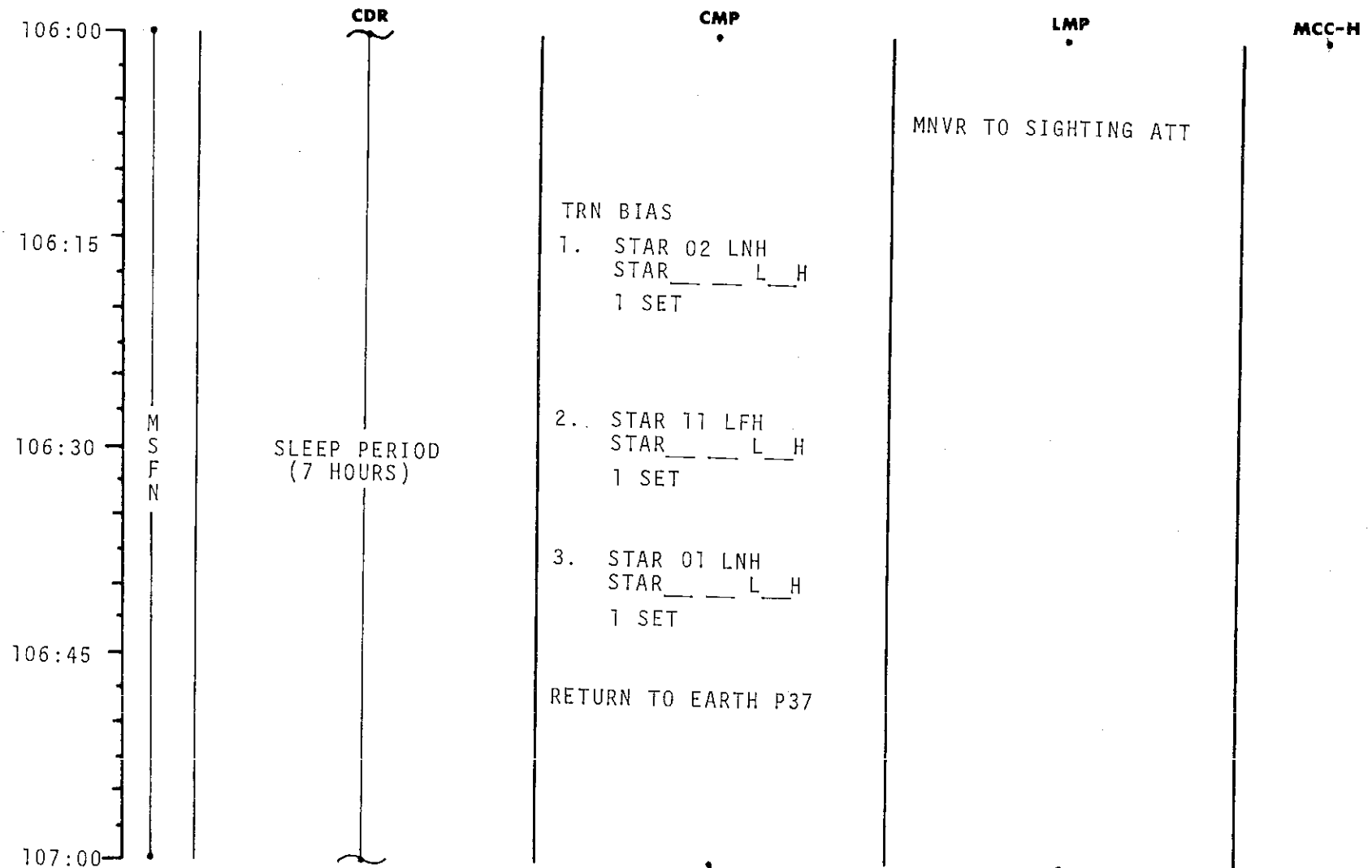
FLIGHT PLANNING BRANCH

# FLIGHT PLAN

|        | CDR              | CMP                                                                   | LMP                                                             | MCC-H |
|--------|------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------|-------|
| 105:00 |                  | LIQH CANISTER CHANGE<br>(CARTRIDGE NO. 12 FROM<br>B6 INTO CANISTER B) | BIOMED SW - CENTER<br>TRANS TO LH COUCH<br>MNVF TO SIGHTING ATT |       |
| 105:15 |                  | TRN BIAS<br>CISLUNAR NAVIGATION P23                                   |                                                                 |       |
| 105:30 | M<br>S<br>F<br>N | SLEEP PERIOD<br>(7 HOURS)                                             |                                                                 |       |
| 105:45 |                  | 1. STAR 22 EFH<br>STAR__ __ E__H<br>2 SETS                            |                                                                 |       |
|        |                  | 2. STAR 26 ENH<br>STAR__ __ E__H<br>1 SET                             |                                                                 |       |
| 106:00 |                  | 3. STAR 31 ENH<br>STAR__ __ E__H<br>2 SETS                            |                                                                 |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 105:00 - 106:00 | 5/TEC   | 2-85 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 106:00 - 107:00 | 5/TEC   | 2-86 |

MSC Form 1810 (Nov 68)

FLIGHT PLANNING BRANCH



# FLIGHT PLAN

|        | CDR                       | CMP                                                                                                                                                                 | LMP                                                                      | MCC-H |
|--------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------|
| 107:00 | SLEEP PERIOD<br>(7 HOURS) | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID _____<br>STAR ANGLE DIFF _____<br><br>TORQUE ANGLES: _____<br>X _____<br>Y _____<br>Z _____<br><br>TRN BIAS _____ | MNVR TO P52 ATTITUDE<br><br><br><br><br><br><br><br>MNVR TO SIGHTING ATT |       |
| 107:15 |                           |                                                                                                                                                                     |                                                                          |       |
| 107:30 |                           |                                                                                                                                                                     |                                                                          |       |
| 107:45 |                           |                                                                                                                                                                     |                                                                          |       |
| 108:00 |                           |                                                                                                                                                                     |                                                                          |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 107:00 - 108:00 | 5/TEC   | 2-87 |

# FLIGHT PLAN

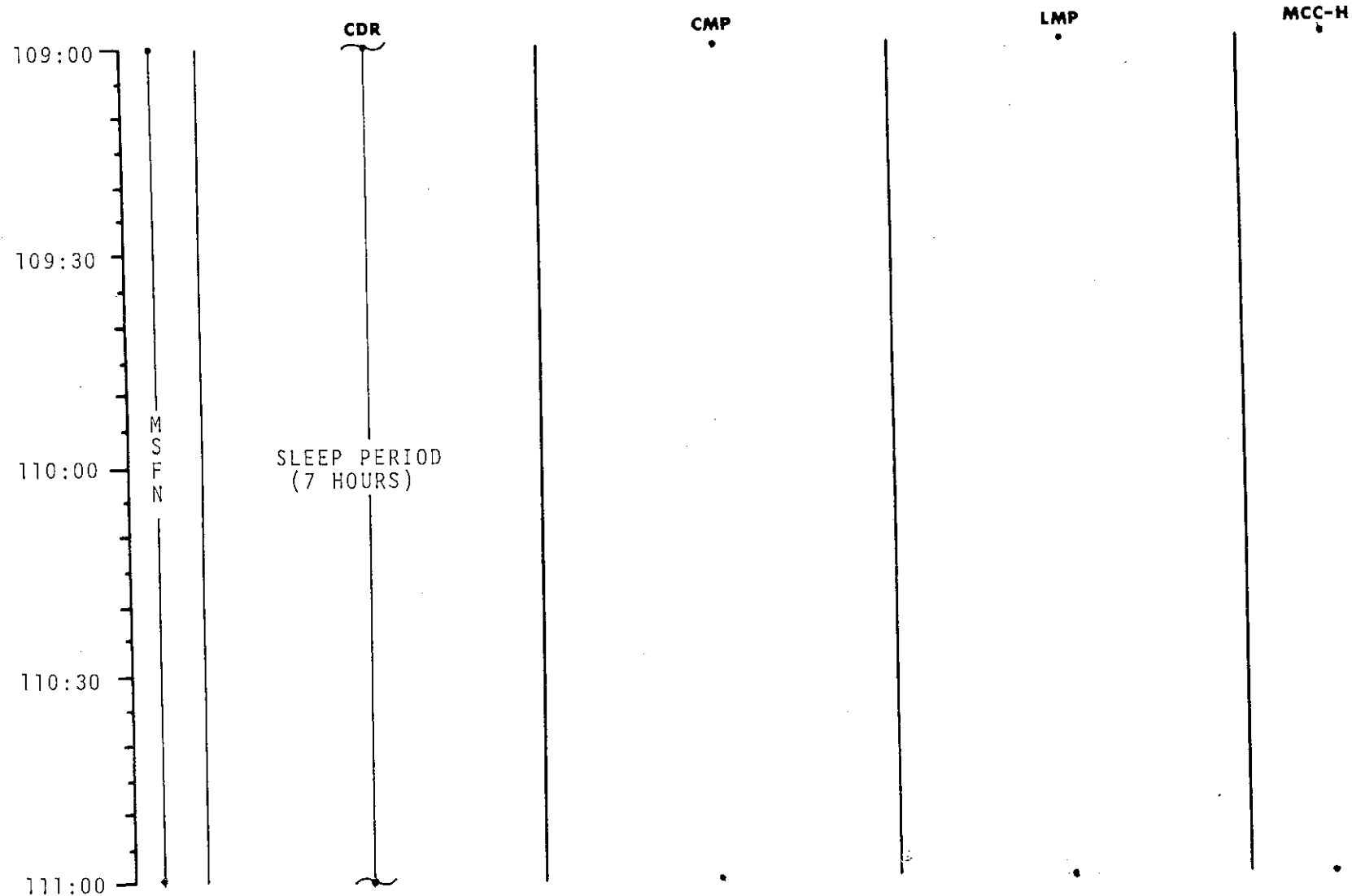
|        | CDR                       | CMP                                          | LMP                                                                                                                           | MCC-H             |
|--------|---------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 108:00 | SLEEP PERIOD<br>(7 HOURS) | CISLUNAR NAVIGATION P23                      |                                                                                                                               | BIOMED Sw - RIGHT |
|        |                           | 1. STAR 22 EFH<br>STAR ___ E ___ H<br>1 SET  |                                                                                                                               |                   |
| 108:15 |                           | 2. STAR 26 ENH<br>STAR ___ E ___ H<br>2 SETS |                                                                                                                               |                   |
| 108:30 |                           | 3. STAR 31 ENH<br>STAR ___ E ___ H<br>2 SETS |                                                                                                                               |                   |
| 108:45 |                           | RETURN TO EARTH P37                          |                                                                                                                               |                   |
| 109:00 |                           |                                              | MNVR TO PTC ATT<br>P 029 Y 315<br>P ___ Y<br>ROLL 0.1°/SEC<br>P&Y FREE<br>REESTABLISH PTC AT +15°<br>IN P OR Y AND RECORD GET |                   |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 108:00 - 109:00 | 5/TEC   | 2-88 |

MSC Form 1910 (Nov 68)

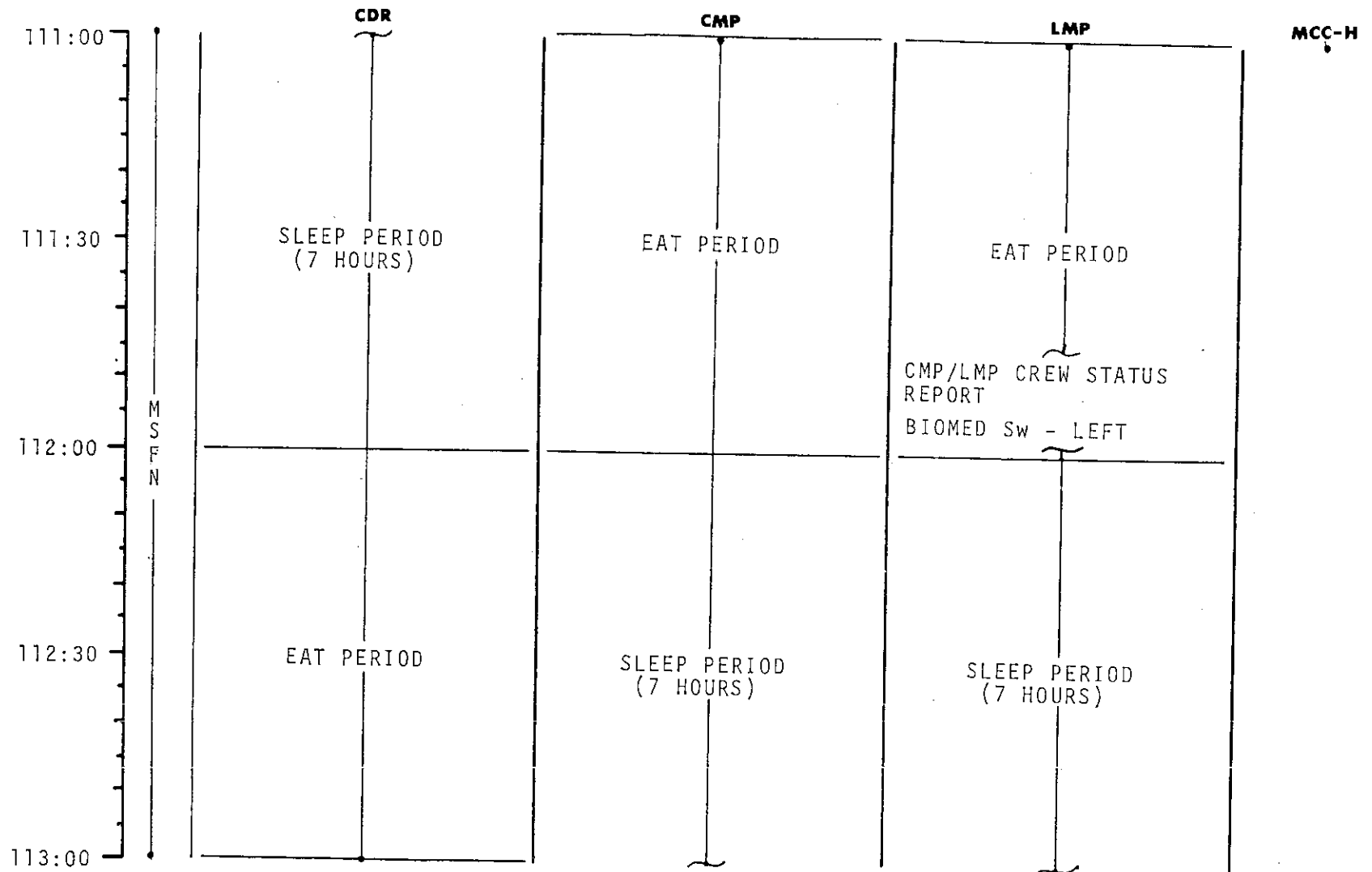
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 109:00 - 111:00 | 5/TEC   | 2-89 |

# FLIGHT PLAN

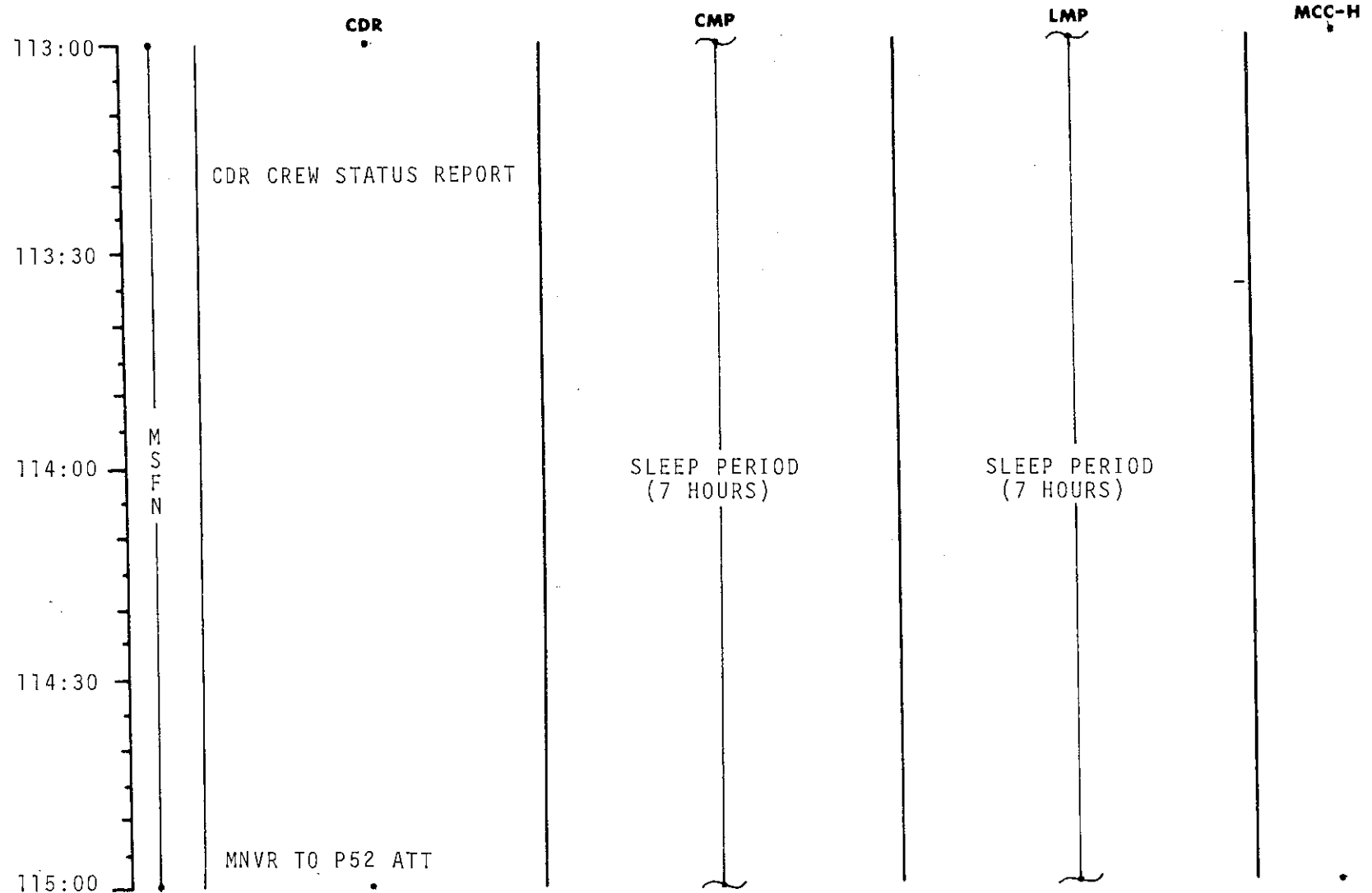


| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 111:00 - 113:00 | 5/TEC   | 2-90 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 113:00 - 115:00 | 5/TEC   | 2-91 |

# FLIGHT PLAN

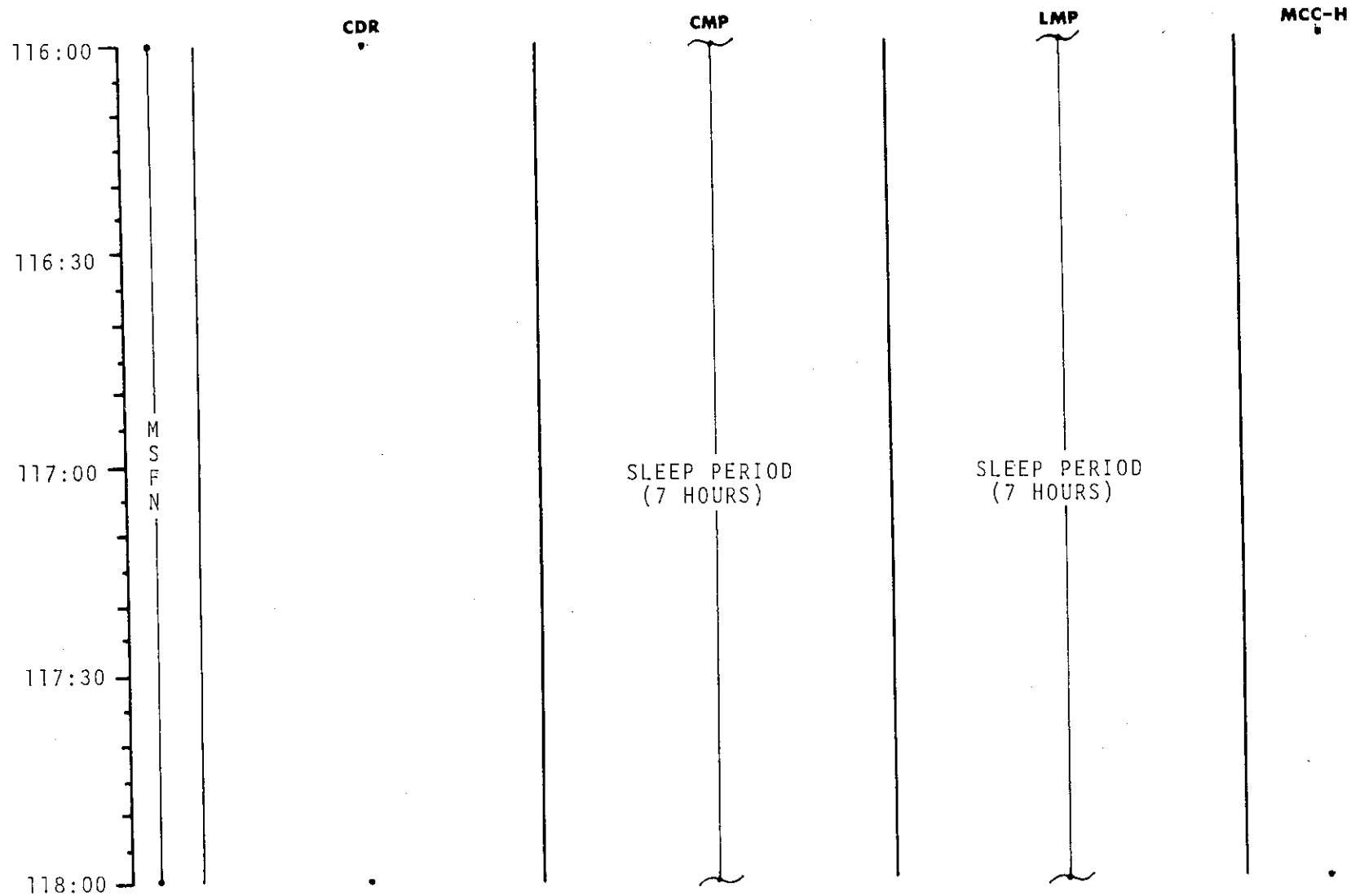
|        | CDR                                                                                                                                                                                                                     | CMP                       | LMP                       | MCC-H |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|-------|
| 115:00 | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID _____<br>STAR ANGLE DIFF _____<br><br>TORQUE ANGLES: _____<br>X _____<br>Y _____<br>Z _____<br><br>MNVR TO PTC ATT<br>P 029 Y 315<br>P _____ Y _____<br>ROLL 0.1°/SEC | SLEEP PERIOD<br>(7 HOURS) | SLEEP PERIOD<br>(7 HOURS) |       |
| 115:15 |                                                                                                                                                                                                                         |                           |                           |       |
| 115:30 |                                                                                                                                                                                                                         |                           |                           |       |
| 115:45 |                                                                                                                                                                                                                         |                           |                           |       |
| 116:00 |                                                                                                                                                                                                                         |                           |                           |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 115:00 - 116:00 | 5/TEC   | 2-92 |

MSC Form 1910 (Nov 68)

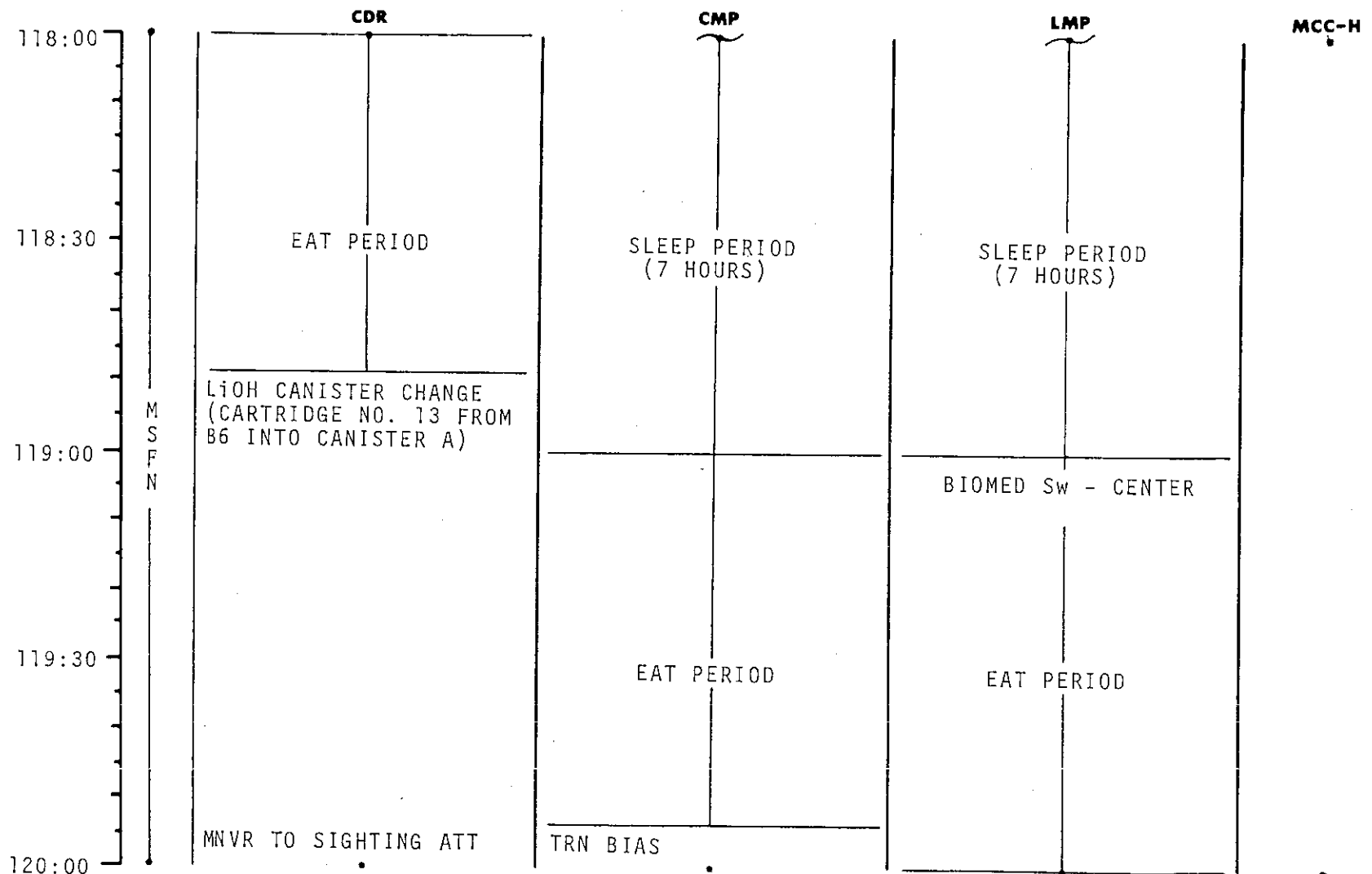
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 116:00 - 118:00 | 5/TEC   | 2-93 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 118:00 - 120:00 | 5/TEC   | 2-94 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH



# FLIGHT PLAN

CDR

CMP

LMP

MCC-H

120:00  
120:15  
120:30  
120:45  
121:00

MSFN

## CISLUNAR NAVIGATION P23

1. STAR 22 EFH  
STAR\_\_ E\_\_H  
1 SET
2. STAR 26 ENH  
STAR\_\_ E\_\_H  
1 SET
3. STAR 31 ENH  
STAR\_\_ E\_\_H  
1 SET

CMP/LMP CREW STATUS  
REPORT

RECORD MNVR PAD

P27 UPDATE:  
STATE  
VECTOR  
TGT LOAD  
VOICE  
UPDATE:  
MNVR PAD

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 120:00 - 121:00 | 6/TEC   | 2-95 |

# BURN STATUS REPORT

X X ☐ : ATIG

X X : BT

☐ V<sub>gx</sub>

TRIM

X X X R

X X X P

X X X Y

☐ V<sub>gx</sub>

☐ V<sub>gy</sub>

☐ V<sub>gz</sub>

☐ ΔV<sub>c</sub>

X X X FUEL

X X X OX

X X X UNBALANCE

REMARKS:

2-95a

MCC'S

## BURN CHART

|          | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME | RESIDUALS       |
|----------|---------------------|------------------|------------------|-----------------|
| MCC(ALL) | 10°/SEC<br>TAKEOVER | 10°<br>TAKEOVER  | B/T +1 SEC       | TRIM TO 0.2 fns |

# FLIGHT PLAN

|              | CDR                                                                                                                                            | CMP                                                                                                                                                                                                    | LMP | MCC-H        |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------|
| 121:00       | MNVR TO P52 ATT                                                                                                                                | <div>IMU REALIGN P52</div> <div>OPTION 3 - REFSMMAT</div> <div>STAR ID _____</div> <div>STAR ANGLE DIFF _____</div> <div>TORQUE ANGLES:</div> <div>X _____</div> <div>Y _____</div> <div>Z _____</div> |     |              |
| 121:15       |                                                                                                                                                |                                                                                                                                                                                                        |     |              |
| 121:30       | <div>V47 TRANS LM STATE</div> <div>VECTOR TO CSM SLOT</div> <div>EXT ΔV P30</div> <div>SPS/RCS THRUST P40/41</div> <div>MNVR TO BURN ATT</div> |                                                                                                                                                                                                        |     |              |
| 121:45       | EMS TEST                                                                                                                                       | <div>SXT STAR CK</div> <div>TRANS TO COUCH</div>                                                                                                                                                       |     | PIPA BIAS CK |
| TEI + 33 HRS | <div>GDC ALIGN</div> <div>MCC<sub>6</sub> ΔV=NOMINALLY ZERO</div>                                                                              | SM RCS MON CK                                                                                                                                                                                          |     |              |
| 122:00       |                                                                                                                                                |                                                                                                                                                                                                        |     |              |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 121:00 - 122:00 | 6/TEC   | 2-96 |

# FLIGHT PLAN

|        | CDR                                      | CMP                                          | LMP                                                        | MCC-H |
|--------|------------------------------------------|----------------------------------------------|------------------------------------------------------------|-------|
| 122:00 | V66 TRANS CSM STATE<br>VECTOR TO LM SLOT | SM RCS MON CK                                | MCC <sub>6</sub> BURN STATUS REPORT<br>INITIATE BAT CHARGE |       |
| 122:15 |                                          |                                              | BIOMED SW - RIGHT                                          |       |
| 122:30 | M<br>S<br>F<br>N<br>MNVR TO SIGHTING ATT | TRN BIAS<br>CISLUNAR NAVIGATION P23          |                                                            |       |
| 122:45 |                                          | 1. STAR 02 LNH<br>STAR ___ L ___ H<br>2 SETS |                                                            |       |
| 123:00 |                                          | 2. STAR 01 LNH<br>STAR ___ L ___ H<br>1 SET  |                                                            |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 122:00 - 123:00 | 6/TEC   | 2-97 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

# FLIGHT PLAN

CDR

CMP

LMP

MCC-H

123:00

MNVR TO SIGHTING ATT

123:30

123:30

M  
S  
F  
N

123:45

124:00

RETURN TO EARTH P37

TRN BIAS

CISLUNAR NAVIGATION P23

1. STAR 22 EFH  
STAR\_\_E\_\_H  
2 SETS

2. STAR 26 ENH  
STAR\_\_E\_\_H  
1 SET

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 123:00 - 124:00 | 6/TEC   | 2-98 |

# FLIGHT PLAN

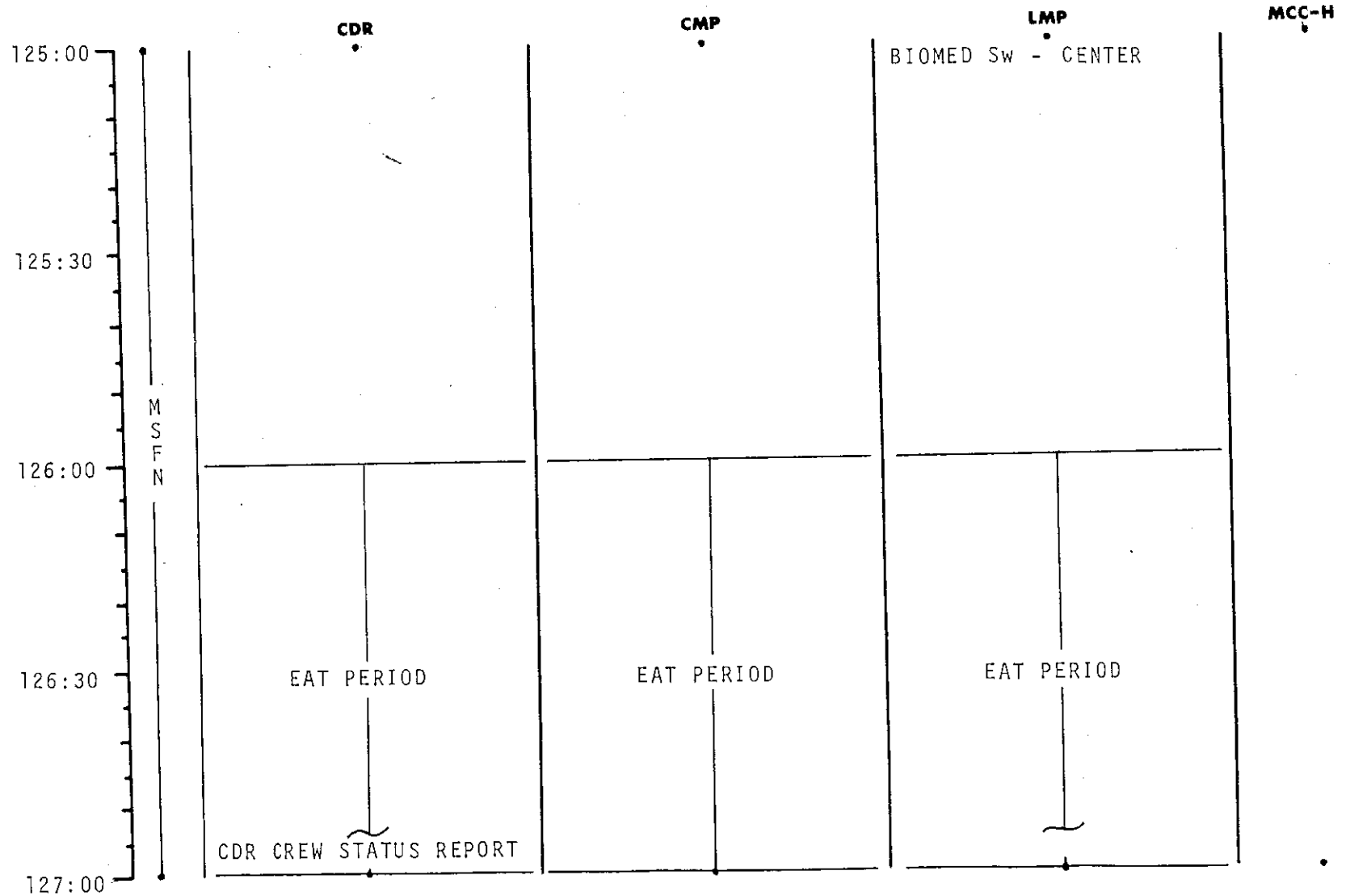
|        | CDR                                                    | CMP                                                        | LMP | MCC-H |
|--------|--------------------------------------------------------|------------------------------------------------------------|-----|-------|
| 124:00 |                                                        |                                                            |     |       |
|        |                                                        | 3. STAR 31 ENH<br>STAR <u>    </u> E <u>  </u> H<br>2 SETS |     |       |
| 124:15 |                                                        |                                                            |     |       |
|        |                                                        |                                                            |     |       |
| 124:30 | MSFN                                                   |                                                            |     |       |
|        | MNVR TO PTC ATT<br>P 029 Y 315<br>P Y<br>ROLL 0.1°/SEC | RETURN TO EARTH P37                                        |     |       |
| 124:45 |                                                        |                                                            |     |       |
|        |                                                        |                                                            |     |       |
| 125:00 |                                                        |                                                            |     |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 124:00 - 125:00 | 6/TEC   | 2-99 |

MSC Form 1910 (Nov 68)

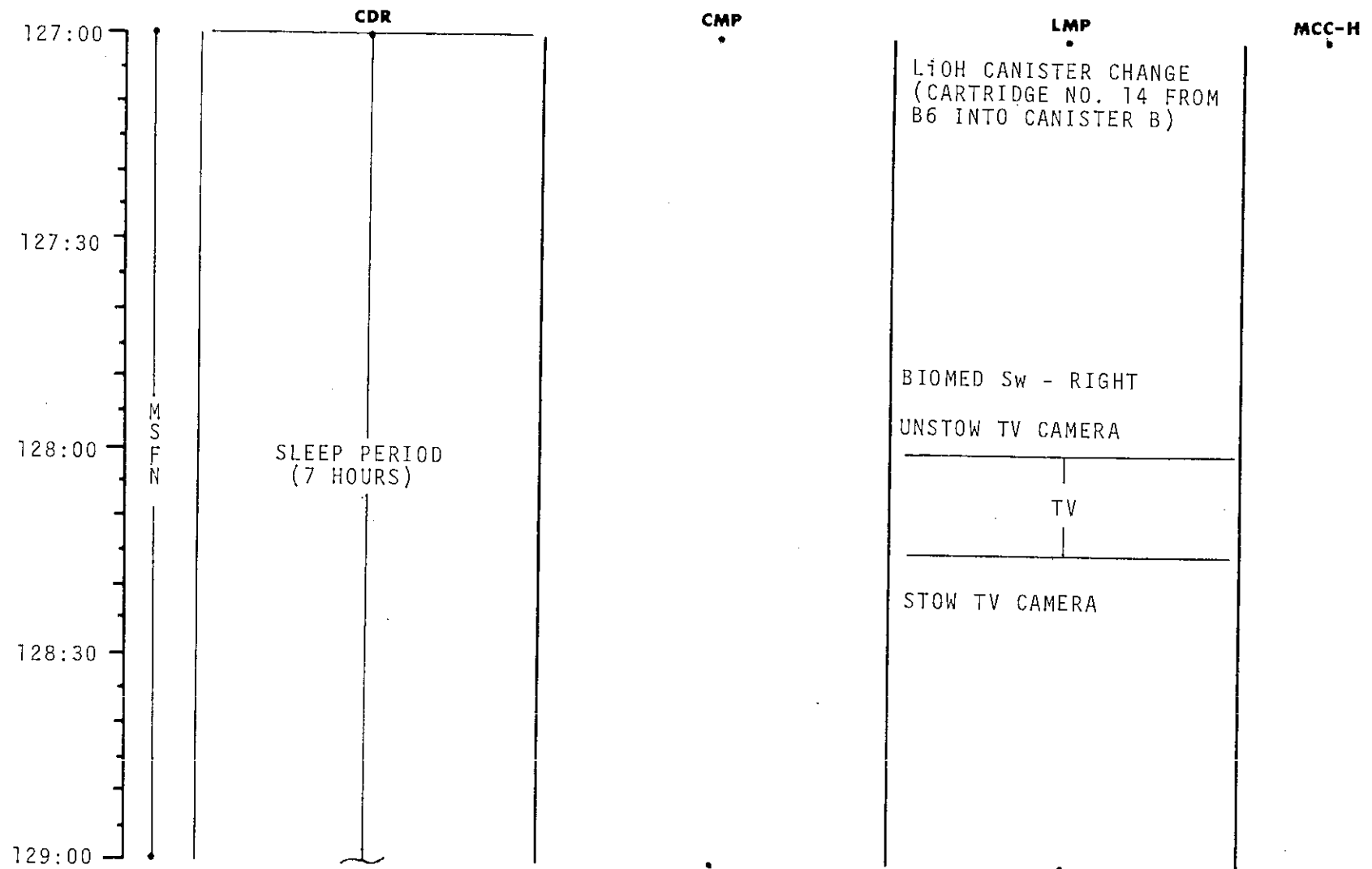
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 125:00 - 127:00 | 6/TEC   | 2-100 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 127:00 - 129:00 | 6/TEC   | 2-101 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH



# FLIGHT PLAN

|        | CDR                       | CMP                                                                                                                                          | LMP                  | MCC-H |
|--------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------|
| 129:00 | SLEEP PERIOD<br>(7 HOURS) | <hr/> IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID<br>STAR ANGLE DIFF<br><hr/> TORQUE ANGLES:<br>X _____<br>Y _____<br>Z _____<br><hr/> | MNVR TO P52 ATT      |       |
| 129:15 |                           |                                                                                                                                              |                      |       |
| 129:30 |                           |                                                                                                                                              |                      |       |
| 129:45 |                           |                                                                                                                                              |                      |       |
| 130:00 |                           |                                                                                                                                              |                      |       |
|        |                           | TRN BIAS                                                                                                                                     | MNVR TO SIGHTING ATT |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 129:00 - 130:00 | 6/TEC   | 2-102 |

# FLIGHT PLAN

|        | CDR                       | CMP                                        | LMP                  | MCC-H |
|--------|---------------------------|--------------------------------------------|----------------------|-------|
| 130:00 | SLEEP PERIOD<br>(7 HOURS) | CISLUNAR NAVIGATION P23                    |                      |       |
| 130:15 |                           | 1. STAR 02 LNH<br>STAR__ _ L__ H<br>2 SETS |                      |       |
| 130:30 |                           |                                            | MNVR TO SIGHTING ATT |       |
| 130:45 |                           | TRN BIAS                                   |                      |       |
| 131:00 |                           | 1. STAR 22 EFH<br>STAR__ _ E__ H<br>2 SETS |                      |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 130:00 - 131:00 | 6/TEC   | 2-103 |

MSC Form 1910 (Nov 68)

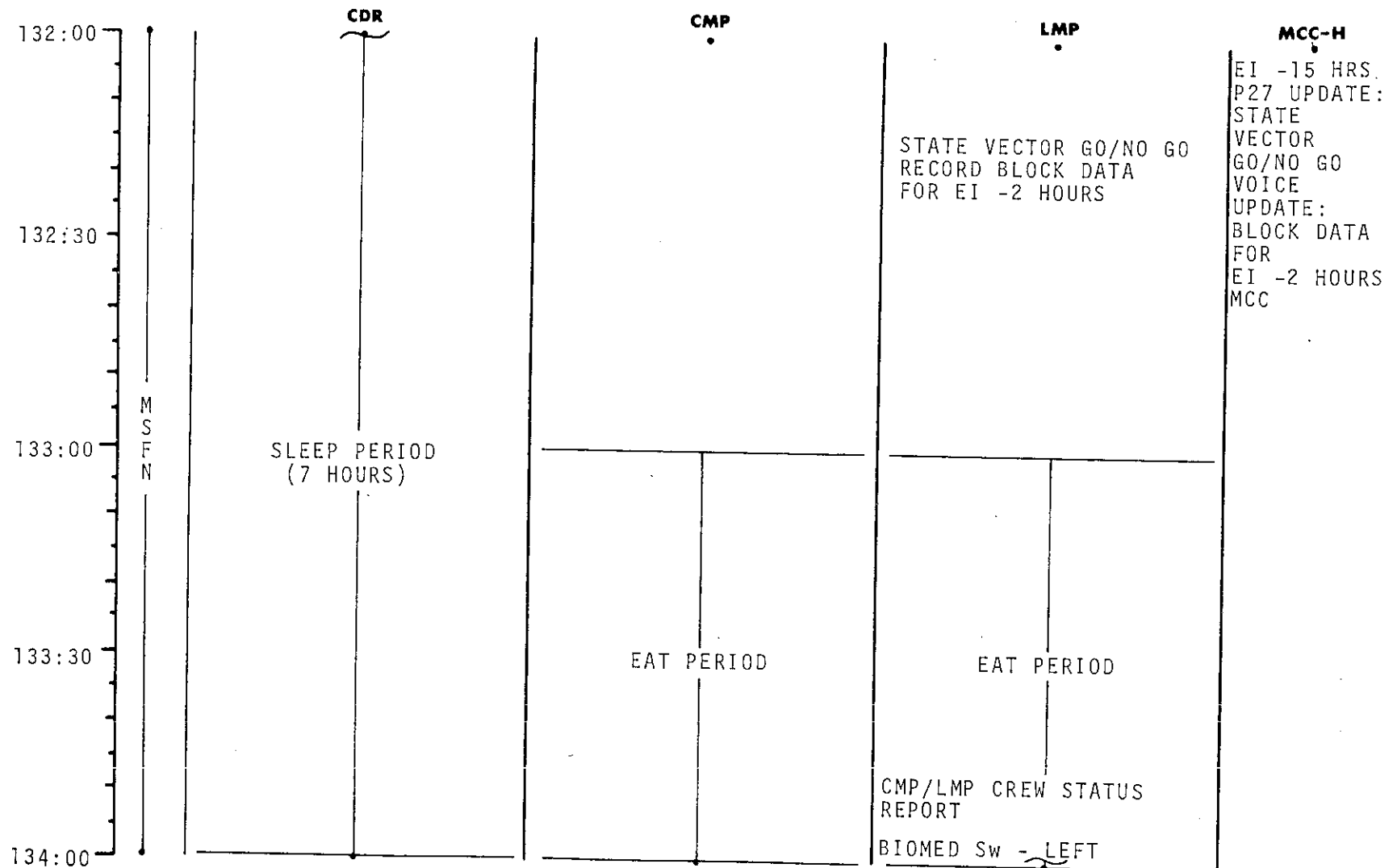
FLIGHT PLANNING BRANCH

# FLIGHT PLAN

|        | CDR                       | CMP                                      | LMP                                                    | MCC-H |
|--------|---------------------------|------------------------------------------|--------------------------------------------------------|-------|
| 131:00 |                           |                                          | BIOMED Sw - CENTER                                     |       |
| 131:15 |                           | 2. STAR 26 ENH<br>STAR___E___H<br>2 SETS |                                                        |       |
| 131:30 | SLEEP PERIOD<br>(7 HOURS) | 3. STAR 31 ENH<br>STAR___E___H<br>2 SETS |                                                        |       |
| 131:45 |                           |                                          |                                                        |       |
| 132:00 |                           | RETURN TO EARTH P37                      | MNVR TO PTC ATT<br>P 029 Y 315<br>P Y<br>ROLL 0.1°/SEC |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 131:00 - 132:00 | 6/TEC   | 2-104 |

# FLIGHT PLAN

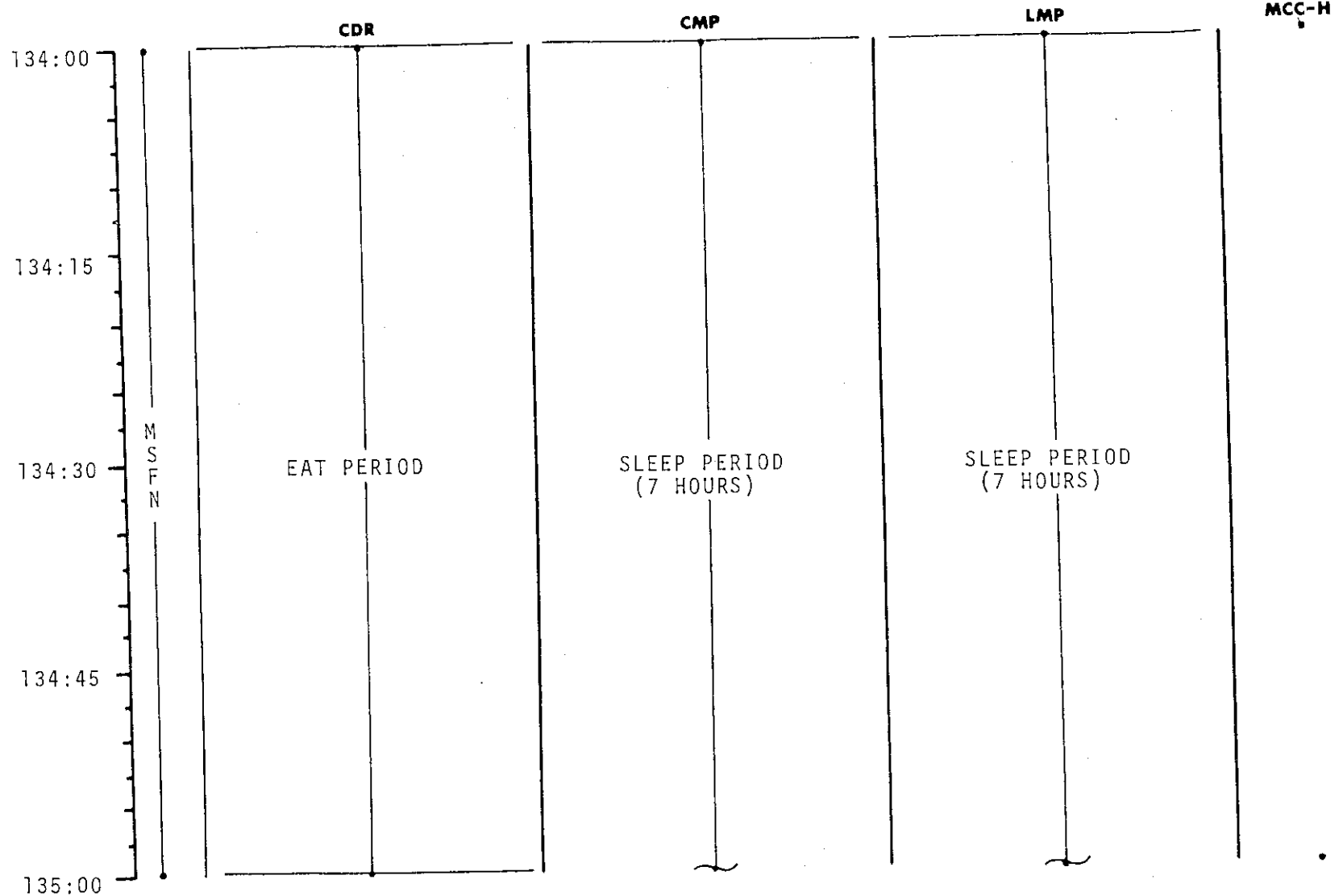


| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 132:00 - 134:00 | 6/TEC   | 2-105 |

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FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 134:00 - 135:00 | 6/TEC   | 2-106 |

# FLIGHT PLAN

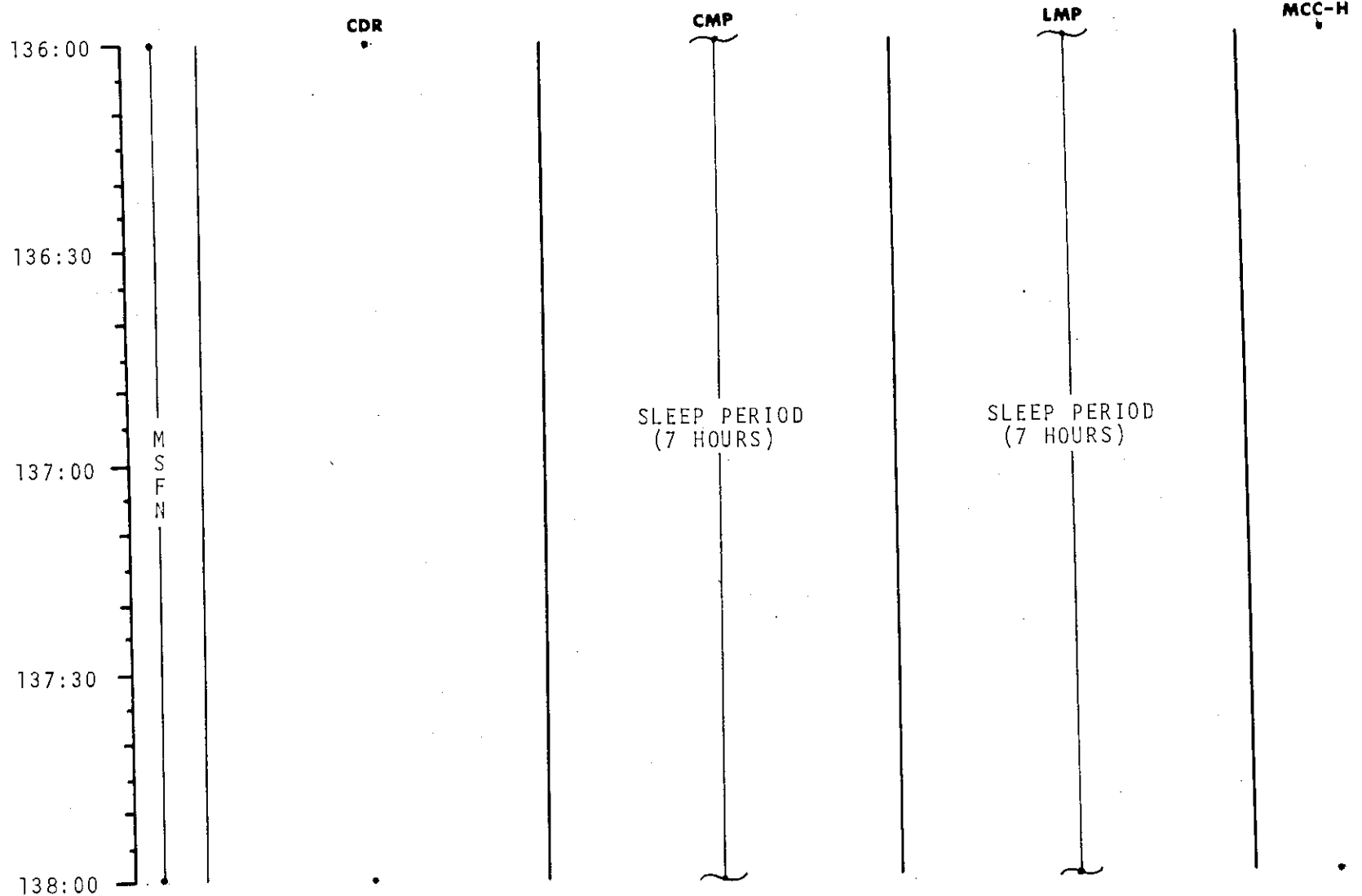
|        | CDR                                                                                                                                                                                            | CMP                       | LMP                       | MCC-H |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|-------|
| 135:00 | BEGIN CABIN COLD SOAK<br>CDR CREW STATUS REPORT                                                                                                                                                |                           |                           |       |
| 135:15 |                                                                                                                                                                                                |                           |                           |       |
| 135:30 | MNVR TO P52 ATT<br><br>IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID<br>STAR ANGLE DIFF<br><br>TORQUE ANGLES:<br>X<br>Y<br>Z<br><br>REESTABLISH PTC<br>P 029 Y 315<br>P Y<br>ROLL 0.1°/SEC | SLEEP PERIOD<br>(7 HOURS) | SLEEP PERIOD<br>(7 HOURS) |       |
| 135:45 |                                                                                                                                                                                                |                           |                           |       |
| 136:00 |                                                                                                                                                                                                |                           |                           |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 135:00 - 136:00 | 6/TEC   | 2-107 |

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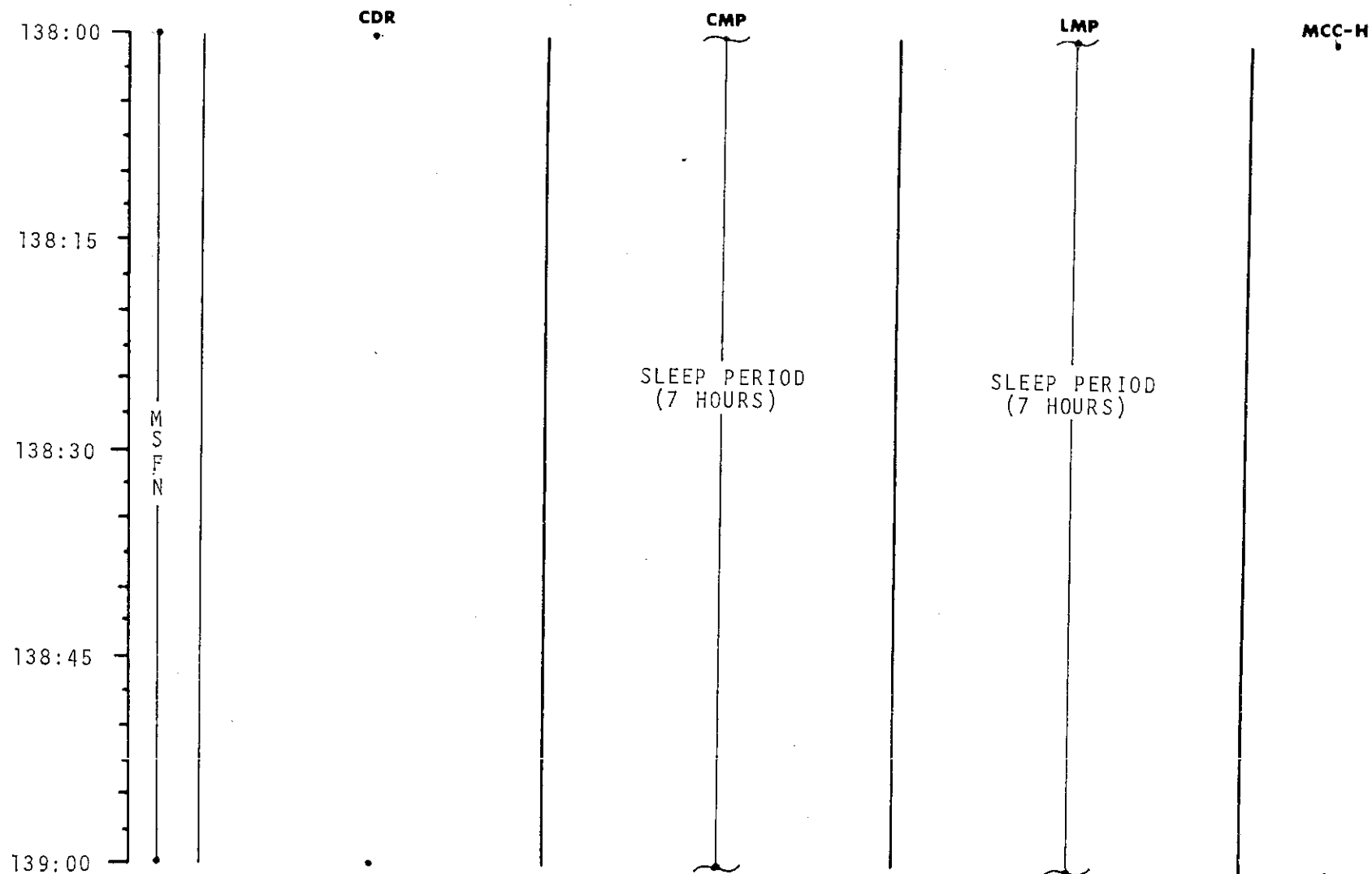
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 136:00 - 138:00 | 6/TEC   | 2-108 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 138:00 - 139:00 | 6/TEC   | 2-109 |

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FLIGHT PLANNING BRANCH

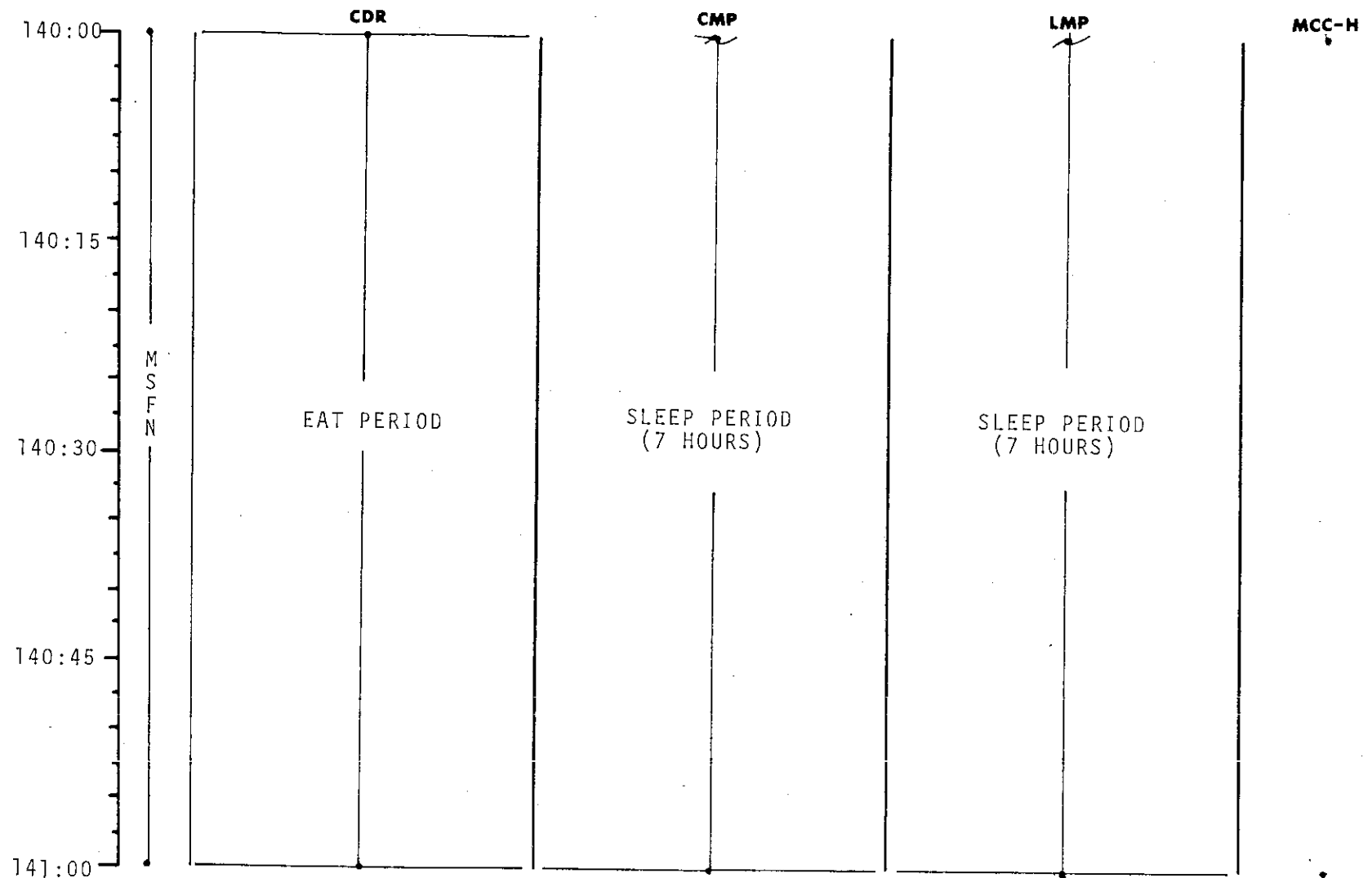


# FLIGHT PLAN

|        | CDR                                                                                                                                 | CMP                       | LMP                       | MCC-H |
|--------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|-------|
| 139:00 | L10H CANISTER CHANGE<br>(CARTRIDGE NO. 15 FROM<br>A2 INTO CANISTER A)                                                               |                           |                           |       |
| 139:15 | MNVR TO P52 ATT                                                                                                                     |                           |                           |       |
| 139:30 | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID _____<br>STAR ANGLE DIFF _____<br>TORQUE ANGLES:<br>X _____<br>Y _____<br>Z _____ | SLEEP PERIOD<br>(7 HOURS) | SLEEP PERIOD<br>(7 HOURS) |       |
| 139:45 | REESTABLISH PTC<br>P 029 Y 315<br>P Y<br>ROLL 0.1°/SEC                                                                              |                           |                           |       |
| 140:00 |                                                                                                                                     |                           |                           |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 139:00 - 140:00 | 6/TEC   | 2-110 |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 140:00 - 141:00 |         | 2-111 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

# FLIGHT PLAN

|        | CDR                  | CMP                                         | LMP               | MCC-H |
|--------|----------------------|---------------------------------------------|-------------------|-------|
| 141:00 | MNVR TO SIGHTING ATT | LOSS OF COMM CONTINGENCY<br>TRN BIAS        | BIOMED SW - RIGHT |       |
|        |                      | CISLUNAR NAVIGATION P23                     |                   |       |
| 141:15 |                      | 1. STAR 22 EFH<br>STAR ___ E ___ H<br>1 SET |                   |       |
|        |                      | 2. STAR 26 ENH<br>STAR ___ E ___ H<br>1 SET |                   |       |
|        |                      | 3. STAR 31 ENH<br>STAR ___ E ___ H<br>1 SET |                   |       |
| 141:30 |                      |                                             |                   |       |
|        |                      |                                             |                   |       |
| 141:45 |                      | EAT PERIOD                                  | EAT PERIOD        |       |
|        |                      |                                             |                   |       |
| 142:00 |                      |                                             |                   |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 141:00 - 142:00 |         | 2-112 |

# FLIGHT PLAN

142:00

CDR

CMP

LMP

MCC-H

142:15

EAT PERIOD

EAT PERIOD

142:30

M  
S  
F  
N

LOSS OF COMM CONTINGENCY

TRN BIAS

CISLUNAR NAVIGATION P23

142:45

1. STAR 22 EFH  
STAR\_\_ \_\_ E\_\_H  
1 SET

2. STAR 26 ENH  
STAR\_\_ \_\_ E\_\_H  
1 SET

143:00

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 142:00 - 143:00 |         | 2-113 |

MSC Form 1910 (Nov 68)

FLIGHT PLANNING BRANCH

# FLIGHT PLAN

|            | CDR             | CMP                                                                                                                                           | LMP                                        | MCC-H                                                             |
|------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|
| 143:00     |                 |                                                                                                                                               |                                            |                                                                   |
| EI-3.5 HRS |                 |                                                                                                                                               |                                            | P27 UPDATE:<br>STATE<br>VECTOR<br>(LM & CSM<br>SLOTS)<br>REFSMMAT |
| 143:30     | MNVR TO P52 ATT |                                                                                                                                               | RECORD MNVR AND ENTRY<br>PAD               | VOICE<br>UPDATE:<br>MNVR PAD<br>ENTRY PAD                         |
| EI-3 HRS   |                 | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>STAR ID _____<br>STAR ANGLE DIFF _____<br><br>TORQUE ANGLES: _____<br>X _____<br>Y _____<br>Z _____ | ECS CK                                     |                                                                   |
| 144:00     |                 | CMC SELF CK<br>DSKY COND LT TEST                                                                                                              | EPS CK<br>SPS CK<br>SM/CM RCS CK<br>C&W CK |                                                                   |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 143:00 - 144:00 | 6/TEC   | 2-114 |

|   |   |  |   |   |                 |
|---|---|--|---|---|-----------------|
| X | X |  | • |   | ΔTIG            |
| X | X |  | • |   | BT              |
|   |   |  |   | • | V <sub>gx</sub> |

| TRIM |   |   |   |                 |
|------|---|---|---|-----------------|
| X    | X | X |   | R               |
| X    | X | X |   | P               |
| X    | X | X |   | Y               |
|      |   |   | . | V <sub>gx</sub> |
|      |   |   | . | V <sub>gy</sub> |
|      |   |   | . | V <sub>gz</sub> |
|      |   |   | . | $\Delta V_c$    |
| X    | X | X |   | FUEL            |
| X    | X | X |   | OX              |
| X    | X | X |   | UNBALANCE       |

MCC 'S

## BURN CHART

|          | P or Y<br>RATES     | ATT<br>DEVIATION | SHUTDOWN<br>TIME | RESIDUALS       |
|----------|---------------------|------------------|------------------|-----------------|
| MCC(ALL) | 10°/SEC<br>TAKEOVER | 10°<br>TAKEOVER  | B/T +1 SEC       | TRIM TO 0.2 fps |

# FLIGHT PLAN

|            | CDR                                                                                                | CMP                | LMP | MCC-H                               |
|------------|----------------------------------------------------------------------------------------------------|--------------------|-----|-------------------------------------|
| 144:00     | EXT ΔV P30                                                                                         |                    |     |                                     |
| EI-2.5 HRS |                                                                                                    |                    |     |                                     |
| 144:30     | SPS/RCS THRUST P40/41<br>MNVR TO BURN ATT                                                          | COAS & SXT STAR CK |     |                                     |
| EI-2 HRS   | GDC ALIGN TO IMU<br>MCC <sub>7</sub> ΔV=NOMINALLY ZERO<br>V66 TRANS CSM STATE<br>VECTOR TO LM SLOT |                    |     | MCC <sub>7</sub> BURN STATUS REPORT |
| 145:00     |                                                                                                    |                    |     |                                     |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 144:00 - 145:00 | 7/TEC   | 2-115 |

# FLIGHT PLAN

|           | CDR                               | CMP                                                                                                   | LMP              | MCC-H           |
|-----------|-----------------------------------|-------------------------------------------------------------------------------------------------------|------------------|-----------------|
| 145:00    |                                   |                                                                                                       |                  |                 |
|           |                                   | LOSS OF COMM CONTINGENCY<br>CISLUNAR NAVIGATION P23<br>1. STAR 33 ENH<br>STAR ____ E ____ H<br>1 SET  | BIOMED SW - LEFT |                 |
|           | MNVR TO ENTRY ATT<br>COAS STAR CK | SXT STAR CK                                                                                           |                  |                 |
| EI-1.5 HR |                                   | IMU REALIGN P52<br>OPTION 3 - REFSMMAT<br>AND GYRO DRIFT TEST<br>STAR ID ____<br>STAR ANGLE DIFF ____ |                  | PIPA BIAS<br>CK |
| 145:30    | MSFN                              | TORQUE ANGLES: ____<br>X ____<br>Y ____<br>Z ____                                                     |                  |                 |
|           | GDC ALIGN TO IMU                  |                                                                                                       |                  |                 |
|           | FINAL STOWAGE                     | FINAL STOWAGE<br>INITIATE CM RCS PREHEAT                                                              | FINAL STOWAGE    |                 |
|           |                                   | WASTE H <sub>2</sub> O DUMP - OFF<br>UR DUMP HT - OFF                                                 |                  |                 |
| EI-1 HR   | EMS CK                            |                                                                                                       |                  |                 |
| 146:00    | FINAL GDC DRIFT CK                |                                                                                                       |                  |                 |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
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FLIGHT PLANNING BRANCH



# FLIGHT PLAN

|           | CDR                                                                  | CMP                                                        | LMP             | MCC-H                                                                          |
|-----------|----------------------------------------------------------------------|------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------|
| 146:00    | SET DET TO EI<br>EMS INITIALIZATION<br>RSI ALIGN TO GDC<br>CM RCS CK | TERM CM RCS PREHEAT<br><br>RECORD ENTRY PAD & RCVY<br>INFO | PYRO BAT CK     | P27 UPDATE:<br>STATE<br>VECTOR<br>VOICE<br>UPDATE:<br>ENTRY PAD<br>& RCVY INFO |
| EI-30 MIN | SEPARATION CK LIST                                                   | SEPARATION CK LIST                                         | ENTRY BATS - ON |                                                                                |
| 146:30    | MNVR TO CM/SM SEP ATT                                                | P61 ENTRY PREP                                             | GO FOR PYRO ARM | GO FOR<br>PYRO ARM                                                             |
| EI-15 MIN | CM/SM SEP<br>MNVR TO ENTRY ATT                                       | P62 ENTRY ATTITUDE                                         |                 |                                                                                |
| 146:50    | EI = 400K                                                            | P63 ENTRY INITIATE                                         |                 |                                                                                |
| 147:00    |                                                                      | P64 ENTRY POST .05G                                        |                 |                                                                                |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
|-----------|---------|-------------------|-----------------|---------|-------|
| AS503/103 | FINAL   | November 22, 1968 | 146:00 - 147:00 | 7/TEC   | 2-117 |

# FLIGHT PLAN

|        | CDR                 | CMP | LMP | MCC-H |
|--------|---------------------|-----|-----|-------|
| 147:00 | EARTH LANDING PHASE |     |     |       |
|        | SPLASHDOWN          |     |     |       |
| 147:30 |                     |     |     |       |
| 148:00 |                     |     |     |       |

| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE  |
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FLIGHT PLANNING BRANCH

SECTION III - CONSUMABLES ANALYSIS

SECTION III



NOTE

Acknowledgement is made to the Consumables Analysis Section (CAS) of the Mission Planning and Analysis Division (MPAD) for their work in the preparation of the RCS and cryogenics consumable analysis presented herein.

### AS503/103 PROPELLANT BUDGET

The results of the SM, CM, and SPS propellant budget analysis are summarized in the following tables and figures:

|              |                                    |      |
|--------------|------------------------------------|------|
| TABLE 3-I    | SM-RCS PROPELLANT LOADING          | 3-3  |
| TABLE 3-II   | SM-RCS USAGE SUMMARY               | 3-4  |
| TABLE 3-III  | SM-RCS PROPELLANT BUDGET           | 3-10 |
| TABLE 3-IV   | CM-RCS PROPELLANT USAGE SUMMARY    | 3-40 |
| TABLE 3-V    | SPS PROPELLANT BUDGET              | 3-41 |
| FIGURE 3-I   | SM RCS PROPELLANT PROFILE          | 3-5  |
| FIGURE 3-II  | SM RCS PROPELLANT PROFILE (QUAD A) | 3-6  |
| FIGURE 3-III | SM RCS PROPELLANT PROFILE (QUAD B) | 3-7  |
| FIGURE 3-IV  | SM RCS PROPELLANT PROFILE (QUAD C) | 3-8  |
| FIGURE 3-V   | SM RCS PROPELLANT PROFILE (QUAD D) | 3-9  |

#### Ground Rules and Assumption

The ground rules and assumptions listed below were used in the construction of the SM-RCS budget.

1. Data Source: Data for SM-RCS engine performance and propellant requirements were obtained from Part 4, Volume I of the CSM/LM Spacecraft Operational Data Book, May 1968.
2. Maneuvers: Since it is impossible to predict in all cases what maneuvers rates or angles will be required, it was assumed that all maneuvers were 3 axis at rates from 0.2 deg/sec to 0.5 deg/sec (unless otherwise noted). Angles were varied as a function of the maneuver requirements. It was also assumed that all IMU alignments

required 3 axis orientations with allowances for minimum impulse control, which may seem conservative, but would allow for unscheduled attitude maneuvers. An increase in passive thermal control (PTC) requirements was made to allow for active control of the non spin axis. Predicted costs are 1.3 lbs/hr translunar and 1.4 lbs/hrs transearth with a total increase of 100 lbs to the budget. Navigation sightings, like the IMU alignments, were predicted at the same cost for all cases. Realizing that some navigation sightings will be easier to accomplish than others, a nominal cost of approximately 1.2 lbs to 1.4 lbs per set was predicted. Lunar orbit maneuvers were budgeted to the lunar orbit attitude timeline. Midcourse translations were budgeted as SM-RCS corrections of 3 fps and 7 fps translunar and 10 fps, 5 fps, and 2 fps transearth. The 7 fps correction could possibly be replaced by a SPS burn since it is 2 fps above the minimum SPS burn time for translunar corrections.

3. Flow Rates: A propellant flow rate of 0.361 lbs/sec/engine was assumed for steady state operation. A minimum impulse burn has been predicted at a maximum 0.005 lbs per pulse.

TABLE 3-I

SM-RCS PROPELLANT LOADING

|                                   | <u>SM-RCS-LBS</u> |
|-----------------------------------|-------------------|
| Nominal Loaded                    | 1347.6            |
| Unusable (trapped)                | <u>26.8</u>       |
| Total Deliverable                 | 1320.8            |
| Loading Uncertainty (temperature) | <u>36.8</u>       |
|                                   | 1284.0            |
| Gauging Inaccuracy                | <u>72.0</u>       |
|                                   | 1212.0            |
| *Mixture Ratio Uncertainty        | <u>82.0</u>       |
|                                   | 1130.0            |
| Total Used                        | <u>655.8</u>      |
| Operational Margin                | 474.2             |

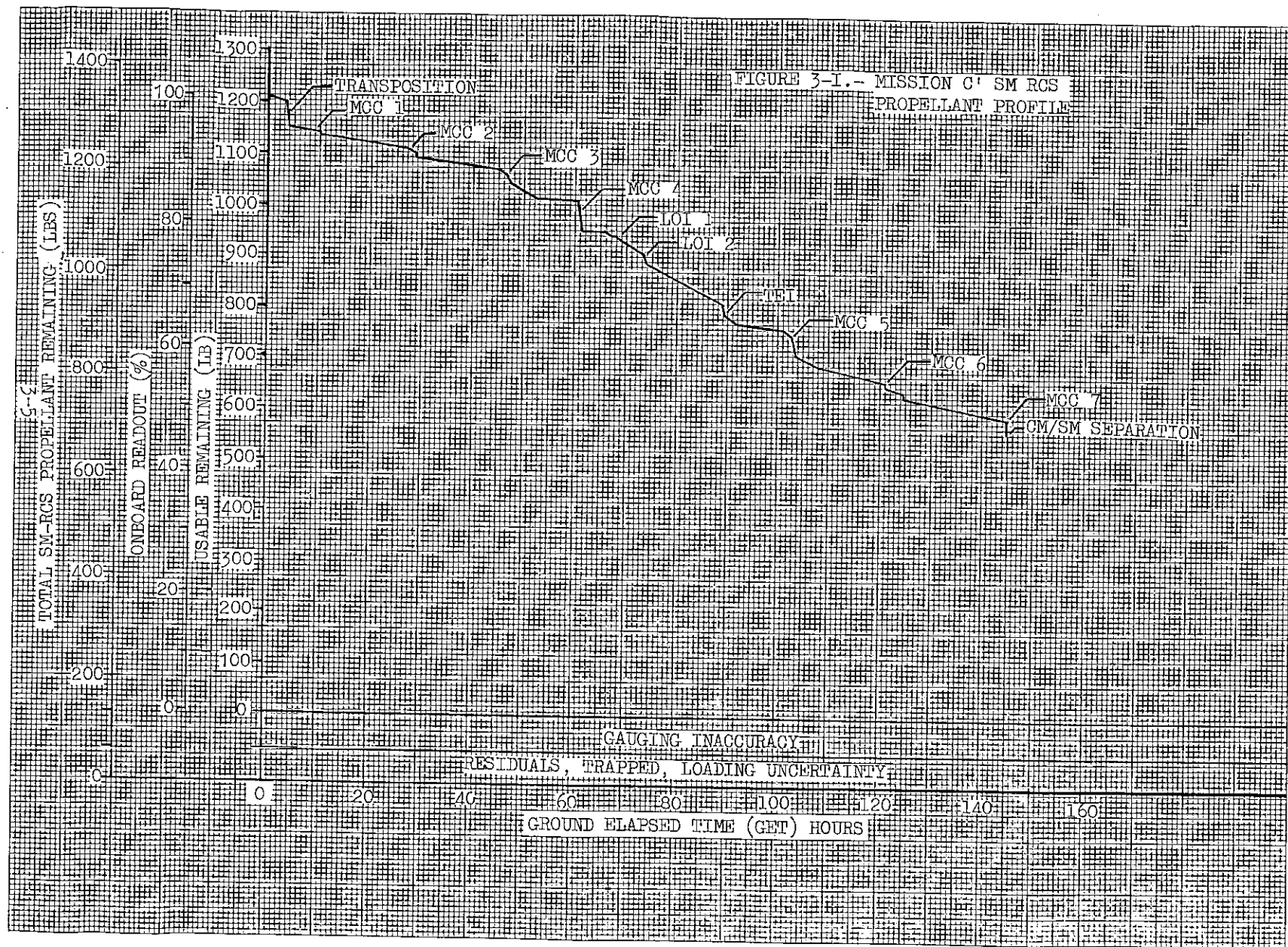
\*This quantity is based on the flight plan and results from the use of approximately 460 lbs of RCS for attitude orientation and control. This is considered unusable for mission planning and represents a dispersion to the nominal profile.

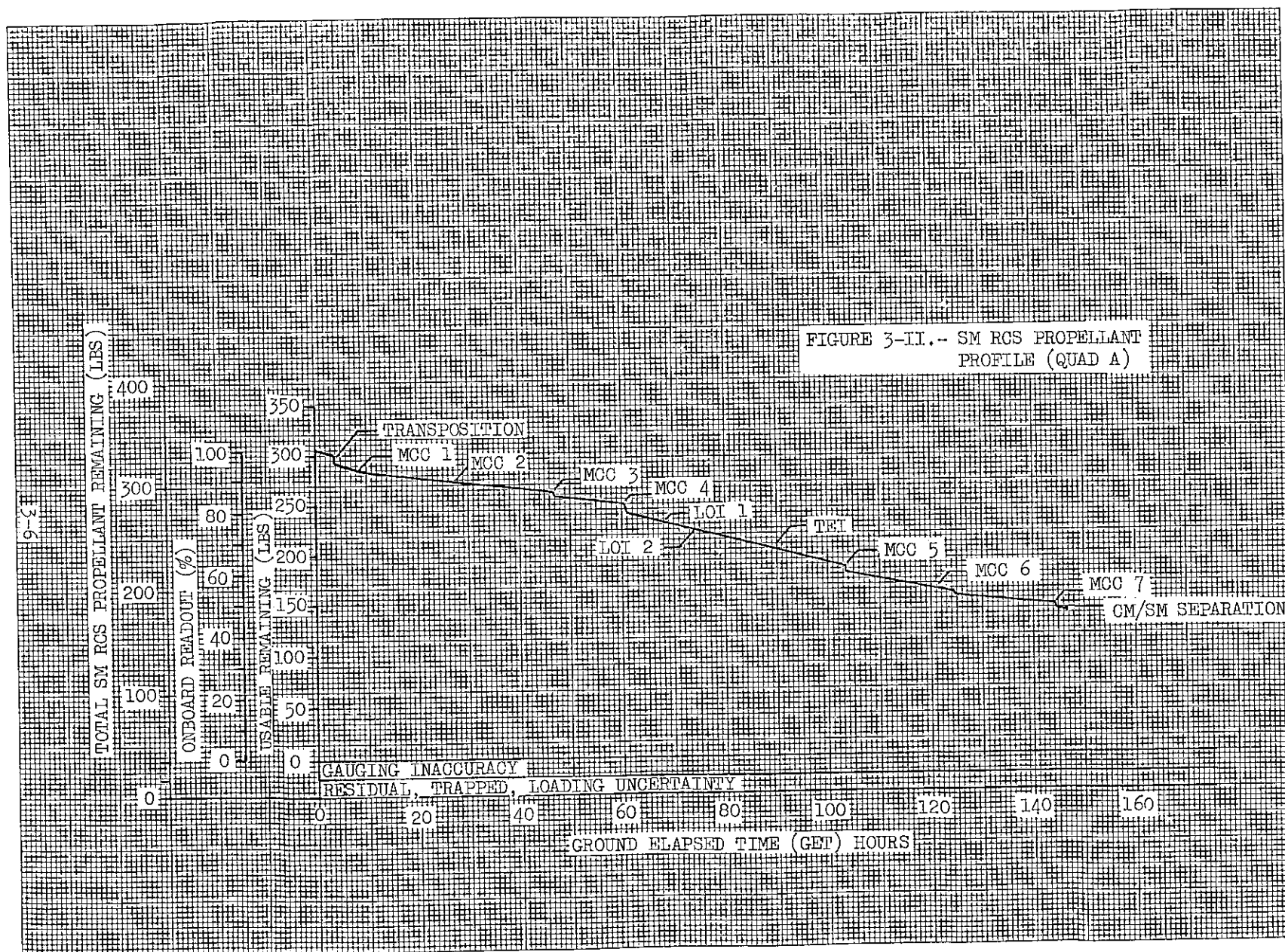


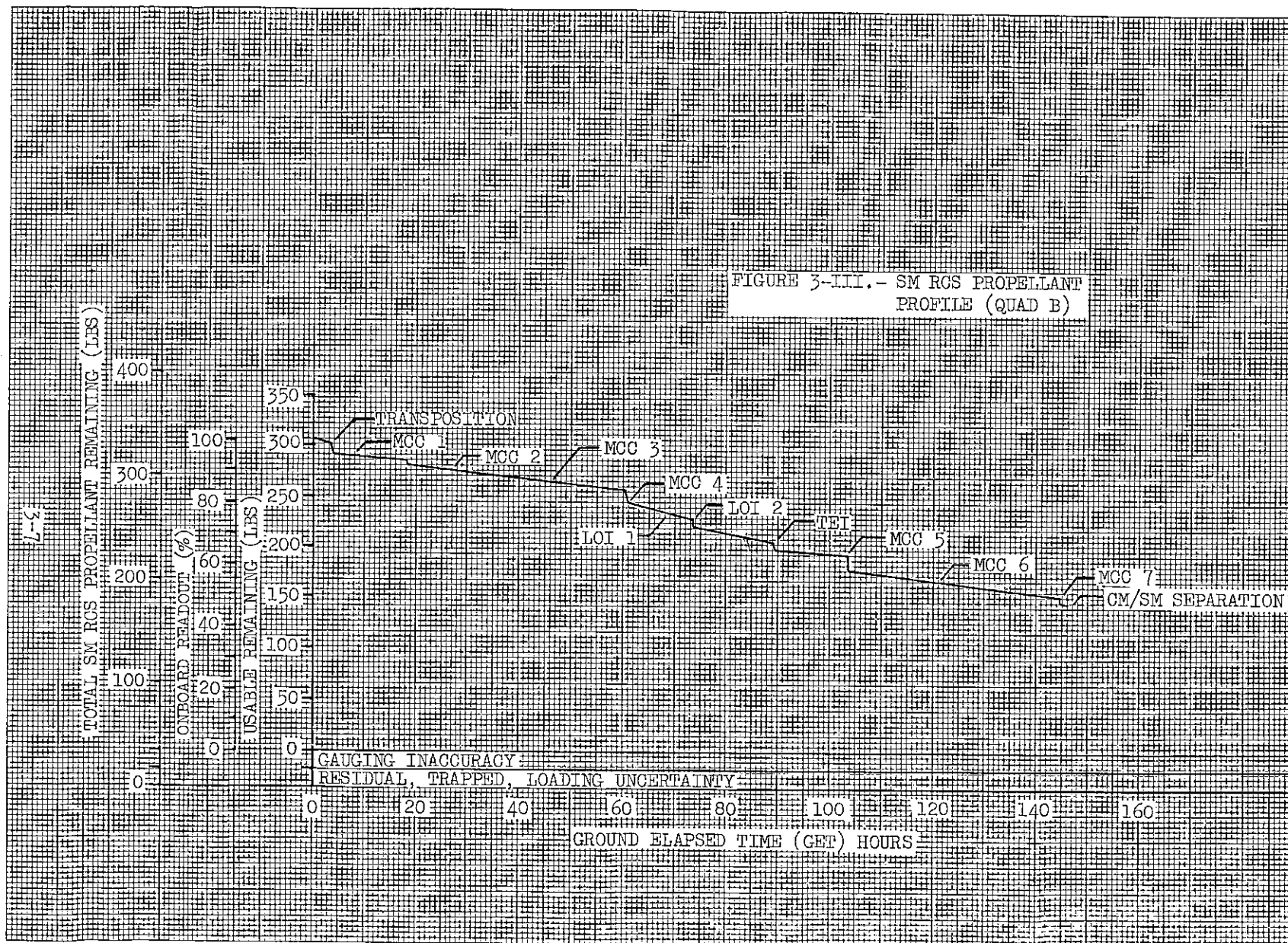
TABLE 3-II

SM RCS USAGE SUMMARY

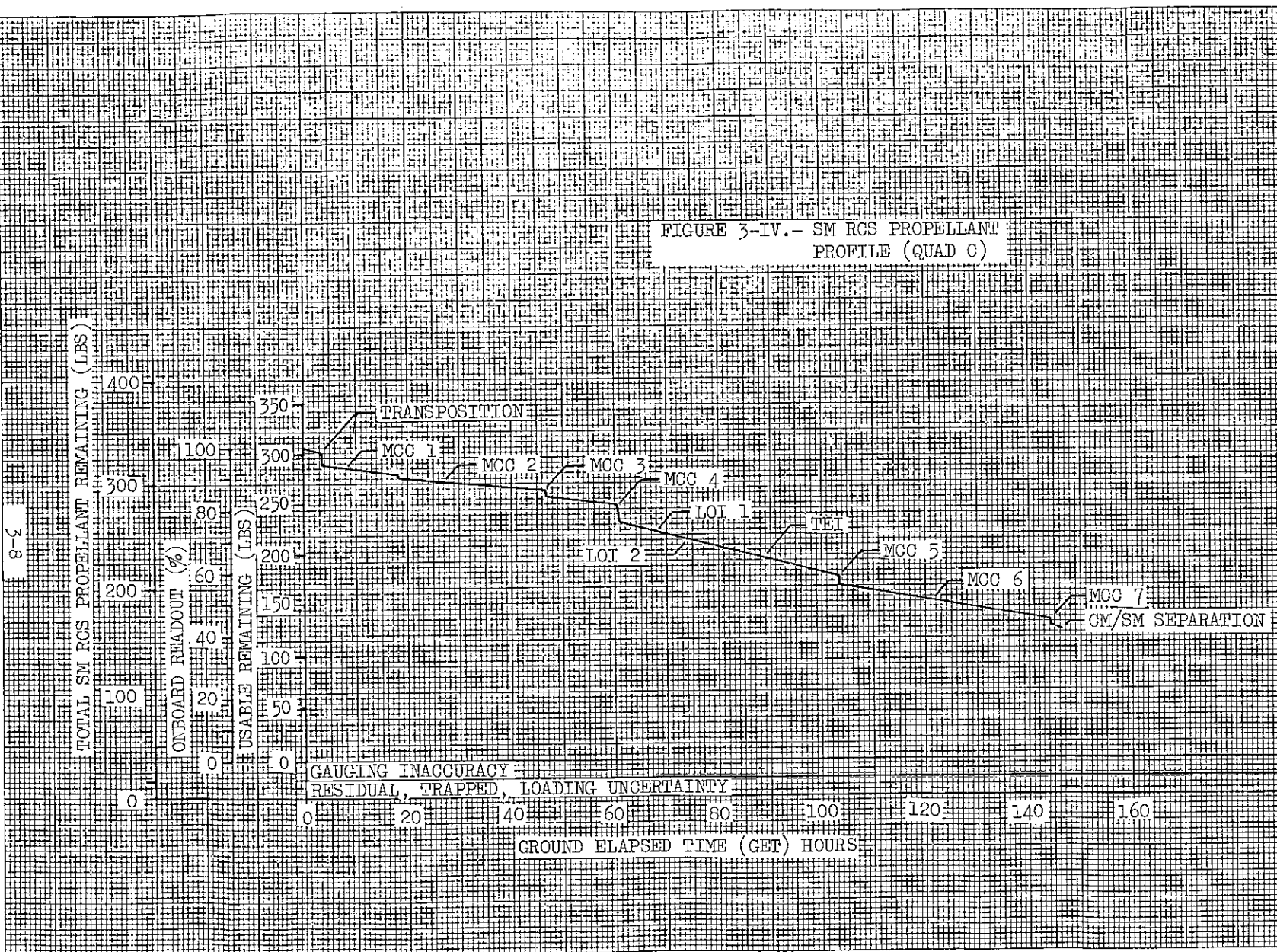
| Day (GET, HRS)      | <u>PROPELLANT USED, LBS</u> |              | <u>PROPELLANT REMAINING, LBS</u> |        |
|---------------------|-----------------------------|--------------|----------------------------------|--------|
|                     | PER DAY                     | ACCUMULATIVE | TOTAL                            | USABLE |
|                     |                             |              | 1347.6                           | 1130.0 |
| PRELAUNCH           | 5.8                         | 5.8          | 1341.8                           | 1124.2 |
| 1 (00:00 - 24:00)   | 99.6                        | 105.4        | 1242.2                           | 1024.6 |
| 2 (24:00 - 48:00)   | 79.9                        | 185.3        | 1162.3                           | 944.7  |
| 3 (48:00 - 72:00)   | 110.3                       | 295.6        | 1052.0                           | 834.4  |
| 4 (72:00 - 96:00)   | 147.0                       | 442.6        | 905.0                            | 687.4  |
| 5 (96:00 - 120:00)  | 109.1                       | 551.7        | 795.9                            | 578.3  |
| 6 (120:00 - 144:00) | 89.3                        | 641.0        | 706.6                            | 489.0  |
| 7 (144:00 - 147:00) | 14.8                        | 655.8        | 691.8                            | 474.2  |











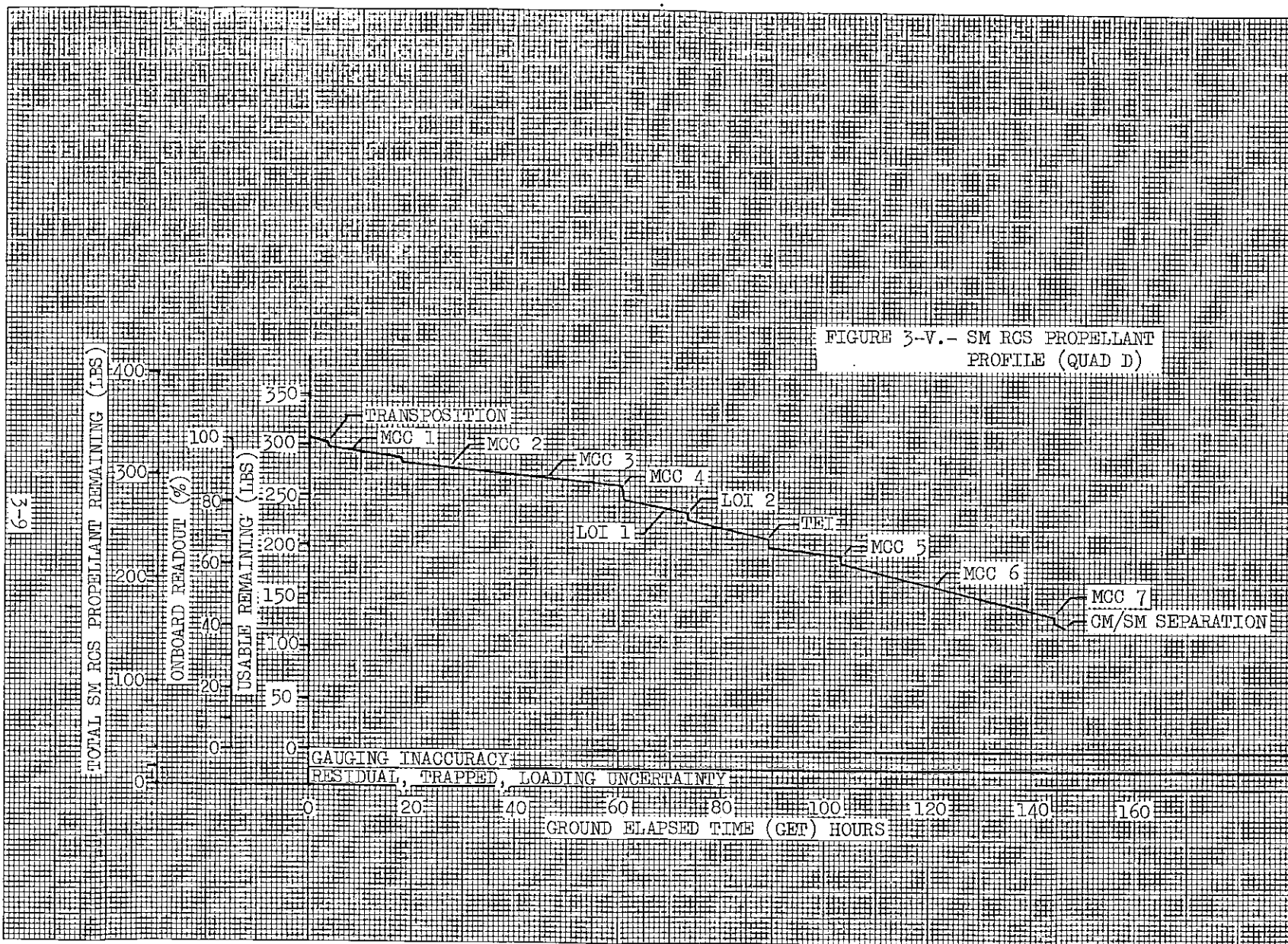


TABLE 3-III

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| .0                       | APOLLO 8 CSM 103                                 | 63878.          | .0                      | 1347.6                  | 100.                      |
| .0                       | SM-RCS CHECKOUT<br>FIRE EACH JET ONE SEC.        | 63872.          | 5.8                     | 1341.8                  | 100.                      |
| 3.5                      | TRANSPOSITION<br>+X TRANS 1 FPS PGHCS            | 63865.          | 7.5                     | 1334.3                  | 99.                       |
| 3.5                      | -X 0.5 FPS                                       | 63860.          | 4.8                     | 1329.5                  | 99.                       |
| 3.5                      | PITCH 4 DEG/SEC                                  | 63853.          | 6.9                     | 1322.7                  | 98.                       |
| 3.5                      | FORMATION FLYING                                 | 63833.          | 20.0                    | 1302.7                  | 97.                       |
| 3.8                      | MANEUVER TO LOCAL VERTICAL                       | 63830.          | 3.2                     | 1299.5                  | 96.                       |
| 3.8                      | ATTITUDE HOLD .5 DEG DB                          | 63830.          | .3                      | 1299.2                  | 96.                       |
| 3.9                      | SEPARATE FROM S-IVB -X 1.5 FPS                   | 63816.          | 13.7                    | 1285.4                  | 95.                       |
| 4.0                      | P52 IMU ALIGN                                    | 63815.          | 1.1                     | 1284.4                  | 95.                       |
| 4.3                      | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 63814.          | .8                      | 1283.5                  | 95.                       |
| 4.3                      | MIN. IMPULSE MARKING                             | 63814.          | .4                      | 1283.1                  | 95.                       |
| 4.3                      | NAV SIGHTING SET 2                               | 63813.          | .8                      | 1282.3                  | 95.                       |
| 4.3                      | MINIMUM IMPULSE MARKING                          | 63812.          | .4                      | 1281.9                  | 95.                       |
| 4.3                      | NAV SIGHTING SET 3                               | 63812.          | .8                      | 1281.0                  | 95.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(#) |
| 4.3                      | MINIMUM IMPULSE MARKING                          | 63811.          | .4                      | 1280.4                  | 95.                       |
| 4.3                      | NAV SIGHTING SET 4                               | 63810.          | .9                      | 1279.7                  | 95.                       |
| 4.3                      | MINIMUM IMPULSE MARKING                          | 63810.          | .4                      | 1279.3                  | 95.                       |
| 8.0                      | PS2 IMU ALIGN                                    | 63809.          | 1.1                     | 1278.2                  | 95.                       |
| 8.5                      | MIDCOURSE CORRECTION<br>3 AXIS ORIENT PCNCS      | 63808.          | 1.1                     | 1277.1                  | 95.                       |
| 8.5                      | ATT HOLD 0.5 DEG DB                              | 63807.          | .6                      | 1276.5                  | 95.                       |
| 9.0                      | DELTA VEL = NOMINALLY ZERO                       | 63807.          | .0                      | 1276.5                  | 95.                       |
| 9.1                      | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 63806.          | .6                      | 1275.7                  | 95.                       |
| 9.1                      | MIN. IMPULSE MARKING                             | 63806.          | .4                      | 1275.3                  | 95.                       |
| 9.1                      | NAV SIGHTING SET 2                               | 63805.          | .8                      | 1274.5                  | 95.                       |
| 9.1                      | MINIMUM IMPULSE MARKING                          | 63805.          | .4                      | 1274.0                  | 95.                       |
| 9.1                      | NAV SIGHTING SET 3                               | 63804.          | .8                      | 1273.2                  | 94.                       |
| 9.1                      | MINIMUM IMPULSE MARKING                          | 63803.          | .4                      | 1272.8                  | 94.                       |
| 9.8                      | ORIENT FOR PTC<br>3 AXIS 0.2 DEG/SEC             | 63803.          | .8                      | 1272.0                  | 94.                       |
| 9.8                      | ATTITUDE HOLD 0.5 DEG DB PGNCS                   | 63802.          | .8                      | 1271.2                  | 94.                       |



| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(8) |
| 9.8                      | EST. 0.3 DEG/SEC ROLL                            | 63802.          | .3                      | 1270.9                  | 94.                       |
| 9.8                      | PITCH AND YAW CONTROL                            | 63792.          | 9.1                     | 1261.8                  | 94.                       |
| 16.6                     | PS2 IMU ALIGN                                    | 63791.          | 1.1                     | 1260.8                  | 94.                       |
| 17.0                     | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 63791.          | .8                      | 1259.9                  | 93.                       |
| 17.0                     | MIN. IMPULSE MARKING                             | 63790.          | .4                      | 1259.5                  | 93.                       |
| 17.0                     | NAV SIGHTING SET 2                               | 63789.          | .8                      | 1258.4                  | 93.                       |
| 17.0                     | MINIMUM IMPULSE MARKING                          | 63789.          | .4                      | 1258.2                  | 93.                       |
| 17.0                     | NAV SIGHTING SET 3                               | 63788.          | .8                      | 1257.4                  | 93.                       |
| 17.0                     | MINIMUM IMPULSE MARKING                          | 63788.          | .4                      | 1257.0                  | 93.                       |
| 17.0                     | NAV SIGHTING SET 4                               | 63787.          | .8                      | 1256.1                  | 93.                       |
| 17.0                     | MINIMUM IMPULSE MARKING                          | 63786.          | .4                      | 1255.7                  | 93.                       |
| 17.0                     | NAV SIGHTING SET 5                               | 63785.          | .8                      | 1254.9                  | 93.                       |
| 17.0                     | MINIMUM IMPULSE MARKING                          | 63785.          | .4                      | 1254.5                  | 93.                       |
| 18.0                     | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC              | 63784.          | .8                      | 1253.7                  | 93.                       |
| 18.0                     | ATT HOLD .5 DEG DB PGNC                          | 63783.          | .8                      | 1252.9                  | 93.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 18.0                     | EST. 0.3 DEG/SEC ROLL                            | 63783.          | .2                      | 1252.6                  | 93.                       |
| 18.0                     | PITCH AND YAW CONTROL                            | 63773.          | 10.4                    | 1242.2                  | 92.                       |
| 26.0                     | P52 IMU ALIGN                                    | 63772.          | 1.0                     | 1241.2                  | 92.                       |
| 26.5                     | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 63771.          | .8                      | 1240.4                  | 92.                       |
| 26.5                     | MIN IMPULSE MARKING                              | 63771.          | .4                      | 1240.0                  | 92.                       |
| 26.5                     | NAV SIGHTING SET 2                               | 63770.          | .9                      | 1239.1                  | 92.                       |
| 26.5                     | MINIMUM IMPULSE MARKING                          | 63769.          | .4                      | 1238.7                  | 92.                       |
| 26.5                     | NAV SIGHTING SET 3                               | 63768.          | .8                      | 1237.8                  | 92.                       |
| 26.5                     | MINIMUM IMPULSE MARKING                          | 63768.          | .4                      | 1237.4                  | 92.                       |
| 27.5                     | MIDCOURSE CORRECTION -<br>MANEUVER TO BURN ATT   | 63767.          | 1.1                     | 1236.3                  | 92.                       |
| 27.5                     | ATT HOLD .5 DEG DB PGNC.                         | 63767.          | .3                      | 1236.1                  | 92.                       |
| 28.0                     | DELTA VEL = NOMINALLY ZERO                       | 63767.          | .0                      | 1236.1                  | 92.                       |
| 28.1                     | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 63766.          | .8                      | 1235.3                  | 92.                       |
| 28.1                     | MIN. IMPULSE MARKING                             | 63765.          | .4                      | 1234.8                  | 92.                       |
| 28.1                     | NAV SIGHTING SET 2                               | 63765.          | .8                      | 1234.0                  | 92.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 28.1                     | MINIMUM IMPULSE MARKING                          | 63764.          | .4                      | 1233.6                  | 92.                       |
| 28.1                     | NAV SIGHTING SET 3                               | 63763.          | .9                      | 1232.7                  | 91.                       |
| 28.1                     | MINIMUM IMPULSE MARKING                          | 63763.          | .4                      | 1232.3                  | 91.                       |
| 28.1                     | NAV SIGHTING SET 4                               | 63762.          | .8                      | 1231.5                  | 91.                       |
| 28.1                     | MINIMUM IMPULSE MARKING                          | 63762.          | .4                      | 1231.0                  | 91.                       |
| 28.9                     | ORIENT FOR PTC<br>3 AXIS 0.2 DEG/SEC             | 63761.          | .8                      | 1230.3                  | 91.                       |
| 28.9                     | ATTITUDE HOLD 0.5 DEG DB PG <sub>NCS</sub>       | 63760.          | .8                      | 1229.5                  | 91.                       |
| 28.9                     | EST. 0.3 DEG/SEC ROLL                            | 63760.          | .3                      | 1229.2                  | 91.                       |
| 28.9                     | PITCH AND YAW CONTROL                            | 63753.          | 6.5                     | 1222.7                  | 91.                       |
| 33.8                     | P52 IMU ALIGN                                    | 63752.          | 1.1                     | 1221.7                  | 91.                       |
| 34.3                     | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 63751.          | .8                      | 1220.8                  | 91.                       |
| 34.3                     | MIN. IMPULSE MARKING                             | 63751.          | .4                      | 1220.4                  | 91.                       |
| 34.3                     | NAV SIGHTING SET 2                               | 63750.          | .8                      | 1219.6                  | 91.                       |
| 34.3                     | MINIMUM IMPULSE MARKING                          | 63750.          | .4                      | 1219.2                  | 90.                       |
| 34.3                     | NAV SIGHTING SET 3                               | 63749.          | .8                      | 1218.3                  | 90.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 34.3                     | MINIMUM IMPULSE MARKING                          | 63749.          | .4                      | 1217.9                  | 90.                       |
| 34.9                     | ORIENT FOR PTC<br>3 AXIS 0.2 DEG/SEC             | 63748.          | .8                      | 1217.1                  | 90.                       |
| 34.9                     | ATTITUDE HOLD 0.5 DEG DB BGNCs                   | 63747.          | .8                      | 1216.4                  | 90.                       |
| 34.9                     | EST. 0.3 DEG/SEC ROLL                            | 63747.          | .2                      | 1216.1                  | 90.                       |
| 34.9                     | PITCH AND YAW CONTROL                            | 63736.          | 11.0                    | 1205.1                  | 89.                       |
| 44.5                     | P52 IMU ALIGN                                    | 63735.          | 1.1                     | 1204.0                  | 89.                       |
| 45.1                     | CISLUNAR NAVIGATION<br>STAR/LUNAR HORIZON ORIENT | 63734.          | .9                      | 1203.1                  | 89.                       |
| 45.1                     | MIN. IMPULSE MARKING                             | 63733.          | .4                      | 1202.7                  | 89.                       |
| 45.1                     | NAV SIGHTING SET 2                               | 63732.          | .8                      | 1201.8                  | 89.                       |
| 45.1                     | MINIMUM IMPULSE MARKING                          | 63732.          | .4                      | 1201.4                  | 89.                       |
| 45.1                     | NAV SIGHTING SET 3                               | 63731.          | .9                      | 1200.4                  | 89.                       |
| 45.1                     | MINIMUM IMPULSE MARKING                          | 63731.          | .4                      | 1200.1                  | 89.                       |
| 45.1                     | NAV SIGHTING SET 4                               | 63730.          | .8                      | 1199.3                  | 89.                       |
| 45.1                     | MINIMUM IMPULSE MARKING                          | 63730.          | .4                      | 1198.9                  | 89.                       |
| 45.1                     | NAV SIGHTING SET 5                               | 63729.          | .8                      | 1198.1                  | 89.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 45.1                     | MINIMUM IMPULSE MARKING                          | 63728.          | .4                      | 1197.7                  | 89.                       |
| 46.4                     | P52 IMU ALIGN                                    | 63727.          | 1.1                     | 1196.6                  | 89.                       |
| 46.6                     | MIDCOURSE CORRECTION<br>3 AXIS ORIENT PGNC       | 63726.          | 1.1                     | 1195.5                  | 89.                       |
| 46.6                     | ATT HOLD 0.5 DEG DB PGNC                         | 63726.          | .4                      | 1195.1                  | 89.                       |
| 47.0                     | RCS +X TRANS PGNC 3 FPS                          | 63704.          | 22.1                    | 1173.0                  | 87.                       |
| 47.2                     | CISLUNAR NAVIGATION<br>STAR EARTH HORIZON ORIENT | 63703.          | .8                      | 1172.2                  | 87.                       |
| 47.2                     | MINIMUM IMPULSE MARKING                          | 63702.          | .4                      | 1171.8                  | 87.                       |
| 47.2                     | NAV SIGHTING SET 2                               | 63702.          | .8                      | 1171.0                  | 87.                       |
| 47.2                     | MINIMUM IMPULSE MARKING                          | 63701.          | .4                      | 1170.4                  | 87.                       |
| 47.2                     | NAV SIGHTING SET 3                               | 63700.          | .8                      | 1169.8                  | 87.                       |
| 47.2                     | MINIMUM IMPULSE MARKING                          | 63700.          | .4                      | 1169.3                  | 87.                       |
| 47.8                     | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC              | 63699.          | .8                      | 1168.5                  | 87.                       |
| 47.8                     | ATTITUDE HOLD 0.5 DEG DB PGNC                    | 63698.          | .8                      | 1167.8                  | 87.                       |
| 47.8                     | EST. 0.3 DEG/SEC ROLL                            | 63698.          | .3                      | 1167.5                  | 87.                       |
| 47.8                     | PITCH AND YAW CONTROL                            | 63693.          | 5.2                     | 1162.3                  | 86.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 51.5                     | P52 IMU ALIGN                                    | 63692.          | .9                      | 1161.4                  | 86.                       |
| 52.3                     | CISLUNAR NAVIGATION<br>STAR/LUNAR HORIZON ORIENT | 63691.          | .8                      | 1160.6                  | 86.                       |
| 52.3                     | MIN. IMPULSE MARKING                             | 63691.          | .4                      | 1160.1                  | 86.                       |
| 52.3                     | NAV SIGHTING SET 2                               | 63690.          | .8                      | 1159.3                  | 86.                       |
| 52.3                     | MINIMUM IMPULSE MARKING                          | 63690.          | .4                      | 1158.9                  | 86.                       |
| 52.3                     | NAV SIGHTING SET 3                               | 63689.          | .8                      | 1158.1                  | 86.                       |
| 52.3                     | MINIMUM IMPULSE MARKING                          | 63688.          | .4                      | 1157.7                  | 86.                       |
| 52.3                     | NAV SIGHTING SET 4                               | 63687.          | .8                      | 1156.9                  | 86.                       |
| 52.3                     | MINIMUM IMPULSE MARKING                          | 63687.          | .4                      | 1156.4                  | 86.                       |
| 52.3                     | NAV SIGHTING SET 5                               | 63686.          | .8                      | 1155.6                  | 86.                       |
| 52.3                     | MINIMUM IMPULSE MARKING                          | 63686.          | .4                      | 1155.2                  | 86.                       |
| 53.3                     | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC              | 63685.          | .8                      | 1154.4                  | 86.                       |
| 53.3                     | ATTITUDE HOLD 0.5 DEG DB PGNCs                   | 63684.          | .8                      | 1153.6                  | 86.                       |
| 53.3                     | EST. 0.3 DEG/SEC ROLL                            | 63684.          | .2                      | 1153.4                  | 86.                       |
| 53.3                     | PITCH AND YAW CONTROL                            | 63675.          | 8.8                     | 1144.6                  | 85.                       |

| SM-RCS PROPELLANT BUDGET |                                             |                 |                         |                         |                           |
|--------------------------|---------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                       | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 60.0                     | P52 IMU ALIGN                               | 63674.          | .9                      | 1143.7                  | 85.                       |
| 60.5                     | MIDCOURSE CORRECTION<br>3 AXIS ORIENT PGNC5 | 63673.          | .9                      | 1142.7                  | 85.                       |
| 60.5                     | ATT HOLD 0.5 DEG DB PGNC5                   | 63673.          | .4                      | 1142.4                  | 85.                       |
| 61.0                     | RCS -X TRANS PGNC5 7 FPS                    | 63622.          | 51.2                    | 1091.2                  | 81.                       |
| 61.2                     | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC         | 63621.          | .8                      | 1090.4                  | 81.                       |
| 61.2                     | ATTITUDE HOLD 0.5 DEG DB PGNC5              | 63620.          | .8                      | 1089.6                  | 81.                       |
| 61.2                     | EST. 0.3 DEG/SEC ROLL                       | 63620.          | .3                      | 1089.3                  | 81.                       |
| 61.2                     | PITCH AND YAW CONTROL                       | 63612.          | 7.5                     | 1081.8                  | 80.                       |
| 66.0                     | P52 IMU ALIGN                               | 63612.          | .9                      | 1080.9                  | 80.                       |
| 66.5                     | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC         | 63611.          | .8                      | 1080.2                  | 80.                       |
| 66.5                     | ATTITUDE HOLD 0.2 DEG DB                    | 63610.          | .8                      | 1079.4                  | 80.                       |
| 66.5                     | EST. 0.3 DEG/SEC ROLL                       | 63610.          | .3                      | 1079.1                  | 80.                       |
| 66.5                     | PITCH AND YAW CONTROL                       | 63609.          | .5                      | 1078.6                  | 80.                       |
| 67.0                     | MANEUVER TO LOI ATT                         | 63608.          | .8                      | 1077.8                  | 80.                       |
| 67.1                     | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC         | 63608.          | .8                      | 1077.1                  | 80.                       |

| SM-RCS PROPELLANT BUDGET |                                                     |                 |                         |                         |                           |
|--------------------------|-----------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                               | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 67.1                     | ATT HOLD .5 DEG DB PGNC5                            | 63607.          | .8                      | 1076.3                  | 80.                       |
| 67.1                     | EST. 0.3 DEG/SEC ROLL                               | 63607.          | .2                      | 1076.1                  | 80.                       |
| 67.1                     | PITCH AND YAW CONTROL                               | 63605.          | 1.3                     | 1074.8                  | 80.                       |
| 68.2                     | P52 IMU ALIGN                                       | 63604.          | .9                      | 1073.9                  | 80.                       |
| ✓68.4                    | LUNAR ORBIT INSERTION BURN 1<br>3-AXIS ORIENT PGNC5 | 63604.          | .9                      | 1073.0                  | 80.                       |
| 68.4                     | ATT HOLD .5 DEG DB PGNC5                            | 63603.          | .4                      | 1072.6                  | 80.                       |
| 68.4                     | START TRANSIENT CONTROL<br>NO ULLAGE                | 63602.          | 1.3                     | 1071.3                  | 79.                       |
| 69.1                     | SPS BURN<br>BUILDUP                                 | 63599.          | .0                      | 1071.3                  | 79.                       |
| 69.1                     | STEADY STATE BURN 245.8 SEC PGNC5                   | 47037.          | .3                      | 1070.9                  | 79.                       |
| 69.1                     | TAILOFF                                             | 46996.          | .0                      | 1070.9                  | 79.                       |
| 69.1                     | DAMP SHUT DOWN TRANSIENT                            | 46995.          | 1.1                     | 1069.8                  | 79.                       |
| 69.1                     | ATTITUDE HOLD 5 DEG DB                              | 46993.          | 2.1                     | 1067.7                  | 79.                       |
| 69.5                     | ROLL 180 DEG                                        | 46992.          | 1.0                     | 1066.7                  | 79.                       |
| 69.5                     | ATTITUDE HOLD 5 DEG DB                              | 46990.          | 2.1                     | 1064.6                  | 79.                       |
| 70.3                     | P52 IMU ALIGN                                       | 46989.          | .8                      | 1063.7                  | 79.                       |



| SM-RCS PROPELLANT BUDGET |                                                |                 |                         |                         |                           |
|--------------------------|------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                          | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 70.6                     | ORIENT TO TRACKING ATTITUDE<br>PITCH 10 ORDEAL | 46988.          | 1.1                     | 1062.6                  | 79.                       |
| 70.6                     | EST ORBITAL RATE                               | 46986.          | 1.5                     | 1061.1                  | 79.                       |
| 71.1                     | MAINTAIN ORB RATE AND COAS GROUND<br>TRACK     | 46986.          | .7                      | 1060.4                  | 79.                       |
| 71.3                     | COAS GROUND TRACK DETERMINATION                | 46985.          | .7                      | 1059.7                  | 79.                       |
| 71.5                     | YAW 45 DEG RT FOR TV ORDEAL                    | 46982.          | 2.9                     | 1056.8                  | 78.                       |
| 71.6                     | YAW 45 DEG LEFT                                | 46979.          | 2.9                     | 1053.9                  | 78.                       |
| 71.9                     | ORIENT FOR PSEUDO LANDING                      | 46978.          | 1.1                     | 1052.7                  | 78.                       |
| 72.0                     | MAINTAIN ORB RATE AND COAS GROUND TR<br>ACK    | 46977.          | .7                      | 1052.0                  | 78.                       |
| 72.4                     | PITCH .2 DEG/SEC TO BURN ATT                   | 46977.          | .3                      | 1051.7                  | 78.                       |
| 72.5                     | ATTITUDE HOLD 5 DEG DB                         | 46973.          | 4.3                     | 1047.4                  | 78.                       |
| 72.6                     | PS2 IMU ALIGN                                  | 46972.          | .8                      | 1046.6                  | 78.                       |
| 73.0                     | ROLL LEFT 180 DEG 2DEG/SEC                     | 46971.          | 1.0                     | 1045.6                  | 78.                       |
| 73.0                     | ATTITUDE HOLD 0.2 DEG DB                       | 46970.          | .8                      | 1044.8                  | 78.                       |
| ✓ 73.5                   | LUNAR ORBIT INSERTION BURN 2<br>ORIENTATION    | 46969.          | .8                      | 1044.0                  | 77.                       |
| 73.5                     | ATT HOLD .5 DEG DB PGNC5                       | 46968.          | 1.1                     | 1042.9                  | 77.                       |

| SM-RCS PROPELLANT BUDGET |                                |                 |                         |                         |                           |
|--------------------------|--------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                          | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LCFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 73.5                     | ULLAGE<br>2 JET 8 AND D        | 46952.          | 15.7                    | 1027.2                  | 76.                       |
| 73.5                     | SPS BURN<br>BUILD UP           | 46949.          | .0                      | 1027.2                  | 76.                       |
| 73.5                     | STEADY STATE BURN 9.7 SEC PGNC | 46296.          | .2                      | 1027.0                  | 76.                       |
| 73.5                     | TAILOFF                        | 46255.          | .0                      | 1027.0                  | 76.                       |
| 73.5                     | DAMP SHUT DOWN TRANSIENT       | 46254.          | 1.1                     | 1025.9                  | 76.                       |
| 73.7                     | ROLL RIGHT 180 DEG             | 46253.          | 1.1                     | 1024.8                  | 76.                       |
| 73.8                     | ORIENT TO TRACKING ATTITUDE    | 46252.          | 1.1                     | 1023.7                  | 76.                       |
| 73.8                     | EST ORBITAL RATE               | 46250.          | 1.5                     | 1022.2                  | 76.                       |
| 74.1                     | ORIENT FOR LANDMARKS           | 46249.          | 1.1                     | 1021.2                  | 76.                       |
| 74.3                     | PITCH TO ACQUIRE MSFN          | 46248.          | 1.3                     | 1019.9                  | 76.                       |
| 74.3                     | ATTITUDE HOLD 5 DEG DB         | 46246.          | 2.1                     | 1017.7                  | 76.                       |
| 74.5                     | P52 IMU ALIGN                  | 46245.          | .8                      | 1016.9                  | 75.                       |
| 75.0                     | PITCH TO VERTICAL .2 DEG/SEC   | 46244.          | .7                      | 1016.2                  | 75.                       |
| 75.2                     | ROLL 180 DEG                   | 46244.          | .3                      | 1015.9                  | 75.                       |
| 75.2                     | EST ORBITAL RATE               | 46243.          | 1.5                     | 1014.4                  | 75.                       |

| SM-RCS PROPELLANT BUDGET |                                                     |                 |                         |                         |                           |
|--------------------------|-----------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                               | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 75.7                     | ROLL RIGHT 180 DEG 2 DEG/SEC                        | 46241.          | 1.1                     | 1013.3                  | 75.                       |
| 75.8                     | MAINTAIN ORB RATE                                   | 46240.          | 1.5                     | 1011.8                  | 75.                       |
| 76.0                     | ORIENT FOR LANDMARKS, PHOTOGRAPHY                   | 46239.          | 1.1                     | 1010.7                  | 75.                       |
| 76.1                     | EST AND STOP SIGHTING RATE                          | 46238.          | 1.3                     | 1009.4                  | 75.                       |
| 76.4                     | ROLL 180 DEG 2 DEG/SEC                              | 46236.          | 1.1                     | 1008.3                  | 75.                       |
| 76.5                     | ATTITUDE HOLD 5 DEG DB                              | 46236.          | .4                      | 1007.9                  | 75.                       |
| 76.6                     | P52 IMU ALIGN.                                      | 46235.          | .8                      | 1007.1                  | 75.                       |
| 77.1                     | ROLL 180 DEG 2 DEG/SEC                              | 46234.          | 1.1                     | 1006.0                  | 75.                       |
| 77.1                     | PITCH DOWN .2 DEG/SEC                               | 46234.          | .4                      | 1005.6                  | 75.                       |
| 77.1                     | EST. ORB RATE                                       | 46232.          | 1.5                     | 1004.1                  | 75.                       |
| 77.2                     | CONTROL POINT SIGHTING                              | 46231.          | 1.1                     | 1003.0                  | 74.                       |
| 77.3                     | PITCH .3 DEG/SEC AT ACQ                             | 46231.          | .5                      | 1002.6                  | 74.                       |
| 77.7                     | PITCH .2 DEG/SEC AT LOSS                            | 46230.          | .7                      | 1001.9                  | 74.                       |
| 77.7                     | EST. ORB RATE                                       | 46229.          | 1.5                     | 1000.4                  | 74.                       |
| 78.0                     | ORIENT FOR PSEUDO LANDING SITE<br>LANDMARK SIGHTING | 46227.          | 1.1                     | 999.3                   | 74.                       |

| SM-RCS PROPELLANT BUDGET |                                                     |                 |                         |                         |                             |
|--------------------------|-----------------------------------------------------|-----------------|-------------------------|-------------------------|-----------------------------|
| TIME<br>(HR)             | EVENT                                               | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(LBS) |
| 78.0                     | PITCH .3 DEG/SEC AT ACQ                             | 46227.          | .5                      | 998.9                   | 74.                         |
| 78.3                     | ROLL 180 DEG AT LOSS                                | 46226.          | 1.1                     | 997.8                   | 74.                         |
| 78.5                     | ATTITUDE HOLD 5 DEG DB                              | 46224.          | 2.1                     | 995.6                   | 74.                         |
| 78.6                     | P52 IMU ALIGN                                       | 46223.          | .8                      | 994.8                   | 74.                         |
| 79.0                     | ROLL 180 DEG                                        | 46222.          | 1.1                     | 993.8                   | 74.                         |
| 79.1                     | PITCH 2 DEG/SEC                                     | 46219.          | 2.4                     | 991.3                   | 74.                         |
| 79.1                     | EST. ORB RATE                                       | 46218.          | 1.5                     | 989.8                   | 73.                         |
| 79.1                     | CONTROL POINT SIGHTING                              | 46217.          | 1.1                     | 988.7                   | 73.                         |
| 79.1                     | MAINTAIN SIGHTING RATE                              | 46217.          | .1                      | 988.6                   | 73.                         |
| 79.5                     | PITCH .3 DEG/SEC AT ACQ                             | 46216.          | .5                      | 988.1                   | 73.                         |
| 79.5                     | PITCH .2 DEG/SEC AT LOSS                            | 46216.          | .3                      | 987.8                   | 73.                         |
| 79.5                     | EST. ORB RATE                                       | 46214.          | 1.5                     | 986.3                   | 73.                         |
| 80.0                     | ORIENT FOR PSEUDO LANDING SITE<br>LANDMARK SIGHTING | 46213.          | 1.1                     | 985.2                   | 73.                         |
| 80.1                     | PITCH .3 DEG/SEC AT ACQ                             | 46213.          | .5                      | 984.7                   | 73.                         |
| 80.2                     | ROLL 180 DEG AT LOSS                                | 46212.          | 1.1                     | 983.7                   | 73.                         |

| SM-RCS PROPELLANT BUDGET |                                                     |                 |                         |                         |                           |
|--------------------------|-----------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                               | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 80.2                     | ATTITUDE HOLD 5 DEG DB                              | 46210.          | 2.1                     | 981.6                   | 73.                       |
| 80.6                     | PS2 IMU ALIGN                                       | 46209.          | .8                      | 980.7                   | 73.                       |
| 80.7                     | ROLL 180 DEG 2 DEG/SEC                              | 46208.          | 1.1                     | 979.6                   | 73.                       |
| 80.8                     | PITCH .2 DEG/SEC                                    | 46207.          | .3                      | 979.3                   | 73.                       |
| 80.8                     | EST. ORB RATE                                       | 46206.          | 1.5                     | 977.8                   | 73.                       |
| 81.1                     | PITCH .3 DEG/SEC AT ACQ                             | 46205.          | .5                      | 977.3                   | 73.                       |
| 81.3                     | PITCH .5 DEG/SEC AT LOSS                            | 46205.          | .7                      | 976.6                   | 72.                       |
| 81.5                     | PITCH .3 DEG/SEC AT ACQ                             | 46204.          | .4                      | 976.1                   | 72.                       |
| 81.6                     | PITCH .5 DEG/SEC AT LOSS                            | 46204.          | .7                      | 975.4                   | 72.                       |
| 81.7                     | PITCH .3 DEG/SEC AT ACQ                             | 46203.          | .5                      | 975.0                   | 72.                       |
| 81.8                     | PITCH .5 DEG/SEC AT LOSS                            | 46202.          | .7                      | 974.3                   | 72.                       |
| 81.9                     | ORIENT FOR PSEUDO LANDING SITE<br>LANDMARK SIGHTING | 46201.          | 1.1                     | 973.2                   | 72.                       |
| 82.1                     | PITCH .3 DEG/SEC AT ACQ                             | 46201.          | .5                      | 972.7                   | 72.                       |
| 82.3                     | ROLL 180 DEG 2 DEG/SEC                              | 46200.          | 1.1                     | 971.6                   | 72.                       |
| 82.3                     | ATTITUDE HOLD 5 DEG DB                              | 46195.          | 4.3                     | 967.3                   | 72.                       |

| SM-RCS PROPELLANT BUDGET |                                                     |                 |                         |                         |                           |
|--------------------------|-----------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                               | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 82.5                     | P52 IMU ALIGN                                       | 46195.          | .8                      | 966.5                   | 72.                       |
| 82.8                     | ROLL 180 DEG 2 DEG/SEC                              | 46194.          | 1.1                     | 965.5                   | 72.                       |
| 83.0                     | PITCH .2 DEG/SEC                                    | 46193.          | .3                      | 965.1                   | 72.                       |
| 83.1                     | PITCH .3 DEG/SEC AT ACQ                             | 46193.          | .5                      | 964.7                   | 72.                       |
| 83.3                     | PITCH .5 DEG/SEC AT LOSS                            | 46192.          | .7                      | 964.0                   | 72.                       |
| 83.4                     | PITCH .3DEG/SEC                                     | 46192.          | .5                      | 963.5                   | 71.                       |
| 83.6                     | PITCH .5 DEG/SEC                                    | 46191.          | .7                      | 962.8                   | 71.                       |
| 83.7                     | PITCH .3DEG/SEC                                     | 46191.          | .5                      | 962.4                   | 71.                       |
| 83.8                     | PITCH .5 DEG/SEC                                    | 46190.          | .7                      | 961.7                   | 71.                       |
| 83.9                     | ORIENT FOR PSEUDO LANDING SITE<br>LANDMARK SIGHTING | 46189.          | 1.1                     | 960.5                   | 71.                       |
| 84.2                     | PITCH .3DEG/SEC                                     | 46188.          | .5                      | 960.1                   | 71.                       |
| 84.2                     | ROLL 180 DEG 2 DEG/SEC                              | 46187.          | 1.0                     | 959.0                   | 71.                       |
| 84.2                     | ATTITUDE HOLD 5 DEG DB                              | 46185.          | 2.1                     | 956.9                   | 71.                       |
| 84.5                     | P52 IMU ALIGN                                       | 46184.          | .8                      | 956.1                   | 71.                       |
| 84.7                     | PITCH .2 DEG/SEC                                    | 46184.          | .3                      | 955.7                   | 71.                       |

| SM-RCS PROPELLANT-BUDGET |                                      |                 |                         |                         |                           |
|--------------------------|--------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 84.7                     | ESTABLISH ORB RATE                   | 46182.          | 1.5                     | 954.2                   | 71.                       |
| 85.1                     | PITCH UP 70 DEG AT .5 DEG/SEC        | 46182.          | .7                      | 953.5                   | 71.                       |
| 85.1                     | MAINTAIN ORB RATE                    | 46182.          | .1                      | 953.4                   | 71.                       |
| 85.5                     | PITCH UP 40 DEG AT .2 DEG/SEC        | 46181.          | .3                      | 953.0                   | 71.                       |
| 85.7                     | ROLL 180 DEG                         | 46180.          | 1.0                     | 952.0                   | 71.                       |
| 86.2                     | ROLL 180 DEG 2DEG/SEC                | 46179.          | 1.1                     | 950.9                   | 71.                       |
| 86.3                     | PITCH 60DEG .2DEG/SEC                | 46179.          | .3                      | 950.6                   | 71.                       |
| 86.2                     | ATTITUDE HOLD 5 DEG DB               | 46177.          | 2.1                     | 948.5                   | 70.                       |
| 86.7                     | PS2 IMU ALIGN                        | 46176.          | .8                      | 947.7                   | 70.                       |
| 86.9                     | PITCH .2 DEG/SEC                     | 46175.          | .3                      | 947.3                   | 70.                       |
| 86.9                     | ESTABLISH ORB RATE                   | 46174.          | 1.5                     | 945.8                   | 70.                       |
| 88.3                     | PS2 IMU ALIGN                        | 46173.          | .8                      | 945.1                   | 70.                       |
| 88.7                     | ROLL 180 DEG                         | 46173.          | .3                      | 944.7                   | 70.                       |
| 89.1                     | TRANS-EARTH INJECTION<br>ORIENTATION | 46172.          | .8                      | 943.9                   | 70.                       |
| 89.1                     | ATT HOLD .5 DEG DB                   | 46171.          | 1.1                     | 942.9                   | 70.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 89.1                     | ULLAGE<br>2 JET B AND D QUADS 21 SEC PGNC5       | 46155.          | 15.7                    | 927.1                   | 69.                       |
| 89.1                     | SPS BURN<br>BUILD UP                             | 46152.          | .0                      | 927.1                   | 69.                       |
| 89.1                     | STEADY STATE BURN 206 SEC PGNC5                  | 32272.          | .2                      | 926.9                   | 69.                       |
| 89.1                     | TAILOFF                                          | 32231.          | .0                      | 926.9                   | 69.                       |
| 89.1                     | DAMP SHUT DOWN TRANSIENT                         | 32230.          | 1.1                     | 925.8                   | 69.                       |
| 89.2                     | PITCH TO VERTICAL<br>ACQUIRE MOON                | 32229.          | .7                      | 925.1                   | 69.                       |
| 90.2                     | P52 IMU ALIGN                                    | 32229.          | .7                      | 924.4                   | 69.                       |
| 90.5                     | CISLUNAR NAVIGATION<br>STAR/LUNAR HORIZON ORIENT | 32228.          | 1.0                     | 923.4                   | 69.                       |
| 90.5                     | MIN. IMPULSE MARKING                             | 32227.          | .4                      | 923.0                   | 68.                       |
| 90.5                     | NAV SIGHTING SET 2                               | 32226.          | 1.0                     | 922.0                   | 68.                       |
| 90.5                     | MINIMUM IMPULSE MARKING                          | 32226.          | .4                      | 921.6                   | 68.                       |
| 90.5                     | NAV SIGHTING SET 3                               | 32225.          | 1.0                     | 920.6                   | 68.                       |
| 90.5                     | MINIMUM IMPULSE MARKING                          | 32225.          | .4                      | 920.2                   | 68.                       |
| 91.0                     | NAV SIGHTING SET 4                               | 32224.          | 1.0                     | 919.2                   | 68.                       |
| 91.0                     | MINIMUM IMPULSE MARKING                          | 32223.          | .4                      | 918.8                   | 68.                       |



| SM-RCS PROPELLANT BUDGET |                                      |                 |                         |                         |                           |
|--------------------------|--------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 91.0                     | NAV SIGHTING SET 5                   | 32222.          | .9                      | 917.9                   | 68.                       |
| 91.0                     | MINIMUM IMPULSE MARKING              | 32222.          | .4                      | 917.4                   | 68.                       |
| 91.0                     | NAV SIGHTING SET 6                   | 32221.          | 1.0                     | 916.5                   | 68.                       |
| 91.0                     | MINIMUM IMPULSE MARKING              | 32220.          | .4                      | 916.0                   | 68.                       |
| 91.0                     | NAV SIGHTING SET 7                   | 32219.          | 1.0                     | 915.1                   | 68.                       |
| 91.0                     | MINIMUM IMPULSE MARKING              | 32219.          | .4                      | 914.6                   | 68.                       |
| 91.0                     | NAV SIGHTING SET 8                   | 32218.          | 1.0                     | 913.7                   | 68.                       |
| 91.0                     | MINIMUM IMPULSE MARKING              | 32218.          | .4                      | 913.2                   | 68.                       |
| 92.0                     | ORIENT FOR PTC (YAW TEST)            | 32217.          | .7                      | 912.5                   | 68.                       |
| 92.0                     | ATT HOLD .5 DEG DB PGNCs             | 32216.          | .8                      | 911.7                   | 68.                       |
| 92.0                     | EST. 0.3 DEG/SEC YAW                 | 32216.          | .5                      | 911.3                   | 68.                       |
| 92.0                     | PITCH AND ROLL CONTROL               | 32210.          | 5.6                     | 905.7                   | 67.                       |
| 96.0                     | P52 IMU ALIGN                        | 32209.          | .7                      | 905.0                   | 67.                       |
| 96.3                     | ORIENT FOR PTC<br>3 AXIS 0.2 DEG/SEC | 32209.          | .7                      | 904.3                   | 67.                       |
| 96.3                     | ATTITUDE HOLD 0.5 DEG DB PGNCs       | 32208.          | .8                      | 903.5                   | 67.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 96.3                     | EST. 0.3 DEG/SEC ROLL                            | 32208.          | .2                      | 903.3                   | 67.                       |
| 96.3                     | PITCH AND YAW CONTROL                            | 32202.          | 5.2                     | 898.1                   | 67.                       |
| 100.0                    | PS2 IMU ALIGN                                    | 32202.          | .7                      | 897.4                   | 67.                       |
| 100.5                    | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 32201.          | 1.0                     | 896.4                   | 67.                       |
| 100.5                    | MIN. IMPULSE MARKING                             | 32200.          | .4                      | 896.0                   | 66.                       |
| 100.5                    | NAV SIGHTING SET 2                               | 32199.          | 1.0                     | 895.0                   | 66.                       |
| 100.5                    | MINIMUM IMPULSE MARKING                          | 32199.          | .4                      | 894.6                   | 66.                       |
| 100.5                    | NAV SIGHTING SET 3                               | 32198.          | 1.0                     | 893.6                   | 66.                       |
| 100.5                    | MINIMUM IMPULSE MARKING                          | 32198.          | .4                      | 893.2                   | 66.                       |
| 101.0                    | MAN TO SIGHTING ATT                              | 32197.          | 1.0                     | 892.2                   | 66.                       |
| 101.5                    | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 32196.          | .9                      | 891.3                   | 66.                       |
| 101.5                    | MIN. IMPULSE MARKING                             | 32195.          | .4                      | 890.8                   | 66.                       |
| 101.5                    | NAV SIGHTING SET 2                               | 32194.          | 1.0                     | 889.9                   | 66.                       |
| 101.5                    | MINIMUM IMPULSE MARKING                          | 32194.          | .4                      | 889.4                   | 66.                       |
| 101.5                    | NAV SIGHTING SET 3                               | 32193.          | 1.0                     | 888.4                   | 66.                       |

| SM-RCS. PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|---------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)              | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 101.5                     | MINIMUM IMPULSE MARKING                          | 32192.          | .4                      | 888.0                   | 66.                       |
| 101.5                     | NAV SIGHTING SET 4                               | 32191.          | 1.0                     | 887.0                   | 66.                       |
| 101.5                     | MINIMUM IMPULSE MARKING                          | 32191.          | .4                      | 886.6                   | 66.                       |
| 102.0                     | NAV SIGHTING SET 5                               | 32190.          | 1.0                     | 885.7                   | 66.                       |
| 102.0                     | MIN. IMPULSE MARKING                             | 32190.          | .4                      | 885.2                   | 66.                       |
| 102.0                     | NAV SIGHTING SET 6                               | 32189.          | 1.0                     | 884.3                   | 66.                       |
| 102.0                     | MINIMUM IMPULSE MARKING                          | 32188.          | .4                      | 883.9                   | 66.                       |
| 102.8                     | PS2 IMU ALIGN                                    | 32188.          | .7                      | 883.1                   | 66.                       |
| 103.5                     | MIDCOURSE CORRECTION<br>3 AXIS ORIENT PGNC       | 32187.          | 1.0                     | 882.2                   | 65.                       |
| 103.5                     | ATTITUDE HOLD .5 DEG DB PGNC                     | 32186.          | .8                      | 881.4                   | 65.                       |
| 103.5                     | RCS +X TRANS PGNC 10 FPS                         | 32149.          | 37.1                    | 844.3                   | 63.                       |
| 105.0                     | PS2 IMU ALIGN                                    | 32148.          | .7                      | 843.6                   | 63.                       |
| 105.2                     | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 32147.          | 1.0                     | 842.6                   | 63.                       |
| 105.2                     | MIN. IMPULSE MARKING                             | 32147.          | .4                      | 842.2                   | 62.                       |
| 105.2                     | NAV SIGHTING SET 2                               | 32146.          | 1.0                     | 841.2                   | 62.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(#) |
| 105.2                    | MINIMUM IMPULSE MARKING                          | 32145.          | .4                      | 840.7                   | 62.                       |
| 105.2                    | NAV SIGHTING SET 3                               | 32144.          | 1.0                     | 839.8                   | 62.                       |
| 105.2                    | MINIMUM IMPULSE MARKING                          | 32144.          | .4                      | 839.3                   | 62.                       |
| 105.2                    | NAV SIGHTING SET 4                               | 32143.          | 1.0                     | 838.4                   | 62.                       |
| 105.2                    | MINIMUM IMPULSE MARKING                          | 32142.          | .4                      | 837.9                   | 62.                       |
| 105.2                    | NAV SIGHTING SET 5                               | 32141.          | 1.0                     | 837.0                   | 62.                       |
| 105.2                    | MINIMUM IMPULSE MARKING                          | 32141.          | .4                      | 836.4                   | 62.                       |
| 106.1                    | HAN TO SIGHTING ATT                              | 32140.          | 1.0                     | 835.6                   | 62.                       |
| 106.3                    | CISLUNAR NAVIGATION<br>STAR/LUNAR HORIZON ORIENT | 32139.          | 1.0                     | 834.6                   | 62.                       |
| 106.3                    | MIN. IMPULSE MARKING                             | 32139.          | .4                      | 834.1                   | 62.                       |
| 106.3                    | NAV SIGHTING SET 2                               | 32138.          | 1.0                     | 833.2                   | 62.                       |
| 106.3                    | MINIMUM IMPULSE MARKING                          | 32137.          | .4                      | 832.7                   | 62.                       |
| 106.3                    | NAV SIGHTING SET 3                               | 32136.          | 1.0                     | 831.8                   | 62.                       |
| 106.3                    | MINIMUM IMPULSE MARKING                          | 32136.          | .4                      | 831.3                   | 62.                       |
| 107.6                    | P52 IMU ALIGN                                    | 32135.          | .7                      | 830.6                   | 62.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 108.0                    | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 32134.          | 1.0                     | 829.7                   | 62.                       |
| 108.0                    | MIN. IMPULSE MARKING                             | 32134.          | .4                      | 829.2                   | 62.                       |
| 108.0                    | NAV SIGHTING SET 2                               | 32133.          | 1.0                     | 828.3                   | 61.                       |
| 108.0                    | MINIMUM IMPULSE MARKING                          | 32132.          | .4                      | 827.8                   | 61.                       |
| 108.0                    | NAV SIGHTING SET 3                               | 32131.          | 1.0                     | 826.9                   | 61.                       |
| 108.0                    | MINIMUM IMPULSE MARKING                          | 32131.          | .4                      | 826.4                   | 61.                       |
| 108.0                    | NAV SIGHTING SET 4                               | 32130.          | 1.0                     | 825.5                   | 61.                       |
| 108.0                    | MINIMUM IMPULSE MARKING                          | 32129.          | .4                      | 825.0                   | 61.                       |
| 108.0                    | NAV SIGHTING SET 5                               | 32128.          | 1.0                     | 824.0                   | 61.                       |
| 108.0                    | MINIMUM IMPULSE MARKING                          | 32128.          | .4                      | 823.4                   | 61.                       |
| 108.8                    | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC              | 32127.          | .7                      | 822.9                   | 61.                       |
| 108.8                    | ATT HOLD .5 DEG DB PGNC                          | 32127.          | .8                      | 822.2                   | 61.                       |
| 108.8                    | EST. 0.3 DEG/SEC ROLL                            | 32126.          | .4                      | 821.7                   | 61.                       |
| 108.8                    | PITCH AND YAW CONTROL                            | 32120.          | 6.0                     | 815.7                   | 61.                       |
| 113.0                    | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC              | 32119.          | .7                      | 815.0                   | 60.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |  |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|--|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |  |
| 113.0                    | ATTITUDE HOLD 0.5 DEG DB PGWCS                   | 32119.          | .8                      | 814.3                   | 60.                       |  |
| 113.0                    | EST. 0.3 DEG/SEC ROLL                            | 32118.          | .2                      | 814.1                   | 60.                       |  |
| 113.0                    | PITCH AND YAW CONTROL                            | 32116.          | 2.8                     | 811.3                   | 60.                       |  |
| 115.1                    | P52 IMU ALIGN                                    | 32115.          | .7                      | 810.6                   | 60.                       |  |
| 115.3                    | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC              | 32114.          | .7                      | 809.9                   | 60.                       |  |
| 115.3                    | ATTITUDE HOLD 0.5 DEG DB PGWCS                   | 32113.          | .8                      | 809.1                   | 60.                       |  |
| 115.3                    | EST. 0.3 DEG/SEC ROLL                            | 32113.          | .2                      | 808.9                   | 60.                       |  |
| 115.3                    | PITCH AND YAW CONTROL                            | 32108.          | 5.0                     | 803.9                   | 60.                       |  |
| 119.0                    | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC              | 32108.          | .7                      | 803.2                   | 60.                       |  |
| 119.0                    | ATT HOLD .5 DEG DB PGWCS                         | 32107.          | .8                      | 802.4                   | 60.                       |  |
| 119.0                    | EST. 0.3 DEG/SEC ROLL                            | 32107.          | .2                      | 802.3                   | 60.                       |  |
| 119.0                    | PITCH AND YAW CONTROL                            | 32105.          | 1.4                     | 800.9                   | 59.                       |  |
| 120.0                    | P52 IMU ALIGN                                    | 32105.          | .7                      | 800.2                   | 59.                       |  |
| 120.0                    | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 32104.          | 1.0                     | 799.2                   | 59.                       |  |
| 120.0                    | MIN. IMPULSE MARKING                             | 32103.          | .4                      | 798.8                   | 59.                       |  |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 120.0                    | NAV SIGHTING SET 2                               | 32102.          | 1.0                     | 797.8                   | 59.                       |
| 120.0                    | MINIMUM IMPULSE MARKING                          | 32102.          | .4                      | 797.3                   | 59.                       |
| 120.0                    | NAV SIGHTING SET 3                               | 32101.          | 1.0                     | 796.4                   | 59.                       |
| 120.0                    | MINIMUM IMPULSE MARKING                          | 32100.          | .4                      | 795.9                   | 59.                       |
| 121.0                    | PS2 IMU ALIGN                                    | 32100.          | .7                      | 795.2                   | 59.                       |
| 121.5                    | MIDCOURSE CORRECTION<br>3 AXIS ORIENT PGNCs      | 32099.          | .7                      | 794.5                   | 59.                       |
| 121.5                    | ATT HOLD .5 DEG DB PGNCs                         | 32098.          | .4                      | 794.1                   | 59.                       |
| 121.5                    | SM-RCS -X TRANS 2FPS                             | 32091.          | 7.5                     | 786.6                   | 58.                       |
| 122.6                    | CISLUNAR NAVIGATION<br>STAR/LUNAR HORIZON ORIENT | 32090.          | 1.0                     | 785.6                   | 58.                       |
| 122.6                    | MIN. IMPULSE MARKING                             | 32090.          | .4                      | 785.2                   | 58.                       |
| 122.6                    | NAV SIGHTING SET 2                               | 32089.          | 1.0                     | 784.2                   | 58.                       |
| 122.6                    | MINIMUM IMPULSE MARKING                          | 32088.          | .4                      | 783.8                   | 58.                       |
| 122.6                    | NAV SIGHTING SET 3                               | 32087.          | 1.0                     | 782.8                   | 58.                       |
| 122.6                    | MINIMUM IMPULSE MARKING                          | 32087.          | .4                      | 782.4                   | 58.                       |
| 123.1                    | MANEUVER TO SIGHTING ATTITUDE                    | 32086.          | .9                      | 781.5                   | 58.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 123.6                    | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 32085.          | 1.0                     | 780.5                   | 58.                       |
| 123.6                    | MIN. IMPULSE MARKING                             | 32084.          | .4                      | 780.1                   | 58.                       |
| 123.6                    | NAV SIGHTING SET 2                               | 32083.          | 1.0                     | 779.1                   | 58.                       |
| 123.6                    | MINIMUM IMPULSE MARKING                          | 32083.          | .4                      | 778.7                   | 58.                       |
| 123.6                    | NAV SIGHTING SET 3                               | 32082.          | 1.0                     | 777.7                   | 58.                       |
| 123.6                    | MINIMUM IMPULSE MARKING                          | 32082.          | .4                      | 777.3                   | 58.                       |
| 123.6                    | NAV SIGHTING SET 4                               | 32081.          | 1.0                     | 776.3                   | 58.                       |
| 123.6                    | MINIMUM IMPULSE MARKING                          | 32080.          | .4                      | 775.9                   | 58.                       |
| 123.6                    | NAV SIGHTING SET 5                               | 32079.          | 1.0                     | 774.9                   | 58.                       |
| 123.6                    | MINIMUM IMPULSE MARKING                          | 32079.          | .4                      | 774.5                   | 57.                       |
| 124.5                    | ORIENT FOR PTC<br>3 AXIS 0.2 DEG/SEC             | 32078.          | .7                      | 773.8                   | 57.                       |
| 124.5                    | ATT HOLD .5 DEG DB PGNC                          | 32077.          | .8                      | 773.0                   | 57.                       |
| 124.5                    | EST. 0.3 DEG/SEC ROLL                            | 32077.          | .2                      | 772.8                   | 57.                       |
| 124.5                    | PITCH AND YAW CONTROL                            | 32070.          | 7.0                     | 765.8                   | 57.                       |
| 129.5                    | PS2 IMU ALIGN                                    | 32069.          | .7                      | 765.1                   | 57.                       |



| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 130.0                    | CISLUNAR NAVIGATION<br>STAR/LUNAR HORIZON ORIENT | 32068.          | 1.0                     | 764.1                   | 57.                       |
| 130.0                    | MIN. IMPULSE MARKING                             | 32068.          | .4                      | 763.7                   | 57.                       |
| 130.0                    | NAV SIGHTING SET 2                               | 32067.          | 1.0                     | 762.7                   | 57.                       |
| 130.0                    | MINIMUM IMPULSE MARKING                          | 32067.          | .4                      | 762.2                   | 57.                       |
| 130.0                    | NAV SIGHTING SET 3                               | 32066.          | 1.0                     | 761.3                   | 56.                       |
| 130.0                    | MINIMUM IMPULSE MARKING                          | 32065.          | .4                      | 760.9                   | 56.                       |
| 130.5                    | MANEUVER TO SIGHTING ATTITUDE                    | 32064.          | 1.0                     | 759.9                   | 56.                       |
| 130.8                    | CISLUNAR NAVIGATION<br>STAR EARTH HORIZON ORIENT | 32063.          | 1.0                     | 758.9                   | 56.                       |
| 130.8                    | MINIMUM IMPULSE MARKING                          | 32063.          | .4                      | 758.5                   | 56.                       |
| 130.8                    | NAV SIGHTING SET 2                               | 32062.          | 1.0                     | 757.5                   | 56.                       |
| 130.8                    | MINIMUM IMPULSE MARKING                          | 32061.          | .4                      | 757.1                   | 56.                       |
| 130.8                    | NAV SIGHTING SET 3                               | 32060.          | 1.0                     | 756.1                   | 56.                       |
| 130.8                    | MINIMUM IMPULSE MARKING                          | 32060.          | .4                      | 755.7                   | 56.                       |
| 131.4                    | NAV SIGHTING SET 4                               | 32059.          | 1.0                     | 754.7                   | 56.                       |
| 131.4                    | MIN. IMPULSE MARKING                             | 32059.          | .4                      | 754.3                   | 56.                       |

| SM-RCS PROPELLANT BUDGET |                                     |                 |                         |                         |                           |
|--------------------------|-------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                               | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(#) |
| 131.4                    | NAV SIGHTING SET 5                  | 32058.          | 1.0                     | 753.3                   | 56.                       |
| 131.4                    | MINIMUM IMPULSE MARKING             | 32057.          | .4                      | 752.8                   | 56.                       |
| 131.4                    | NAV SIGHTING SET 6                  | 32056.          | 1.0                     | 751.9                   | 56.                       |
| 131.4                    | MINIMUM IMPULSE MARKING             | 32056.          | .4                      | 751.5                   | 56.                       |
| 131.8                    | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC | 32055.          | .7                      | 750.7                   | 56.                       |
| 131.8                    | ATT HOLD .5 DEG DB PGNC             | 32054.          | .8                      | 749.9                   | 56.                       |
| 131.8                    | EST. 0.3 DEG/SEC ROLL               | 32054.          | .2                      | 749.7                   | 56.                       |
| 131.8                    | PITCH AND YAW CONTROL               | 32048.          | 6.4                     | 743.3                   | 55.                       |
| 135.5                    | P52 IMU ALIGN                       | 32047.          | .7                      | 742.6                   | 55.                       |
| 135.7                    | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC | 32046.          | .7                      | 741.9                   | 55.                       |
| 135.7                    | ATT HOLD .5 DEG DB PGNC             | 32046.          | .8                      | 741.1                   | 55.                       |
| 135.7                    | EST. 0.3 DEG/SEC ROLL               | 32045.          | .2                      | 741.0                   | 55.                       |
| 135.7                    | PITCH AND YAW CONTROL               | 32040.          | 5.0                     | 736.0                   | 55.                       |
| 139.4                    | P52 IMU ALIGN                       | 32040.          | .7                      | 735.2                   | 55.                       |
| 139.5                    | ORIENT FOR PTC<br>3AXIS 0.2 DEG/SEC | 32039.          | .8                      | 734.5                   | 55.                       |

| SM-RCS PROPELLANT BUDGET |                                                  |                 |                         |                         |                           |  |
|--------------------------|--------------------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|--|
| TIME<br>(HR)             | EVENT                                            | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |  |
| 139.5                    | ATTITUDE HOLD 0.5 DEG DB PG <sub>NCS</sub>       | 32038.          | .8                      | 733.7                   | 54.                       |  |
| 139.5                    | EST. 0.3 DEG/SEC ROLL                            | 32038.          | .2                      | 733.5                   | 54.                       |  |
| 139.5                    | PITCH AND YAW CONTROL                            | 32034.          | 3.9                     | 729.6                   | 54.                       |  |
| 142.2                    | CISLUNAR NAVIGATION<br>STAR/EARTH HORIZON ORIENT | 32033.          | 1.0                     | 728.7                   | 54.                       |  |
| 142.2                    | MIN. IMPULSE MARKING                             | 32033.          | .4                      | 728.2                   | 54.                       |  |
| 142.2                    | NAV SIGHTING SET 2                               | 32032.          | .9                      | 727.3                   | 54.                       |  |
| 142.2                    | MINIMUM IMPULSE MARKING                          | 32031.          | .4                      | 726.9                   | 54.                       |  |
| 143.7                    | P52 IMU ALIGN                                    | 32031.          | .7                      | 726.1                   | 54.                       |  |
| 144.4                    | MIDCOURSE CORRECTION                             | 32030.          | .7                      | 725.4                   | 54.                       |  |
| 144.4                    | ATT HOLD .5 DEG DB PG <sub>NCS</sub>             | 32030.          | .3                      | 725.2                   | 54.                       |  |
| 144.4                    | RCS +X TRANS PG <sub>NCS</sub> 5 FPS             | 32011.          | 18.6                    | 706.6                   | 52.                       |  |
| 145.0                    | CISLUNAR NAVIGATION<br>STAR EARTH HORIZON ORIENT | 32011.          | .0                      | 706.6                   | 52.                       |  |
| 145.0                    | MINIMUM IMPULSE MARKING                          | 32010.          | .4                      | 706.1                   | 52.                       |  |
| 145.7                    | P52 IMU ALIGN                                    | 32010.          | .7                      | 705.4                   | 52.                       |  |
| 146.1                    | MANEUVER TO REENTRY ATTITUDE                     | 32009.          | 1.0                     | 704.4                   | 52.                       |  |

| SM-RCS PROPELLANT BUDGET |                                     |                 |                         |                         |                           |
|--------------------------|-------------------------------------|-----------------|-------------------------|-------------------------|---------------------------|
| TIME<br>(HR)             | EVENT                               | S/C WT<br>(LBS) | SM-RCS<br>USED<br>(LBS) | SM-RCS<br>LEFT<br>(LBS) | SM-<br>RCS<br>LEFT<br>(%) |
| 146.5                    | PITCH TO ACQUIRE HORIZON            | 32008.          | .7                      | 703.7                   | 52.                       |
| 146.5                    | YAW 45 DEG                          | 32007.          | .7                      | 703.0                   | 52.                       |
| 146.5                    | ATT HOLD .5 DEG DB PGNC             | 32007.          | .4                      | 702.6                   | 52.                       |
| 146.6                    | CM/SM SEPARATION<br>DELTA VEL=3 FPS | 19526.          | 10.8                    | 691.8                   | 51.                       |

TABLE 3-IV  
CM RCS PROPELLANT USAGE SUMMARY

|                                                 |      |             |
|-------------------------------------------------|------|-------------|
| Loaded, lb.                                     |      | 270.0       |
| Less                                            |      |             |
| Trapped, lb.                                    | 32.4 |             |
| Temperature variation allowance, lb.            | 6.4  |             |
| Available for mission planning, lb.             |      | 231.2       |
| Nominal Usage                                   |      |             |
| Separation and attitude hold<br>prior to 0.05 g | 12.1 |             |
| Guidance commands for remainder<br>of reentry   | 19.5 |             |
| Operational Reserve, lb.                        |      | <hr/> 199.6 |

TABLE 3-V  
Mission C Prime SPS Propellant Budget

| <u>Item</u>               | <u>Propellant (Lbs)</u> |
|---------------------------|-------------------------|
| Loaded                    | 40785                   |
| Trapped                   | -441                    |
| Outage                    | - 18                    |
| Unbalance Meter           | <u>-100</u>             |
| Available for $\Delta V$  | 40226                   |
| Required for $\Delta V^1$ | <u>-31253</u>           |
| Nominal Remaining         | 8973                    |

<sup>1</sup>Includes 14.4 lbs per start for start losses

$$I_{SP} = 314.25 \pm 1.593$$

$$MR = 1.595 \pm 0.0067$$

Vehicle Wt

CSM 22955.7

| <u>Burn</u>       | <u><math>\Delta V</math></u> | <u>Propellant Required</u> |
|-------------------|------------------------------|----------------------------|
| TLMC <sup>2</sup> | 120.                         | 780.0                      |
| LOI               | 2991.                        | 16074.7                    |
| Circularization   | 138.5                        | 648.5                      |
| TEI               | 3531.7                       | 13537.7                    |
| TEMC              | 62                           | <u>211.9</u>               |
|                   |                              | 31252.7                    |

<sup>2</sup>2 starts

TABLE 3--VI

CRYOGENIC CONSUMPTION ANALYSIS SUMMARY

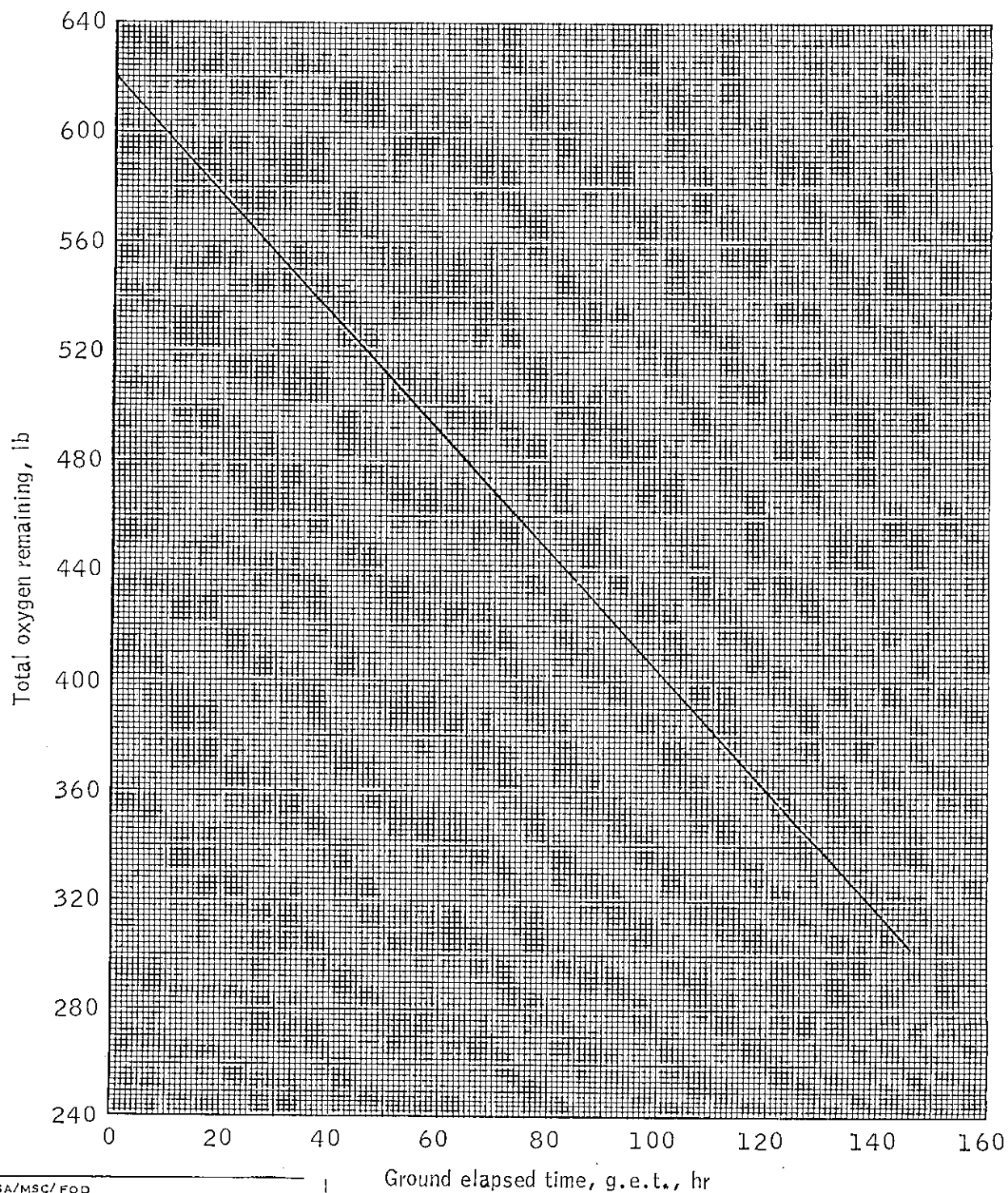
|                                  | <u>O<sub>2</sub> (lbs)</u> | <u>H<sub>2</sub> (lbs)</u> |
|----------------------------------|----------------------------|----------------------------|
| Loaded                           | 653.0                      | 58.4                       |
| Residual & Instrumentation Error | 30.0                       | 3.9                        |
| Available for mission            | 623.0                      | 54.5                       |
| Mission requirements             | 344.0                      | 33.9                       |
| Margin                           | 279.0                      | 20.6                       |

The results of the cryogenic consumption analysis are summarized in the following figures:

1. Figure - Nominal Mission O<sub>2</sub> Profile.
2. Figure - Nominal Mission H<sub>2</sub> Profile.

The following ground rules and assumptions were used for the construction of the mission H<sub>2</sub> and O<sub>2</sub> profiles.

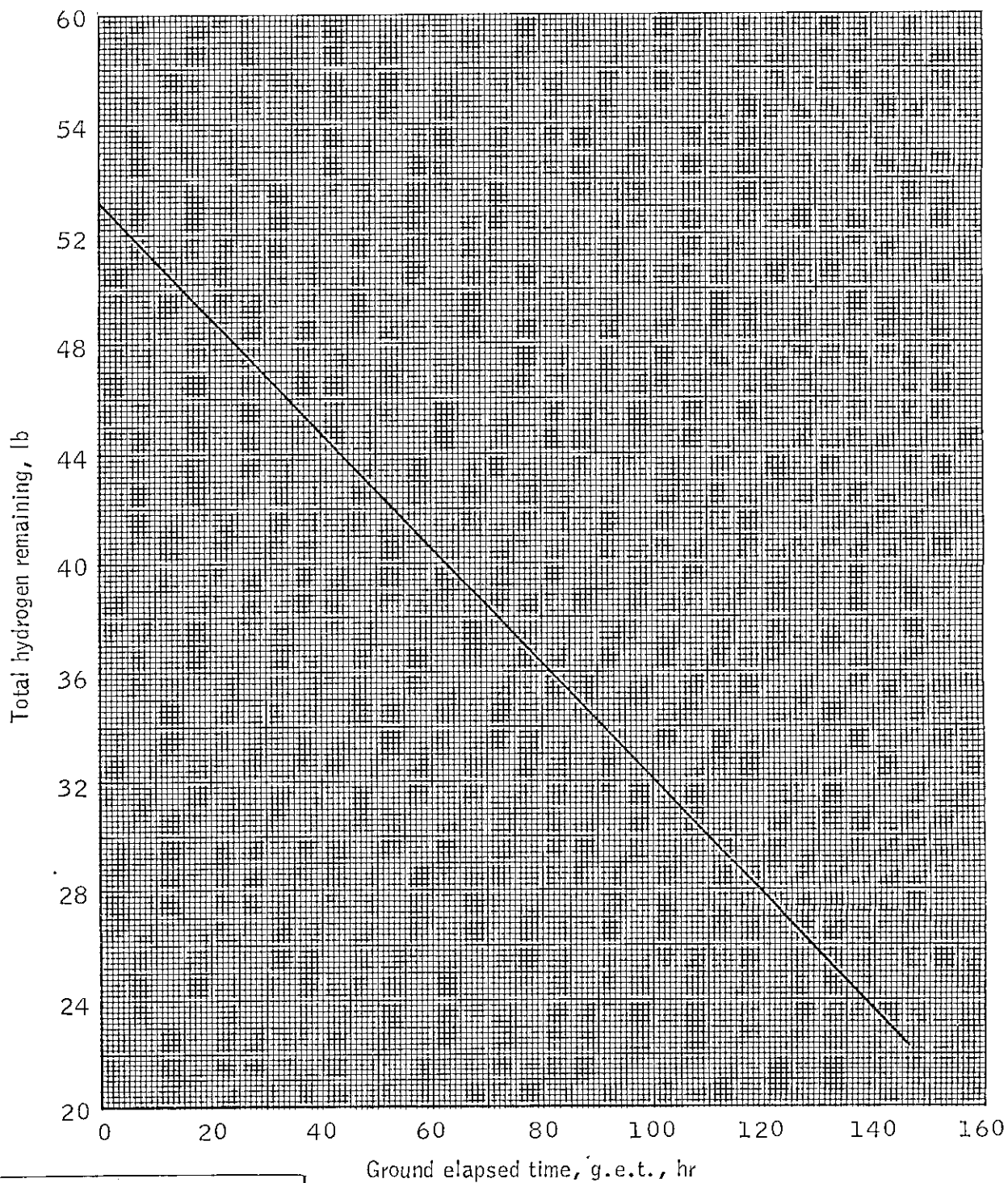
1. 10 fuel cell purges.
2. Cabin O<sub>2</sub> leak rate of 0.2 lb/hr.
3. Metabolic O<sub>2</sub> rate of 0.23 lb/hr for 3 crewmen.
4. Waste management O<sub>2</sub> rate of 0.051 lb/hr.
5. Water tank O<sub>2</sub> purge rate of 0.056 lb/hr.
6. The hydrogen consumption rate is .00257 lb/amp/hr.
7. The oxygen consumption rate is 7.936 times the hydrogen consumption rate.



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 BRANCH GPB DATE 11-29-68  
 BY Scott PLOT NO. 386

Figure 3-VI.- Oxygen profile for Apollo 8.





NASA/MSC/FOD  
MISSION PLANNING AND ANALYSIS DIVISION  
BRANCH GFB DATE 11-29-68  
BY Scott PLOT NO. 387

Figure 3-VII. - Hydrogen profile for Apollo 8.

TABLE 3-VII

CREW CHECKLIST PROCEDURE/CONSUMABLES DATA SUMMARY  
(AVERAGE ELECTRICAL POWER LOADS)

| CREW<br>CHECKLIST<br>PROCEDURE               | AVERAGE POWER |        | BUS<br>VOLTAGE | BATTERY<br>CURRENT | BATTERY<br>POWER | FUEL<br>CELL<br>CURRENT | BUS<br>POWER | CRYO CONSUMPTION |                | H <sub>2</sub> O<br>PRODUCTION |
|----------------------------------------------|---------------|--------|----------------|--------------------|------------------|-------------------------|--------------|------------------|----------------|--------------------------------|
|                                              | AC            | DC     |                |                    |                  |                         |              | H <sub>2</sub>   | O <sub>2</sub> |                                |
|                                              | WATTS         | WATTS  |                | AMPS               | WATTS            | AMPS                    | WATTS        | LB/HR            | LB/HR          | LB/HR                          |
| Contingency<br>Drift Flight                  |               |        |                |                    |                  |                         |              |                  |                |                                |
| 1. Normal                                    | 676.6         | 1067.5 | 28.75          | -                  | -                | 74.0                    | 2130         | 0.189            | 1.50           | 1.69                           |
| 2. Power Down                                | 676.6         | 905.9  | 28.90          | -                  | -                | 67.0                    | 1940         | 0.174            | 1.38           | 1.55                           |
| Boost thru Separation and Formation Flying   |               |        |                |                    |                  |                         |              |                  |                |                                |
| 1. Lift-Off                                  | 770.0         | 1085.0 | 29.30          | 19.0               | 556.0            | 58.0                    | 2260         | 0.149            | 1.18           | 1.34                           |
| 2. 1st Stage Separation                      | 779.1         | 1812.5 | 28.65          | 33.0               | 945.0            | 73.5                    | 3060         | 0.189            | 1.50           | 1.69                           |
| 3. Insertion (Earth Orbit)                   | 734.9         | 1202.3 | 28.30          | -                  | -                | 81.0                    | 2290         | 0.208            | 1.65           | 1.86                           |
| 4. TLI & Separation Preparation              | 744.6         | 1179.5 | 28.30          | -                  | -                | 81.0                    | 2290         | 0.207            | 1.65           | 1.85                           |
| 5. Separation and Formation Flying           | 764.4         | 1193.6 | 28.25          | -                  | -                | 82.0                    | 2320         | 0.210            | 1.67           | 1.88                           |
| Power Up                                     |               |        |                |                    |                  |                         |              |                  |                |                                |
| 1. Stabilization and Control                 | 767.0         | 1128.0 | 28.35          | -                  | -                | 80.0                    | 2270         | 0.206            | 1.64           | 1.84                           |
| 2. G&N Optics                                | 681.8         | 1156.6 | 28.45          | -                  | -                | 77.5                    | 2210         | 0.199            | 1.59           | 1.77                           |
| 3. Stabilization and Control, and G&N Optics | 772.2         | 1217.1 | 28.25          | -                  | -                | 82.5                    | 2330         | 0.213            | 1.68           | 1.90                           |
| G&N/SPS Orbit Change                         |               |        |                |                    |                  |                         |              |                  |                |                                |
| 1. Initial Conditions                        | 774.8         | 1156.8 | 28.30          | -                  | -                | 81.0                    | 2290         | 0.208            | 1.65           | 1.86                           |
| 2. Ignition Preparation                      | 788.2         | 1897.5 | 28.60          | -                  | -                | 74.0                    | 3080         | 0.192            | 1.52           | 1.71                           |
| 3. Ignition                                  | 868.5         | 2458   | 28.05          | 45.0               | 1260             | 86.5                    | 3690         | 0.224            | 1.78           | 2.00                           |
| Battery Charging                             | 724.6         | 1116.5 | 28.5           | -                  | -                | 77.0                    | 2190         | 0.197            | 1.57           | 1.76                           |
| G&N SM RCS Orbit Change Thrusting (P41)      |               |        |                |                    |                  |                         |              |                  |                |                                |
| 1. Initial Conditions                        | 774.8         | 1170.2 | 28.3           | -                  | -                | 82.0                    | 2320         | 0.210            | 1.66           | 1.87                           |

CREW CHECKLIST PROCEDURE/CONSUMABLES DATA SUMMARY  
(AVERAGE ELECTRICAL POWER LOADS)

| CREW<br>CHECKLIST<br>PROCEDURE | AVERAGE POWER |        | BUS<br>VOLTAGE | BATTERY<br>CURRENT | BATTERY<br>POWER | FUEL<br>CELL<br>CURRENT | BUS<br>POWER | CRYO CONSUMPTION |                | H <sub>2</sub> O<br>PRODUCTION |
|--------------------------------|---------------|--------|----------------|--------------------|------------------|-------------------------|--------------|------------------|----------------|--------------------------------|
|                                | AC            | DC     |                |                    |                  |                         |              | H <sub>2</sub>   | O <sub>2</sub> |                                |
|                                | WATTS         | WATTS  | VOLTS          | AMPS               | WATTS            | AMPS                    | WATTS        | LB/HR            | LB/HR          | LB/HR                          |
| 2. Ignition                    | 774.8         | 1602.2 | 27.6           | -                  | -                | 97.0                    | 2680         | 0.250            | 1.99           | 2.24                           |
| Entry to Landing               |               |        |                |                    |                  |                         |              |                  |                |                                |
| 1. Preparation                 | 684.4         | 1089.2 | 29.6           | 23.0               | 680.0            | 52.0                    | 2220         | 0.134            | 1.06           | 1.20                           |
| 2. CM RCS Htrs                 | 684.4         | 1659.2 | 29.2           | 34.0               | 993.0            | 62.0                    | 2800         | 0.159            | 1.26           | 1.43                           |
| 3. CM SM Separation            | 501.8         | 760.6  | 28.5           | 55.0               | 1565.0           | -                       | 1565.0       | -                | -              | -                              |
| 4. Entry                       | 492.3         | 812.7  | 28.6           | 56.5               | 1605.0           | -                       | 1605.0       | -                | -              | -                              |
| Post Landing                   | -             | 86.0   | 28.0           | 3.07               | 86.0             | -                       | 86.0         | -                | -              | -                              |



SECTION IV - DETAILED TEST OBJECTIVES



## SECTION 4

### TEST OBJECTIVE ACTIVITIES

This section contains the activity summaries which reflect the test objectives for Mission C' as described in "Mission Requirements AS 503/CSM 103 C' Type Mission (Lunar Orbit)" dated November 16, 1968. These activity summaries are presented in the approximate sequence in which they are planned to occur during the mission. In the case of activities which are repeated in the mission plan, they are described only once in this summary. Those test objectives which do not relate to specific mission activities are grouped at the end of this section as "Miscellaneous Tests" and "Telecommunications". Test objective requirements which are not scheduled in this flight plan are identified as being not implemented, "N.I."

Each activity summary provides the following information:

- A. TEST OBJECTIVES. This is the listing of the Functional Test Objectives (complete or partial) which relate to the particular activity;
- B. TEST REQUIREMENTS. Here the special test prerequisites are presented in addition to brief statements of the requirements for performing the activity;
- C. TEST PROCEDURES/CHECKLISTS. These are the procedural references for the performance of the activity as far as the test objectives are concerned; and
- D. DATA REQUIREMENTS. This part of the summary identifies the gross data which are needed for evaluation of test results in terms of flight crew and ground support requirements.

A cross reference for relating Detailed and Functional Test Objectives with the activity summaries is provided as the initial part of this section.

The following ground rules are to be used in implementing data requirements:

- A. The collection of highly desirable (HD) data should not constrain the timeline or the crew procedures.
- B. CSM data storage equipment (DSE) HBR recording is needed only when MSFN coverage is not available and when mandatory data are required.
- C. Data collected by the crew which are required only for postflight analysis, need not be voiced to MSFN in real time.

TEST OBJECTIVE/MISSION ACTIVITY  
CROSS REFERENCE

| DTO/FTO<br>NUMBER                                            | TEST OBJECTIVE                                                                                                                                                                                                                                                                                                                                    | MISSION ACTIVITY                                                                                                                                                       | SECTION<br>PAGE NO.                           |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| SL.27<br>SL.27-1<br>SL.27-2                                  | GNCS Boost Monitor/Saturn V<br>Boost Monitor - GNCS Performance<br>Boost Monitor - Display Adequacy                                                                                                                                                                                                                                               | Launch<br>Launch                                                                                                                                                       | 4-8<br>4-8                                    |
| SL.30<br>SL.30-1<br>SL.30-2<br>SL.30-3                       | IMU Performance<br>GNCS Accelerometer Biases During Coasting Flight<br>GNCS/IRIG Drift Rates During Coasting Flight<br>Overall GNCS Errors During Thrusting Maneuvers                                                                                                                                                                             | PIPA Bias Check<br>IMU Realignment<br>Lunar Orbit Insertion                                                                                                            | 4-18<br>4-15<br>4-8,27                        |
| PL.31<br>PL.31-1<br>PL.31-2                                  | GNCS Entry Lunar Return<br>GNCS Performance During Entry - Lunar Return<br>EMS Capability to Monitor Entry - Lunar Return                                                                                                                                                                                                                         | Entry<br>Entry                                                                                                                                                         | 4-12,35<br>4-12,35                            |
| SL.32<br>SL.32-1<br>SL.32-2<br>SL.32-3<br>SL.32-4<br>SL.32-5 | Midcourse Navigation/Star Earth - Landmark<br>Star/Earth Landmark Navigation Sighting Accuracy<br>Verify and Update Landmark Lighting Constraints<br>OSS Adequacy for Navigation Sightings in Deep Space<br>Crew Capability to Identify Landmarks and Coordinate Optics/Maneuvers<br>RCS Propellant Use and Time to Accomplish Onboard Navigation | Star/Earth Landmark Navigation<br>Star/Earth Landmark Navigation<br>Star/Earth Landmark Navigation<br>Star/Earth Landmark Navigation<br>Star/Earth Landmark Navigation | 4-11,20<br>4-11,20<br>4-11,20<br>4-20<br>4-20 |
| PL.33<br>PL.33-1<br>PL.33-2<br>PL.33-3<br>PL.33-4            | Midcourse Navigation/Star-Lunar Horizon<br>Star/Lunar Horizon Navigation Sighting Accuracy<br>OSS Adequacy for Navigation Sightings in Deep Space<br>Crew Capability to Coordinate Optics/Maneuvers<br>RCS Propellant Use and Time to Accomplish Onboard Navigation                                                                               | Star/Lunar Horizon Navigation<br>Star/Lunar Horizon Navigation<br>Star/Lunar Horizon Navigation<br>Star/Lunar Horizon Navigation                                       | 4-24<br>4-24<br>4-24<br>4-24                  |



TEST OBJECTIVE/MISSION ACTIVITY  
CROSS REFERENCE

| DTO/FTO<br>NUMBER | TEST OBJECTIVE                                                           | MISSION ACTIVITY              | SECTION<br>PAGE NO. |
|-------------------|--------------------------------------------------------------------------|-------------------------------|---------------------|
| P1.34             | Midcourse Navigation/Star-Earth Horizon                                  |                               |                     |
| P1.34-1           | Star/Earth Horizon Navigation Sighting Accuracy                          | Star/Earth Horizon Navigation | 4-16                |
| P1.34-2           | Verify and Update Horizon Lighting Constraints                           | Star/Earth Horizon Navigation | 4-16                |
| P1.34-3           | OSS Adequacy for Navigation Sightings in Deep Space                      | Star/Earth Horizon Navigation | 4-16                |
| P1.34-4           | Crew Ability to Identify Horizon Locator and Coordinate Optics/Maneuvers | Star/Earth Horizon Navigation | 4-16                |
| P1.34-5           | RCS Propellant Use and Time to Accomplish Onboard Navigation             | Star/Earth Horizon Navigation | 4-16                |
| S1.35             | IMU Orientation Determination/Visibility                                 |                               | 4-31                |
| S1.35-1           | IMU Realignment in Daylight                                              | IMU Realignment               | 4-15                |
| S1.35-2           | Obtain Data - Star Visibility During TL and TE Coast                     | Star Visibility               | 4-26                |
| S1.35-3           | Degradation of Navigation/IMU Align by Vented/Ejected Particles          | Star Visibility               | 4-26                |
| S3.21             | SPS Evaluation                                                           |                               |                     |
| S3.21-1           | SPS Isp - Adequacy of Conversion - Ground to Vacuum Results              | LOI and TEI                   | 4-27,33             |
| S3.21-2           | SPS Performance for LOI and TEI Burns                                    | LOI and TEI                   | 4-27,33             |
| S3.21-3           | SPS PUGS in Auxiliary Mode - Relative Accuracy Aux/Pri                   | TEI                           | 4-33                |
| S3.21-4           | Thermal Effects - Long SPS Burn - Heat Protection System                 | LOI and TEI                   | 4-27,33             |
| S4.5              | ECS Lunar Return Entry                                                   |                               |                     |
| S4.5-1            | ECS Performance During Manned Lunar Return Entry                         | Entry                         | 4-35                |
| S4.5-2            | Compare ECS Data with Developed Model and Unmanned Results               | Entry                         | 4-35                |
| S6.10             | CSM Omni Antennas Lunar Distance                                         |                               |                     |
| S6.10-1           | S-Band Performance with Omni Antennas at Lunar Distance                  | Telecommunications            | 4-38                |

TEST OBJECTIVE/MISSION ACTIVITY  
CROSS REFERENCE

| DTO/FTO<br>NUMBER | TEST OBJECTIVE                                                | MISSION ACTIVITY               | SECTION<br>PAGE NO. |
|-------------------|---------------------------------------------------------------|--------------------------------|---------------------|
| P6.11             | CSM/MSFN Communications Lunar Distance                        |                                |                     |
| P6.11-1           | S-Band Performance with High Gain Antenna - TL Coast          | Telecommunications             | 4-38                |
| P6.11-2           | S-Band Performance with High Gain Antenna - Lunar Distance    | Telecommunications             | 4-38                |
| P6.11-3           | CSM HGA Oper. in Potential S/C Reflectivity Region            | Not Implemented                |                     |
| S7.30             | Heat Shield Lunar Return                                      |                                |                     |
| S7.30-1           | Block II Thermal Protection System During Manned Lunar Return | Entry                          | 4-35                |
| S7.30-2           | TPS Surface Recession and Char Data Comparison                | Entry                          | 4-35                |
| P7.31             | Spacecraft Environment Thermal Control                        |                                |                     |
| P7.31-1           | Evaluate Thermal Control System During TL and TE Coast        | Miscellaneous Tests            | 4-20,36             |
| P7.31-2           | Thermal Control System During Lunar Orbit                     | Miscellaneous Tests            | 4-20,36             |
| P7.32             | Spacecraft Dynamic Environment                                |                                |                     |
| P7.32-1           | CSM Loads/Vibrations - Launch, TLI and Separation             | Launch, TLI, CSM/S-IVB Sep.    | 4-8,11,12           |
| P7.32-2           | SLA Dynamic Response - Launch, TLI and Separation             | Launch, TLI, CSM/S-IVB Sep.    | 4-8,11,12           |
| P7.32-3           | S-Band HGA Response During Deployment and SPS Burns           | CSM/S-IVB Sep., Midcourse, LOI | 4-19,12,27          |
| P7.33             | SLA Panel Jettison                                            |                                | 4-14                |
| P7.33-1           | SLA Panel Jettison Demonstration                              | CSM/S-IVB Separation           | 4-14                |
| S20.104           | Transposition                                                 |                                |                     |
| S20.104-1         | Separation and Transposition Maneuver                         | CSM/S-IVB Sep., Transposition  | 4-12,13,14          |
| S20.104-2         | Separation and Transposition Procedures and Timeline Adequacy | CSM/S-IVB Sep., Transposition  | 4-12,13             |
| S20.104-3         | S-Band Performance During Transposition                       | Telecommunications             | 4-38                |

TEST OBJECTIVE/MISSION ACTIVITY  
CROSS REFERENCE

| DTO/FTO<br>NUMBER | TEST OBJECTIVE                                                 | MISSION ACTIVITY              | SECTION<br>PAGE NO. |
|-------------------|----------------------------------------------------------------|-------------------------------|---------------------|
| P20.105           | LOI Maneuver                                                   | Lunar Orbit Insertion         | 4-27                |
| P20.105-1         | Crew/Spacecraft/MSFN Preparation and Execution of LOI Maneuver | Lunar Orbit Insertion         | 4-27                |
| P20.105-2         | Procedures and Timeline Adequacy for LOI Maneuver              | Lunar Orbit Insertion         | 4-27                |
| P20.106           | TEI Maneuver                                                   | Transearth Insertion          | 4-33                |
| P20.106-1         | Crew/Spacecraft/MSFN Preparation and Execution of TEI Maneuver | Transearth Insertion          | 4-33                |
| P20.106-2         | Procedures and Timeline Adequacy for TEI Maneuver              | Transearth Insertion          | 4-33                |
| P20.107           | Crew Activities Lunar Distance                                 | Miscellaneous Tests           | 4-36                |
| P20.107-1         | Crew Procedures for Lunar Orbit Mission                        | Miscellaneous Tests           | 4-36                |
| P20.107-2         | Crew Procedures and Timeline Adequacy -- Lunar Mission         | Miscellaneous Tests           | 4-36                |
| S20.108           | CSM Consumables Lunar Mission                                  | Miscellaneous Tests           | 4-36                |
| S20.108-1         | EPS Use, Performance and Fuel Cell Management                  | Miscellaneous Tests           | 4-36                |
| S20.108-2         | Water and Oxygen Requirements Data                             | Miscellaneous Tests           | 4-36                |
| S20.108-3         | LiOH Cartridge Requirements Data                               | Miscellaneous Tests           | 4-36                |
| S20.108-4         | SPS, SM/RCS and CM/RCS Propellant Requirements Data            | Miscellaneous Tests           | 4-36                |
| S20.108-5         | Crew Food Requirements Data                                    | Miscellaneous Tests           | 4-36                |
| P20.109           | Passive Thermal Control Modes                                  | Passive Thermal Control Modes | 4-22                |
| P20.109-1         | PTC Procedures and RCS Use -- Roll with Attitude Hold          | Passive Thermal Control Modes | 4-22                |
| P20.109-2         | PTC Procedures and RCS Use -- Roll without Attitude Hold       | Telecommunications            | 4-22, 38            |
| P20.109-3         | Communication Procedures Adequacy for PTC Mode                 | Passive Thermal Control Modes | 4-22                |
| P20.109-4         | PTC Procedures and RCS Use -- Yaw without Attitude Hold        | Passive Thermal Control Modes | 4-22                |
| P20.110           | Ground Support Lunar Distance                                  | Miscellaneous Tests           | 4-36                |
| P20.110-1         | Ground Operation Support for Lunar Mission Without LM          | Miscellaneous Tests           | 4-36                |

TEST OBJECTIVE/MISSION ACTIVITY  
CROSS REFERENCE

| DTO/FTO<br>NUMBER | TEST OBJECTIVE                                               | MISSION ACTIVITY                               | SECTION<br>PAGE NO. |
|-------------------|--------------------------------------------------------------|------------------------------------------------|---------------------|
| P20.111           | Lunar Landmark Tracking                                      |                                                |                     |
| P20.111-1         | Error Uncertainties in Lunar Landing Site Location           | Lunar Landmark Tracking                        | 4-29                |
| P20.111-2         | Obtain Data to Calibrate MSFN at Lunar Distance              | Lunar Landmark Tracking                        | 4-29                |
| P20.111-3         | Determine Minimum Sun Angle to Clearly Identify Landmarks    | Lunar Landmark Tracking                        | 4-29                |
| P20.111-4         | Lunar Landmark Tracking in Earthshine                        | Lunar Landmark Tracking                        | 4-29                |
| P20.111-5         | Adequacy of CSS and OSS for Obtaining Landmark Sightings     | Lunar Landmark Tracking                        | 4-29                |
| P20.111-6         | Crew Ability to Coordinate Sightings and Vehicle Maneuvers   | Lunar Landmark Tracking                        | 4-29                |
| P20.111-7         | RCS Propellant and Time Required for Landmark Sighting       | Lunar Landmark Tracking                        | 4-29                |
| P20.112           | TLI Burn                                                     |                                                |                     |
| P20.112-1         | Crew/Ground Capability for TLI at First Opportunity          | Pre-TLI and TLI                                | 4-10                |
| P20.112-2         | TLI Preparation and Execution Procedure Demonstration        | Pre-TLI and TLI                                | 4-10, 11            |
| P20.112-3         | TLI Procedures and Timeline Adequacy                         | Pre-TLI and TLI                                | 4-10, 11            |
| P20.112-4         | CSM Cabin Vibration Environment during TLI Burn              | Translunar Injection                           | 4-11                |
| P20.112-5         | Crew Monitoring of GNCS and LV Displays - TLI Burn           | Translunar Injection                           | 4-11                |
| P20.114           | Midcourse Correction Capability                              |                                                |                     |
| P20.114-1         | Procedures and Timeline Adequacy for MCC'S                   | Midcourse Corrections                          | 4-19                |
| P20.114-2         | GNCS, SPS and RCS Performance on MCC'S                       | Midcourse Corrections                          | 4-19                |
| P20.114-3         | Ground Capability to Update State Vector and Target for MCC  | Midcourse Corrections                          | 4-19                |
| S20.115           | Lunar Mission Photography from the CSM                       |                                                |                     |
| S20.115-1         | Overlapping Photos from Terminator to Terminator-Lunar Orbit | Photography                                    | 4-31                |
| S20.115-2         | Photos of Earth and Lunar Landmarks                          | Photo, S/E Landmark Nav.                       | 4-20                |
| S20.115-3         | Photos of Earth and Lunar Horizons                           | Photo, S/E Landmark Nav., S/L<br>Landmark Nav. | 4-16, 24, 31        |
| S20.115-4         | Photos of General and Scientific Interest                    | Photography                                    | 4-31                |

TEST OBJECTIVE/MISSION ACTIVITY  
CROSS REFERENCE

| DTO/FTO<br>NUMBER                 | TEST OBJECTIVE                                                                                                       | MISSION ACTIVITY              | SECTION<br>PAGE NO. |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------|
| S20.116<br>S20.116-1<br>S20.116-2 | Exhaust Effects/CSM Windows<br>Changes in Visual Acuity Through Windows<br>Changes in Total Transmittance of Windows | Launch Transposition<br>Entry | 4-8,13<br>4-35      |

## LAUNCH

### A. TEST OBJECTIVES

- S1.27-1 Boost Monitor - GNCS Performance
- S1.27-2 Boost Monitor - Display Adequacy
- S1.30-3 Overall GNCS Errors During Thrusting Maneuvers
- P7.32-1 CSM Loads/Vibrations - S-V Boost
- P7.32-2 SLA Dynamic Response - S-V Boost
- S20.116-1 Changes in Visual Acuity Through Windows

### B. TEST REQUIREMENTS

- 1. CSM insertion into earth orbit by a Saturn V vehicle [1.27, 1.30, 7.32 & 20.116]
- 2. FQTR operation throughout Saturn V boost [7.32]
- 3. Crew observations of changes in visual acuity through the windows as the result of TJM firing & S-II retro firing [20.116]

### C. TEST PROCEDURES/CHECKLISTS

- 1. AOH paragraph 4.3.1 "Boost and Insertion"
- 2. FCAC L-1, L-2, & L-3 "Boost-Insertion"

### D. DATA REQUIREMENTS

- 1. Flight Crew Reports/Logs
  - a. FDAI or DSKY display inadequacies [1.27] (M)
  - b. Apparent GNCS trajectory errors [1.27] (M)
  - c. Unexpected structural loads or vibrations [7.32] (M)
  - d. Log changes in visual acuity through windows ("Window Visibility" log) [20.116] (M)
- 2. Ground Support
  - a. CSM TM HBR [1.27, 7.32] (M)
  - b. Continuous tracking [1.27] (M)
  - c. BET [7.32] (M)
  - d. Dynamic pressure & angle of attack during S-IC operation [7.32] (M)
  - e. Wind data from 0 to 60,000 ft at launch time [7.32] (M)

f. Saturn V thrust variations vs. time-MSFC [7.32] (M)

g. Saturn V accel & IU gyro data - MSFC [7.32] (M)

A. TEST OBJECTIVES

P20.112-1 Crew/Ground Capability for TLI at First Injection Opportunity  
P20.112-2 TLI Preparation and Execution Procedure Demonstration  
P20.112-3 TLI Procedures and Timeline Adequacy

B. TEST REQUIREMENTS

1. Following earth orbit insertion, flight crew accomplishment of all pre-TLI activities required for a Pacific injection on the second pass
2. Ground operational support personnel evaluation of all spacecraft systems and making of the go/no-go decision for the TLI burn

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.3.2 "Post Orbital Insertion Check"
2. AOH paragraph 4.4.1 "Translunar Injection"
3. FCAC L-4, L-5, L-1, & L-2 "Post Insertion"
4. FCAC L-6 & L-7 "System Verification & Monitoring"
5. FCAC L-8 & L-9 "TLI Preparation" to GET 02:50:38

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Procedural and/or timeline difficulties or inadequacies (M)
  - b. Comments regarding feasibility of first Atlantic injection opportunity during the second revolution (M)
2. Ground Support
  - a. Flight Director reports of timeline and/or procedural difficulties (M)



A. TEST OBJECTIVES

- P7.32-1 CSM Loads/Vibrations - TLI
- P7.32-2 SLA Dynamic Response - TLI
- P20.112-2 TLI Preparation and Execution Procedure Demo.
- P20.112-3 TLI Procedures and Timeline Adequacy
- P20.112-4 CSM Cabin Vibration Environment During TLI Burn
- P20.112-5 Crew Monitoring of GNCS & LV Displays During TLI

B. TEST REQUIREMENTS

- 1. Ground operational support and flight crew capability to execute the TLI burn [20.112]
- 2. FQTR operation throughout the S-IV burn [7.32]

C. TEST PROCEDURES/CHECKLISTS

- 1. AOH paragraph 4.4.1 "Translunar Injection"
- 2. FCAC L-9 & L-10 "TLI Preparation"

D. DATA REQUIREMENTS

- 1. Flight Crew Reports/Logs
  - a. Procedural and/or timeline difficulties or inadequacies [20.112] (M)
  - b. FDAI, DSKY & LV Tank Pressure display inadequacies [20.112] (M)
  - c. Unexpected induced resonances in cabin (Visual, audible and tactile) [20.112] (M)
  - d. Comments on adequacy of TLI monitoring [20.112] (M)
- 2. Ground Support
  - a. CSM TM HBR [7.32] (M)
  - b. Flight Director reports of procedural and/or timeline difficulties or inadequacies [20.112] (M)

## CSM/S-IVB SEPARATION

### A. TEST OBJECTIVES

- P7.32-1 CSM Loads/Vibrations - CSM/S-IVB Separation
- P7.32-2 SLA Dynamic Response - CSM/S-IVB Separation
- P7.32-3 S-Band High Gain Antenna Dynamic Response During Deployment
- S20.104-1 Separation and Transposition Maneuver
- S20.104-2 Separation and Transposition Procedures & Timeline Adequacy

### B. TEST REQUIREMENTS

- 1. CSM/S-IVB separation, SLA panel jettison and S-band high gain antenna deployment [20.104]
- 2. FQTR operation during the period of the CSM/S-IVB separation, SLA panel jettison and S-band high gain antenna deployment [7.32]

### C. TEST PROCEDURES/CHECKLISTS

- 1. AOH paragraph 4.5.1.1, "CSM/SLA Separation Through Capture Latch" through step 5
- 2. FCAC Separation TBD

### D. DATA REQUIREMENTS

- 1. Flight Crew Reports/Logs
  - a. Procedural and/or timeline difficulties or inadequacies [20.104] (M)
- 2. Ground Support
  - a. CSM TM HBR [7.32] (M)
  - b. Flight Director reports of procedural and/or timeline difficulties or inadequacies [20.104] (M)

A. TEST OBJECTIVES

- S20.104-1 Separation and Transposition Maneuver
- S20.104-2 Transposition Procedures and Timeline Adequacy
- S20.116-1 Changes in Visual Acuity Through Windows

B. TEST REQUIREMENTS

1. After separation, a (+X) translation from the S-IVB for three seconds, a coast for one minute to approximately 70 ft., a (-X) translation for 1 1/2 seconds, and then a pitch up at 4°/sec until the spacecraft (+X) axis points at the S-IVB [20.104]
2. Determination if there are any changes in visual acuity through the windows as the result of SM RCS use [20.116]

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.5.1.1, "CSM/SLA Separation Through Capture Latch" step 6
2. FCAC Transposition TBD

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Procedural and/or timeline difficulties or inadequacies [20.104] (M)
  - b. Inadequacies in communications with ground operational support [20.104] (M)
  - c. Log changes in visual acuity through the windows from SM RCS use ("Window Visibility" log) [20.116] (M)
2. Ground Support
  - a. CSM TM HBR [20.104] (M)
  - b. USB tracking data processor output recording [20.104] (M)
  - c. USB TM bit stream recording [20.104] (M)
  - d. MSFN tape records of S-band received signal strength [20.104] (M)
  - e. MSFN tape recordings of CSM to MSFN and MSFN to CSM voice [20.104] (M)
  - f. Flight Director reports of procedural and/or timeline difficulties or inadequacies [20.104] (M)
  - g. BET [20.104] (M)

A. TEST OBJECTIVES

P7.33-1 SLA Panel Jettison Demonstration  
S20.104-1 Separation and Transposition Maneuver

B. TEST REQUIREMENTS

1. Formation flying with the S-IVB to acquire the following photographs:
  - a. Sequence camera photographs of the S-IVB following transposition [20.104] (HD)
  - b. Photographs of the SLA aft portion and LTA-B showing that the panels jettisoned and of any anomaly observed [7.33] (M)

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.5.1.2 "Formation Flight With S-IVB (General)"
2. Photo checklist TBD

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Observed anomalies in the jettison of the SLA panels or launch vehicle instability [7.33] (M)
  - b. Disposition of any observed SLA panels [7.33] (M)
  - c. Sequence photographs of the S-IVB following transposition [20.104] (HD)
  - d. Photographs of the SLA aft portion and the LTA-B showing that the panels jettisoned and of any anomalies observed [7.33] (M)
2. Ground Support
  - a. None

## IMU REALIGNMENT

### A. TEST OBJECTIVES

S1.30-2 GNCS IRIG Drift Rates During Coasting Flight  
S1.35-1 IMU Realignment in Daylight

### B. TEST REQUIREMENTS

1. At least two sets of back-to-back IMU realignments (at least one hour apart without an intervening  $\Delta V$  maneuver) to determine IRIG drift rates [1.30]
2. One IMU realignment in daylight when at an altitude of greater than 2000 NM [1.35]

### C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.11.1 "IMU Orientation Determination (P51)" or
2. AOH paragraph 4.11.2 "IMU Realign (P52)"
3. FCAC "P51 - IMU Orientation" or
4. FCAC "P52 - IMU Realign"

### D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs - None
2. Ground Support - CSM TM [1.30] (M)

A. TEST OBJECTIVES

- Pl.34-1 Star/Earth Horizon Navigation Sighting Accuracy
- Pl.34-2 Verify and Update Horizon Lighting Constraints
- Pl.34-3 OSS Adequacy for Navigation Sightings in Deep Space
- Pl.34-4 Crew Ability to Identify Horizon Locator and Coordinate Optics/Maneuvers
- Pl.34-5 RCS Propellant Use and Time to Accomplish Onboard Navigation
- S20.115-3 Photographs of Earth and Lunar Horizons

B. TEST REQUIREMENTS

1. Use of the trunnion calibration option of P23 preceding a set of sightings if more than about 30 minutes has elapsed since the previous P23 trunnion calibration [1.34]
2. Translunar - Near Earth
  - a. Five sets of star/earth horizon sightings at more than 15,000 NM and less than 50,000 NM from the center of the earth [1.34]
  - b. At least 10 sequence photographs through the special sextant camera adapter while duplicating star/horizon navigation sightings when between 10,000 and 30,000 NM from the earth on TL phase. It is desirable that these photographs be taken at varying distances [20.115] - Not Implemented
  - c. Update of the onboard state vector utilizing the navigation sighting data following each set of sightings [1.34]
3. Translunar - Midcourse
  - a. Eighteen sets of star/earth horizon sightings at more than 50,000 NM from both the earth and the moon [1.34]
  - b. Update of the onboard state vector utilizing the navigation sighting data following each set of sightings [1.34]
4. Transearth
  - a. Star/earth horizon sightings under the following conditions [1.34]
    - (1) Seventeen sets at more than 50,000 NM from the earth and moon
    - (2) Three sets at less than 50,000 NM from the center of the earth
  - b. Update of the onboard state vector utilizing the navigation sighting data following each set of sightings [1.34]

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.12.3 "Cislunar Midcourse Navigation (P23) including "Trunnion Calibration" option
2. FCAC G-20A "P23 Cislunar Midcourse Navigation Measurement"
3. Photo checklist TBD

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Difficulties and/or inadequacies noted in handling the optics and the spacecraft [1.34] (M)
  - b. Log navigation data on Flight Plan timeline [1.34] (M)
  - c. Photo log TBD [20.115] (M)
2. Ground Support
  - a. CSM TM HBR [1.34] (M)
  - b. BET [1.34] (M)

PIPA BIAS CHECK

A. TEST OBJECTIVES

SL30-1 GNCS Accelerometer Biases During Coasting Flight

B. TEST REQUIREMENTS

1. MSFN collection of PIPA bias data at least five times during the mission using uninterrupted TM for at least five minutes as follows:
  - a. Prior to and following an SPS burn of at least 40 seconds
  - b. As close to entry as possible
  - c. Any two other periods in the mission
2. Onboard test performed only if communications are lost

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.10.1.13 "Measurement and Loading of PIPA Bias"
2. FCAC G-51 "Measure & Load PIPA Bias"

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Log PIPA bias data on Flight Plan timeline if PIPA bias tests are performed onboard
2. Ground Support
  - a. CSM TM (M)



A. TEST OBJECTIVES

- P7.32-3 S-Band High Gain Antenna Dynamic Response During SPS burns
- P20.114-1 Procedures and Timeline Adequacy for Midcourse Corrections
- P20.114-2 GNCS, SPS & RCS Performance on Midcourse Corrections
- P20.114-3 Ground Capability to Update State Vector and Target for MCC

B. TEST REQUIREMENTS

1. Ground update of the CSM state vector and of the MCC target [20.114]
2. MCC maneuver performance [20.114]
3. FQTR operation during SPS ignition, cutoff and for at least five seconds during the burn [7.32]

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.10.1.5 "CSM-CMC Update P27"
2. AOH paragraph 4.14.2 "G&N/SPS Orbit Change Thrusting (P40)" or
3. AOH paragraph 4.14.3 "G&N SM RCS Orbit Change Thrusting (P41)"
4. FCAC G-21 "P27 CMC Update
5. FCAC G-30 "P40 SPS Thrust" or
6. FCAC G-37 "P41 RCS Thrust"

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Procedural and/or timeline difficulties or inadequacies [20.114] (M)
2. Ground Support
  - a. CSM TM HBR [20.114] (M)
  - b. BET [20.114] (M)
  - c. Flight Director reports of procedural and/or timeline difficulties or inadequacies [20.114] (M)

## STAR/EARTH LANDMARK NAVIGATION

### A. TEST OBJECTIVES

- Sl.32-1 Star/Earth Landmark Navigation Sighting Accuracy
- Sl.32-2 Verify and Update Landmark Sighting Constraints
- Sl.32-3 OSS Adequacy for Navigation Sightings in Deep Space
- Sl.32-4 Crew Capability to Identify Landmarks and Coordinate Optics/Maneuvers
- Sl.32-5 RCS Propellant Use and Time to Accomplish Onboard Navigation
- S20.115-2 Photographs of Earth and Lunar Landmarks

### B. TEST REQUIREMENTS

1. Use of the trunion calibration option of P23 preceding these sets of sightings
2. Star/earth landmark sightings under the following conditions [1.32]:
  - a. Three sets after TLI while the spacecraft is less than 50,000 NM from the surface of the earth (TL or TE). These sightings may be made on the same landmark but three different stars will be used
  - b. One set at a sun elevation angle of less than ten degrees. One of the above sightings can be used to satisfy this requirement

Note - Performing sightings on one landmark and one star, then making from three to five "marks" constitutes one set of sightings
3. Approximately ten sequence photographs through the special sextant camera adapter immediately after performing each group of star/earth landmark sightings. This photography must not jeopardize any nav sightings [20.115] - Not Implemented
4. Identification of the landmark closest to sunrise or sunset that is considered acceptable for landmark tracking [1.32]

### C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.12.3 "Cislunar Midcourse Navigation (P23)" including "trunion calibration" option
2. FCAC G-20A "P23 Cislunar Midcourse Nav Measurement"
3. Photo checklist TBD

### D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Difficulties and/or inadequacies noted in handling the optics and the spacecraft [1.32] (M)

b. Log navigation data on Flight Plan timeline [1.32] (M)

c. Photo log TBD [20.115] (HD)

2. Ground Support

a. CSM TM HBR [1.32] (M)

b. BET [1.32] (M)

## PASSIVE THERMAL CONTROL MODES

### A. TEST OBJECTIVES

- PL09-1 PTC Procedures & RCS Use - Roll With Att. Hold
- PL09-2 PTC Procedures & RCS Use - Roll W/O Att. Hold
- PL09-3 Communications Procedures Adequacy for PTC Mode
- PL09-4 PTC Procedures & RCS Use - Yaw W/O Att. Hold

### B. TEST REQUIREMENTS

1. At least six uninterrupted hours of PTC in the translunar phase with the following conditions:
  - a. CSM (X) axis oriented perpendicular to the sun-spacecraft line and with the (-X) axis approximately perpendicular to the CSM-earth line of sight
  - b. Spacecraft attitude control in maximum deadband in pitch and yaw and roll free and rolling from 0.1 to 0.5 deg/sec.
2. At least six uninterrupted hours of PTC in the transearth phase with the following conditions:
  - a. CSM (X) axis oriented perpendicular to the sun-spacecraft line and with the (-X) axis approximately perpendicular to the CSM-earth line of sight
  - b. Spacecraft attitude control free in all axes with a roll rate of from 0.1 to 0.5 deg/sec
  - c. Deviation (coning) of (-X) axis from original orientation to be maintained to less than 15 degrees
3. If the above two modes are found to be inadequate during real time, at least six uninterrupted hours of PTC in the transearth phase with the following conditions:
  - a. CSM (+X) axis toward the sun with the high gain antenna boom on the earth side in the sun-earth-moon plane.
  - b. Spacecraft attitude control free with rotation about the major inertia axis at a rate of between 0.1 and 0.5 deg/sec
  - c. Deviations of rotational axis from original orientation to be maintained less than 7.5 degrees
4. Normal communications procedures evaluation during PTC

### C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph TBD - meanwhile use "Special PTC Procedure"
2. "Passive Thermal Control Mode 1" (X Axis Roll, Pitch and Yaw Hold) or

3. "Passive Thermal Control Mode 2" (X Axis Roll W/O Pitch and Yaw Hold) or
4. "Passive Thermal Control Mode 3" (Yaw - Z Axis with Pitch and Roll Free)

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs

- a. Procedural difficulties in establishing, maintaining or terminating PTC (M)
- b. Narration of procedure and results when establishing or correcting the PTC Mode (M)
- c. Difficulties in establishing and maintaining S-band communications during the PTC Mode (M)
- d. Log PTC maneuver correction times & difficulties on the Flight Plan timeline where they occur (M)

2. Ground Support

- a. CSM TM HBR (M)
- b. BET (M)
- c. MSFN records of received S-band signal strength (M)
- d. Flight Director reports of procedural and/or timeline difficulties or inadequacies (M)

## STAR/LUNAR HORIZON NAVIGATION

### A. TEST OBJECTIVES

- Pl.33-1 Star/Lunar Horizon Navigation Sighting Accuracy
- Pl.33-2 OSS Adequacy for Navigation Sightings in Deep Space
- Pl.33-3 Crew Capability to Coordinate Optics/Maneuvers
- Pl.33-4 RCS Propellant Use and Time to Accomplish Onboard Navigation
- S20.115-3 Photographs of Earth and Lunar Horizons

### B. TEST REQUIREMENTS

1. Use of the trunnion calibration option of P23 preceding a set of sightings if more than about 30 minutes has elapsed since the previous P23 trunnion calibration [1.33]
2. Translunar - Near Earth
  - a. Five sets of star/lunar horizon sightings at more than 15,000 NM and less than 50,000 NM from the center of the earth [1.33]
  - b. Update of the onboard state vector utilizing the navigation sighting data following each set of sightings [1.33]
3. Transearth - Near Moon
  - a. Fourteen sets of star/lunar horizon sightings at less than 50,000 NM from the center of the moon [1.33]
  - b. Update of the onboard state vector utilizing the navigation sighting data following each set of sightings [1.33]
  - c. At least 10 sequence photographs through the special sextant camera adapter while duplicating star/horizon navigation sightings when between 10,000 and 20,000 NM from the moon on the TE phase. It is desirable that these photographs be taken at varying distances [20.115] - Not Implemented
4. Transearth - Midcourse
  - a. Nineteen sets of star/lunar horizon sightings at more than 50,000 NM from the center of the moon [1.33]
  - b. Update of the onboard state vector utilizing the navigation sighting data following each set of sightings [1.33]

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.12.3 "Cislunar Midcourse Navigation (P23)" including "trunnion calibration" option
2. FCAC G-20A "P23 Cislunar Midcourse Navigation Measurement"
3. Photo checklist TBD

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Difficulties and/or inadequacies noted in handling the optics and the spacecraft [1.33] (M)
  - b. Log navigation data on Flight Plan timeline [1.33] (M)
  - c. Photo log TBD [20.115] (HD)
2. Ground Support
  - a. CSM TM HBR [1.33] (M)
  - b. BET [1.33] (M)

## STAR VISIBILITY

### A. TEST OBJECTIVES

- Sl.35-2 Obtain Data - Star Visibility During TL and TE Coast
- Sl.35-3 Degradation of Navigation or IMU Alignment by Vented or Ejected Materials

### B. TEST REQUIREMENTS

1. Star visibility tests by observing the star field surrounding the navigation star targets of opportunity through the SCT and identifying the dimmest star in the star pattern under at least the following conditions:
  - a. Sun - SCT LOS approximately 120°
  - b. Sun - SCT LOS approximately 70°
2. Light adaptation period required prior to performing visibility tests
3. Evaluation of the effect of vented particles on the use of the optics during IMU realignments and navigation sightings

### C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.11.2 "IMU Realign (P52)" or
2. AOH paragraph 4.12.3 "Cislunar Midcourse Navigation (P23)"
3. FCAC G-39 "P52 IMU Realign" or
4. FCAC G-20A "P23 Cislunar Midcourse Navigation Measurement"

### D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs  
Identify and record dimmest star during each visibility test
2. Ground Support  
BET during star visibility tests (M)



A. TEST OBJECTIVES

- S1.30-3 Overall GNCS Errors During Thrusting Maneuvers
- S3.21-1 SPS Isp - Adequacy of Conversion of Gnd. to Vacuum Results
- S3.21-2 SPS Performance for LOI and TEI Burns
- S3.21-4 Thermal Effects - Long SPS Burn - Heat Protection System
- P7.32-3 S-Band High Gain Antenna Dynamic Response During SPS Burns
- P20.105-1 Crew/Spacecraft/MSFN Preparation and Execution of LOI Maneuver
- P20.105-2 Procedures and Timeline Adequacy for LOI Maneuver

B. TEST REQUIREMENTS

1. SPS operation to insert the CSM into lunar orbit in two stages as follows [20.105]:
  - a. The first burn of approximately 245 seconds in duration will insert the CSM into a 60 x 170 NM orbit
  - b. The second burn of approximately 10 seconds in duration will circularize the orbit to 60 NM
2. CSM fully loaded at liftoff [20.105]
3. FQTR ON for SPS thrust buildup, 5 sec at steady state and for thrust tailoff, during first LOI-1 only [7.32]
4. DSE recording immediately before, during, and after both burns [3.21]
5. DSE recording or TM for at least five seconds duration at approximately 15 minute intervals for about three hours after LOI-1 [3.21]
6. After LOI-1 it is HD to orient the CSM to an attitude that will result in a near constant external thermal environment to the SPS aft bulkhead during the subsequent 45 minute period. The aft bulkhead should be shielded from solar radiation during this period by orienting the (+X) axis toward the sun [3.21] N.I.
7. IMU realignment performed as soon prior to LOI-1 as practicable [20.105]

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.14.2 "G&N SPS Orbit Change Thrusting (P40)"
2. FCAC G-32 "SPS Thrusting (P40)"

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Procedural and/or timeline difficulties or inadequacies [20.105] (M)

- b. Feasibility of monitoring abort parameters on FDAI's [20.105] (M)
  - c. Adequacy of contact with ground operational support [20.105] (HD)
  - d. If the PU valve position is changed, record GET  $\pm$  1 sec and new valve position [3.21] (M)
2. Ground Support
- a. CSM TM LBR [20.105] (M)
  - b. BET [3.21] (HD), [20.105] (M)
  - c. Flight Director reports of procedural and/or timeline difficulties or inadequacies [20.105] (M)

## LUNAR LANDMARK TRACKING

### A. TEST OBJECTIVES

- P20.111-1 Error Uncertainties in Lunar Landing Site Location
- P20.111-2 Obtain Data to Calibrate MSFN at Lunar Distance
- P20.111-3 Determine Minimum Sun Angle to Clearly Identify Lunar Landmarks
- P20.111-4 Lunar Landmark Tracking in Earthshine
- P20.111-5 Adequacy of CSS and OSS for Obtaining Landmark Sightings
- P20.111-6 Crew Ability to Coordinate Landmark Sightings and Vehicle Maneuvers
- P20.111-7 RCS Propellant and Time Required for Landmark Sighting

### B. TEST REQUIREMENTS

1. At least 12 sets of four marks each obtained on four selected landmarks according to the following schedule:
  - a. Track a pre-selected pseudo landing site (same terrain and lighting as the lunar landing site) for four consecutive revolutions. The sun elevation angle to be five degrees on the first tracking orbit
  - b. Track three pre-selected landmarks (or others deemed acceptable by the crew). The same three landmarks to be tracked in a four orbit tracking sequence. It is highly desirable that the sun angle be greater than 60 degrees.
  - c. An attempt made to track a landmark at a sun elevation angle of three degrees
2. On one revolution, CMP verbal description of the terrain near the sunrise terminator as observed through the scanning telescope. The observation to be made from about 10° preceding the terminator to about 15° past the terminator in the earthlit portion of the dark side. The optics angle to be fixed and the spacecraft in an orbital rate mode.
3. If the observations made of the earthlit landmarks reveals that it is feasible, it is highly desirable that the crew track two landmarks (four marks each) on two successive earthlit passes.
4. IMU realigned during the dark pass preceding each tracking sequence
5. DSE Tape Recorder ON for 20 seconds following each set of marks while N49 ( $\Delta R$ ,  $\Delta V$ ) displayed
6. ORDEAL functioning

7. Photographic records of each of the four tracked landmarks using the special sextant camera adapter and D. A. camera at 12 Fps and taking the photographs at the same time as making the MARK during sighting. Take one photograph of each landmark on two consecutive passes (i.e., eight pictures)

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.12.2, "Orbital Navigation (P22)"
2. FCAC G-18 "P22 Orbital Navigation"
3. FCAC "Lunar Landmark Tracking" TBD
4. Reference Photographic Operation's Plan, Apollo 8

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Verbal description of lunar terrain during earthshine landmark lighting evaluation (M)
  - b. Comments on the effect of earthshine while tracking lunar landmarks (M)
  - c. Procedural and/or timeline difficulties or inadequacies (M)
  - d. "Landmark Tracking Log" (Flight Plan)
  - e. Photo log
    1. Film mag IP
    2. GET camera - ON
    3. Initial shutter speed
    4. Final shutter speed
    5. Estimated remaining film at beginning and end of a sequence

## PHOTOGRAPHY

### A. TEST OBJECTIVES

- 1.35-3 Photographs of Vented Particles Through Windows
- 20.115-1 Overlapping Photographs from Terminator to Terminator -  
Lunar Orbit
- 20.115-3 Photographs of Earth and Lunar Landmarks
- 20.115-4 Photographs of General and Scientific Interest

### B. TEST REQUIREMENTS

1. Terminator-to-terminator photographs on at least two passes as follows:
  - a. The first pass consisting of vertical photos with 55-60% forward overlap taken at 20 sec intervals
  - b. The subsequent pass consisting of photos taken with the camera axis  $20^\circ$  from local vertical. From the far side terminator to  $90^\circ$  sun elevation at the nadir, the camera axis will be inclined  $20^\circ$  foreward along the orbital plane. At  $90^\circ$ , the camera axis will be aligned  $20^\circ$  aft of the nadir in the orbital plane
2. Sequence camera photographs if possible of the following targets of opportunity:
  - a. Vertical zero phase (250 mm lens)
  - b. High angle zero phase: Apollo landing sites and geologic units (250 mm lens)
  - c. Vertical terminator (250 mm lens)
  - d. Oblique terminator (250 mm lens)
  - e. Specific feature near vertical and oblique (250 mm)
  - f. Specific area near vertical (80 mm)
  - g. Lunar exploration site:
    - (1) Near vertical or oblique as required (250 mm lens)
    - (2) Approach (250 mm lens)
  - h. Surveyor landing sites (250 mm lens)
  - i. Other features and phenomena as observed (250 mm lens)
  - j. Image motion compensation (250 mm lens)

3. Dim light and astronomical photos attempted either during TL or TE coast or during dark portion of lunar orbits
  - a. Total darkness preferred -- S/C lighting also as low as possible
  - b. Specific star fields should be taken during TL and TE coast and in lunar orbit during earthshine
4. Photos taken of the lunar surface during earthshine
5. Solar corona photos taken at sunrise or sunset in lunar orbit using the moon as an occultating disc
6. Earth photos during TL coast when the S/C is within 40,000 NM of the earth. The S/C shall be oriented with the (+X) axis toward earth for approximately 10 minutes for these photos
7. Sequence camera photographs of the lunar terrain as viewed through the rendezvous window while duplicating the Post LOI-2 approach attitude of missions F and G.
8. Photographs of vented particles through windows when observed

C. TEST PROCEDURES/CHECKLISTS

1. Photographic Operation Plan for Apollo Mission C'

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Photos of lunar surface per Test Requirement No. 1 above (M)
  - b. Photos of scientific and general interest subjects as defined by photo plan (HD)
2. Ground Support  
BET (M)

A. TEST OBJECTIVES

- S3.21-1 SPS Isp - Adequacy of Conversion of Ground to Vacuum Results
- S3.21-2 SPS Performance for LOI and TEI Burns
- S3.21-3 SPS PUGS in Auxiliary Mode - Relative Accuracy of Aux and Pri
- S3.21-4 Thermal Effects - Long SPS Burn - Heat Protection System
- P20.106-1 Crew/Spacecraft/MSFN Preparation and Execution of TEI Maneuver
- P20.106-2 Procedures and Timeline Adequacy for TEI Maneuver

B. TEST REQUIREMENTS

1. SPS use for injection of the CSM into transearth coast [3.21]
2. DSE recording of TM for the period 40 seconds prior to, during, and two minutes after the TEI burn [3.21, 20.116]
3. DSE recording or TM of at least five seconds duration at approximately 15 minute intervals for about three hours after TEI [3.21]
4. PUGS mode switch in the AUX position during the SPS burn [3.21]
5. After the burn it is HD to orient the CSM to an attitude that will result in a near constant external thermal environment to the SPS during the subsequent 45 minute period. The aft bulkhead should be shielded from solar radiation during this period by orienting the (+X) axis toward the sun [3.21] - N.I.
6. IMU realignment performed as soon before TEI as practicable [20.106]
7. CMC update provided on a timeline similar to the LLM such that errors at  $T_{ig}$  can be determined and compared with predicted estimates [20.106]

C. TEST PROCEDURES/CHECKLISTS

1. AOH paragraph 4.14-2 "G&N SPS Orbit Change Thrusting (P40)"
2. FCAC G-32 "SPS Thrusting (P40)"

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs
  - a. Adequacy of contact with ground operational support [20.106] (HD)
  - b. If the PU valve position is changed, record GET  $\pm 1$  sec and new valve position [3.21] (M)

2. Ground Support

1. CSM TM LBR [3.21, 20.106] (M)
2. BET [3.21] (HD), [20.106] (M)
3. Flight Director reports of procedural and/or timeline difficulties or inadequacies [20.106] (M)
4. Flight Operations Plan [20.106] (M)



A. TEST OBJECTIVES

- 1.31-1 GNCS Performance During Entry - Lunar Return
- 1.31-2 EMS Capability to Monitor Entry - Lunar Return
- 4.5-1 ECS Performance During Manned Lunar Return Entry
- 4.5-2 Compare ECS Data With Development Model & Unmanned Results
- 7.30-1 Block II Thermal Protection System - Manned Lunar Return Entry
- 7.30-2 TPS Surface Recession and Char Data Comparison With Models
- 20.116-2 Changes in Total Transmittance of Windows

B. TEST REQUIREMENTS

- 1. Entry velocity at 400,000 ft. altitude at least 36,000 feet per second [7.30]
- 2. Automatic lunar return entry - nominal range 1350 NM and 2500 NM maximum for inclement weather [1.31]
- 3. DSE ON during entry "blackout" - recorder fully rewound and on HBR starting at CSM/SM separation [1.31]
- 4. Crew monitor the EMS and record voice on the DSE during entry [1.31]

C. TEST PROCEDURES/CHECKLISTS

- 1. AOH paragraph 4.17.7, "G and N Entry"
- 2. FCAC "Entry"

D. DATA REQUIREMENTS

- 1. Flight Crew Reports/Logs
  - a. Comments on adequacy of DSKY, FDAI and EMS displays to evaluate entry conditions [1.31] (HD)
  - b. Comments on crew comfort [4.5] (HD)
- 2. Ground Support
  - a. CSM TM HBR [1.31, 4.5] (M)
  - b. BET [1.31, 7.30] (M)
  - c. EMS Scroll Recovery [1.31] (M)
  - d. CM Recovery [7.30] (M)
  - e. Heat Shield core sample after recovery [7.30] (M)
  - f. All S/C window outer panes after recovery [20.116] (M)

## MISCELLANEOUS TESTS

### A. TEST OBJECTIVES

- P7.31-1 Thermal Control System During PTC
- P7.31-2 Thermal Control System During Lunar Orbit
- P20.107-1 Crew Procedures for Lunar Orbit Mission
- P20.107-2 Crew Procedures Timeline Adequacy - Lunar Mission
- S20.108-1 EPS Use, Performance and Fuel Cell Management
- S20.108-2 Water and Oxygen Requirements Data
- S20.108-3 LiOH Cartridge Requirements Data
- S20.108-4 SPS, SM/RCS & CM/RCS Propellant Requirements Data -  
Translation and Attitude Maneuvers
- S20.108-5 Crew Food Requirements Data
- 20.110-1 Ground Operational Support for Lunar Mission Without LM

### B. TEST REQUIREMENTS

1. Operation of the CSM in lunar orbit, undocked to determine the thermal properties of the spacecraft insulation [7.31]
2. Determination of the effectiveness of the thermal control system to provide a safe environment for operation of spacecraft equipment and consumables [7.31]
3. Accomplishment of all CM procedures peculiar to the lunar orbit mission in the same manner, where possible, as required for the LLM [20.107]
4. Management of the EPS during the lunar mission per the flight plan [20.108]
5. Utilization of the potable water produced during EPS operation as necessary [20.108]
6. Utilization of the oxygen supply system during the lunar mission as required [20.108]
7. Management of the replacement of LiOH cartridges during the mission per the flight plan [20.108]
8. Management of the consumption of the SPS and RCS propellants during the lunar mission as required [20.108]
9. Utilization of the food supply per the flight plan [20.108]
10. Performance of operational ground support of the spacecraft throughout the mission with emphasis on those phases being accomplished for the first time as part of a lunar orbit mission [20.110]

C. TEST PROCEDURES/CHECKLISTS

1. AOH CSM 103 SM2A-03-SC103-2
2. FCAC CSM 103

D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs

- a. Narrative describing any recommended changes in procedures and/or equipment required to accomplish the procedures [20.107] (M)
- b. Comments on any procedure requiring repeating or any procedure not completed [20.107] (M)
- c. Water Count Log [20.108] (M)
- d. LiOH Log [20.108] (M)
- e. Astronaut Menus [20.108] (HD)
- f. Comments on ECS environment comfort level [20.108] (HD)

2. Ground Support

- a. CSM TM HBR [7.31, 20.108] (M)
- b. BET [7.31, 20.108] (M)
- c. MSFN voice recording of CSM/MSFN communications [20.107] (HD)
- d. Chemical analysis of recovered LiOH cartridges [20.108] (M)
- e. Measurement of recovered water quantity [20.108] (M)
- f. Measurement of energy remaining in recovered Entry and Post-Landing Batteries [20.108] (M)
- g. Flight Director reports of mission support activities [20.110] (M)

A. TEST OBJECTIVES

- S6.10-1 S-Band Performance with Omni Antennas at Lunar Distance (85' MSFN)
- P6.11-1 S-Band Performance with High Gain Antenna During TL Coast
- P6.11-2 S-Band Performance with High Gain Antenna at Lunar Distance
- S20.104-3 S-Band Communications Performance During Transposition
- S20.109-3 Communications Procedures During PTC

B. TEST REQUIREMENTS

1. Continuous CSM/MSFN S-band communication during transposition using CSM omni and 85' ground antennas [20.104]

| <u>Signal<br/>Combination</u> | <u>Communications Mode</u>                     |                 |
|-------------------------------|------------------------------------------------|-----------------|
|                               | <u>Uplink</u>                                  | <u>Downlink</u> |
| 4.2                           | Carrier, PRN, Voice - Carrier, PRN, Voice, HBR |                 |

2. MSFN performance of various designated S-band communication mode tests of at least two minutes each as soon as practical after high gain antenna deployment (near earth) using the 85' ground antenna and the CSM high gain antenna as follows: [6.11]

| <u>Signal<br/>Combination</u> | <u>Communications Mode</u>                                |                     |
|-------------------------------|-----------------------------------------------------------|---------------------|
|                               | <u>Uplink</u>                                             | <u>Downlink</u>     |
| 4.2                           | Carrier, PRN; Voice - Carrier, PRN, Voice, HBR            |                     |
| 6.2                           | Carrier, PRN, Voice, - Carrier, PRN, Voice, HBR<br>Updata |                     |
| .2                            |                                                           | Recorded voice, LBR |
| 4.3                           | Carrier, PRN, Voice - Carrier, PRN, Voice, LBR            |                     |
| 6.3                           | Carrier, PRN, Voice, - Carrier, PRN, Voice, LBR<br>Updata |                     |
| 5.2                           | Carrier, PRN, Updata - Carrier, PRN, Voice, HBR           |                     |
| 8.1                           | Carrier, BU Voice - Carrier, Voice, LBR                   |                     |
| .4                            |                                                           | CSM television      |

3. MSFN performance of various designated S-band communication mode tests at lunar distance using 85' ground antenna or equivalent lunar distance using 30' ground antenna\* and the CSM omni antennas [6.10]

\*NOTE: The test can be accomplished under any one of the three following conditions which represent lunar distance or equivalent:

- a. Greater than 200 K NM with 85-foot dish (preferred)
- b. At 100 K  $\pm$  10% NM with cooled 30-foot dish (HAW,CWM,ASC,CRO)
- c. At 70 K  $\pm$  10% NM with uncooled 30-foot dish (others)

| <u>Signal<br/>Combination</u> | <u>Communications Mode</u> |                         |
|-------------------------------|----------------------------|-------------------------|
|                               | <u>Uplink</u>              | <u>Downlink</u>         |
| .5                            |                            | Carrier, LBR            |
| .6                            |                            | Carrier, Key Subcarrier |
| 1.7                           | Carrier, PRN               | - Carrier, PRN          |
| .8                            |                            | Carrier, BU Voice, LBR  |
| .10                           |                            | Carrier, BU Voice       |

4. MSFN performance of various designated S-band communication mode tests at lunar distance using 85' ground antenna or equivalent lunar distance using 30' ground antenna\* and CSM Hi gain antenna [6.11]

| <u>Signal<br/>Combination</u> | <u>Communications Mode</u> |                            |
|-------------------------------|----------------------------|----------------------------|
|                               | <u>Uplink</u>              | <u>Downlink</u>            |
| 4.2                           | Carrier, PRN, Voice        | - Carrier, PRN, Voice, HBR |
| 5.2                           | Carrier, PRN, Updata       | - Carrier, PRN, Voice, HBR |
| 4.3                           | Carrier, PRN, Voice        | - Carrier, PRN, Voice, LBR |
| 8.1                           | Carrier, BU Voice          | - Carrier, Voice, LBR      |

5. During steerable S-band antenna tests, demonstration of manual acquisition and automatic tracking using wide, medium and narrow beamwidths

#### C. TEST PROCEDURES/CHECKLISTS

1. AOH Section 4.7.6, "Telecommunications", paragraphs 4.7.6.1 through 4.7.6.10
2. FCAC communications checklist TBD

#### D. DATA REQUIREMENTS

1. Flight Crew Reports/Logs  
Comments on voice quality in flight plan [6.10, 6.11 & 20.104] (M)
2. Ground Support
  - a. CSM TM HBR [6.10, 6.11] (M)
  - b. CSM MSFN Voice records [6.10, 6.11, 20.104]

\*NOTE: The test can be accomplished under any one of the three following conditions which represent lunar distance or equivalent:

- a. Greater than 200 K NM with 85-foot dish (preferred)
- b. At 100 K  $\pm$  10% NM with cooled 30-foot dish (HAW, CWM, ASC, CRO)
- c. At 70 K  $\pm$  10% NM with uncooled 30-foot dish (others)

- c. Flight Director reports of procedural and/or timeline difficulties or inadequacies in implementing all required communication tests [6.10, 6.11, 20.104] (HD)
- d. USB TM bit stream [6.10, 6.11, 20.104] (M)
- e. USB tracking data processor output [6.10, 6.11] (HD), [20.104] (M)
- f. MSFN records of received S-band carrier strength [6.10, 6.11, 20.104] (M)
- g. MSFN tape recordings of CSM to MSFN emergency key mode [6.10] (M)

## APOLLO 8 CREW STATUS REPORTS

### Purpose:

1. To enhance crew safety.
2. To acquire data on food, water, sleep and exercise requirements for future mission planning.
3. To obtain medical data required for analysis and correlation with the inflight bioenvironmental data and the pre- and postflight physical examination results.

### Requirements:

To obtain food, water, sleep and radiation data during the Apollo 8 mission and exercise data postflight.

### Procedures:

During the Apollo 8 mission, two crew status reports via air-to-ground communications will be made by the flight crew during each 24-hours ground elapsed time (GET). The first report will be given after the first meal of the work day and will concern the sleep obtained during the previous sleep period. The second report will be given following the final meal of the work day and will concern the food and water consumed, and the radiation dose received during the previous 24 hours GET. The following information should be transmitted:

1. Food - a daily record of each crew member reflecting the meal identification code together with the description or serial number(s) of only those items not consumed from a scheduled meal.
2. Water - a daily log containing the number of aliquots from the water gun each astronaut drank. Water used for food rehydration will not be recorded.
3. Sleep - a daily notation of each crewman's best estimate as to sleep quantity and quality.
4. Radiation - a daily record of the integrated dose each crewman receives.
5. Exercise (postflight analysis only) - a daily record of the time of exercise, the type, and the duration of exercise performed by each astronaut.



Q



W

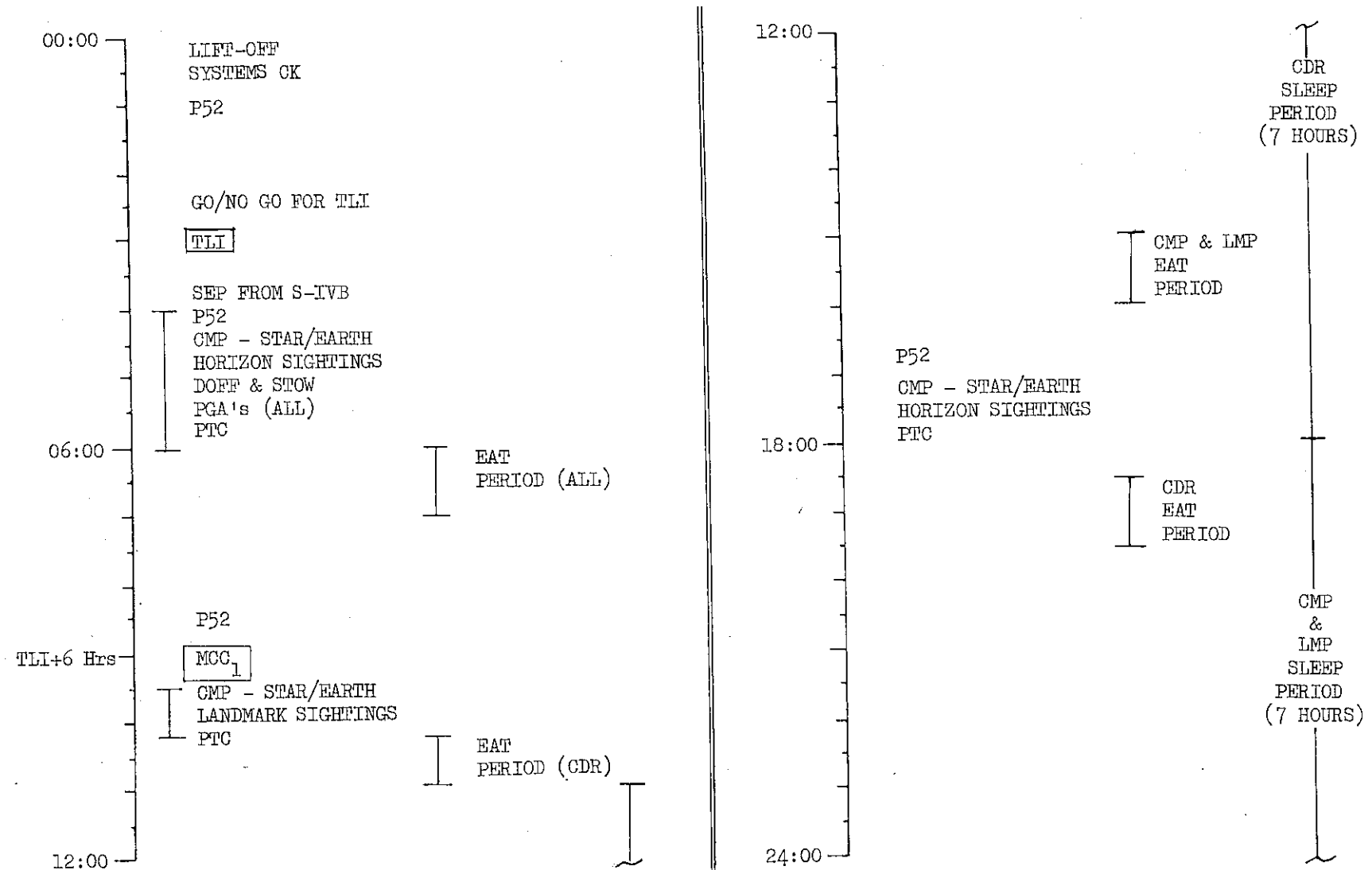




SECTION V - SUMMARY FLIGHT PLAN

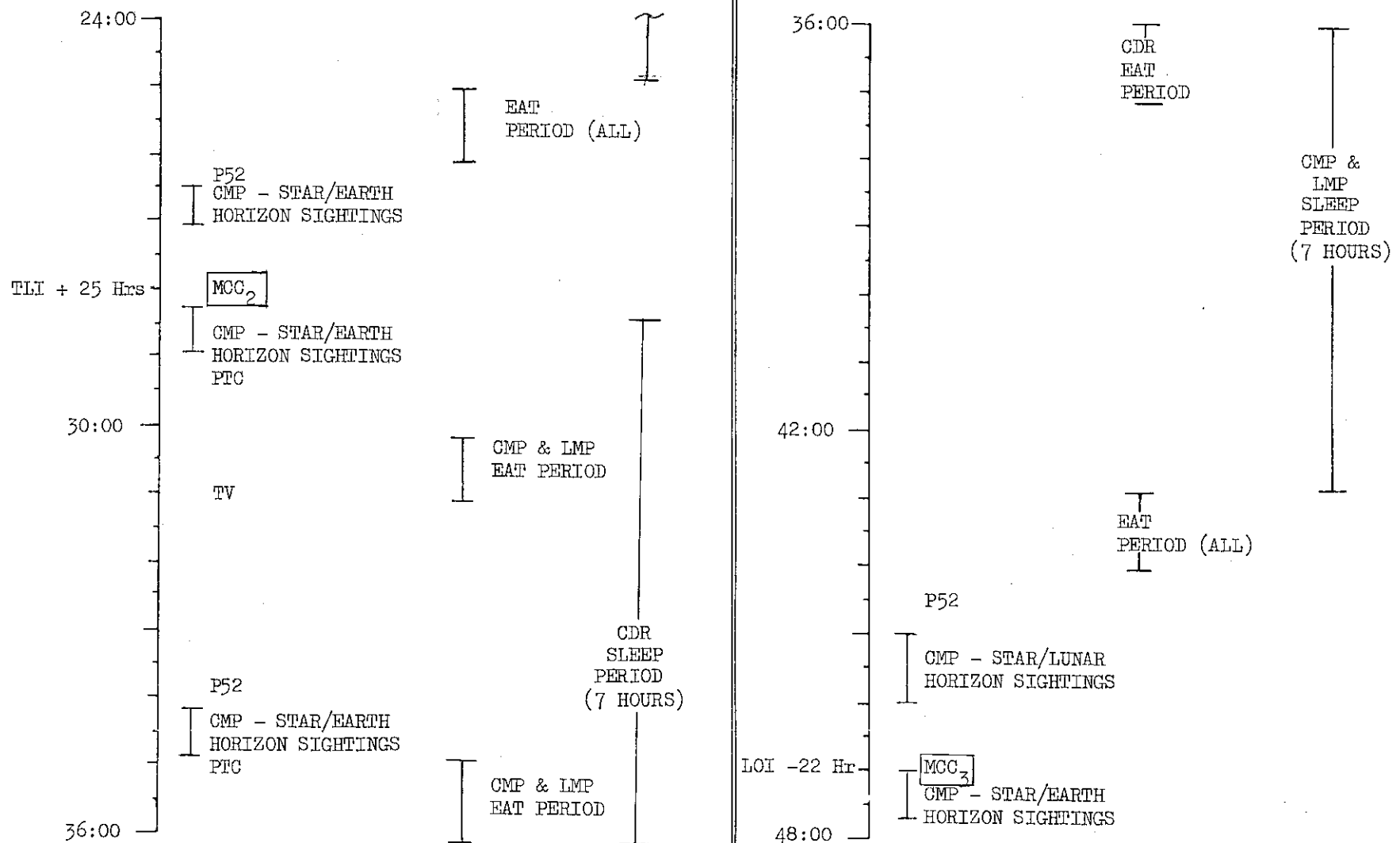


# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME         | DAY/REV | PAGE |
|-----------|---------|-------------------|--------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 00:00 -24:00 | 1       | 5-1  |

# FLIGHT PLAN

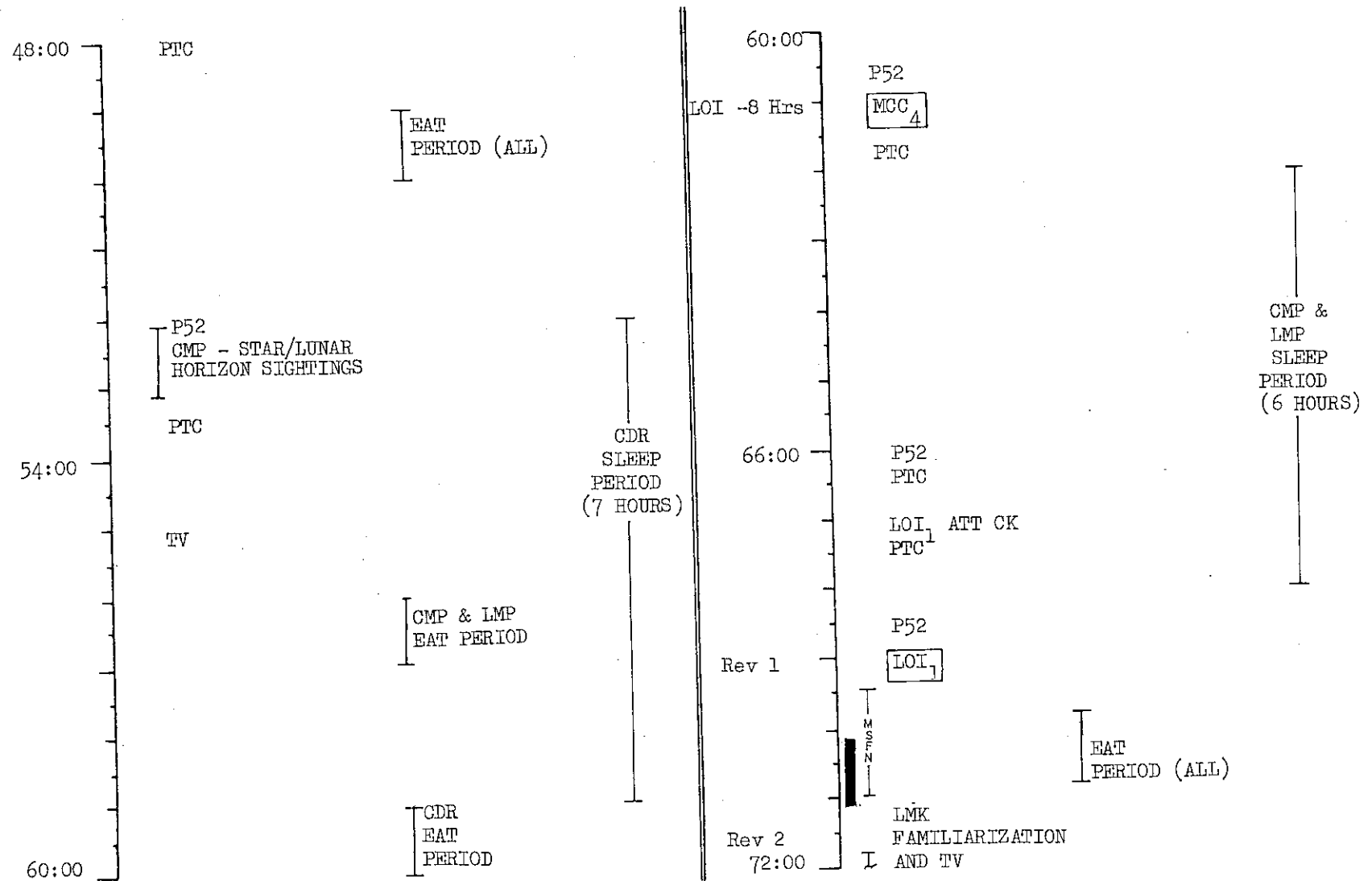


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 24:00 - 48:00 | 2       | 5-2  |

MSC FORM 1186 (SEP 67)

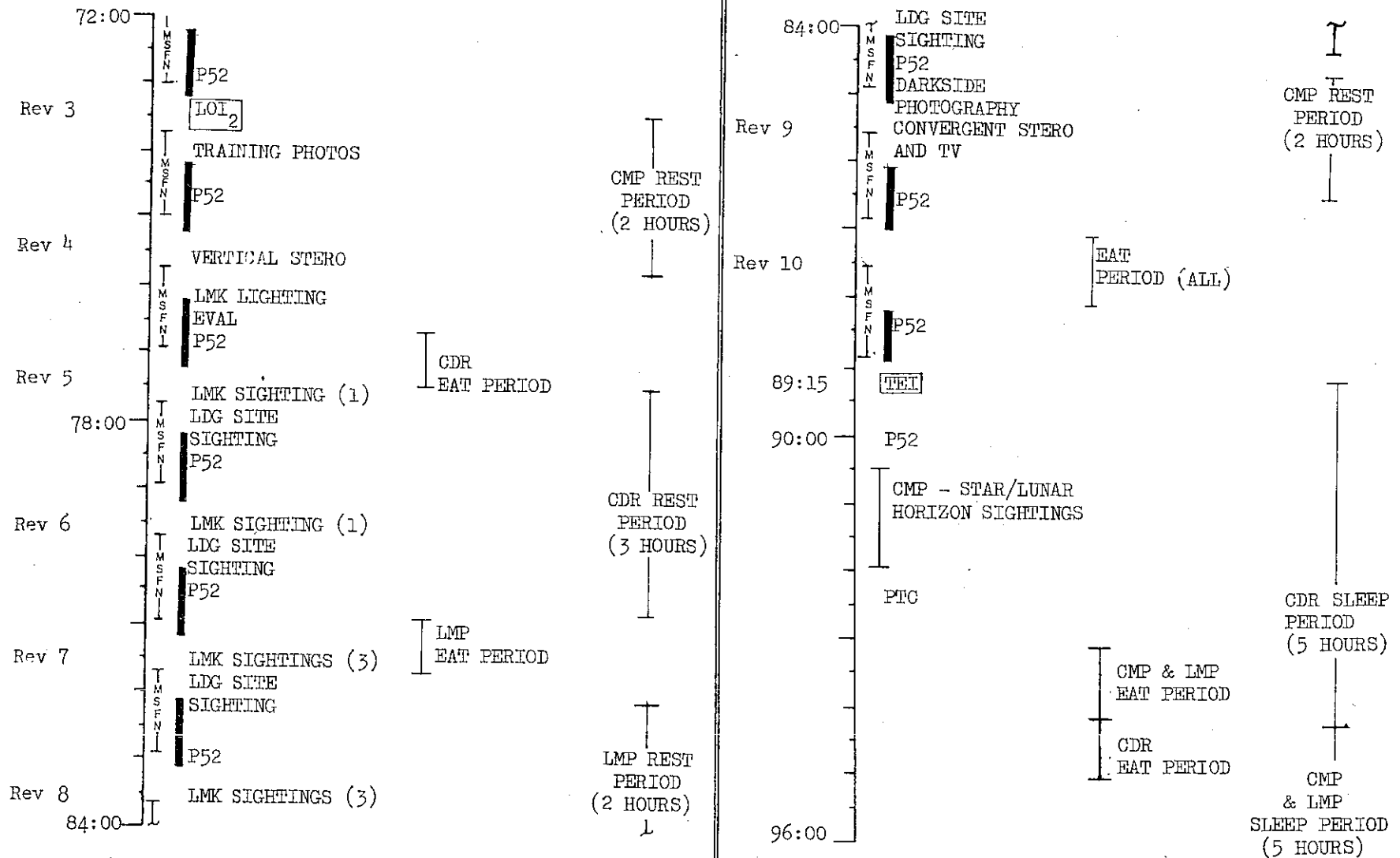
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 48:00 - 72:00 | 3       | 5-3  |

# FLIGHT PLAN

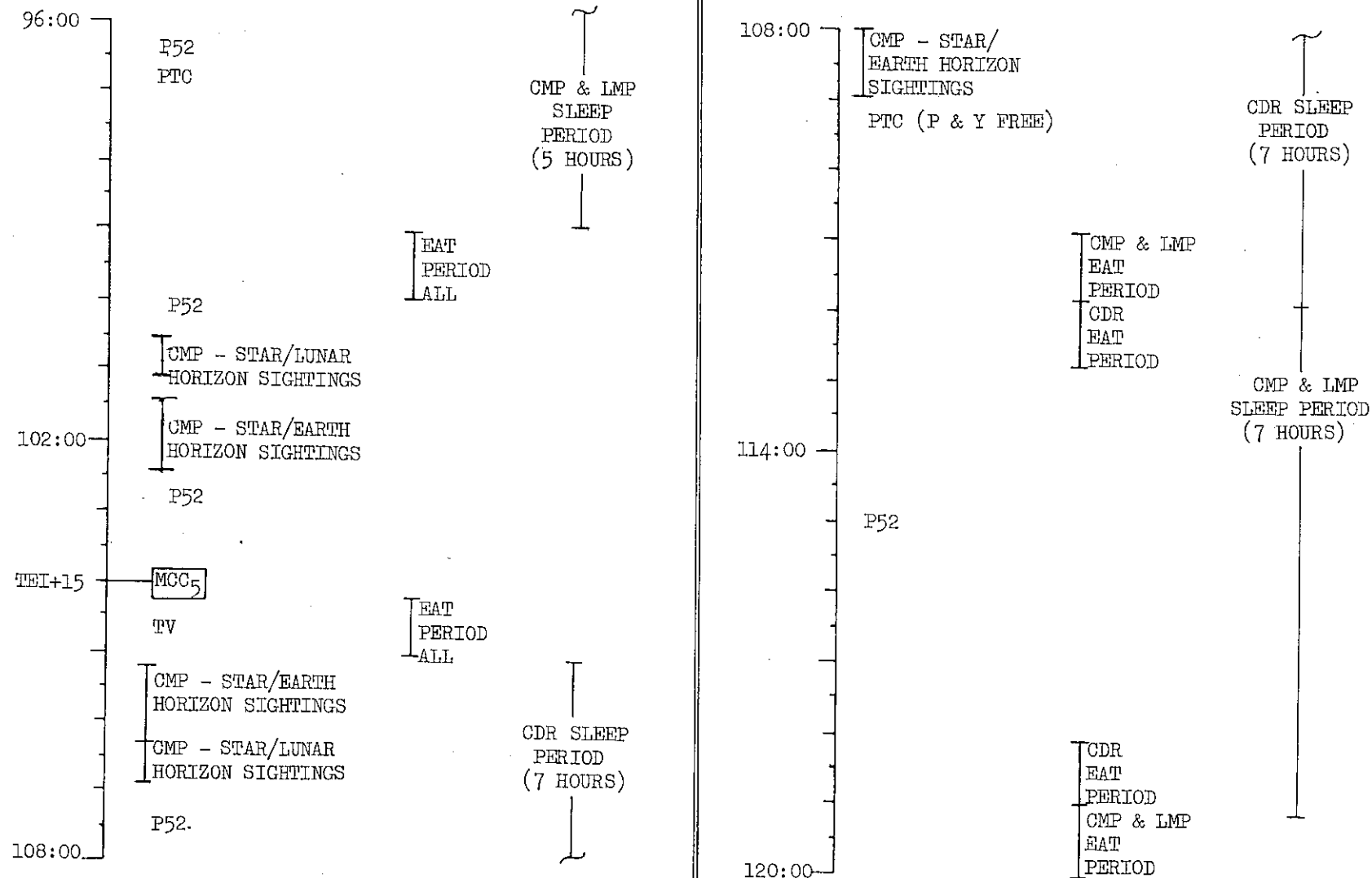


| MISSION   | EDITION | DATE              | TIME          | DAY/REV | PAGE |
|-----------|---------|-------------------|---------------|---------|------|
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MSC FORM 1186 (SEP 67)

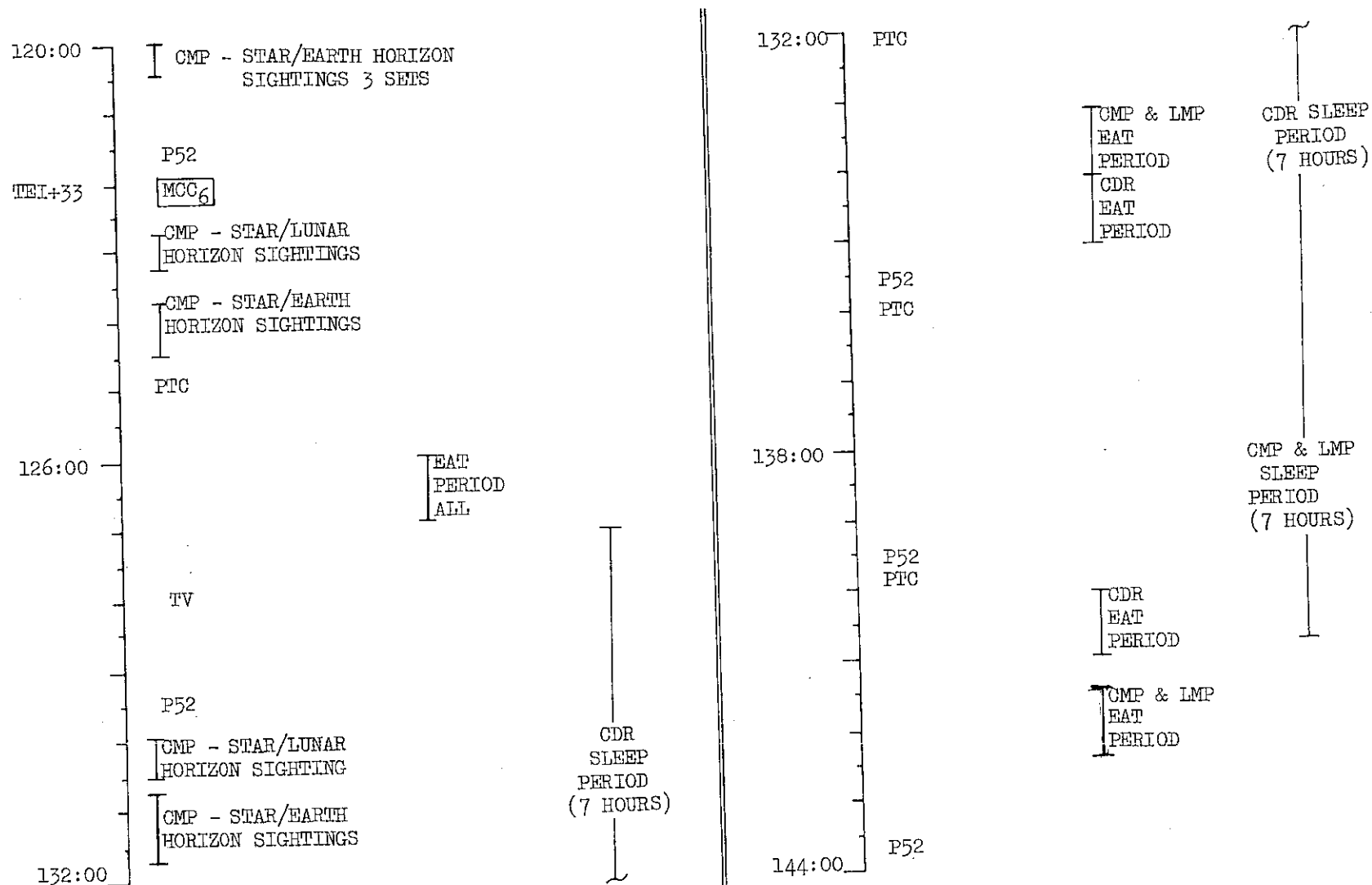
FLIGHT PLANNING BRANCH

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME           | DAY/REV | PAGE |
|-----------|---------|-------------------|----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 96:00 - 120:00 | 5       | 5-5  |

# FLIGHT PLAN



| MISSION   | EDITION | DATE              | TIME            | DAY/REV | PAGE |
|-----------|---------|-------------------|-----------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 120:00 - 144:00 | 6       | 5-6  |

MSC FORM 1186 (SEP 67)

FLIGHT PLANNING BRANCH



# FLIGHT PLAN

144:00

ET -2 HRS

MCC<sub>7</sub>

P52

OM/SM SEP

ET = 400K

147:10

SPLASHDOWN

148:00

| MISSION   | EDITION | DATE              | TIME             | DAY/REV | PAGE |
|-----------|---------|-------------------|------------------|---------|------|
| AS503/103 | FINAL   | November 22, 1968 | 144:00 to 146:50 | 6       | 5-7  |

MSC FORM 1186 (SEP 67)

FLIGHT PLANNING BRANCH

